

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:28:41

Source: data/json/noise_experiments/v2/noise8i_nocoplayer_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_nocoplayer/noise8i_nocoplayer_memprimary_game_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_nocoplayer
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
dyadic_pairing_schedule: {
  "num_turns": 10,
  "role_targets": {
    "Agent_1": 5,
    "Agent_2": 5,
    "Agent_3": 5,
    "Agent_4": 5,
    "Agent_5": 5,
    "Agent_6": 5,
    "Agent_7": 5,
    "Agent_8": 5
  },
  "strategy": "balanced_random_roles_v1"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths,

follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

=====

ROUND 1

=====

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Sender, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_6=Sender, Agent_1=Receiver
dyad_2: Agent_3=Sender, Agent_2=Receiver
dyad_3: Agent_7=Sender, Agent_5=Receiver
dyad_4: Agent_4=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_6 sent \$3; received \$9; Agent_1 returned \$3.5; payoffs Agent_6=\$5.5, Agent_1=\$5.5;
noise/visible: sent visible=\$2.71, received visible=\$8.13, returned visible=\$2.95
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_3 sent \$4; received \$12; Agent_2 returned \$6; payoffs Agent_3=\$7, Agent_2=\$6;
noise/visible: sent visible=\$3.82, received visible=\$11.46, returned visible=\$6.95
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_3: Agent_7 sent \$3; received \$9; Agent_5 returned \$6; payoffs Agent_7=\$8, Agent_5=\$3;
noise/visible: sent visible=\$3.66, received visible=\$10.98, returned visible=\$5.68
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_4 sent \$4; received \$12; Agent_8 returned \$5; payoffs Agent_4=\$6, Agent_8=\$7;
noise/visible: sent visible=\$3.45, received visible=\$10.35, returned visible=\$5.46
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 12.0, "min": 0, "raw": 5.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 5.5,
"Agent_2": 6.0,
"Agent_3": 7.0,
"Agent_4": 6.0,
"Agent_5": 3.0,
"Agent_6": 5.5,
"Agent_7": 8.0,
"Agent_8": 7.0

}

Visible balances after round: {

"Agent_1": 4.6299999999999999,
"Agent_2": 5.4599999999999999,
"Agent_3": 7.95,
"Agent_4": 6.46,
"Agent_5": 4.98,
"Agent_6": 4.95,
"Agent_7": 7.68,
"Agent_8": 5.3500000000000001

}

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-08-12T09:14:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_6",
    "Agent_1"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_1": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust-building with prudent risk management. Since this is a multi-round game, establishing cooperation early could be beneficial for future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive. I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts seen may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive. Common strategies:

- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything. Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them

...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:14:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_1"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_1": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

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- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining
- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with: $\$2.29 + \$3.50 = \$5.79$ (net gain of \$0.79)
- I would end with: $\$8.13 - \$3.50 = \$4.63$ (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

{"return": 3.5}

Reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since $\$2.71 \times 3 = \8.13)
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has: $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system: $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ($\$8.13 - \5.21)
2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
 - Sender ends with: $\$2.29 + \$3 = \$5.29$
 - I end with: $\$8.13 - \$3 = \$5.13$
 - Both profit slightly
- If I return \$4:
 - Sender ends with: $\$2.29 + \$4 = \$6.29$
 - I end with: $\$8.13 - \$4 = \$4.13$
 - Sender profits more, I profit less
- If I return \$2.92 (equal split):
 - Sender ends with: $\$2.29 + \$2.92 = \$5.21$
 - I end with: $\$8.13 - \$2.92 = \$5.21$
 - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk

Usage: {

"input_tokens": 537,
"output_tokens": 1000,
"reasoning_tokens": 0

}

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T09:15:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_2"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_2": "trustee",
      "Agent_3": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

****{"send": 4}****

Reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground-it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

Usage: {

```
"input_tokens": 500,  
"output_tokens": 922,  
"reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:15:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_2"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_3": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with: $\$1.18 + \$6.00 = \$7.18$ (better than their initial \$5)
- I end with: $\$11.46 - \$6.00 = \$5.46$

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ($3.82 \times 3 = 11.46$)
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about what would be fair:

The sender:

- Started with \$5
- Sent \$3.82
- Currently has: $\$5 - \$3.82 = \$1.18$

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with: $\$1.18 + \$5.73 = \$6.91$
- I end with: $\$11.46 - \$5.73 = \$5.73$

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with: $\$1.18 + \$6.82 = \$8.00$
- I end with: $\$11.46 - \$6.82 = \$4.64$

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with: $\$1.18 + \$7.64 = \$8.82$
- I end with: $\$11.46 - \$7.64 = \$3.82$

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64
To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

```
Usage: {  
  "input_tokens": 537,  
  "output_tokens": 927,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-08-12T09:15:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_5"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter
5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit
4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3

- Receiver gets $\$3 \times 3 = \9
- I keep $\$5 - \$3 = \$2$ initially
- If receiver returns $\$3$, I end with $\$2 + \$3 = \$5$ (same as start, receiver gets $\$6$)
- ...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T09:16:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_5"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_5": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

Reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is 3.66×3)
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has $5 - 3.66 = 1.34$ left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:
 - One approach: We could split the total gains equally
 - Total money in the system: \$5 (sender initial) + \$10.98 (what I received) = \$15.98
 - If we split equally: $15.98 / 2 = 7.99$ each
 - Sender currently has: \$1.34
 - So I would return: $7.99 - 1.34 = 6.65$
5. Another approach: Return roughly the same proportion they sent
 - They sent 73.2% of their endowment
 - 73.2% of \$10.98 = \$8.04
 - But this seems too much and would leave me with only \$2.94
6. A middle ground approach:
 - The sender showed high trust by sending 73.2%
 - I should return enough to reward that trust
 - If I return around \$5-6, that would:
 - * Give the sender a good return on their trust
 - * Still leave me with a reasonable amount
7. Let me calculate what happens with different return amounts:
 - If I return \$5: Sender ends with \$6.34 (profit of \$1.34), I end with \$5.98 (profit of \$5.98)
 - If I return \$6: Sender ends with \$7.34 (profit of \$2.34), I end with \$4.98 (profit of \$4.98)

98)
- If I return \$6.65: Sender ends with \$7.99 (profit of \$2.99), I end with \$4.33 (profit of \$4.33)

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 1097,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_4

Timestamp: 2026-08-12T09:16:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_8"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

Key considerations:

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send **\$4** to signal strong willingness to cooperate while maintaining a small hedge. If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. **Risk**: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. However, my behavior here might set a precedent for future rounds when I'm a receiver.
4. **Multi-round consideration**: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.
5. **Communication noise**: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 1133,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T09:16:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with: \$5 - \$3.45 + \$5.00 = \$6.55
- I end with: \$10.35 - \$5.00 = \$5.35
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

Reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets \$5 - \$3.45 + \$0 = \$1.55, I get \$10.35

- If I return half (\$5.175): Sender gets $\$5 - \$3.45 + \$5.175 = \6.725 , I get $\$10.35 - \$5.175 = \$5.175$
- If I return \$6.90 (2x what they sent): Sender gets $\$5 - \$3.45 + \$6.90 = \8.45 , I get $\$10.35 - \$6.90 = \$3.45$

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has $\$5 - \$3.45 = \$1.55$ remaining, I have $\$10.35$ total = $\$11.90$ total
4. Equal split would be $\$5.95$ each

To give sender $\$5.95$ total: $\$5.95 = \$5 - \$3.45 + \text{return}$, so $\text{return} = \$5.95 - \$1.55 = \$4.40$

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return

...

```
Usage: {
  "input_tokens": 537,
  "output_tokens": 807,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Receiver, Agent_8=Sender

Pairings:

```
dyad_1: Agent_8=Sender, Agent_6=Receiver
dyad_2: Agent_1=Sender, Agent_3=Receiver
dyad_3: Agent_2=Sender, Agent_7=Receiver
dyad_4: Agent_5=Sender, Agent_4=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_8 sent $3; received $9; Agent_6 returned $3.2; payoffs Agent_8=$5.2, Agent_6=$5.8;
noise/visible: sent visible=$2.11, received visible=$6.33, returned visible=$3.93
  Action validation: {"returned": {"amount": 3.2, "clamped": false, "max": 9.0, "min": 0, "raw": 3.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_1 sent $2.5; received $7.5; Agent_3 returned $4; payoffs Agent_1=$6.5, Agent_3=$3.5;
noise/visible: sent visible=$2.34, received visible=$7.02, returned visible=$3.01
  Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 7.5, "min": 0, "raw": 4.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_3: Agent_2 sent $4; received $12; Agent_7 returned $6.5; payoffs Agent_2=$7.5, Agent_7=$5.5;
noise/visible: sent visible=$4.04, received visible=$12.12, returned visible=$6.29
  Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_5 sent $3; received $9; Agent_4 returned $4.1; payoffs Agent_5=$6.1, Agent_4=$4.9;
noise/visible: sent visible=$2.73, received visible=$8.19, returned visible=$4.8
  Action validation: {"returned": {"amount": 4.1, "clamped": false, "max": 9.0, "min": 0, "raw": 4.1}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 12.0,
"Agent_2": 13.5,
"Agent_3": 10.5,
"Agent_4": 10.9,
"Agent_5": 9.1,
"Agent_6": 11.3,
"Agent_7": 13.5,
"Agent_8": 12.2
```

}

Visible balances after round: {

```
"Agent_1": 10.139999999999999,
"Agent_2": 12.75,
"Agent_3": 10.969999999999999,
"Agent_4": 10.55,
"Agent_5": 11.780000000000001,
"Agent_6": 8.08,
"Agent_7": 13.3,
"Agent_8": 11.280000000000001
```



```
}
```

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:17:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_6"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_6": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with: $\$5 - \$3.45 + \$5.00 = \6.55
- I end with: $\$10.35 - \$5.00 = \$5.35$
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets $\$5 - \$3.45 + \$0 = \1.55 , I get \$10.35
- If I return half (\$5.175): Sender gets $\$5 - \$3.45 + \$5.175 = \6.725 , I get $\$10.35 - \$5.175 = \$5.175$
- If I return \$6.90 (2x what they sent): Sender gets $\$5 - \$3.45 + \$6.90 = \8.45 , I get $\$10.35 - \$6.90 = \$3.45$

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has $\$5 - \$3.45 = \$1.55$ remaining, I have \$10.35 total = \$11.90 total
4. Equal split would be \$5.95 each

To give sender \$5.95 total: $\$5.95 = \$5 - \$3.45 + \text{return}$, so $\text{return} = \$5.95 - \$1.55 = \$4.40$

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return
- ...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 807,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.0. Your visible payoff was \$5.350000000000001.

Your total visible earnings across all rounds are \$5.350000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- In the previous round, my co-player showed significant trust by sending 69% of their endowment
- I reciprocated fairly by returning a substantial amount
- Now with a new co-player, I want to establish cooperation

I'll send a generous amount to signal trust and cooperation, while being slightly more conservative since this is a new pairing.

{'send': 3.00}

Reasoning:

I'm now the SENDER in Round 2. Let me think about this situation:

- In Round 1, I was the RECEIVER and someone sent me \$3.45 (69% of their endowment)
- I returned \$5.00, which was a fair reciprocation
- I kept \$5.35, and the sender ended with \$6.55

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. In the previous round, my co-player sent 69% of their endowment, showing significant trust
2. I rewarded that trust by returning about 48% of what I received
3. This is a new co-player, so I can't assume they'll behave the same way
4. However, I should probably maintain a cooperative stance to build trust in the experiment
5. There's communication noise, so amounts may differ from what was actually sent

Given that:

- The previous sender was generous (69%)
- I want to establish cooperation with this new co-player
- Being moderately generous but not overly risky seems reasonable

I'll send around 60% of my endowment, which would be \$3.00. This shows trust and cooperation while being slightly more conservative than what I received in the previous round.

Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I want to maintain a cooperative equilibrium in this multi-agent setting, I should probably match that level of trust or be similarly generous.

I'll send \$3.00 (60% of my endowment), which is a solid show of trust and cooperation.

Usage: {

"input_tokens": 843,
"output_tokens": 473,
"reasoning_tokens": 0

}

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-08-12T09:17:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_8",
"pairing": {
"agents": [

```

    "Agent_8",
    "Agent_6"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_6": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management. Since this is a multi-round game, establishing cooperation early could be beneficial for

or future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive. I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts seen may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive. Common strategies:

- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything. Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$4.95.

Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

****Fairness considerations:****

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

****Decision:****

If I return \$3.20:

- Sender ends with: $\$5 - \$2.11 + \$3.20 = \6.09 (gain of ~\$1.09)
- I keep: $\$6.33 - \$3.20 = \$3.13$ (gain of ~\$3.13)

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER:

- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.95 to me
- My payoff: $\$5 - \$3 + \$2.95 = \4.95
- The receiver kept: $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:

- My co-player (a different agent) sent me \$2.11
- I received \$6.33 (which is $\$2.11 \times 3$)
- I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33
- If I want to be reciprocal and fair, I should return enough that:
 1. The sender ends up better off than if they had sent nothing
 2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with: $\$5 - \$2.11 + \$3.165 = \6.055
- I keep: $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:

- Sender ends up with: $\$5 - \$2.11 + \$2.09 = \4.98
- I keep: $\$6.33 - \$2.09 = \$4.24$

The sender invested 42% of their endowment, so I should consider whether they break even or better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us....

```
Usage: {  
  "input_tokens": 780,  
  "output_tokens": 1431,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-08-12T09:17:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_3"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_3": "trustee"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining
- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with: $\$2.29 + \$3.50 = \$5.79$ (net gain of \$0.79)
- I would end with: $\$8.13 - \$3.50 = \$4.63$ (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

{"return": 3.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since $\$2.71 \times 3 = \8.13)
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has: $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system: $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ($\$8.13 - \5.21)
2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
 - Sender ends with: $\$2.29 + \$3 = \$5.29$
 - I end with: $\$8.13 - \$3 = \$5.13$
 - Both profit slightly

- If I return \$4:
 - Sender ends with: $\$2.29 + \$4 = \$6.29$
 - I end with: $\$8.13 - \$4 = \$4.13$
 - Sender profits more, I profit less
- If I return \$2.92 (equal split):
 - Sender ends with: $\$2.29 + \$2.92 = \$5.21$
 - I end with: $\$8.13 - \$2.92 = \$5.21$
 - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1000,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit
- This opening move signals cooperation

```
{"send": 2.5}
```

Reasoning:

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50
3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive: $\$2.50 \times 3 = \7.50
- If they return around 40-50% of what they receive (\$3-\$3.75), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

```
Usage: {
  "input_tokens": 852,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T09:17:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_3"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

****{"send": 4}****

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground-it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Round 1 context:**** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

****Current situation (Round 2):****

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

```
{'return': 4.00}
```

Reasoning:

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0
- They returned \$6.95
- My payoff: $\$5 - \$4 + \$6.95 = \7.95
- Their payoff: $\$12.0 - \$6.95 = \$5.05$

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 (2.34×3)
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have $\$2.66 + \$2.18 = \$4.84$, I'd have $\$7.02 - \$2.18 = \$4.84$
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

```
Usage: {
  "input_tokens": 783,
  "output_tokens": 979,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-08-12T09:18:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_7"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with: $\$1.18 + \$6.00 = \$7.18$ (better than their initial \$5)
- I end with: $\$11.46 - \$6.00 = \$5.46$

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ($3.82 \times 3 = 11.46$)
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about

t what would be fair:

The sender:

- Started with \$5
- Sent \$3.82
- Currently has: $\$5 - \$3.82 = \$1.18$

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with: $\$1.18 + \$5.73 = \$6.91$
- I end with: $\$11.46 - \$5.73 = \$5.73$

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with: $\$1.18 + \$6.82 = \$8.00$
- I end with: $\$11.46 - \$6.82 = \$4.64$

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with: $\$1.18 + \$7.64 = \$8.82$
- I end with: $\$11.46 - \$7.64 = \$3.82$

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64

To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 927,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER

- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

```
Usage: {
  "input_tokens": 855,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-08-12T09:18:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_7"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter

5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit
4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3
- Receiver gets $\$3 \times 3 = \9
- I keep $\$5 - \$3 = \$2$ initially
- If receiver returns \$3, I end with $\$2 + \$3 = \$5$ (same as start, receiver gets \$6)

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received \$12.120000000000001. How much do you return? (0-12.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: $\$0.96 + \$6.50 = \$7.46$
- I end with: $\$12.12 - \$6.50 = \$5.62$

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: $\$5 - \$3 + \$5.68 = \7.68
- The receiver kept $\$9 - \$5.68 = \$3.32$

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received $\$4.04 \times 3 = \12.12
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be \$7.27 - \$7.88
- That would leave them with $\$0.96 + \text{return}$, and me with $\$12.12 - \text{return}$

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with $\$0.96 + \$6 = \$6.96$, I end with $\$12.12 - \$6 = \$6.12$
- If I return \$7, they end with \$7.96, I end with \$5.12

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: $\$0.96 + \$6.50 = \$7.46$

...

```
Usage: {  
  "input_tokens": 754,  
  "output_tokens": 665,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:18:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_4"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_5": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

Message 3 role=assistant**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is $\$3.66 \times 3$)
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has $\$5 - \$3.66 = \$1.34$ left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:
 - One approach: We could split the total gains equally
 - Total money in the system: $\$5$ (sender initial) + $\$10.98$ (what I received) = $\$15.98$
 - If we split equally: $\$15.98 / 2 = \7.99 each
 - Sender currently has: $\$1.34$
 - So I would return: $\$7.99 - \$1.34 = \$6.65$
5. Another approach: Return roughly the same proportion they sent
 - They sent 73.2% of their endowment
 - 73.2% of $\$10.98 = \8.04
 - But this seems too much and would leave me with only $\$2.94$
6. A middle ground approach:
 - The sender showed high trust by sending 73.2%
 - I should return enough to reward that trust
 - If I return around \$5-6, that would:
 - * Give the sender a good return on their trust
 - * Still leave me with a reasonable amount
7. Let me calculate what happens with different return amounts:
 - If I return \$5: Sender ends with $\$6.34$ (profit of $\$1.34$), I end with $\$5.98$ (profit of $\$5.98$)
 - If I return \$6: Sender ends with $\$7.34$ (profit of $\$2.34$), I end with $\$4.98$ (profit of $\$4.98$)
 - If I return \$6.65: Sender ends with $\$7.99$ (profit of $\$2.99$), I end with $\$4.33$ (profit of $\$4.33$)

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1097,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98
- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)
2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)
2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

```
Usage: {
  "input_tokens": 806,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-08-12T09:18:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_4"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_4": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
```

```

"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

Key considerations:

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send **\$4** to signal strong willingness to cooperate while maintaining a small hedge. If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. **Risk**: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. However, my behavior here might set a precedent for future rounds when I'm a receiver.
4. **Multi-round consideration**: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.
5. **Communication noise**: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1133,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Round 2 Analysis (as RECEIVER):****

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

****Reciprocity consideration:****

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

****Decision:****

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: $\$5 - \$4 + \$5.46 = \6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 (2.73×3)
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ($\$2.73/\$5 = 54.6\%$), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game
3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

```
Usage: {
  "input_tokens": 790,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Sender, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Receiver

Pairings:

dyad_1: Agent_3=Sender, Agent_7=Receiver
 dyad_2: Agent_2=Sender, Agent_5=Receiver
 dyad_3: Agent_1=Sender, Agent_8=Receiver
 dyad_4: Agent_6=Sender, Agent_4=Receiver

Parsed game decisions:

dyad_1: Agent_3 sent \$4; received \$12; Agent_7 returned \$7; payoffs Agent_3=\$8, Agent_7=\$5;
 noise/visible: sent visible=\$4.38, received visible=\$13.14, returned visible=\$7.98
 Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_2: Agent_2 sent \$4; received \$12; Agent_5 returned \$6.5; payoffs Agent_2=\$7.5, Agent_5=\$5.5;
 noise/visible: sent visible=\$3.99, received visible=\$11.97, returned visible=\$7.32
 Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_3: Agent_1 sent \$2.5; received \$7.5; Agent_8 returned \$3.2; payoffs Agent_1=\$5.7, Agent_8=\$4.3; noise/visible: sent visible=\$2.14, received visible=\$6.42, returned visible=\$2.68
 Action validation: {"returned": {"amount": 3.2, "clamped": false, "max": 7.5, "min": 0, "raw": 3.2}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
 dyad_4: Agent_6 sent \$3; received \$9; Agent_4 returned \$5.5; payoffs Agent_6=\$7.5, Agent_4=\$3.5;
 noise/visible: sent visible=\$3.55, received visible=\$10.65, returned visible=\$4.68
 Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 17.7,
 "Agent_2": 21.0,
 "Agent_3": 18.5,
 "Agent_4": 14.4,
 "Agent_5": 14.6,
 "Agent_6": 18.8,
 "Agent_7": 18.5,
 "Agent_8": 16.5

}

Visible balances after round: {

"Agent_1": 15.319999999999999,
 "Agent_2": 21.07,
 "Agent_3": 19.95,
 "Agent_4": 15.7,
 "Agent_5": 17.25,
 "Agent_6": 14.76,
 "Agent_7": 19.44,
 "Agent_8": 14.5

}

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:19:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_7",

```

"pairing": {
  "agents": [
    "Agent_3",
    "Agent_7"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

```
**{"send": 4}**
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground—it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 1 context:**** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

****Current situation (Round 2):****

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

{'return': 4.00}

reasoning:

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0
- They returned \$6.95
- My payoff: $\$5 - \$4 + \$6.95 = \7.95
- Their payoff: $\$12.0 - \$6.95 = \$5.05$

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 (2.34×3)
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have $\$2.66 + \$2.18 = \$4.84$, I'd have $\$7.02 - \$2.18 = \$4.84$
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 979,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

****Round 1:**** I sent \$4 and received a generous return (\$6.95)

****Round 2:**** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

****Current situation (Round 3):****

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2.

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

{'send': 4}

Reasoning:

Let me analyze the situation:

****Round 1 (I was SENDER):****

- I sent \$4.0
- They received \$12.0
- They returned \$6.95
- My payoff: \$1.0 + \$6.95 = \$7.95

****Round 2 (I was RECEIVER):****

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff: \$5 - \$2.34 + \$4.0 = \$6.66

****Round 3 (I am SENDER):****

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation
- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

```
Usage: {  
  "input_tokens": 1097,  
  "output_tokens": 541,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-08-12T09:19:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_7"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter
5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit

4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3
- Receiver gets $\$3 \times 3 = \9
- I keep $\$5 - \$3 = \$2$ initially
- If receiver returns \$3, I end with $\$2 + \$3 = \$5$ (same as start, receiver gets \$6)

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received $\$12.120000000000001$. How much do you return? (0-12.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: $\$0.96 + \$6.50 = \$7.46$
- I end with: $\$12.12 - \$6.50 = \$5.62$

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: $\$5 - \$3 + \$5.68 = \7.68
- The receiver kept $\$9 - \$5.68 = \$3.32$

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received $\$4.04 \times 3 = \12.12
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be \$7.27 - \$7.88
- That would leave them with \$0.96 + return, and me with \$12.12 - return

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with \$0.96 + \$6 = \$6.96, I end with \$12.12 - \$6 = \$6.12
- If I return \$7, they end with \$7.96, I end with \$5.12

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: \$0.96 + \$6.50 = \$7.46

...

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible payoff was \$5.620000000000001.

Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with: \$0.62 + \$7.00 = \$7.62
- I end with: \$13.14 - \$7.00 = \$6.14

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

Reasoning:

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.

I received \$13.14

They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:

- They end with: \$0.62 + \$7.00 = \$7.62
- I end with: \$13.14 - \$7.00 = \$6.14

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:

1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:

- Their total: $\$0.62 + \$7.00 = \$7.62$ (profit of \$2.62)
- My total: $\$13.14 - \$7.00 = \$6.14$ (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:

- Their total: $\$0.62 + \$7.50 = \$8.12$ (profit of \$3.12)
- My total: $\$13.14 - \$7.50 = \$5.64$ (profit of \$0.64)

I think \$7.00 is reasonable. It gives them a nice profit while I also benefit from the cooperation.

```
Usage: {  
  "input_tokens": 1061,  
  "output_tokens": 567,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-08-12T09:19:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_5"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with: $\$1.18 + \$6.00 = \$7.18$ (better than their initial \$5)
- I end with: $\$11.46 - \$6.00 = \$5.46$

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ($3.82 \times 3 = 11.46$)
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about what would be fair:

The sender:

- Started with \$5
- Sent \$3.82

- Currently has: \$5 - \$3.82 = \$1.18

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with: \$1.18 + \$5.73 = \$6.91
- I end with: \$11.46 - \$5.73 = \$5.73

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with: \$1.18 + \$6.82 = \$8.00
- I end with: \$11.46 - \$6.82 = \$4.64

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with: \$1.18 + \$7.64 = \$8.82
- I end with: \$11.46 - \$7.64 = \$3.82

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64

To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 927,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER
- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

usage:

```
{
  "input_tokens": 855,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings

3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

```
Usage: {  
  "input_tokens": 1089,  
  "output_tokens": 383,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_2

Timestamp: 2026-08-12T09:19:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_5"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is $\$3.66 \times 3$)
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has $\$5 - \$3.66 = \$1.34$ left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:
 - One approach: We could split the total gains equally
 - Total money in the system: $\$5$ (sender initial) + $\$10.98$ (what I received) = $\$15.98$
 - If we split equally: $\$15.98 / 2 = \7.99 each
 - Sender currently has: $\$1.34$
 - So I would return: $\$7.99 - \$1.34 = \$6.65$
5. Another approach: Return roughly the same proportion they sent
 - They sent 73.2% of their endowment
 - 73.2% of $\$10.98 = \8.04
 - But this seems too much and would leave me with only $\$2.94$
6. A middle ground approach:
 - The sender showed high trust by sending 73.2%
 - I should return enough to reward that trust
 - If I return around \$5-6, that would:
 - * Give the sender a good return on their trust
 - * Still leave me with a reasonable amount

7. Let me calculate what happens with different return amounts:
- If I return \$5: Sender ends with \$6.34 (profit of \$1.34), I end with \$5.98 (profit of \$5.98)
 - If I return \$6: Sender ends with \$7.34 (profit of \$2.34), I end with \$4.98 (profit of \$4.98)
 - If I return \$6.65: Sender ends with \$7.99 (profit of \$2.99), I end with \$4.33 (profit of \$4.33)

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1097,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98
- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)
2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)
2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)
- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)
- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)

- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

```
Usage: {
  "input_tokens": 1092,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-08-12T09:19:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_8"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining
- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with: $\$2.29 + \$3.50 = \$5.79$ (net gain of \$0.79)
- I would end with: $\$8.13 - \$3.50 = \$4.63$ (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

{"return": 3.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since $\$2.71 \times 3 = \8.13)
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has: $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system: $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ($\$8.13 - \5.21)

2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
 - Sender ends with: $\$2.29 + \$3 = \$5.29$
 - I end with: $\$8.13 - \$3 = \$5.13$
 - Both profit slightly
- If I return \$4:
 - Sender ends with: $\$2.29 + \$4 = \$6.29$
 - I end with: $\$8.13 - \$4 = \$4.13$
 - Sender profits more, I profit less
- If I return \$2.92 (equal split):
 - Sender ends with: $\$2.29 + \$2.92 = \$5.21$
 - I end with: $\$8.13 - \$2.92 = \$5.21$
 - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1000,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit
- This opening move signals cooperation

```
{"send": 2.5}
```

reasoning:

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50

3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive: $\$2.50 \times 3 = \7.50
- If they return around 40-50% of what they receive (\$3-\$3.75), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50% has proven effective and signals trust.

```
{"send": 2.5}
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50
- They received \$7.50
- They returned \$3.01
- My payoff: $\$2.50 + \$3.01 = \$5.51$

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation.

```
Usage: {
  "input_tokens": 1113,
  "output_tokens": 353,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:19:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_8"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with: $\$5 - \$3.45 + \$5.00 = \6.55
- I end with: $\$10.35 - \$5.00 = \$5.35$
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets $\$5 - \$3.45 + \$0 = \1.55 , I get \$10.35
- If I return half (\$5.175): Sender gets $\$5 - \$3.45 + \$5.175 = \6.725 , I get $\$10.35 - \$5.175 = \$5.175$
- If I return \$6.90 (2x what they sent): Sender gets $\$5 - \$3.45 + \$6.90 = \8.45 , I get $\$10.35 - \$6.90 = \$3.45$

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has $\$5 - \$3.45 = \$1.55$ remaining, I have \$10.35 total = \$11.90 total
4. Equal split would be \$5.95 each

To give sender \$5.95 total: $\$5.95 = \$5 - \$3.45 + \text{return}$, so $\text{return} = \$5.95 - \$1.55 = \$4.40$

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 807,
```



```

    "reasoning_tokens": 0
  }
Message 4 role=user
content:
  Round 2

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $3.45, it became $10.350000000000001, and you returned $5.0. Your visible
  payoff was $5.350000000000001.
  Your total visible earnings across all rounds are $5.350000000000001.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
  Looking at this situation:

  - In the previous round, my co-player showed significant trust by sending 69% of their end
  owment
  - I reciprocated fairly by returning a substantial amount
  - Now with a new co-player, I want to establish cooperation

  I'll send a generous amount to signal trust and cooperation, while being slightly more con
  servative since this is a new pairing.

  {'send': 3.00}
reasoning:
  I'm now the SENDER in Round 2. Let me think about this situation:

  - In Round 1, I was the RECEIVER and someone sent me $3.45 (69% of their endowment)
  - I returned $5.00, which was a fair reciprocation
  - I kept $5.35, and the sender ended with $6.55

  Now I'm the SENDER with a new co-player. I need to decide how much to send.

  Key considerations:
  1. In the previous round, my co-player sent 69% of their endowment, showing significant tr
  ust
  2. I rewarded that trust by returning about 48% of what I received
  3. This is a new co-player, so I can't assume they'll behave the same way
  4. However, I should probably maintain a cooperative stance to build trust in the experime
  nt
  5. There's communication noise, so amounts may differ from what was actually sent

  Given that:
  - The previous sender was generous (69%)
  - I want to establish cooperation with this new co-player
  - Being moderately generous but not overly risky seems reasonable

  I'll send around 60% of my endowment, which would be $3.00. This shows trust and cooperati
  on while being slightly more conservative than what I received in the previous round.

  Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I wan
  t to maintain a cooperative equilibrium in this multi-agent setting, I should probably m
  atch that level of trust or be similarly generous.

  I'll send $3.00 (60% of my endowment), which is a solid show of trust and cooperation.
usage:
  {
    "input_tokens": 843,
    "output_tokens": 473,
    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 3

  In your most recent previous game round, you were the SENDER against another agent. You se
  nt $3.0, it became $9.0, and you saw them return $3.93 to you. Your visible payoff was $

```

5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:

If I return \$3.20:

- Sender ends with: $\$5 - \$2.14 + \$3.20 = \6.06
- I end with: $\$6.42 - \$3.20 = \$3.22$
- This creates relatively balanced outcomes and rewards their trust

{'return': 3.20}

Reasoning:

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust
2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:

- Sender ends with: $\$5 - \$2.14 + \$3.00 = \5.86
- I end with: $\$6.42 - \$3.00 = \$3.42$
- This gives them a decent return on their trust

If I return \$3.50:

- Sender ends with: $\$5 - \$2.14 + \$3.50 = \6.36
- I end with: $\$6.42 - \$3.50 = \$2.92$
- This is more generous but leaves me with less than them

If I return \$3.20:

- Sender ends with: $\$5 - \$2.14 + \$3.20 = \6.06
- I end with: $\$6.42 - \$3.20 = \$3.22$
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

Usage: {

"input_tokens": 1075,
"output_tokens": 588,
"reasoning_tokens": 0

}

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-08-12T09:20:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_4",

```

"pairing": {
  "agents": [
    "Agent_6",
    "Agent_4"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_4": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management . Since this is a multi-round game, establishing cooperation early could be beneficial for future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive . I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts sent may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive . Common strategies:

- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything . Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them
- ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$

4.95.

Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

****Fairness considerations:****

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

****Decision:****

If I return \$3.20:

- Sender ends with: $\$5 - \$2.11 + \$3.20 = \6.09 (gain of ~\$1.09)
- I keep: $\$6.33 - \$3.20 = \$3.13$ (gain of ~\$3.13)

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER:

- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.95 to me
- My payoff: $\$5 - \$3 + \$2.95 = \4.95
- The receiver kept: $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:

- My co-player (a different agent) sent me \$2.11
- I received \$6.33 (which is $\$2.11 \times 3$)
- I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33
- If I want to be reciprocal and fair, I should return enough that:
 1. The sender ends up better off than if they had sent nothing
 2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with: $\$5 - \$2.11 + \$3.165 = \6.055
- I keep: $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:

- Sender ends up with: $\$5 - \$2.11 + \$2.09 = \4.98
- I keep: $\$6.33 - \$2.09 = \$4.24$

The sender invested 42% of their endowment, so I should consider whether they break even or

r better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here.

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 1431,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

****Considerations:****

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

****This round:****

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

****Considerations:****

1. This is a different co-player than in rounds 1 and 2
2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

****Strategy:****

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

```
Usage: {
  "input_tokens": 1133,
  "output_tokens": 554,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-08-12T09:20:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_4"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_4": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

Key considerations:

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send **\$4** to signal strong willingness to cooperate while maintaining a small hedge. If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. **Risk**: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decision on.

ons on. However, my behavior here might set a precedent for future rounds when I'm a receiver.

4. ****Multi-round consideration****: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.

5. ****Communication noise****: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1133,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis (as RECEIVER):****

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

****Reciprocity consideration:****

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

****Decision:****

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: $\$5 - \$4 + \$5.46 = \6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 (2.73×3)
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ($\$2.73/\$5 = 54.6\%$), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game
3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

usage:

```
{
  "input_tokens": 790,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier

- This is more trust than the previous co-player showed in Round 2

****Reciprocity:****

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion th
at rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

```
'''json
{"return": 5.50}
'''
```

Reasoning:

Let me analyze this situation:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

****Past behavior:****

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

****Reciprocity consideration:****

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). Th
is shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high t
rust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \5.54

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50
- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

```
Usage: {
  "input_tokens": 1157,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

=====

ROUND 4

=====

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Sender, Agent_5=Sender,
Agent_6=Receiver, Agent_7=Sender, Agent_8=Sender

Pairings:

- dyad_1: Agent_4=Sender, Agent_3=Receiver
- dyad_2: Agent_8=Sender, Agent_2=Receiver
- dyad_3: Agent_7=Sender, Agent_1=Receiver
- dyad_4: Agent_5=Sender, Agent_6=Receiver

Parsed game decisions:

- dyad_1: Agent_4 sent \$4; received \$12; Agent_3 returned \$8; payoffs Agent_4=\$9, Agent_3=\$4;
- noise/visible: sent visible=\$4.21, received visible=\$12.63, returned visible=\$8.39

```

    Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw":
    8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_8 sent $3; received $9; Agent_2 returned $6; payoffs Agent_8=$8, Agent_2=$3;
noise/visible: sent visible=$3.35, received visible=$10.05, returned visible=$6.24
    Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw":
    6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_7 sent $4; received $12; Agent_1 returned $6.5; payoffs Agent_7=$7.5, Agent_1=$5.5;
noise/visible: sent visible=$4.71, received visible=$14.13, returned visible=$7.06
    Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw":
    6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_5 sent $3.5; received $10.5; Agent_6 returned $6.8; payoffs Agent_5=$8.3,
Agent_6=$3.7; noise/visible: sent visible=$4.14, received visible=$12.42, returned visible=$6.99
    Action validation: {"returned": {"amount": 6.8, "clamped": false, "max": 10.5, "min": 0, "raw":
    6.8}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
Balances after round: {
  "Agent_1": 23.2,
  "Agent_2": 24.0,
  "Agent_3": 22.5,
  "Agent_4": 23.4,
  "Agent_5": 22.9,
  "Agent_6": 22.5,
  "Agent_7": 26.0,
  "Agent_8": 24.5
}
Visible balances after round: {
  "Agent_1": 22.949999999999996,
  "Agent_2": 25.12,
  "Agent_3": 24.58,
  "Agent_4": 25.09,
  "Agent_5": 25.740000000000002,
  "Agent_6": 20.38,
  "Agent_7": 27.5,
  "Agent_8": 22.740000000000002
}

```

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-08-12T09:20:38
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

Key considerations:

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send **\$4** to signal strong willingness to cooperate while maintaining a small hedge. If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5

- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

Strategic considerations:

1. ****Trust and cooperation****: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. ****Risk****: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. ****First round dynamics****: This is the first round, so there's no history to base decisions on. However, my behavior here might set a precedent for future rounds when I'm a receiver.
4. ****Multi-round consideration****: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.
5. ****Communication noise****: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1133,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis (as RECEIVER):****

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

****Reciprocity consideration:****

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

****Decision:****

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: $\$5 - \$4 + \$5.46 = \6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 (2.73×3)
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ($\$2.73/\$5 = 54.6\%$), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game
3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

usage:

```
{
  "input_tokens": 790,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

}

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier
- This is more trust than the previous co-player showed in Round 2

****Reciprocity:****

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion that rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

****Past behavior:****

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

****Reciprocity consideration:****

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). This shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high trust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \5.54

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50
- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

usage:


```
{
  "input_tokens": 1157,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.55, it became \$10.649999999999999, and you returned \$5.5. Your visible payoff was \$5.149999999999999.

Your total visible earnings across all rounds are \$15.7.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 4 Analysis (as SENDER):****

Looking at my experience so far:

- Round 1: Sent \$4, got good reciprocation (\$5.46 back)
- Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
- Total earnings: \$15.70 (averaging \$5.23/round)

****New co-player consideration:****

This is a different co-player than previous rounds, so no direct history. However, the cooperative approach has been working well overall.

****Decision:****

I'll send ****\$4**** again because:

- The 3x multiplier creates strong mutual benefit
- It signals trust and willingness to cooperate
- It keeps \$1 as insurance against complete defection
- This amount worked well in Round 1

```
'''json
{"send": 4}
'''
```

Reasoning:

****Round 4 Analysis (as SENDER):****

Let me review the pattern so far:

- Round 1: I sent \$4, received back \$5.46 (good return, ~45.5% of \$12)
- Round 2: As receiver, got \$2.73 (sent), I returned \$4.10 (~50% of \$8.19)
- Round 3: As receiver, got \$3.55 (sent), I returned \$5.50 (~52% of \$10.65)

Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized each round).

Key considerations:

1. I have no history with this specific co-player
2. My previous experiences have been fairly positive:
 - Round 1: Good reciprocation when I sent high
 - Round 2 & 3: I reciprocated fairly as receiver
3. The game mechanics reward cooperation through the 3x multiplier
4. There's communication noise, so exact amounts may vary

Strategy:

- Since cooperation has worked well so far, I should lean toward sending a substantial amount
- However, this is a new co-player with no history
- I've been playing fairly cooperatively, which might be building reputation (though names are hidden, the pattern of fair play might matter in multi-agent settings)

I think sending \$3.50-\$4.50 would be reasonable. Let me go with \$4 again - it's worked before, shows strong trust without being all-in, and leaves some hedge.

Actually, looking at my total earnings (\$15.70 over 3 rounds), I'm doing reasonably well averaging about \$5.23 per round. The cooperative strategy is working.

```
I'll send $4 again.  
Usage: {  
  "input_tokens": 1463,  
  "output_tokens": 607,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-08-12T09:20:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

****{"send": 4}****

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground—it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 1 context:**** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

****Current situation (Round 2):****

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

{ 'return': 4.00 }

reasoning:

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0
- They returned \$6.95
- My payoff: $\$5 - \$4 + \$6.95 = \7.95
- Their payoff: $\$12.0 - \$6.95 = \$5.05$

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 (2.34×3)
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have $\$2.66 + \$2.18 = \$4.84$, I'd have $\$7.02 - \$2.18 = \$4.84$
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 979,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

****Round 1:**** I sent \$4 and received a generous return (\$6.95)

****Round 2:**** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

****Current situation (Round 3):****

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2.

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

```
{'send': 4}
```

reasoning:

Let me analyze the situation:

****Round 1 (I was SENDER):****

- I sent \$4.0
- They received \$12.0

- They returned \$6.95
- My payoff: $\$1.0 + \$6.95 = \$7.95$

****Round 2 (I was RECEIVER):****

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff: $\$5 - \$2.34 + \$4.0 = \6.66

****Round 3 (I am SENDER):****

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation
- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Round 3 context:**** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with $\$0.79 + \$8.00 = \$8.79$ total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

```
{'return': 8.00}
```

Reasoning:

Let me analyze this situation:

****Round 3:**** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - very high trust!
- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21
- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with \$0.79 + \$8.00 = \$8.79
- I keep \$12.63 - \$8.00 = \$4.63
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left. If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total.
They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received

.
...

```
Usage: {  
  "input_tokens": 1419,  
  "output_tokens": 685,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-08-12T09:20:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_2"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with: \$5 - \$3.45 + \$5.00 = \$6.55
- I end with: \$10.35 - \$5.00 = \$5.35
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets \$5 - \$3.45 + \$0 = \$1.55, I get \$10.35
- If I return half (\$5.175): Sender gets \$5 - \$3.45 + \$5.175 = \$6.725, I get \$10.35 - \$5.175 = \$5.175
- If I return \$6.90 (2x what they sent): Sender gets \$5 - \$3.45 + \$6.90 = \$8.45, I get \$10.35 - \$6.90 = \$3.45

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has \$5 - \$3.45 = \$1.55 remaining, I have \$10.35 total = \$11.90 total
4. Equal split would be \$5.95 each

To give sender \$5.95 total: \$5.95 = \$5 - \$3.45 + return, so return = \$5.95 - \$1.55 = \$4.40

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 807,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.0. Your visible payoff was \$5.350000000000001.

Your total visible earnings across all rounds are \$5.350000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In the previous round, my co-player showed significant trust by sending 69% of their endowment
- I reciprocated fairly by returning a substantial amount
- Now with a new co-player, I want to establish cooperation

I'll send a generous amount to signal trust and cooperation, while being slightly more conservative since this is a new pairing.

```
{'send': 3.00}
```

reasoning:

I'm now the SENDER in Round 2. Let me think about this situation:

- In Round 1, I was the RECEIVER and someone sent me \$3.45 (69% of their endowment)
- I returned \$5.00, which was a fair reciprocation
- I kept \$5.35, and the sender ended with \$6.55

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. In the previous round, my co-player sent 69% of their endowment, showing significant trust
2. I rewarded that trust by returning about 48% of what I received
3. This is a new co-player, so I can't assume they'll behave the same way
4. However, I should probably maintain a cooperative stance to build trust in the experiment
5. There's communication noise, so amounts may differ from what was actually sent

Given that:

- The previous sender was generous (69%)
- I want to establish cooperation with this new co-player
- Being moderately generous but not overly risky seems reasonable

I'll send around 60% of my endowment, which would be \$3.00. This shows trust and cooperation while being slightly more conservative than what I received in the previous round.

Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I want to maintain a cooperative equilibrium in this multi-agent setting, I should probably match that level of trust or be similarly generous.

I'll send \$3.00 (60% of my endowment), which is a solid show of trust and cooperation.

usage:

```
{
  "input_tokens": 843,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.93 to you. Your visible payoff was \$5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:

If I return \$3.20:

- Sender ends with: \$5 - \$2.14 + \$3.20 = \$6.06
- I end with: \$6.42 - \$3.20 = \$3.22
- This creates relatively balanced outcomes and rewards their trust

{'return': 3.20}

reasoning:

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust
2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:

- Sender ends with: $\$5 - \$2.14 + \$3.00 = \5.86
- I end with: $\$6.42 - \$3.00 = \$3.42$
- This gives them a decent return on their trust

If I return \$3.50:

- Sender ends with: $\$5 - \$2.14 + \$3.50 = \6.36
- I end with: $\$6.42 - \$3.50 = \$2.92$
- This is more generous but leaves me with less than them

If I return \$3.20:

- Sender ends with: $\$5 - \$2.14 + \$3.20 = \6.06
- I end with: $\$6.42 - \$3.20 = \$3.22$
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.14, it became \$6.42, and you returned \$3.2. Your visible payoff was \$3.2199999999999998.

Your total visible earnings across all rounds are \$14.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- When I sent \$3.00 in Round 2, I received a fair return of \$3.93
- I've been playing cooperatively and seeing reasonable reciprocation

I'll continue with a cooperative strategy and send a generous amount to this new co-player.

```
{'send': 3.00}
```

Reasoning:

Looking at my history:

- Round 1: As receiver, got \$10.35, returned \$5.00, earned \$5.35
- Round 2: As sender, sent \$3.00, got back \$3.93, earned \$5.93
- Round 3: As receiver, got \$6.42, returned \$3.20, earned \$3.22

In Round 2, I sent \$3.00 (60%) and my co-player returned \$3.93 out of \$9.00 (43.7%), which was decent reciprocation.

Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reasonable reciprocation. I should continue with a cooperative strategy.

I'll send around \$3.00 again to maintain consistency and signal trust.

Usage: {

```
"input_tokens": 1348,
"output_tokens": 269,
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:21:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_2"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_2": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with: $\$1.18 + \$6.00 = \$7.18$ (better than their initial \$5)
- I end with: $\$11.46 - \$6.00 = \$5.46$

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ($3.82 \times 3 = 11.46$)
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about what would be fair:

The sender:

- Started with \$5
- Sent \$3.82
- Currently has: $\$5 - \$3.82 = \$1.18$

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with: $\$1.18 + \$5.73 = \$6.91$
- I end with: $\$11.46 - \$5.73 = \$5.73$

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with: $\$1.18 + \$6.82 = \$8.00$
- I end with: $\$11.46 - \$6.82 = \$4.64$

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with: $\$1.18 + \$7.64 = \$8.82$
- I end with: $\$11.46 - \$7.64 = \$3.82$

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64

To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 927,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER
- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

usage:

```
{
  "input_tokens": 855,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings
3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

usage:

```
{
  "input_tokens": 1089,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.

Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This rewards their trust while both of us benefit from the exchange.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has $\$5 - \$3.35 = \$1.65$ left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust (67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with: $\$1.65 + \$5.00 = \$6.65$ (profit of \$1.65)
- I end with: $\$10.05 - \$5.00 = \$5.05$

Or if I return \$6.50:

- Sender ends with: $\$1.65 + \$6.50 = \$8.15$ (profit of \$3.15)
- I end with: $\$10.05 - \$6.50 = \$3.55$

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

```
Usage: {
  "input_tokens": 1315,
  "output_tokens": 602,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-08-12T09:21:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_1"
    ],
    "investor": "Agent_7",
    "roles": {
```



```

    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter
5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit
4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3
- Receiver gets $\$3 \times 3 = \9
- I keep $\$5 - \$3 = \$2$ initially
- If receiver returns \$3, I end with $\$2 + \$3 = \$5$ (same as start, receiver gets \$6)
- ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received \$12.120000000000001. How much do you return? (0-12.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: $\$0.96 + \$6.50 = \$7.46$
- I end with: $\$12.12 - \$6.50 = \$5.62$

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: $\$5 - \$3 + \$5.68 = \7.68
- The receiver kept $\$9 - \$5.68 = \$3.32$

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received $\$4.04 \times 3 = \12.12
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be \$7.27 - \$7.88
- That would leave them with $\$0.96 + \text{return}$, and me with $\$12.12 - \text{return}$

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with $\$0.96 + \$6 = \$6.96$, I end with $\$12.12 - \$6 = \$6.12$
- If I return \$7, they end with \$7.96, I end with \$5.12

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: $\$0.96 + \$6.50 = \$7.46$

...

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible payoff was \$5.620000000000001.

Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with: $\$0.62 + \$7.00 = \$7.62$
- I end with: $\$13.14 - \$7.00 = \$6.14$

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.
I received \$13.14
They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:

- They end with: $\$0.62 + \$7.00 = \$7.62$
- I end with: $\$13.14 - \$7.00 = \$6.14$

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:

1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:

- Their total: $\$0.62 + \$7.00 = \$7.62$ (profit of \$2.62)
- My total: $\$13.14 - \$7.00 = \$6.14$ (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:

- Their total: $\$0.62 + \$7.50 = \$8.12$ (profit of \$3.12)
- My total: $\$13.14 - \$7.50 = \$5.64$ (profit of \$0.64)

I think \$7.00 is reasonable. It gives them a nice profit while I also benefit from the cooperation.

usage:

```
{
  "input_tokens": 1061,
  "output_tokens": 567,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.38, it became \$13.14, and you returned \$7.0. Your visible payoff was \$6

.140000000000001.

Your total visible earnings across all rounds are \$19.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- In Round 1, I sent \$3 and received good reciprocation (\$5.68 back)
- In Rounds 2 and 3, co-players showed high trust by sending \$4.04 and \$4.38
- I've been reciprocating fairly, returning about half of what I received

This is a new co-player, but the general pattern suggests cooperation pays off. I'll send a high amount to match the trust I've seen from others.

```
'''json
{"send": 4}
'''
```

Reasoning:

I'm now the SENDER in Round 4. Let me think about what I've learned:

Round 1: I sent \$3, received \$5.68 back, ended with \$7.68

Round 2: I received \$12.12 (they sent \$4.04), returned \$6.50, ended with \$5.62

Round 3: I received \$13.14 (they sent \$4.38), returned \$7.00, ended with \$6.14

My recent experiences as a receiver show co-players sending high amounts (\$4.04 and \$4.38). When I was a sender in Round 1, I sent \$3 and got \$5.68 back, which was good reciprocation.

Given that I've seen high trust from others and I've been reciprocating fairly, I should continue to show trust. This is a new co-player this round (since pairings are randomized), so I don't have specific history with them, but the general pattern suggests cooperation is beneficial.

I could send:

- \$3: Moderate trust, similar to Round 1
- \$4: High trust, similar to what I've received from others
- \$5: Maximum trust

Given that I've been receiving high amounts from others (around \$4+), and I've been reciprocating fairly well, I should send a similarly high amount to maintain the cooperative dynamic. Sending \$4 shows significant trust while not being overly risky.

If I send \$4:

- They receive \$12
- If they return around 50-60%, I'd get back \$6-7.20
- I'd end with roughly \$7-8.20

This seems reasonable and aligns with the cooperative pattern I've been experiencing.

```
Usage: {
  "input_tokens": 1349,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-08-12T09:21:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_1"
    ],
    "investor": "Agent_7",
  }
}
```

```

"roles": {
  "Agent_1": "trustee",
  "Agent_7": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining

- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with: $\$2.29 + \$3.50 = \$5.79$ (net gain of \$0.79)
- I would end with: $\$8.13 - \$3.50 = \$4.63$ (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

```
{"return": 3.5}
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since $\$2.71 \times 3 = \8.13)
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has: $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system: $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ($\$8.13 - \5.21)
2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
 - Sender ends with: $\$2.29 + \$3 = \$5.29$
 - I end with: $\$8.13 - \$3 = \$5.13$
 - Both profit slightly
- If I return \$4:
 - Sender ends with: $\$2.29 + \$4 = \$6.29$
 - I end with: $\$8.13 - \$4 = \$4.13$
 - Sender profits more, I profit less
- If I return \$2.92 (equal split):
 - Sender ends with: $\$2.29 + \$2.92 = \$5.21$
 - I end with: $\$8.13 - \$2.92 = \$5.21$
 - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1000,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit
- This opening move signals cooperation

{"send": 2.5}

reasoning:

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50
3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive: $\$2.50 \times 3 = \7.50
- If they return around 40-50% of what they receive ($\$3-\3.75), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50%

has proven effective and signals trust.

```
{"send": 2.5}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50
- They received \$7.50
- They returned \$3.01
- My payoff: $\$2.50 + \$3.01 = \$5.51$

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation

usage:

```
{  
  "input_tokens": 1113,  
  "output_tokens": 353,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

```
{"return": 6.5}
```

Reasoning:

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has: $\$5 - \$4.71 = \$0.29$ remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with $\$0.29 + \$6.00 = \$6.29$ (net gain of \$1.29), I keep \$8.13
- If I return \$7.00: Sender ends with $\$0.29 + \$7.00 = \$7.29$ (net gain of \$2.29), I keep \$7.13
- If I return \$5.00: Sender ends with $\$0.29 + \$5.00 = \$5.29$ (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

```
Usage: {
  "input_tokens": 1345,
  "output_tokens": 549,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:21:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    }
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is 3×3.66)
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has $5 - 3.66 = 1.34$ left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:

- One approach: We could split the total gains equally
 - Total money in the system: \$5 (sender initial) + \$10.98 (what I received) = \$15.98
 - If we split equally: $\$15.98 / 2 = \7.99 each
 - Sender currently has: \$1.34
 - So I would return: $\$7.99 - \$1.34 = \$6.65$
5. Another approach: Return roughly the same proportion they sent
- They sent 73.2% of their endowment
 - 73.2% of \$10.98 = \$8.04
 - But this seems too much and would leave me with only \$2.94
6. A middle ground approach:
- The sender showed high trust by sending 73.2%
 - I should return enough to reward that trust
 - If I return around \$5-6, that would:
 - * Give the sender a good return on their trust
 - * Still leave me with a reasonable amount
7. Let me calculate what happens with different return amounts:
- If I return \$5: Sender ends with \$6.34 (profit of \$1.34), I end with \$5.98 (profit of \$5.98)
 - If I return \$6: Sender ends with \$7.34 (profit of \$2.34), I end with \$4.98 (profit of \$4.98)
 - If I return \$6.65: Sender ends with \$7.99 (profit of \$2.99), I end with \$4.33 (profit of \$4.33)

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1097,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98

- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)
2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)
2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)
- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)
- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
```

'''

reasoning:

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)
- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

usage:

```
{
  "input_tokens": 1092,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.4700000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern. I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

Reasoning:

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving. This signals strong trust while being neither too conservative nor too aggressive.

```
Usage: {  
  "input_tokens": 1363,  
  "output_tokens": 473,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:21:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant**content:**

Looking at this first round, I need to balance trust-building with prudent risk management. Since this is a multi-round game, establishing cooperation early could be beneficial for future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive. I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts seen may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive

- . Common strategies:
- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything
 . Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$4.95.

Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

****Fairness considerations:****

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

****Decision:****

If I return \$3.20:

- Sender ends with: $\$5 - \$2.11 + \$3.20 = \6.09 (gain of $\sim\$1.09$)
- I keep: $\$6.33 - \$3.20 = \$3.13$ (gain of $\sim\$3.13$)

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER:

- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.95 to me
- My payoff: $\$5 - \$3 + \$2.95 = \4.95
- The receiver kept: $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:

- My co-player (a different agent) sent me \$2.11
- I received \$6.33 (which is $\$2.11 \times 3$)
- I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33
- If I want to be reciprocal and fair, I should return enough that:
 1. The sender ends up better off than if they had sent nothing
 2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with: $\$5 - \$2.11 + \$3.165 = \6.055
- I keep: $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:

- Sender ends up with: $\$5 - \$2.11 + \$2.09 = \4.98
- I keep: $\$6.33 - \$2.09 = \$4.24$

The sender invested 42% of their endowment, so I should consider whether they break even or better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here.

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us...

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 1431,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Analysis:****

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

****Considerations:****

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

****This round:****

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

****Considerations:****

1. This is a different co-player than in rounds 1 and 2
2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

****Strategy:****

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

usage:

```
{
  "input_tokens": 1133,
```

```
"output_tokens": 554,  
"reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

****Considerations:****

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

****Calculation:****

If I return \$6.80:

- Sender ends with: $\$5 - \$4.14 + \$6.80 = \7.66 (gain of \$2.66)
- I keep: $\$12.42 - \$6.80 = \$5.62$ (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json  
{"return": 6.80}  
'''
```

Reasoning:

Let me analyze this situation:

****What happened:****

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

****Considerations:****

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

****Calculation options:****

If I return \$6.00:

- Sender ends with: $\$5 - \$4.14 + \$6.00 = \6.86 (gain of \$1.86)
- I keep: $\$12.42 - \$6.00 = \$6.42$ (gain of \$6.42)

If I return \$7.00:

- Sender ends with: $\$5 - \$4.14 + \$7.00 = \7.86 (gain of \$2.86)
- I keep: $\$12.42 - \$7.00 = \$5.42$ (gain of \$5.42)

If I return \$6.50:

- Sender ends with: $\$5 - \$4.14 + \$6.50 = \7.36 (gain of \$2.36)
- I keep: $\$12.42 - \$6.50 = \$5.92$ (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.

Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd gain \$3.36 while...

```
Usage: {
  "input_tokens": 1425,
  "output_tokens": 847,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Sender, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_3=Sender, Agent_8=Receiver
dyad_2: Agent_7=Sender, Agent_4=Receiver
dyad_3: Agent_2=Sender, Agent_6=Receiver
dyad_4: Agent_1=Sender, Agent_5=Receiver

Parsed game decisions:

dyad_1: Agent_3 sent \$4; received \$12; Agent_8 returned \$6.5; payoffs Agent_3=\$7.5, Agent_8=\$5.5;
noise/visible: sent visible=\$4.44, received visible=\$13.32, returned visible=\$6.84
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_7 sent \$4; received \$12; Agent_4 returned \$5.2; payoffs Agent_7=\$6.2, Agent_4=\$6.8;
noise/visible: sent visible=\$3.45, received visible=\$10.35, returned visible=\$5.12
Action validation: {"returned": {"amount": 5.2, "clamped": false, "max": 12.0, "min": 0, "raw": 5.2}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_2 sent \$4; received \$12; Agent_6 returned \$5.8; payoffs Agent_2=\$6.8, Agent_6=\$6.2;
noise/visible: sent visible=\$3.63, received visible=\$10.89, returned visible=\$5.93
Action validation: {"returned": {"amount": 5.8, "clamped": false, "max": 12.0, "min": 0, "raw": 5.8}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_1 sent \$3; received \$9; Agent_5 returned \$4.5; payoffs Agent_1=\$6.5, Agent_5=\$4.5;
noise/visible: sent visible=\$2.79, received visible=\$8.37, returned visible=\$3.69
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 29.7,
"Agent_2": 30.8,
"Agent_3": 30.0,
"Agent_4": 30.2,
"Agent_5": 27.4,
"Agent_6": 28.7,
"Agent_7": 32.2,
"Agent_8": 30.0

}

Visible balances after round: {

"Agent_1": 28.639999999999993,
"Agent_2": 32.05,
"Agent_3": 32.42,
"Agent_4": 30.240000000000002,
"Agent_5": 29.610000000000003,
"Agent_6": 25.47,
"Agent_7": 33.62,
"Agent_8": 29.560000000000002

}

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:21:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_8",
"pairing": {
 "agents": [
 "Agent_3",

```

    "Agent_8"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

```
**{"send": 4}**
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground—it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You se

nt \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 1 context:**** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

****Current situation (Round 2):****

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

{'return': 4.00}

reasoning:

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0
- They returned \$6.95
- My payoff: $\$5 - \$4 + \$6.95 = \7.95
- Their payoff: $\$12.0 - \$6.95 = \$5.05$

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 (2.34×3)
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have $\$2.66 + \$2.18 = \$4.84$, I'd have $\$7.02 - \$2.18 = \$4.84$
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 979,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

****Round 1:**** I sent \$4 and received a generous return (\$6.95)

****Round 2:**** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

****Current situation (Round 3):****

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2.

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

{'send': 4}

reasoning:

Let me analyze the situation:

****Round 1 (I was SENDER):****

- I sent \$4.0
- They received \$12.0
- They returned \$6.95
- My payoff: \$1.0 + \$6.95 = \$7.95

****Round 2 (I was RECEIVER):****

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff: \$5 - \$2.34 + \$4.0 = \$6.66

****Round 3 (I am SENDER):****

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation
- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Round 3 context:**** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with \$0.79 + \$8.00 = \$8.79 total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

{'return': 8.00}

reasoning:

Let me analyze this situation:

****Round 3:**** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - very high trust!
- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21

- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with $\$0.79 + \$8.00 = \$8.79$
- I keep $\$12.63 - \$8.00 = \$4.63$
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left. If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total. They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received.

...

usage:

```
{
  "input_tokens": 1419,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.21, it became \$12.629999999999999, and you returned \$8.0. Your visible payoff was \$4.629999999999999. Your total visible earnings across all rounds are \$24.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

****Pattern as SENDER:**** I've sent \$4 in both previous sender rounds (R1 and R3), receiving generous returns (\$6.95 and \$7.98)

****Pattern as RECEIVER:**** I've returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 5):****

- New co-player (different from previous rounds)
- I'm the sender with \$5

My consistent strategy of sending \$4 has worked well:

- It signals strong cooperation
- Creates substantial mutual benefit potential (\$12 after 3x)
- Has been rewarded with good returns

I'll maintain this approach with my new partner.

```
{'send': 4}
```

Reasoning:

Looking at my history:

****Round 1:**** I sent \$4, received \$6.95 back ? Payoff \$7.95

****Round 2:**** Received \$7.02, returned \$4.00 ? Payoff \$3.02

****Round 3:**** I sent \$4, received \$7.98 back ? Payoff \$8.98

****Round 4:**** Received \$12.63, returned \$8.00 ? Payoff \$4.63

****Current situation (Round 5):****

- New co-player (no history with this specific agent)
- I'm the sender again

My pattern has been:

- Send \$4 consistently when I'm the sender
- Return fairly (57-63% of what I receive) when I'm the receiver
- This has worked well, with generous returns when I've been the sender

The strategy seems successful. I should continue sending \$4 to:

1. Maintain consistency
2. Signal cooperation from the start
3. Create opportunities for mutual benefit
4. The high amounts have been reciprocated well so far

I'll send \$4 again.

```
Usage: {  
  "input_tokens": 1757,  
  "output_tokens": 427,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_1

Timestamp: 2026-08-12T09:22:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_8"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with: \$5 - \$3.45 + \$5.00 = \$6.55
- I end with: \$10.35 - \$5.00 = \$5.35
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets \$5 - \$3.45 + \$0 = \$1.55, I get \$10.35
- If I return half (\$5.175): Sender gets \$5 - \$3.45 + \$5.175 = \$6.725, I get \$10.35 - \$5.175 = \$5.175
- If I return \$6.90 (2x what they sent): Sender gets \$5 - \$3.45 + \$6.90 = \$8.45, I get \$10.35 - \$6.90 = \$3.45

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has \$5 - \$3.45 = \$1.55 remaining, I have \$10.35 total = \$11.90 total

4. Equal split would be \$5.95 each

To give sender \$5.95 total: $\$5.95 = \$5 - \$3.45 + \text{return}$, so $\text{return} = \$5.95 - \$1.55 = \$4.40$

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 807,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.0. Your visible payoff was \$5.350000000000001.

Your total visible earnings across all rounds are \$5.350000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In the previous round, my co-player showed significant trust by sending 69% of their endowment
- I reciprocated fairly by returning a substantial amount
- Now with a new co-player, I want to establish cooperation

I'll send a generous amount to signal trust and cooperation, while being slightly more conservative since this is a new pairing.

```
{'send': 3.00}
```

reasoning:

I'm now the SENDER in Round 2. Let me think about this situation:

- In Round 1, I was the RECEIVER and someone sent me \$3.45 (69% of their endowment)
- I returned \$5.00, which was a fair reciprocation
- I kept \$5.35, and the sender ended with \$6.55

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. In the previous round, my co-player sent 69% of their endowment, showing significant trust
2. I rewarded that trust by returning about 48% of what I received
3. This is a new co-player, so I can't assume they'll behave the same way
4. However, I should probably maintain a cooperative stance to build trust in the experiment
5. There's communication noise, so amounts may differ from what was actually sent

Given that:

- The previous sender was generous (69%)
- I want to establish cooperation with this new co-player
- Being moderately generous but not overly risky seems reasonable

I'll send around 60% of my endowment, which would be \$3.00. This shows trust and cooperation while being slightly more conservative than what I received in the previous round.

Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I want to maintain a cooperative equilibrium in this multi-agent setting, I should probably match that level of trust or be similarly generous.

I'll send \$3.00 (60% of my endowment), which is a solid show of trust and cooperation.

usage:

```
{
  "input_tokens": 843,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.93 to you. Your visible payoff was \$5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:

If I return \$3.20:

- Sender ends with: $\$5 - \$2.14 + \$3.20 = \6.06
- I end with: $\$6.42 - \$3.20 = \$3.22$
- This creates relatively balanced outcomes and rewards their trust

{'return': 3.20}

reasoning:

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust
2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:

- Sender ends with: $\$5 - \$2.14 + \$3.00 = \5.86
- I end with: $\$6.42 - \$3.00 = \$3.42$
- This gives them a decent return on their trust

If I return \$3.50:

- Sender ends with: $\$5 - \$2.14 + \$3.50 = \6.36
- I end with: $\$6.42 - \$3.50 = \$2.92$
- This is more generous but leaves me with less than them

If I return \$3.20:

- Sender ends with: $\$5 - \$2.14 + \$3.20 = \6.06
- I end with: $\$6.42 - \$3.20 = \$3.22$
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

usage:

```
{
```

```
"input_tokens": 1075,
"output_tokens": 588,
"reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.14, it became \$6.42, and you returned \$3.2. Your visible payoff was \$3.2199999999999998.

Your total visible earnings across all rounds are \$14.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- When I sent \$3.00 in Round 2, I received a fair return of \$3.93
- I've been playing cooperatively and seeing reasonable reciprocation

I'll continue with a cooperative strategy and send a generous amount to this new co-player

```
{'send': 3.00}
```

reasoning:

Looking at my history:

- Round 1: As receiver, got \$10.35, returned \$5.00, earned \$5.35
- Round 2: As sender, sent \$3.00, got back \$3.93, earned \$5.93
- Round 3: As receiver, got \$6.42, returned \$3.20, earned \$3.22

In Round 2, I sent \$3.00 (60%) and my co-player returned \$3.93 out of \$9.00 (43.7%), which was decent reciprocation.

Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reasonable reciprocation. I should continue with a cooperative strategy.

I'll send around \$3.00 again to maintain consistency and signal trust.

usage:

```
{
  "input_tokens": 1348,
  "output_tokens": 269,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.24 to you. Your visible payoff was \$8.24.

Your total visible earnings across all rounds are \$22.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$4.44 (88.8% of their endowment) - this is exceptional trust!
- I received \$13.32 after the 3x multiplication
- Total resources created: \$13.88

This sender showed tremendous cooperation. I should reciprocate strongly:

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This rewards their high trust with balanced outcomes


```
{'return': 6.50}
```

Reasoning:

The sender sent \$4.44, which is 88.8% of their endowment - this is very high trust! I received \$13.32 after multiplication.

Total resources created: \$5 - \$4.44 + \$13.32 = \$13.88

The sender showed exceptional trust. I should reciprocate generously.

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This creates a fairly balanced outcome and rewards their high trust

If I return \$7.00:

- Sender ends with: \$5 - \$4.44 + \$7.00 = \$7.56
- I end with: \$13.32 - \$7.00 = \$6.32
- This is also reasonable

Let me go with \$6.50 as it provides good reciprocation while maintaining balance.

```
Usage: {  
  "input_tokens": 1564,  
  "output_tokens": 392,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:22:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_4"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter
5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit
4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3
- Receiver gets $\$3 \times 3 = \9
- I keep $\$5 - \$3 = \$2$ initially
- If receiver returns \$3, I end with $\$2 + \$3 = \$5$ (same as start, receiver gets \$6)
- ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received \$12.120000000000001. How much do you return? (0-12.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: $\$0.96 + \$6.50 = \$7.46$
- I end with: $\$12.12 - \$6.50 = \$5.62$

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: $\$5 - \$3 + \$5.68 = \7.68
- The receiver kept $\$9 - \$5.68 = \$3.32$

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received $\$4.04 \times 3 = \12.12
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be \$7.27 - \$7.88
- That would leave them with $\$0.96 + \text{return}$, and me with $\$12.12 - \text{return}$

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with $\$0.96 + \$6 = \$6.96$, I end with $\$12.12 - \$6 = \$6.12$
- If I return \$7, they end with \$7.96, I end with \$5.12

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: $\$0.96 + \$6.50 = \$7.46$

...

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible payoff was \$5.620000000000001.

Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with: $\$0.62 + \$7.00 = \$7.62$
- I end with: $\$13.14 - \$7.00 = \$6.14$

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.
I received \$13.14
They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:
- They end with: $\$0.62 + \$7.00 = \$7.62$
- I end with: $\$13.14 - \$7.00 = \$6.14$

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:
1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:
- Their total: $\$0.62 + \$7.00 = \$7.62$ (profit of \$2.62)
- My total: $\$13.14 - \$7.00 = \$6.14$ (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:
- Their total: $\$0.62 + \$7.50 = \$8.12$ (profit of \$3.12)
- My total: $\$13.14 - \$7.50 = \$5.64$ (profit of \$0.64)

I think \$7.00 is reasonable. It gives them a nice profit while I also benefit from the cooperation.

usage:

```
{
  "input_tokens": 1061,
  "output_tokens": 567,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.38, it became \$13.14, and you returned \$7.0. Your visible payoff was \$6.1400000000000001.
Your total visible earnings across all rounds are \$19.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- In Round 1, I sent \$3 and received good reciprocation (\$5.68 back)
- In Rounds 2 and 3, co-players showed high trust by sending \$4.04 and \$4.38
- I've been reciprocating fairly, returning about half of what I received

This is a new co-player, but the general pattern suggests cooperation pays off. I'll send a high amount to match the trust I've seen from others.

```
'''json
{"send": 4}
'''
```

reasoning:

I'm now the SENDER in Round 4. Let me think about what I've learned:

Round 1: I sent \$3, received \$5.68 back, ended with \$7.68
Round 2: I received \$12.12 (they sent \$4.04), returned \$6.50, ended with \$5.62
Round 3: I received \$13.14 (they sent \$4.38), returned \$7.00, ended with \$6.14

My recent experiences as a receiver show co-players sending high amounts (\$4.04 and \$4.38). When I was a sender in Round 1, I sent \$3 and got \$5.68 back, which was good reciprocation.

Given that I've seen high trust from others and I've been reciprocating fairly, I should continue to show trust. This is a new co-player this round (since pairings are randomized), so I don't have specific history with them, but the general pattern suggests cooperation is beneficial.

I could send:

- \$3: Moderate trust, similar to Round 1
- \$4: High trust, similar to what I've received from others
- \$5: Maximum trust

Given that I've been receiving high amounts from others (around \$4+), and I've been reciprocating fairly well, I should send a similarly high amount to maintain the cooperative dynamic. Sending \$4 shows significant trust while not being overly risky.

If I send \$4:

- They receive \$12
- If they return around 50-60%, I'd get back \$6-7.20
- I'd end with roughly \$7-8.20

This seems reasonable and aligns with the cooperative pattern I've been experiencing.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.06 to you. Your visible payoff was \$8.059999999999999.

Your total visible earnings across all rounds are \$27.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent rounds:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back - excellent reciprocation

The cooperation has been strong, with both co-players returning generously. This suggests high trust sends are being rewarded.

With a new co-player this round, I'll continue the cooperative strategy that's been working well.

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at my history:

- Round 1: Sent \$3, received \$5.68 back (good return)
- Round 2: Received \$12.12, returned \$6.50
- Round 3: Received \$13.14, returned \$7.00
- Round 4: Sent \$4, received \$7.06 back (very good return - they returned about 59% of what they got)

The cooperation has been strong across all rounds. People are sending high amounts and returning generously. My earnings have been good (\$27.50 total across 4 rounds with visible payoffs).

This is Round 5 with a new co-player. Given the positive cooperation pattern, I should continue

e with a high trust strategy. Sending \$4 worked well last round.

I'll send \$4 again to show trust and likely get good reciprocation.

```
Usage: {  
  "input_tokens": 1602,  
  "output_tokens": 314,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:22:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_4"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

Key considerations:

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send **\$4** to signal strong willingness to cooperate while maintaining a small hedge. If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. **Risk**: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. However, my behavior here might set a precedent for future rounds when I'm a receiver.
4. **Multi-round consideration**: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.
5. **Communication noise**: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1133,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis (as RECEIVER):****

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

****Reciprocity consideration:****

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

****Decision:****

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: $\$5 - \$4 + \$5.46 = \6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 (2.73×3)
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ($\$2.73/\$5 = 54.6\%$), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game
3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

usage:

```
{
  "input_tokens": 790,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier
- This is more trust than the previous co-player showed in Round 2

****Reciprocity:****

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion that rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

****Past behavior:****

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

****Reciprocity consideration:****

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). This shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high trust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \5.54

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50
- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

usage:

```
{
  "input_tokens": 1157,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.55, it became \$10.649999999999999, and you returned \$5.5. Your visible payoff was \$5.149999999999999.

Your total visible earnings across all rounds are \$15.7.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Round 4 Analysis (as SENDER):****

Looking at my experience so far:

- Round 1: Sent \$4, got good reciprocation (\$5.46 back)
- Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
- Total earnings: \$15.70 (averaging \$5.23/round)

****New co-player consideration:****

This is a different co-player than previous rounds, so no direct history. However, the coo

perative approach has been working well overall.

****Decision:****

I'll send ****\$4**** again because:

- The 3x multiplier creates strong mutual benefit
- It signals trust and willingness to cooperate
- It keeps \$1 as insurance against complete defection
- This amount worked well in Round 1

'''json

{"send": 4}

'''

reasoning:

****Round 4 Analysis (as SENDER):****

Let me review the pattern so far:

- Round 1: I sent \$4, received back \$5.46 (good return, ~45.5% of \$12)
- Round 2: As receiver, got \$2.73 (sent), I returned \$4.10 (~50% of \$8.19)
- Round 3: As receiver, got \$3.55 (sent), I returned \$5.50 (~52% of \$10.65)

Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized each round).

Key considerations:

1. I have no history with this specific co-player
2. My previous experiences have been fairly positive:
 - Round 1: Good reciprocation when I sent high
 - Round 2 & 3: I reciprocated fairly as receiver
3. The game mechanics reward cooperation through the 3x multiplier
4. There's communication noise, so exact amounts may vary

Strategy:

- Since cooperation has worked well so far, I should lean toward sending a substantial amount
- However, this is a new co-player with no history
- I've been playing fairly cooperatively, which might be building reputation (though names are hidden, the pattern of fair play might matter in multi-agent settings)

I think sending \$3.50-\$4.50 would be reasonable. Let me go with \$4 again - it's worked before, shows strong trust without being all-in, and leaves some hedge.

Actually, looking at my total earnings (\$15.70 over 3 rounds), I'm doing reasonably well averaging about \$5.23 per round. The cooperative strategy is working.

I'll send \$4 again.

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.39 to you. Your visible payoff was \$9.39.

Your total visible earnings across all rounds are \$25.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their endowment) - strong trust
- I received \$10.35 after the multiplier
- Previous round showed excellent reciprocation when I was sender (\$8.39 back!)

****Reciprocity:****

My co-player demonstrated high trust by sending nearly 70%. I should maintain the fair reciprocation pattern that's been working well.

Returning approximately 50% (\$5.20) would:

- Give my co-player a total of \$6.75 (\$1.55 kept + \$5.20 returned)
- Leave me with \$5.15
- Reward their trust appropriately

```
'''json
{"return": 5.20}
'''
```

Reasoning:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their \$5 endowment) - showing high trust
- I received \$10.35 after the multiplier
- This is similar trust level to Round 3 (where co-player sent \$3.55)

****Previous round performance:****

- Round 4: I sent \$4, got back \$8.39 (70% return rate!) - excellent reciprocation
- This suggests cooperative behavior is being rewarded

****Reciprocity consideration:****

My co-player sent 69% of their endowment, showing strong trust. I should return a fair proportion.

Returning approximately 50-52% seems appropriate:

- 50% = \$5.18
- 52% = \$5.38

If I return \$5.20:

- My co-player gets total: \$1.55 (kept) + \$5.20 (returned) = \$6.75
- I keep: \$5.15
- Both benefit well

This is consistent with my previous fair reciprocation strategy and should maintain cooperation.

```
Usage: {
  "input_tokens": 1801,
  "output_tokens": 448,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_3

Timestamp: 2026-08-12T09:22:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_6"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with: \$1.18 + \$6.00 = \$7.18 (better than their initial \$5)
- I end with: \$11.46 - \$6.00 = \$5.46

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```json

```
 {"return": 6.0}
 """
```

**reasoning:**

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ( $3.82 \times 3 = 11.46$ )
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about what would be fair:

The sender:

- Started with \$5
- Sent \$3.82
- Currently has:  $\$5 - \$3.82 = \$1.18$

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with:  $\$1.18 + \$5.73 = \$6.91$
- I end with:  $\$11.46 - \$5.73 = \$5.73$

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with:  $\$1.18 + \$6.82 = \$8.00$
- I end with:  $\$11.46 - \$6.82 = \$4.64$

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with:  $\$1.18 + \$7.64 = \$8.82$
- I end with:  $\$11.46 - \$7.64 = \$3.82$

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64

To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 927,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage

age reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER
- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

**usage:**

```
{
 "input_tokens": 855,
 "output_tokens": 480,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.



```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings
3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

**usage:**

```
{
 "input_tokens": 1089,
 "output_tokens": 383,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.

Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with:  $\$1.65 + \$6.00 = \$7.65$  (profit of \$2.65)
- I end with:  $\$10.05 - \$6.00 = \$4.05$

This rewards their trust while both of us benefit from the exchange.

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has  $\$5 - \$3.35 = \$1.65$  left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust (67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with:  $\$1.65 + \$6.00 = \$7.65$  (profit of \$2.65)
- I end with:  $\$10.05 - \$6.00 = \$4.05$

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with:  $\$1.65 + \$5.00 = \$6.65$  (profit of \$1.65)
- I end with:  $\$10.05 - \$5.00 = \$5.05$

Or if I return \$6.50:

- Sender ends with:  $\$1.65 + \$6.50 = \$8.15$  (profit of \$3.15)
- I end with:  $\$10.05 - \$6.50 = \$3.55$

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

usage:

```
{
 "input_tokens": 1315,
 "output_tokens": 602,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.35, it became \$10.05, and you returned \$6.0. Your visible payoff was \$4.050000000000001.

Your total visible earnings across all rounds are \$25.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- When I sent \$4.0 in previous rounds, I received back \$6.29 and \$7.32 - both profitable outcomes
- My cooperative strategy has been working well, building trust

I'll continue with the same successful approach of sending a substantial amount to encourage reciprocity.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my track record:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.0, kept \$5.46
- Round 2 (SENDER): Sent \$4.0, received back \$6.29, kept \$7.29
- Round 3 (SENDER): Sent \$4.0, received back \$7.32, kept \$8.32
- Round 4 (RECEIVER): Received \$10.05, returned \$6.0, kept \$4.05

Total visible earnings: \$25.12

My cooperative strategy of sending \$4.0 has been working well - both times I sent \$4.0, I received good returns (\$6.29 and \$7.32), both yielding solid profits. As a receiver, I've been fair by returning \$6.0 when others showed trust.

I'm now paired with a new co-player for round 5 as sender. Given my successful track record with cooperation, I should continue this strategy.

Sending \$4.0 again makes sense - it shows significant trust (80% of endowment) and has consist

ently led to profitable outcomes for me.

```
Usage: {
 "input_tokens": 1608,
 "output_tokens": 357,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_6 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:22:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_6"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with prudent risk management . Since this is a multi-round game, establishing cooperation early could be beneficial for future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive . I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts seen may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive . Common strategies:

- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything . Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them
- ...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$4.95.

Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

**\*\*Fairness considerations:\*\***

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

**\*\*Decision:\*\***

If I return \$3.20:

- Sender ends with:  $\$5 - \$2.11 + \$3.20 = \$6.09$  (gain of ~\$1.09)
- I keep:  $\$6.33 - \$3.20 = \$3.13$  (gain of ~\$3.13)

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER:

- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.95 to me
- My payoff:  $\$5 - \$3 + \$2.95 = \$4.95$
- The receiver kept:  $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:

- My co-player (a different agent) sent me \$2.11
- I received \$6.33 (which is  $\$2.11 \times 3$ )
- I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33

- If I want to be reciprocal and fair, I should return enough that:
  1. The sender ends up better off than if they had sent nothing
  2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with:  $\$5 - \$2.11 + \$3.165 = \$6.055$
- I keep:  $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:

- Sender ends up with:  $\$5 - \$2.11 + \$2.09 = \$4.98$
- I keep:  $\$6.33 - \$2.09 = \$4.24$

The sender invested 42% of their endowment, so I should consider whether they break even or better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here.

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us....

usage:

```
{
 "input_tokens": 780,
 "output_tokens": 1431,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

**\*\*Analysis:\*\***

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

**\*\*Considerations:\*\***

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

**\*\*This round:\*\***

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

**\*\*Considerations:\*\***

1. This is a different co-player than in rounds 1 and 2
2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

**\*\*Strategy:\*\***

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

**usage:**

```
{
 "input_tokens": 1133,
 "output_tokens": 554,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

**\*\*Considerations:\*\***

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

**\*\*Calculation:\*\***

If I return \$6.80:

- Sender ends with: \$5 - \$4.14 + \$6.80 = \$7.66 (gain of \$2.66)
- I keep: \$12.42 - \$6.80 = \$5.62 (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 6.80}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

**\*\*Considerations:\*\***

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

**\*\*Calculation options:\*\***

If I return \$6.00:

- Sender ends with:  $\$5 - \$4.14 + \$6.00 = \$6.86$  (gain of \$1.86)
- I keep:  $\$12.42 - \$6.00 = \$6.42$  (gain of \$6.42)

If I return \$7.00:

- Sender ends with:  $\$5 - \$4.14 + \$7.00 = \$7.86$  (gain of \$2.86)
- I keep:  $\$12.42 - \$7.00 = \$5.42$  (gain of \$5.42)

If I return \$6.50:

- Sender ends with:  $\$5 - \$4.14 + \$6.50 = \$7.36$  (gain of \$2.36)
- I keep:  $\$12.42 - \$6.50 = \$5.92$  (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.

Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd gain \$3.36 while...

**usage:**

```
{
 "input_tokens": 1425,
 "output_tokens": 847,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$6.8. Your visible payoff was \$5.619999999999998.

Your total visible earnings across all rounds are \$20.38.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

**\*\*Analysis:\*\***

My co-player sent \$3.63 (73% of their endowment), showing substantial trust. I received \$10.89 after the 3x multiplier.

**\*\*Considerations:\*\***

- They sent a significant portion of their endowment
- Both of us should benefit from this cooperation
- I want to reward their trust appropriately

**\*\*Calculation:\*\***

If I return \$5.80:



- Sender ends with:  $\$5 - \$3.63 + \$5.80 = \$7.17$  (gain of  $\$2.17$ )
- I keep:  $\$10.89 - \$5.80 = \$5.09$  (gain of  $\$5.09$ )

Both of us gain substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 5.80}
'''
```

#### Reasoning:

**\*\*Analysis:\*\***

My co-player sent  $\$3.63$  (72.6% of their endowment), showing substantial trust.  
I received  $\$10.89$  after the 3x multiplier.

**\*\*Considerations:\*\***

- They sent a good amount, demonstrating trust
- I want to reward this appropriately
- Both of us should benefit from the cooperation

**\*\*Calculation:\*\***

If I return around  $\$5.50$ :

- Sender ends with:  $\$5 - \$3.63 + \$5.50 = \$6.87$  (gain of  $\$1.87$ )
- I keep:  $\$10.89 - \$5.50 = \$5.39$  (gain of  $\$5.39$ )

Both benefit, sender is rewarded for trust.

Let me think about a slightly different amount. The sender sent 72.6% of their endowment, which is quite high. I should return enough that they benefit well.

If I return  $\$5.80$ :

- Sender ends with:  $\$5 - \$3.63 + \$5.80 = \$7.17$  (gain of  $\$2.17$ )
- I keep:  $\$10.89 - \$5.80 = \$5.09$  (gain of  $\$5.09$ )

This seems fair - both of us gain substantially, and the sender is well-rewarded for their high trust.

```
Usage: {
 "input_tokens": 1758,
 "output_tokens": 514,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Sender - dyad\_4

Timestamp: 2026-08-12T09:22:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user****content:**

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

**Message 3 role=assistant****content:**

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining
- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with: \$2.29 + \$3.50 = \$5.79 (net gain of \$0.79)
- I would end with: \$8.13 - \$3.50 = \$4.63 (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

```
{"return": 3.5}
```

**reasoning:**

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since  $2.71 \times 3 = 8.13$ )
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has:  $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system:  $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ( $\$8.13 - \$5.21$ )
2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
  - Sender ends with:  $\$2.29 + \$3 = \$5.29$
  - I end with:  $\$8.13 - \$3 = \$5.13$
  - Both profit slightly
- If I return \$4:
  - Sender ends with:  $\$2.29 + \$4 = \$6.29$
  - I end with:  $\$8.13 - \$4 = \$4.13$
  - Sender profits more, I profit less
- If I return \$2.92 (equal split):
  - Sender ends with:  $\$2.29 + \$2.92 = \$5.21$
  - I end with:  $\$8.13 - \$2.92 = \$5.21$
  - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1000,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit

- This opening move signals cooperation

```
{"send": 2.5}
```

**reasoning:**

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50
3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive:  $\$2.50 \times 3 = \$7.50$
- If they return around 40-50% of what they receive ( $\$3-\$3.75$ ), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

**usage:**

```
{
 "input_tokens": 852,
 "output_tokens": 441,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50% has proven effective and signals trust.

```
{"send": 2.5}
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50

- They received \$7.50
- They returned \$3.01
- My payoff:  $\$2.50 + \$3.01 = \$5.51$

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation

**usage:**

```
{
 "input_tokens": 1113,
 "output_tokens": 353,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:

- Sender ends with:  $\$0.29 + \$6.50 = \$6.79$  (net gain of \$1.79)
- I end with:  $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

```
{"return": 6.5}
```

**reasoning:**

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has:  $\$5 - \$4.71 = \$0.29$  remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with  $\$0.29 + \$6.00 = \$6.29$  (net gain of \$1.29), I keep \$8.13

- If I return \$7.00: Sender ends with  $\$0.29 + \$7.00 = \$7.29$  (net gain of \$2.29), I keep \$7.13
- If I return \$5.00: Sender ends with  $\$0.29 + \$5.00 = \$5.29$  (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:

- Sender ends with:  $\$0.29 + \$6.50 = \$6.79$  (net gain of \$1.79)
- I end with:  $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

usage:

```
{
 "input_tokens": 1345,
 "output_tokens": 549,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.71, it became \$14.129999999999999, and you returned \$6.5. Your visible payoff was \$7.629999999999999.

Your total visible earnings across all rounds are \$22.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

I've had positive results with cooperative play. In Round 4, I benefited from a sender who showed very high trust (\$4.71), and I reciprocated well. My sending strategy of around \$2.50 has worked reasonably, but I could signal stronger cooperation.

With a new co-player, I'll increase my trust signal slightly to \$3.00 (60% of endowment), which is more than my previous sends but not as extreme as the \$4.71 I received.

```
{"send": 3.0}
```

Reasoning:

Looking at my history:

- Round 1 (Receiver): Sender sent \$2.71, I returned \$3.50
- Round 2 (Sender): I sent \$2.50, got back \$3.01
- Round 3 (Sender): I sent \$2.50, got back \$2.68
- Round 4 (Receiver): Sender sent \$4.71, I returned \$6.50

The pattern shows:

- When I send ~50% (\$2.50), I get modest returns (around \$2.68-\$3.01)
- When someone sent me a lot (\$4.71), I reciprocated generously (\$6.50)
- Overall, cooperation is working reasonably well

Given that I've been relatively successful with moderate cooperation, and I've seen high trust rewarded (the \$4.71 sender got \$6.50 back from me), I could consider sending more this round to signal strong cooperation.

However, I've also learned that sending around 50% tends to get decent returns.

With a new co-player, I could:

1. Continue with \$2.50 (safe, proven strategy)
2. Send more to signal higher trust (like the \$4.71 example)
3. Send slightly more than before to test responsiveness

I think I'll send a bit more than my previous rounds to signal stronger cooperation - maybe \$3.00 or so. This is 60% of my endowment, showing more trust than before but not as extreme as the 94% I saw.

```
Let me go with $3.00.
Usage: {
 "input_tokens": 1645,
 "output_tokens": 495,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_5 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:22:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

**reasoning:**

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is  $\$3.66 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has  $\$5 - \$3.66 = \$1.34$  left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:
  - One approach: We could split the total gains equally
  - Total money in the system:  $\$5$  (sender initial) +  $\$10.98$  (what I received) =  $\$15.98$
  - If we split equally:  $\$15.98 / 2 = \$7.99$  each
  - Sender currently has:  $\$1.34$
  - So I would return:  $\$7.99 - \$1.34 = \$6.65$
5. Another approach: Return roughly the same proportion they sent
  - They sent 73.2% of their endowment
  - 73.2% of  $\$10.98 = \$8.04$
  - But this seems too much and would leave me with only  $\$2.94$
6. A middle ground approach:
  - The sender showed high trust by sending 73.2%
  - I should return enough to reward that trust
  - If I return around \$5-6, that would:
    - \* Give the sender a good return on their trust
    - \* Still leave me with a reasonable amount
7. Let me calculate what happens with different return amounts:
  - If I return \$5: Sender ends with  $\$6.34$  (profit of  $\$1.34$ ), I end with  $\$5.98$  (profit of  $\$5.98$ )
  - If I return \$6: Sender ends with  $\$7.34$  (profit of  $\$2.34$ ), I end with  $\$4.98$  (profit of  $\$4.98$ )



- If I return \$6.65: Sender ends with \$7.99 (profit of \$2.99), I end with \$4.33 (profit of \$4.33)

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1097,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98
- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)
2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)

2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)
- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)
- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)
- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

**usage:**

```
{
 "input_tokens": 1092,
 "output_tokens": 630,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern. I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

**reasoning:**

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving. This signals strong trust while being neither too conservative nor too aggressive.

**usage:**

```
{
 "input_tokens": 1363,
 "output_tokens": 473,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.99 to you. Your visible payoff was \$8.49.

Your total visible earnings across all rounds are \$25.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is moderate trust - not as high as other rounds, but still showing willingness to cooperate by sending more than half their endowment.

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This fairly rewards their trust

```
'''json
{"return": 4.50}
'''
```

**Reasoning:**

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is lower trust than I've seen in previous rounds (73.2%, 79.8% in earlier rounds). However, it's still a majority of their endowment, showing moderate trust.

I should still reciprocate fairly. Let me think about what would be appropriate:

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This is fair and rewards their trust

If I return \$4.00:

- The sender ends with \$6.21 (profit of \$1.21)
- I keep \$4.37
- This is also reasonable

Given that they sent a moderate amount (not as high as previous senders), I should return a moderate but fair amount. I'll return \$4.50 to maintain a cooperative reputation while being proportional to their trust level.

```
Usage: {
 "input_tokens": 1640,
 "output_tokens": 387,
 "reasoning_tokens": 0
}
```

## ROUND 6

Roles: Agent\_1=Receiver, Agent\_2=Receiver, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Sender, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Sender

### Pairings:

dyad\_1: Agent\_8=Sender, Agent\_2=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_1=Receiver  
dyad\_3: Agent\_6=Sender, Agent\_7=Receiver  
dyad\_4: Agent\_5=Sender, Agent\_3=Receiver

### Parsed game decisions:

dyad\_1: Agent\_8 sent \$3; received \$9; Agent\_2 returned \$6; payoffs Agent\_8=\$8, Agent\_2=\$3;  
noise/visible: sent visible=\$3.46, received visible=\$10.38, returned visible=\$5.16  
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_4 sent \$4; received \$12; Agent\_1 returned \$7; payoffs Agent\_4=\$8, Agent\_1=\$5;  
noise/visible: sent visible=\$4.81, received visible=\$14.43, returned visible=\$6.08  
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_6 sent \$3; received \$9; Agent\_7 returned \$6.5; payoffs Agent\_6=\$8.5, Agent\_7=\$2.5;  
noise/visible: sent visible=\$3.98, received visible=\$11.94, returned visible=\$6.95  
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 9.0, "min": 0, "raw": 6.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_5 sent \$3.5; received \$10.5; Agent\_3 returned \$5; payoffs Agent\_5=\$6.5, Agent\_3=\$5.5; noise/visible: sent visible=\$2.87, received visible=\$8.61, returned visible=\$4.17  
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

### Balances after round: {

"Agent\_1": 34.7,  
"Agent\_2": 33.8,  
"Agent\_3": 35.5,  
"Agent\_4": 38.2,  
"Agent\_5": 33.9,  
"Agent\_6": 37.2,  
"Agent\_7": 34.7,  
"Agent\_8": 38.0

}

### Visible balances after round: {

"Agent\_1": 36.069999999999999,  
"Agent\_2": 36.429999999999999,  
"Agent\_3": 36.03,  
"Agent\_4": 37.32,  
"Agent\_5": 35.28,  
"Agent\_6": 34.42,  
"Agent\_7": 39.059999999999995,  
"Agent\_8": 36.72

}

## LLM CALLS

### GAME - Agent\_8 - Sender - dyad\_1

Timestamp: 2026-08-12T09:23:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",  
"pairing": {

```

"agents": [
 "Agent_8",
 "Agent_2"
],
"investor": "Agent_8",
"roles": {
 "Agent_2": "trustee",
 "Agent_8": "investor"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with:  $\$5 - \$3.45 + \$5.00 = \$6.55$
- I end with:  $\$10.35 - \$5.00 = \$5.35$
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

**reasoning:**

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets  $\$5 - \$3.45 + \$0 = \$1.55$ , I get \$10.35
- If I return half (\$5.175): Sender gets  $\$5 - \$3.45 + \$5.175 = \$6.725$ , I get  $\$10.35 - \$5.175 = \$5.175$
- If I return \$6.90 (2x what they sent): Sender gets  $\$5 - \$3.45 + \$6.90 = \$8.45$ , I get  $\$10.35 - \$6.90 = \$3.45$

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has  $\$5 - \$3.45 = \$1.55$  remaining, I have \$10.35 total = \$11.90 total
4. Equal split would be \$5.95 each

To give sender \$5.95 total:  $\$5.95 = \$5 - \$3.45 + \text{return}$ , so  $\text{return} = \$5.95 - \$1.55 = \$4.40$

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return

...

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 807,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.0. Your visible payoff was \$5.350000000000001.

Your total visible earnings across all rounds are \$5.350000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- In the previous round, my co-player showed significant trust by sending 69% of their endowment
- I reciprocated fairly by returning a substantial amount
- Now with a new co-player, I want to establish cooperation

I'll send a generous amount to signal trust and cooperation, while being slightly more conservative since this is a new pairing.

```
{'send': 3.00}
```

**reasoning:**

I'm now the SENDER in Round 2. Let me think about this situation:

- In Round 1, I was the RECEIVER and someone sent me \$3.45 (69% of their endowment)
- I returned \$5.00, which was a fair reciprocation
- I kept \$5.35, and the sender ended with \$6.55

Now I'm the SENDER with a new co-player. I need to decide how much to send.

**Key considerations:**

1. In the previous round, my co-player sent 69% of their endowment, showing significant trust
2. I rewarded that trust by returning about 48% of what I received
3. This is a new co-player, so I can't assume they'll behave the same way
4. However, I should probably maintain a cooperative stance to build trust in the experiment
5. There's communication noise, so amounts may differ from what was actually sent

Given that:

- The previous sender was generous (69%)
- I want to establish cooperation with this new co-player
- Being moderately generous but not overly risky seems reasonable

I'll send around 60% of my endowment, which would be \$3.00. This shows trust and cooperation while being slightly more conservative than what I received in the previous round.

Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I want to maintain a cooperative equilibrium in this multi-agent setting, I should probably match that level of trust or be similarly generous.

I'll send \$3.00 (60% of my endowment), which is a solid show of trust and cooperation.

**usage:**

```
{
 "input_tokens": 843,
 "output_tokens": 473,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.93 to you. Your visible payoff was \$5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:



If I return \$3.20:  
- Sender ends with:  $\$5 - \$2.14 + \$3.20 = \$6.06$   
- I end with:  $\$6.42 - \$3.20 = \$3.22$   
- This creates relatively balanced outcomes and rewards their trust

`{'return': 3.20}`

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust
2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:  
- Sender ends with:  $\$5 - \$2.14 + \$3.00 = \$5.86$   
- I end with:  $\$6.42 - \$3.00 = \$3.42$   
- This gives them a decent return on their trust

If I return \$3.50:  
- Sender ends with:  $\$5 - \$2.14 + \$3.50 = \$6.36$   
- I end with:  $\$6.42 - \$3.50 = \$2.92$   
- This is more generous but leaves me with less than them

If I return \$3.20:  
- Sender ends with:  $\$5 - \$2.14 + \$3.20 = \$6.06$   
- I end with:  $\$6.42 - \$3.20 = \$3.22$   
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

**usage:**

```
{
 "input_tokens": 1075,
 "output_tokens": 588,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.14, it became \$6.42, and you returned \$3.2. Your visible payoff was \$3.2199999999999998.

Your total visible earnings across all rounds are \$14.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- When I sent \$3.00 in Round 2, I received a fair return of \$3.93
- I've been playing cooperatively and seeing reasonable reciprocation

I'll continue with a cooperative strategy and send a generous amount to this new co-player

.

`{'send': 3.00}`

**reasoning:**

Looking at my history:

- Round 1: As receiver, got \$10.35, returned \$5.00, earned \$5.35
- Round 2: As sender, sent \$3.00, got back \$3.93, earned \$5.93
- Round 3: As receiver, got \$6.42, returned \$3.20, earned \$3.22

In Round 2, I sent \$3.00 (60%) and my co-player returned \$3.93 out of \$9.00 (43.7%), which was decent reciprocation.

Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reasonable reciprocation. I should continue with a cooperative strategy.

I'll send around \$3.00 again to maintain consistency and signal trust.

**usage:**

```
{
 "input_tokens": 1348,
 "output_tokens": 269,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.24 to you. Your visible payoff was \$8.24.

Your total visible earnings across all rounds are \$22.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$4.44 (88.8% of their endowment) - this is exceptional trust!
- I received \$13.32 after the 3x multiplication
- Total resources created: \$13.88

This sender showed tremendous cooperation. I should reciprocate strongly:

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This rewards their high trust with balanced outcomes

```
{'return': 6.50}
```

**reasoning:**

The sender sent \$4.44, which is 88.8% of their endowment - this is very high trust! I received \$13.32 after multiplication.

Total resources created: \$5 - \$4.44 + \$13.32 = \$13.88

The sender showed exceptional trust. I should reciprocate generously.

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This creates a fairly balanced outcome and rewards their high trust

If I return \$7.00:

- Sender ends with: \$5 - \$4.44 + \$7.00 = \$7.56
- I end with: \$13.32 - \$7.00 = \$6.32
- This is also reasonable

Let me go with \$6.50 as it provides good reciprocation while maintaining balance.

**usage:**

```
{
 "input_tokens": 1564,
 "output_tokens": 392,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.44, it became \$13.32, and you returned \$6.5. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$29.560000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

- When I sent \$3.00, I've seen returns of \$3.93 and \$6.24
- The game has shown good cooperation overall
- I want to maintain a cooperative strategy

I'll continue with a consistent, generous approach.

{'send': 3.00}

**Reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received back \$3.93
- Round 4: I sent \$3.00, received back \$6.24 (very generous!)
- Round 5: Partner sent me \$4.44 (very trusting), I returned \$6.50

The cooperation has been generally good, with some partners being very generous. I've been sending \$3.00 consistently and seeing good returns. I could continue with that strategy, or perhaps be slightly more generous given the positive experiences.

Let me continue with \$3.00 as it's been working well and represents 60% of my endowment - a reasonable cooperative level.

Usage: {

"input\_tokens": 1837,  
"output\_tokens": 233,  
"reasoning\_tokens": 0

}

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:23:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 1,  
"opponent\_id": "Agent\_8",  
"pairing": {  
  "agents": [  
    "Agent\_8",  
    "Agent\_2"  
  ],  
  "investor": "Agent\_8",  
  "roles": {  
    "Agent\_2": "trustee",  
    "Agent\_8": "investor"  
  },  
  "trustee": "Agent\_2"  
},

"role": "trustee",  
"role\_label": "Receiver",  
"round": 6,  
"task": "game",  
"task\_index": 0

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with: \$1.18 + \$6.00 = \$7.18 (better than their initial \$5)
- I end with: \$11.46 - \$6.00 = \$5.46

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```
'''json
{"return": 6.0}
'''
```

#### reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ( $3.82 \times 3 = 11.46$ )
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about what would be fair:

The sender:

- Started with \$5
- Sent \$3.82
- Currently has:  $\$5 - \$3.82 = \$1.18$

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with:  $\$1.18 + \$5.73 = \$6.91$
- I end with:  $\$11.46 - \$5.73 = \$5.73$

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with:  $\$1.18 + \$6.82 = \$8.00$
- I end with:  $\$11.46 - \$6.82 = \$4.64$

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with:  $\$1.18 + \$7.64 = \$8.82$
- I end with:  $\$11.46 - \$7.64 = \$3.82$

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64

To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 927,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER
- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

**usage:**

```
{
 "input_tokens": 855,
 "output_tokens": 480,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.

```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings
3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

**usage:**

```
{
 "input_tokens": 1089,
 "output_tokens": 383,
 "reasoning_tokens": 0
}
```

**Message 8 role=user****content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.

Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant****content:**

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with: \$1.65 + \$6.00 = \$7.65 (profit of \$2.65)
- I end with: \$10.05 - \$6.00 = \$4.05

This rewards their trust while both of us benefit from the exchange.

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$5 - \$3.35 = \$1.65 left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust (67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with:  $\$1.65 + \$6.00 = \$7.65$  (profit of \$2.65)
- I end with:  $\$10.05 - \$6.00 = \$4.05$

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with:  $\$1.65 + \$5.00 = \$6.65$  (profit of \$1.65)
- I end with:  $\$10.05 - \$5.00 = \$5.05$

Or if I return \$6.50:

- Sender ends with:  $\$1.65 + \$6.50 = \$8.15$  (profit of \$3.15)
- I end with:  $\$10.05 - \$6.50 = \$3.55$

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

usage:

```
{
 "input_tokens": 1315,
 "output_tokens": 602,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.35, it became \$10.05, and you returned \$6.0. Your visible payoff was \$4.050000000000001.

Your total visible earnings across all rounds are \$25.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

- When I sent \$4.0 in previous rounds, I received back \$6.29 and \$7.32 - both profitable outcomes
- My cooperative strategy has been working well, building trust

I'll continue with the same successful approach of sending a substantial amount to encourage reciprocity.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my track record:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.0, kept \$5.46
- Round 2 (SENDER): Sent \$4.0, received back \$6.29, kept \$7.29
- Round 3 (SENDER): Sent \$4.0, received back \$7.32, kept \$8.32
- Round 4 (RECEIVER): Received \$10.05, returned \$6.0, kept \$4.05

Total visible earnings: \$25.12

My cooperative strategy of sending \$4.0 has been working well - both times I sent \$4.0, I received good returns (\$6.29 and \$7.32), both yielding solid profits. As a receiver, I've been fair by returning \$6.0 when others showed trust.

I'm now paired with a new co-player for round 5 as sender. Given my successful track record with cooperation, I should continue this strategy.

Sending \$4.0 again makes sense - it shows significant trust (80% of endowment) and has consistently led to profitable outcomes for me.

usage:



```
{
 "input_tokens": 1608,
 "output_tokens": 357,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.93 to you. Your visible payoff was \$6.93.

Your total visible earnings across all rounds are \$32.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

Consistent with my previous cooperative approach, I'll return a fair amount that rewards their trust.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (good profit)
- I end with: \$10.38 - \$6.00 = \$4.38

```
'''json
{"return": 6.0}
'''
```

**Reasoning:**

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

My pattern has been to return around \$6.00 when I receive amounts in this range (around \$10-11). This has been fair and consistent.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (profit of \$2.54)
- I end with: \$10.38 - \$6.00 = \$4.38

This is consistent with my previous behavior and rewards their trust appropriately.

Usage: {

```
"input_tokens": 1837,
"output_tokens": 332,
"reasoning_tokens": 0
```

}

**GAME - Agent\_4 - Sender - dyad\_2**

Timestamp: 2026-08-12T09:23:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_4",
 "Agent_1"
],
"investor": "Agent_4",
"roles": {
 "Agent_1": "trustee",
```

```

 "Agent_4": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

**Key considerations:**

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send \*\*\$4\*\* to signal strong willingness to cooperate while maintaining a small hedge . If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

#### reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

#### Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

#### Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. **Risk**: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. However, my behavior here might set a precedent for future rounds when I'm a receiver.
4. **Multi-round consideration**: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.
5. **Communication noise**: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

#### Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange....

#### usage:

```
{
 "input_tokens": 500,
```

```
"output_tokens": 1133,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Round 2 Analysis (as RECEIVER):\*\***

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

**\*\*Reciprocity consideration:\*\***

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

**\*\*Decision:\*\***

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: \$5 - \$4 + \$5.46 = \$6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 ( $2.73 \times 3$ )
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ( $\$2.73/\$5 = 54.6\%$ ), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game

3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

usage:

```
{
 "input_tokens": 790,
 "output_tokens": 926,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

**\*\*Round 3 Analysis (as RECEIVER):\*\***

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier
- This is more trust than the previous co-player showed in Round 2

**\*\*Reciprocity:\*\***

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion that rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

**\*\*Round 3 Analysis (as RECEIVER):\*\***

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

**\*\*Past behavior:\*\***

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

**\*\*Reciprocity consideration:\*\***

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). This shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high trust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \$5.54$

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50
- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

**usage:**

```
{
 "input_tokens": 1157,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.55, it became \$10.649999999999999, and you returned \$5.5. Your visible payoff was \$5.149999999999999.

Your total visible earnings across all rounds are \$15.7.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Round 4 Analysis (as SENDER):\*\***

Looking at my experience so far:

- Round 1: Sent \$4, got good reciprocation (\$5.46 back)
- Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
- Total earnings: \$15.70 (averaging \$5.23/round)

**\*\*New co-player consideration:\*\***

This is a different co-player than previous rounds, so no direct history. However, the cooperative approach has been working well overall.

**\*\*Decision:\*\***

I'll send **\*\*\$4\*\*** again because:

- The 3x multiplier creates strong mutual benefit
- It signals trust and willingness to cooperate
- It keeps \$1 as insurance against complete defection
- This amount worked well in Round 1

'''json

{"send": 4}

'''

**reasoning:**

**\*\*Round 4 Analysis (as SENDER):\*\***

Let me review the pattern so far:

- Round 1: I sent \$4, received back \$5.46 (good return, ~45.5% of \$12)
- Round 2: As receiver, got \$2.73 (sent), I returned \$4.10 (~50% of \$8.19)
- Round 3: As receiver, got \$3.55 (sent), I returned \$5.50 (~52% of \$10.65)

Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized each round).

Key considerations:

1. I have no history with this specific co-player
2. My previous experiences have been fairly positive:

- Round 1: Good reciprocation when I sent high
  - Round 2 & 3: I reciprocated fairly as receiver
3. The game mechanics reward cooperation through the 3x multiplier
4. There's communication noise, so exact amounts may vary

Strategy:

- Since cooperation has worked well so far, I should lean toward sending a substantial amount
- However, this is a new co-player with no history
- I've been playing fairly cooperatively, which might be building reputation (though names are hidden, the pattern of fair play might matter in multi-agent settings)

I think sending \$3.50-\$4.50 would be reasonable. Let me go with \$4 again - it's worked before, shows strong trust without being all-in, and leaves some hedge.

Actually, looking at my total earnings (\$15.70 over 3 rounds), I'm doing reasonably well averaging about \$5.23 per round. The cooperative strategy is working.

I'll send \$4 again.

usage:

```
{
 "input_tokens": 1463,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.39 to you. Your visible payoff was \$9.39.

Your total visible earnings across all rounds are \$25.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

**\*\*Round 5 Analysis (as RECEIVER):\*\***

- My co-player sent \$3.45 (69% of their endowment) - strong trust
- I received \$10.35 after the multiplier
- Previous round showed excellent reciprocation when I was sender (\$8.39 back!)

**\*\*Reciprocity:\*\***

My co-player demonstrated high trust by sending nearly 70%. I should maintain the fair reciprocation pattern that's been working well.

Returning approximately 50% (\$5.20) would:

- Give my co-player a total of \$6.75 (\$1.55 kept + \$5.20 returned)
- Leave me with \$5.15
- Reward their trust appropriately

```
'''json
{"return": 5.20}
'''
```

reasoning:

**\*\*Round 5 Analysis (as RECEIVER):\*\***

- My co-player sent \$3.45 (69% of their \$5 endowment) - showing high trust
- I received \$10.35 after the multiplier
- This is similar trust level to Round 3 (where co-player sent \$3.55)

**\*\*Previous round performance:\*\***

- Round 4: I sent \$4, got back \$8.39 (70% return rate!) - excellent reciprocation
- This suggests cooperative behavior is being rewarded

**\*\*Reciprocity consideration:\*\***

My co-player sent 69% of their endowment, showing strong trust. I should return a fair proportion.

Returning approximately 50-52% seems appropriate:

- 50% = \$5.18
- 52% = \$5.38

If I return \$5.20:

- My co-player gets total: \$1.55 (kept) + \$5.20 (returned) = \$6.75
- I keep: \$5.15
- Both benefit well

This is consistent with my previous fair reciprocation strategy and should maintain cooperation.

**usage:**

```
{
 "input_tokens": 1801,
 "output_tokens": 448,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.2. Your visible payoff was \$5.150000000000001.

Your total visible earnings across all rounds are \$30.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

**\*\*Round 6 Analysis (as SENDER):\*\***

Performance review:

- As sender: Sent \$4 twice, received good returns both times (avg \$6.93 back)
- As receiver: Been consistent with fair ~50% returns
- Average earnings: \$6.05/round

**\*\*New co-player:\*\***

Different partner again, but the strategy of sending \$4 has been working well.

**\*\*Decision:\*\***

Sending **\*\*\$4\*\*** because:

- Proven successful in previous sender rounds
- Demonstrates strong trust
- Creates significant value through 3x multiplier
- Maintains small hedge (\$1)

```
'''json
{"send": 4}
'''
```

**Reasoning:**

**\*\*Round 6 Analysis (as SENDER):\*\***

Looking at my track record:

- Round 1: Sent \$4, got \$5.46 back ? Profit of \$6.46
- Round 4: Sent \$4, got \$8.39 back ? Profit of \$9.39

Both times I sent \$4, I received good to excellent returns. Average earnings per round is now \$6.05 (\$30.24/5 rounds).

As receiver:

- Round 2: Received \$8.19, returned \$4.10
- Round 3: Received \$10.65, returned \$5.50
- Round 5: Received \$10.35, returned \$5.20

I've been consistently fair as a receiver, returning around 50%.

**\*\*New co-player:\*\***

This is another new pairing. The pattern of sending \$4 has been successful, showing trust while keeping some insurance.



```
Decision:
I'll continue with $4 because:
- It's been consistently successful
- Shows strong cooperation
- Keeps $1 as hedge
- The multiplier effect makes this beneficial for both parties if reciprocated
```

Let me send \$4.

```
Usage: {
 "input_tokens": 2113,
 "output_tokens": 432,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:23:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_1"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_1": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining
- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with:  $\$2.29 + \$3.50 = \$5.79$  (net gain of \$0.79)
- I would end with:  $\$8.13 - \$3.50 = \$4.63$  (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

{"return": 3.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since  $\$2.71 \times 3 = \$8.13$ )
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has:  $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system:  $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ( $\$8.13 - \$5.21$ )
2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
  - Sender ends with:  $\$2.29 + \$3 = \$5.29$
  - I end with:  $\$8.13 - \$3 = \$5.13$
  - Both profit slightly

- If I return \$4:

- Sender ends with:  $\$2.29 + \$4 = \$6.29$
- I end with:  $\$8.13 - \$4 = \$4.13$
- Sender profits more, I profit less
- If I return  $\$2.92$  (equal split):
  - Sender ends with:  $\$2.29 + \$2.92 = \$5.21$
  - I end with:  $\$8.13 - \$2.92 = \$5.21$
  - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1000,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit
- This opening move signals cooperation

```
{"send": 2.5}
```

reasoning:

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50
3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive:  $\$2.50 \times 3 = \$7.50$
- If they return around 40-50% of what they receive (\$3-\$3.75), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

**usage:**

```
{
 "input_tokens": 852,
 "output_tokens": 441,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50% has proven effective and signals trust.

```
{"send": 2.5}
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50
- They received \$7.50
- They returned \$3.01
- My payoff: \$2.50 + \$3.01 = \$5.51

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation.

**usage:**

```
{
 "input_tokens": 1113,
 "output_tokens": 353,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:

- Sender ends with:  $\$0.29 + \$6.50 = \$6.79$  (net gain of \$1.79)

- I end with:  $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

{"return": 6.5}

**reasoning:**

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has:  $\$5 - \$4.71 = \$0.29$  remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with  $\$0.29 + \$6.00 = \$6.29$  (net gain of \$1.29), I keep \$8.13

- If I return \$7.00: Sender ends with  $\$0.29 + \$7.00 = \$7.29$  (net gain of \$2.29), I keep \$7.13

- If I return \$5.00: Sender ends with  $\$0.29 + \$5.00 = \$5.29$  (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:

- Sender ends with:  $\$0.29 + \$6.50 = \$6.79$  (net gain of \$1.79)

- I end with:  $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

**usage:**

```
{
 "input_tokens": 1345,
 "output_tokens": 549,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.71, it became \$14.129999999999999, and you returned \$6.5. Your visible

payoff was \$7.629999999999999.  
Your total visible earnings across all rounds are \$22.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience:

I've had positive results with cooperative play. In Round 4, I benefited from a sender who showed very high trust (\$4.71), and I reciprocated well. My sending strategy of around \$2.50 has worked reasonably, but I could signal stronger cooperation.

With a new co-player, I'll increase my trust signal slightly to \$3.00 (60% of endowment), which is more than my previous sends but not as extreme as the \$4.71 I received.

```
{"send": 3.0}
```

**reasoning:**

Looking at my history:

- Round 1 (Receiver): Sender sent \$2.71, I returned \$3.50
- Round 2 (Sender): I sent \$2.50, got back \$3.01
- Round 3 (Sender): I sent \$2.50, got back \$2.68
- Round 4 (Receiver): Sender sent \$4.71, I returned \$6.50

The pattern shows:

- When I send ~50% (\$2.50), I get modest returns (around \$2.68-\$3.01)
- When someone sent me a lot (\$4.71), I reciprocated generously (\$6.50)
- Overall, cooperation is working reasonably well

Given that I've been relatively successful with moderate cooperation, and I've seen high trust rewarded (the \$4.71 sender got \$6.50 back from me), I could consider sending more this round to signal strong cooperation.

However, I've also learned that sending around 50% tends to get decent returns.

With a new co-player, I could:

1. Continue with \$2.50 (safe, proven strategy)
2. Send more to signal higher trust (like the \$4.71 example)
3. Send slightly more than before to test responsiveness

I think I'll send a bit more than my previous rounds to signal stronger cooperation - maybe \$3.00 or so. This is 60% of my endowment, showing more trust than before but not as extreme as the 94% I saw.

Let me go with \$3.00.

**usage:**

```
{
 "input_tokens": 1645,
 "output_tokens": 495,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$28.639999999999993.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

The sender showed exceptional trust by sending \$4.81 (96.2% of their endowment) - nearly all of it! This is an extremely strong cooperative signal.

They now have only \$0.19 remaining, and I received \$14.43.

This is similar to Round 4 where I received \$4.71 and returned \$6.50. Given this sender showed even more trust, I should reciprocate generously.

If I return \$7.00:

- Sender ends with:  $\$0.19 + \$7.00 = \$7.19$  (net gain of \$2.19)
- I end with:  $\$14.43 - \$7.00 = \$7.43$

This rewards their exceptional trust while ensuring we both profit substantially.

```
{"return": 7.0}
```

#### Reasoning:

The sender sent \$4.81, which is 96.2% of their endowment - extremely high trust!

I received \$14.43 after the 3x multiplication.

They now have:  $\$5 - \$4.81 = \$0.19$  remaining

This is even higher trust than the \$4.71 I saw in Round 4 (where I returned \$6.50).

I should reward this exceptional trust with a generous return. Let me think about fair amounts :

If I return \$7.00:

- Sender ends with:  $\$0.19 + \$7.00 = \$7.19$  (net gain of \$2.19)
- I end with:  $\$14.43 - \$7.00 = \$7.43$

This gives them a good return on their trust and we both profit nicely. This is similar proportionally to what I did in Round 4.

Actually, given they sent even more than Round 4, I could be slightly more generous. Let me consider \$7.50:

- Sender ends with:  $\$0.19 + \$7.50 = \$7.69$  (net gain of \$2.69)
- I end with:  $\$14.43 - \$7.50 = \$6.93$

That works well too. I'll go with \$7.00 as a strong but reasonable reciprocation.

```
Usage: {
 "input_tokens": 1915,
 "output_tokens": 511,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_6 - Sender - dyad\_3

Timestamp: 2026-08-12T09:23:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_7"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with prudent risk management. Since this is a multi-round game, establishing cooperation early could be beneficial for future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive. I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:



1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts seen may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive. Common strategies:

- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything. Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$4.95.

Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Analysis:\*\***

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

**\*\*Fairness considerations:\*\***

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

**\*\*Decision:\*\***

If I return \$3.20:

- Sender ends with:  $\$5 - \$2.11 + \$3.20 = \$6.09$  (gain of  $\sim\$1.09$ )
- I keep:  $\$6.33 - \$3.20 = \$3.13$  (gain of  $\sim\$3.13$ )

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I was the SENDER:

- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.95 to me
- My payoff:  $\$5 - \$3 + \$2.95 = \$4.95$
- The receiver kept:  $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:

- My co-player (a different agent) sent me \$2.11
- I received \$6.33 (which is  $\$2.11 \times 3$ )
- I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33
- If I want to be reciprocal and fair, I should return enough that:
  1. The sender ends up better off than if they had sent nothing
  2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with:  $\$5 - \$2.11 + \$3.165 = \$6.055$
- I keep:  $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:

- Sender ends up with:  $\$5 - \$2.11 + \$2.09 = \$4.98$
- I keep:  $\$6.33 - \$2.09 = \$4.24$

The sender invested 42% of their endowment, so I should consider whether they break even or better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here.

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us....

**usage:**

```
{
 "input_tokens": 780,
 "output_tokens": 1431,
 "reasoning_tokens": 0
}
```

```
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

**\*\*Analysis:\*\***

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

**\*\*Considerations:\*\***

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

**\*\*This round:\*\***

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

**\*\*Considerations:\*\***

1. This is a different co-player than in rounds 1 and 2
2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

**\*\*Strategy:\*\***

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

**usage:**

```
{
```

```
"input_tokens": 1133,
"output_tokens": 554,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

**\*\*Analysis:\*\***

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

**\*\*Considerations:\*\***

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

**\*\*Calculation:\*\***

If I return \$6.80:

- Sender ends with:  $\$5 - \$4.14 + \$6.80 = \$7.66$  (gain of \$2.66)
- I keep:  $\$12.42 - \$6.80 = \$5.62$  (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 6.80}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

**\*\*Considerations:\*\***

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

**\*\*Calculation options:\*\***

If I return \$6.00:

- Sender ends with:  $\$5 - \$4.14 + \$6.00 = \$6.86$  (gain of \$1.86)
- I keep:  $\$12.42 - \$6.00 = \$6.42$  (gain of \$6.42)

If I return \$7.00:

- Sender ends with:  $\$5 - \$4.14 + \$7.00 = \$7.86$  (gain of \$2.86)
- I keep:  $\$12.42 - \$7.00 = \$5.42$  (gain of \$5.42)

If I return \$6.50:

- Sender ends with:  $\$5 - \$4.14 + \$6.50 = \$7.36$  (gain of \$2.36)
- I keep:  $\$12.42 - \$6.50 = \$5.92$  (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.  
Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd gain \$3.36 while...

usage:

```
{
 "input_tokens": 1425,
 "output_tokens": 847,
 "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$6.8. Your visible payoff was \$5.619999999999998.

Your total visible earnings across all rounds are \$20.38.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

**\*\*Analysis:\*\***

My co-player sent \$3.63 (73% of their endowment), showing substantial trust.  
I received \$10.89 after the 3x multiplier.

**\*\*Considerations:\*\***

- They sent a significant portion of their endowment
- Both of us should benefit from this cooperation
- I want to reward their trust appropriately

**\*\*Calculation:\*\***

If I return \$5.80:

- Sender ends with:  $\$5 - \$3.63 + \$5.80 = \$7.17$  (gain of \$2.17)
- I keep:  $\$10.89 - \$5.80 = \$5.09$  (gain of \$5.09)

Both of us gain substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 5.80}
'''
```

reasoning:

**\*\*Analysis:\*\***

My co-player sent \$3.63 (72.6% of their endowment), showing substantial trust.  
I received \$10.89 after the 3x multiplier.

**\*\*Considerations:\*\***

- They sent a good amount, demonstrating trust
- I want to reward this appropriately
- Both of us should benefit from the cooperation

**\*\*Calculation:\*\***

If I return around \$5.50:

- Sender ends with:  $\$5 - \$3.63 + \$5.50 = \$6.87$  (gain of \$1.87)
- I keep:  $\$10.89 - \$5.50 = \$5.39$  (gain of \$5.39)

Both benefit, sender is rewarded for trust.

Let me think about a slightly different amount. The sender sent 72.6% of their endowment, which is quite high. I should return enough that they benefit well.

If I return \$5.80:

- Sender ends with:  $\$5 - \$3.63 + \$5.80 = \$7.17$  (gain of \$2.17)
- I keep:  $\$10.89 - \$5.80 = \$5.09$  (gain of \$5.09)

This seems fair - both of us gain substantially, and the sender is well-rewarded for their

```

 high trust.
usage:
{
 "input_tokens": 1758,
 "output_tokens": 514,
 "reasoning_tokens": 0
}
Message 12 role=user
content:
 Round 6

 In your most recent previous game round, you were the RECEIVER against another agent. You
 saw them send $3.63, it became $10.89, and you returned $5.8. Your visible payoff was $5
 .0900000000000001.
 Your total visible earnings across all rounds are $25.47.

 This round, you are the SENDER against your current co-player. You have $5. How much do yo
 u send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Assistant output:
 Analysis:

 I'm facing a new co-player as sender. My previous sender experiences:
 - Round 1: Sent $3, received $2.95 back (payoff $4.95)
 - Round 3: Sent $3, received $4.68 back (payoff $6.68)

 Both times, sending $3 led to positive outcomes with reasonable reciprocation.

 Strategy:
 - $3 creates $9 for the receiver via the 3x multiplier
 - This amount signals willingness to cooperate
 - It balances trust-building with prudent risk management
 - Consistency may help establish a cooperative reputation

 I'll continue with my established approach.

 '''json
 {"send": 3}
 '''
Reasoning:
 Analysis:

 I'm the sender this round with a new co-player. Let me review my experience so far:

 Sender rounds:
 - Round 1: Sent $3, got back $2.95 (payoff $4.95)
 - Round 3: Sent $3, got back $4.68 (payoff $6.68)

 Receiver rounds:
 - Round 2: Received $6.33 from $2.11 sent, returned $3.20 (payoff $3.13)
 - Round 4: Received $12.42 from $4.14 sent, returned $6.80 (payoff $5.62)
 - Round 5: Received $10.89 from $3.63 sent, returned $5.80 (payoff $5.09)

 As sender, I've consistently sent $3 and received reasonable returns. My co-players as receive
 rs have varied in how much they sent initially.

 This is round 6, so I'm establishing patterns. I want to continue building cooperative norms.
 Sending $3 has worked well for me in the past - it's enough to create value through the 3x m
 ultipplier while not being overly risky.

 I'll stick with my strategy of sending $3.
Usage: {
 "input_tokens": 2077,
 "output_tokens": 470,
 "reasoning_tokens": 0
}
GAME - Agent_7 - Receiver - dyad_3
Timestamp: 2026-08-12T09:23:44
Model: anthropic/claude-sonnet-4.5

```

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_7"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

**\*\*My decision:\*\***

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter
5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit
4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3
- Receiver gets  $3 \times 3 = 9$
- I keep  $5 - 3 = 2$  initially
- If receiver returns \$3, I end with  $2 + 3 = 5$  (same as start, receiver gets \$6)

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 647,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**



Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received \$12.120000000000001. How much do you return? (0-12.120000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: \$0.96 + \$6.50 = \$7.46
- I end with: \$12.12 - \$6.50 = \$5.62

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: \$5 - \$3 + \$5.68 = \$7.68
- The receiver kept \$9 - \$5.68 = \$3.32

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received \$4.04 × 3 = \$12.12
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be \$7.27 - \$7.88
- That would leave them with \$0.96 + return, and me with \$12.12 - return

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with \$0.96 + \$6 = \$6.96, I end with \$12.12 - \$6 = \$6.12
- If I return \$7, they end with \$7.96, I end with \$5.12

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: \$0.96 + \$6.50 = \$7.46

...

**usage:**

```
{
```

```
"input_tokens": 754,
"output_tokens": 665,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible payoff was \$5.620000000000001.

Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with:  $\$0.62 + \$7.00 = \$7.62$
- I end with:  $\$13.14 - \$7.00 = \$6.14$

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

**reasoning:**

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.

I received \$13.14

They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:

- They end with:  $\$0.62 + \$7.00 = \$7.62$
- I end with:  $\$13.14 - \$7.00 = \$6.14$

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:

1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:

- Their total:  $\$0.62 + \$7.00 = \$7.62$  (profit of \$2.62)
- My total:  $\$13.14 - \$7.00 = \$6.14$  (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:

- Their total:  $\$0.62 + \$7.50 = \$8.12$  (profit of \$3.12)
- My total:  $\$13.14 - \$7.50 = \$5.64$  (profit of \$0.64)

I think \$7.00 is reasonable. It gives them a nice profit while I also benefit from the coo

```

 peration.
usage:
{
 "input_tokens": 1061,
 "output_tokens": 567,
 "reasoning_tokens": 0
}
Message 8 role=user
content:
 Round 4

 In your most recent previous game round, you were the RECEIVER against another agent. You
 saw them send $4.38, it became $13.14, and you returned $7.0. Your visible payoff was $6
 .1400000000000001.
 Your total visible earnings across all rounds are $19.44.

 This round, you are the SENDER against your current co-player. You have $5. How much do yo
 u send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
content:
 Looking at my experience so far:

 - In Round 1, I sent $3 and received good reciprocation ($5.68 back)
 - In Rounds 2 and 3, co-players showed high trust by sending $4.04 and $4.38
 - I've been reciprocating fairly, returning about half of what I received

 This is a new co-player, but the general pattern suggests cooperation pays off. I'll send
 a high amount to match the trust I've seen from others.

 '''json
 {"send": 4}
 '''
reasoning:
 I'm now the SENDER in Round 4. Let me think about what I've learned:

 Round 1: I sent $3, received $5.68 back, ended with $7.68
 Round 2: I received $12.12 (they sent $4.04), returned $6.50, ended with $5.62
 Round 3: I received $13.14 (they sent $4.38), returned $7.00, ended with $6.14

 My recent experiences as a receiver show co-players sending high amounts ($4.04 and $4.38)
 . When I was a sender in Round 1, I sent $3 and got $5.68 back, which was good reciproca
 tion.

 Given that I've seen high trust from others and I've been reciprocating fairly, I should c
 ontinue to show trust. This is a new co-player this round (since pairings are randomized
), so I don't have specific history with them, but the general pattern suggests cooperat
 ion is beneficial.

 I could send:
 - $3: Moderate trust, similar to Round 1
 - $4: High trust, similar to what I've received from others
 - $5: Maximum trust

 Given that I've been receiving high amounts from others (around $4+), and I've been recipr
 ocating fairly well, I should send a similarly high amount to maintain the cooperative d
 ynamic. Sending $4 shows significant trust while not being overly risky.

 If I send $4:
 - They receive $12
 - If they return around 50-60%, I'd get back $6-7.20
 - I'd end with roughly $7-8.20

 This seems reasonable and aligns with the cooperative pattern I've been experiencing.
usage:
{
 "input_tokens": 1349,
 "output_tokens": 524,
 "reasoning_tokens": 0
}

```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.06 to you. Your visible payoff was \$8.059999999999999.

Your total visible earnings across all rounds are \$27.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my recent rounds:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back - excellent reciprocation

The cooperation has been strong, with both co-players returning generously. This suggests high trust sends are being rewarded.

With a new co-player this round, I'll continue the cooperative strategy that's been working well.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.68 back (good return)
- Round 2: Received \$12.12, returned \$6.50
- Round 3: Received \$13.14, returned \$7.00
- Round 4: Sent \$4, received \$7.06 back (very good return - they returned about 59% of what they got)

The cooperation has been strong across all rounds. People are sending high amounts and returning generously. My earnings have been good (\$27.50 total across 4 rounds with visible payoffs).

This is Round 5 with a new co-player. Given the positive cooperation pattern, I should continue with a high trust strategy. Sending \$4 worked well last round.

I'll send \$4 again to show trust and likely get good reciprocation.

**usage:**

```
{
 "input_tokens": 1602,
 "output_tokens": 314,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.12 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$33.62.

This round, you are the RECEIVER against your current co-player. They sent you \$3.98, so you received \$11.94. How much do you return? (0-11.94)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing strong trust
- I received \$11.94
- They kept only \$1.02

This is solid trust that deserves fair reciprocation. Based on my previous patterns as receive

r, returning about 50-55% of what I received has worked well.

If I return \$6.50:

- They end with:  $\$1.02 + \$6.50 = \$7.52$
- I end with:  $\$11.94 - \$6.50 = \$5.44$

Both of us gain from cooperation.

```
'''json
{"return": 6.5}
'''
```

#### Reasoning:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing good trust
- I received \$11.94
- They kept only \$1.02

Previous rounds have shown:

- When I was receiver in Round 2, I got \$12.12 and returned \$6.50
- When I was receiver in Round 3, I got \$13.14 and returned \$7.00
- These were fair returns that both parties benefited from

For this round:

If I return \$6.50:

- They end with:  $\$1.02 + \$6.50 = \$7.52$
- I end with:  $\$11.94 - \$6.50 = \$5.44$

Both benefit from cooperation. This seems like a fair reciprocation for their trust.

```
Usage: {
 "input_tokens": 1846,
 "output_tokens": 360,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_5 - Sender - dyad\_4

Timestamp: 2026-08-12T09:23:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_3"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_3": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

**reasoning:**

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is  $3 \times \$3.66$ )
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has  $\$5 - \$3.66 = \$1.34$  left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:
  - One approach: We could split the total gains equally
  - Total money in the system:  $\$5$  (sender initial) +  $\$10.98$  (what I received) =  $\$15.98$
  - If we split equally:  $\$15.98 / 2 = \$7.99$  each
  - Sender currently has:  $\$1.34$
  - So I would return:  $\$7.99 - \$1.34 = \$6.65$
5. Another approach: Return roughly the same proportion they sent
  - They sent 73.2% of their endowment
  - 73.2% of  $\$10.98 = \$8.04$
  - But this seems too much and would leave me with only  $\$2.94$
6. A middle ground approach:
  - The sender showed high trust by sending 73.2%
  - I should return enough to reward that trust
  - If I return around \$5-6, that would:
    - \* Give the sender a good return on their trust
    - \* Still leave me with a reasonable amount
7. Let me calculate what happens with different return amounts:
  - If I return \$5: Sender ends with  $\$6.34$  (profit of  $\$1.34$ ), I end with  $\$5.98$  (profit of  $\$5.98$ )
  - If I return \$6: Sender ends with  $\$7.34$  (profit of  $\$2.34$ ), I end with  $\$4.98$  (profit of  $\$4.98$ )
  - If I return \$6.65: Sender ends with  $\$7.99$  (profit of  $\$2.99$ ), I end with  $\$4.33$  (profit of  $\$4.33$ )

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 1097,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

**reasoning:**

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98
- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)
2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)
2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)
- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)



- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)
- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

**usage:**

```
{
 "input_tokens": 1092,
 "output_tokens": 630,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.4700000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern

. I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

**reasoning:**

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving. This signals strong trust while being neither too conservative nor too aggressive.

**usage:**

```
{
 "input_tokens": 1363,
 "output_tokens": 473,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.99 to you. Your visible payoff was \$8.49.

Your total visible earnings across all rounds are \$25.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is moderate trust - not as high as other rounds, but still showing willingness to cooperate by sending more than half their endowment.

If I return \$4.50:  
- The sender ends with \$6.71 (profit of \$1.71)  
- I keep \$3.87  
- This fairly rewards their trust

```
'''json
{"return": 4.50}
'''
```

**reasoning:**

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is lower trust than I've seen in previous rounds (73.2%, 79.8% in earlier rounds). However, it's still a majority of their endowment, showing moderate trust.

I should still reciprocate fairly. Let me think about what would be appropriate:

If I return \$4.50:  
- The sender ends with \$6.71 (profit of \$1.71)  
- I keep \$3.87  
- This is fair and rewards their trust

If I return \$4.00:  
- The sender ends with \$6.21 (profit of \$1.21)  
- I keep \$4.37  
- This is also reasonable

Given that they sent a moderate amount (not as high as previous senders), I should return a moderate but fair amount. I'll return \$4.50 to maintain a cooperative reputation while being proportional to their trust level.

**usage:**

```
{
 "input_tokens": 1640,
 "output_tokens": 387,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.79, it became \$8.370000000000001, and you returned \$4.5. Your visible payoff was \$3.870000000000001.

Your total visible earnings across all rounds are \$29.610000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history:

- Round 2: I sent \$3.00 (60%), received \$4.80 back - good cooperation
- Round 4: I sent \$3.50 (70%), received \$6.99 back - excellent cooperation
- Overall, higher sends have been rewarded

With a new co-player, I'll maintain a cooperative approach. Sending \$3.50 (70%) worked very well last time, so I'll repeat that strategy.

```
'''json
{"send": 3.50}
'''
```

**Reasoning:**

Let me review my history:

Round 1: Receiver - received from sender who sent \$3.66 (73.2%), I returned \$6.00, earned \$4.98

Round 2: Sender - sent \$3.00 (60%), received \$4.80 back, earned \$6.80

Round 3: Receiver - received from sender who sent \$3.99 (79.8%), I returned \$6.50, earned \$5.47  
Round 4: Sender - sent \$3.50 (70%), received \$6.99 back, earned \$8.49  
Round 5: Receiver - received from sender who sent \$2.79 (55.8%), I returned \$4.50, earned \$3.87

My sending history:

- Round 2: sent \$3.00 (60%), got back \$4.80 - good return
- Round 4: sent \$3.50 (70%), got back \$6.99 - excellent return

Both times I've sent, I've received good cooperation in return. The pattern suggests that sending around 60-70% works well.

Round 6 now - new co-player (since pairings are randomized each round). I should continue with a cooperative strategy. I'll send around the same range, maybe \$3.25 to be between my previous sends.

Actually, let me stick with \$3.50 since that worked very well last time.

```
Usage: {
 "input_tokens": 1907,
 "output_tokens": 457,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_3 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:24:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_3"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_3": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

**\*\*{"send": 4}\*\***

##### reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground-it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 922,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Round 1 context:\*\*** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

**\*\*Current situation (Round 2):\*\***

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

```
{'return': 4.00}
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0

- They returned \$6.95
- My payoff:  $\$5 - \$4 + \$6.95 = \$7.95$
- Their payoff:  $\$12.0 - \$6.95 = \$5.05$

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 ( $2.34 \times 3$ )
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have  $\$2.66 + \$2.18 = \$4.84$ , I'd have  $\$7.02 - \$2.18 = \$4.84$
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

usage:

```
{
 "input_tokens": 783,
 "output_tokens": 979,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

**\*\*Round 1:\*\*** I sent \$4 and received a generous return (\$6.95)

**\*\*Round 2:\*\*** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

**\*\*Current situation (Round 3):\*\***

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

{'send': 4}

**reasoning:**

Let me analyze the situation:

**\*\*Round 1 (I was SENDER):\*\***

- I sent \$4.0
- They received \$12.0
- They returned \$6.95
- My payoff:  $\$1.0 + \$6.95 = \$7.95$

**\*\*Round 2 (I was RECEIVER):\*\***

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff:  $\$5 - \$2.34 + \$4.0 = \$6.66$

**\*\*Round 3 (I am SENDER):\*\***

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation
- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

**usage:**

```
{
 "input_tokens": 1097,
 "output_tokens": 541,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:



**\*\*Round 3 context:\*\*** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

**\*\*Current situation (Round 4):\*\***

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with \$0.79 + \$8.00 = \$8.79 total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

{'return': 8.00}

**reasoning:**

Let me analyze this situation:

**\*\*Round 3:\*\*** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

**\*\*Current situation (Round 4):\*\***

- My co-player sent \$4.21 (84% of their endowment) - very high trust!
- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21
- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with \$0.79 + \$8.00 = \$8.79
- I keep \$12.63 - \$8.00 = \$4.63
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left. If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total. They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received.

...

**usage:**

```
{
 "input_tokens": 1419,
 "output_tokens": 685,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.21, it became \$12.629999999999999, and you returned \$8.0. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$24.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my history:

**\*\*Pattern as SENDER:\*\*** I've sent \$4 in both previous sender rounds (R1 and R3), receiving generous returns (\$6.95 and \$7.98)

**\*\*Pattern as RECEIVER:\*\*** I've returned 57-63% of what I received, maintaining reciprocity

**\*\*Current situation (Round 5):\*\***

- New co-player (different from previous rounds)
- I'm the sender with \$5

My consistent strategy of sending \$4 has worked well:

- It signals strong cooperation
- Creates substantial mutual benefit potential (\$12 after 3x)
- Has been rewarded with good returns

I'll maintain this approach with my new partner.

{'send': 4}

**reasoning:**

Looking at my history:

**\*\*Round 1:\*\*** I sent \$4, received \$6.95 back ? Payoff \$7.95

**\*\*Round 2:\*\*** Received \$7.02, returned \$4.00 ? Payoff \$3.02

**\*\*Round 3:\*\*** I sent \$4, received \$7.98 back ? Payoff \$8.98

**\*\*Round 4:\*\*** Received \$12.63, returned \$8.00 ? Payoff \$4.63

**\*\*Current situation (Round 5):\*\***

- New co-player (no history with this specific agent)
- I'm the sender again

My pattern has been:

- Send \$4 consistently when I'm the sender
- Return fairly (57-63% of what I receive) when I'm the receiver
- This has worked well, with generous returns when I've been the sender

The strategy seems successful. I should continue sending \$4 to:

1. Maintain consistency
2. Signal cooperation from the start
3. Create opportunities for mutual benefit
4. The high amounts have been reciprocated well so far

I'll send \$4 again.

**usage:**

```
{
 "input_tokens": 1757,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$32.42.

This round, you are the RECEIVER against your current co-player. They sent you \$2.87, so you received \$8.61. How much do you return? (0-8.61)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Round 5 context:\*\*** I sent \$4.00 and received \$6.84 back (57% return rate)

**\*\*Current situation (Round 6):\*\***

- My co-player sent \$2.87 (57% of their endowment) - moderate trust
- I received \$8.61
- They kept \$2.13

**\*\*My return pattern as RECEIVER:\*\***

- Round 2: 57% return rate
- Round 4: 63% return rate

My co-player showed moderate but meaningful trust by sending 57% of their endowment. I'll return about 58% of what I received:

Returning \$5.00:

- They'd end with \$2.13 + \$5.00 = \$7.13 (good profit)
- I keep \$3.61 (also profitable)
- Maintains my cooperative pattern

{'return': 5.00}

**Reasoning:**

Let me analyze the situation:

**\*\*Round 5 context:\*\*** I sent \$4.00 and received \$6.84 back (57% of the \$12 they received)

**\*\*Current situation (Round 6):\*\***

- My co-player sent \$2.87 (57% of their endowment)
- I received \$8.61
- They kept \$2.13

**\*\*History of my returns as RECEIVER:\*\***

- Round 2: Returned \$4.00 out of \$7.02 (57%)
- Round 4: Returned \$8.00 out of \$12.63 (63%)

My co-player sent about 57% of their endowment, which is a moderate level of trust - less than the very high trust shown in Round 4 (84%) but still meaningful.

If I apply a similar return rate of around 55-60% of what I received:

- 55% of \$8.61 = \$4.74
- 60% of \$8.61 = \$5.17

Let me consider what makes sense:

- Returning \$5.00 would give them \$2.13 + \$5.00 = \$7.13 total (better than their initial \$5)
- I'd keep \$3.61 (less than \$5 but still profitable)
- This represents 58% return rate, consistent with my previous behavior

I think \$5.00 is a good, fair return that:

1. Rewards their trust
2. Gives them a meaningful profit
3. Maintains my cooperative reputation
4. Is consistent with my 55-60% return pattern

Usage: {

```

 "input_tokens": 2055,
 "output_tokens": 596,
 "reasoning_tokens": 0
 }
```

=====

**ROUND 7**

=====

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Receiver, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Receiver

**Pairings:**

- dyad\_1: Agent\_2=Sender, Agent\_5=Receiver
- dyad\_2: Agent\_6=Sender, Agent\_7=Receiver
- dyad\_3: Agent\_3=Sender, Agent\_8=Receiver
- dyad\_4: Agent\_4=Sender, Agent\_1=Receiver

**Parsed game decisions:**

- dyad\_1: Agent\_2 sent \$4; received \$12; Agent\_5 returned \$5.5; payoffs Agent\_2=\$6.5, Agent\_5=\$6.5;
- noise/visible: sent visible=\$3.28, received visible=\$9.84, returned visible=\$5.91

```

 Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_6 sent $4; received $12; Agent_7 returned $6; payoffs Agent_6=$7, Agent_7=$6;
noise/visible: sent visible=$3.72, received visible=$11.16, returned visible=$6.79
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_3 sent $4; received $12; Agent_8 returned $7; payoffs Agent_3=$8, Agent_8=$5;
noise/visible: sent visible=$4.73, received visible=$14.19, returned visible=$7.55
 Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_4 sent $4; received $12; Agent_1 returned $4.5; payoffs Agent_4=$5.5, Agent_1=$7.5;
noise/visible: sent visible=$3.29, received visible=$9.87, returned visible=$5.44
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 12.0, "min": 0, "raw": 4.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
 "Agent_1": 42.2,
 "Agent_2": 40.3,
 "Agent_3": 43.5,
 "Agent_4": 43.7,
 "Agent_5": 40.4,
 "Agent_6": 44.2,
 "Agent_7": 40.7,
 "Agent_8": 43.0
}
Visible balances after round: {
 "Agent_1": 41.44,
 "Agent_2": 43.33999999999999,
 "Agent_3": 44.58,
 "Agent_4": 43.76,
 "Agent_5": 39.620000000000005,
 "Agent_6": 42.21,
 "Agent_7": 44.22,
 "Agent_8": 43.91
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:24:10  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {

```

 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_5"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
 }

```

### Exact input messages sent to model:

#### Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.82, that is 76.4% of their total endowment. You received \$11.459999999999999. How much do you return to the sender? (0-11.459999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.82 (76.4% of their endowment)
- I received \$11.46 after the 3x multiplier
- The sender currently has only \$1.18 left

I want to reciprocate this trust appropriately. The sender took a risk and trusted me with most of their endowment.

If I return \$6.00:

- The sender ends with:  $\$1.18 + \$6.00 = \$7.18$  (better than their initial \$5)
- I end with:  $\$11.46 - \$6.00 = \$5.46$

This rewards their trust while still leaving both of us better off than if no exchange had occurred.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.82 (which is 76.4% of their \$5 endowment)
- I received \$11.46 ( $3.82 \times 3 = 11.46$ )
- I need to decide how much to return (between \$0 and \$11.46)

The sender showed significant trust by sending 76.4% of their endowment. Let me think about what would be fair:

The sender:

- Started with \$5
- Sent \$3.82
- Currently has:  $\$5 - \$3.82 = \$1.18$

If I return nothing:

- Sender ends with: \$1.18
- I end with: \$11.46

If I return half (\$5.73):

- Sender ends with:  $\$1.18 + \$5.73 = \$6.91$
- I end with:  $\$11.46 - \$5.73 = \$5.73$

If I return \$6.82 (so sender gets back double what they sent):

- Sender ends with:  $\$1.18 + \$6.82 = \$8.00$
- I end with:  $\$11.46 - \$6.82 = \$4.64$

If I return \$7.64 (so we both end up with equal amounts):

- Sender ends with:  $\$1.18 + \$7.64 = \$8.82$
- I end with:  $\$11.46 - \$7.64 = \$3.82$

Let me think about fairness and reciprocity. The sender showed high trust. A common approach in trust games is to:

1. Ensure the sender doesn't lose money from trusting
2. Share the gains reasonably

The total pool is now \$1.18 (what sender kept) + \$11.46 (what I have) = \$12.64

To split equally, each of us should get \$6.32, which means returning \$5.14. I could also aim for a solid return on their investment-returning \$5-6 would give them a reasonable gain while I keep a meaningful amount. I'll return \$5.50 as a balanced choice that reciprocates their trust without leaving me with too little....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 927,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```json

```
{"send": 4.0}
'''
```

reasoning:

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER
- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

usage:

```
{
  "input_tokens": 855,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings
3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

usage:

```
{
  "input_tokens": 1089,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.

Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This rewards their trust while both of us benefit from the exchange.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has $\$5 - \$3.35 = \$1.65$ left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust

(67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with: $\$1.65 + \$5.00 = \$6.65$ (profit of \$1.65)
- I end with: $\$10.05 - \$5.00 = \$5.05$

Or if I return \$6.50:

- Sender ends with: $\$1.65 + \$6.50 = \$8.15$ (profit of \$3.15)
- I end with: $\$10.05 - \$6.50 = \$3.55$

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

usage:

```
{
  "input_tokens": 1315,
  "output_tokens": 602,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.35, it became \$10.05, and you returned \$6.0. Your visible payoff was \$4.050000000000001.

Your total visible earnings across all rounds are \$25.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

- When I sent \$4.0 in previous rounds, I received back \$6.29 and \$7.32 - both profitable outcomes
- My cooperative strategy has been working well, building trust

I'll continue with the same successful approach of sending a substantial amount to encourage reciprocity.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my track record:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.0, kept \$5.46
- Round 2 (SENDER): Sent \$4.0, received back \$6.29, kept \$7.29
- Round 3 (SENDER): Sent \$4.0, received back \$7.32, kept \$8.32
- Round 4 (RECEIVER): Received \$10.05, returned \$6.0, kept \$4.05

Total visible earnings: \$25.12

My cooperative strategy of sending \$4.0 has been working well - both times I sent \$4.0, I received good returns (\$6.29 and \$7.32), both yielding solid profits. As a receiver, I've been fair by returning \$6.0 when others showed trust.

I'm now paired with a new co-player for round 5 as sender. Given my successful track record with cooperation, I should continue this strategy.

Sending \$4.0 again makes sense - it shows significant trust (80% of endowment) and has consistently led to profitable outcomes for me.

usage:

```
{
  "input_tokens": 1608,
```

```
    "output_tokens": 357,  
    "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.93 to you. Your visible payoff was \$6.93.

Your total visible earnings across all rounds are \$32.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

Consistent with my previous cooperative approach, I'll return a fair amount that rewards their trust.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (good profit)
- I end with: \$10.38 - \$6.00 = \$4.38

```
'''json  
{"return": 6.0}  
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

My pattern has been to return around \$6.00 when I receive amounts in this range (around \$10-11). This has been fair and consistent.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (profit of \$2.54)
- I end with: \$10.38 - \$6.00 = \$4.38

This is consistent with my previous behavior and rewards their trust appropriately.

usage:

```
{  
  "input_tokens": 1837,  
  "output_tokens": 332,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.46, it became \$10.379999999999999, and you returned \$6.0. Your visible payoff was \$4.379999999999999.

Your total visible earnings across all rounds are \$36.429999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent strategy:

- I've been sending \$4.0 as sender and it has consistently resulted in profitable returns
- My cooperative approach has been working well across different co-players

I'll maintain this successful strategy.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my history:

- I've been consistently sending \$4.0 as sender
- When receiving, I've been consistently returning \$6.0
- My strategy has been working well overall

My recent returns as sender:

- Round 2: sent \$4.0, received back \$6.29 (profit: \$2.29)
- Round 3: sent \$4.0, received back \$7.32 (profit: \$3.32)
- Round 5: sent \$4.0, received back \$5.93 (profit: \$1.93)

All profitable, showing that the cooperative strategy is paying off. I should continue with the same approach.

```
Usage: {
  "input_tokens": 2121,
  "output_tokens": 225,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-08-12T09:24:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_5"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.66, that is 73.2% of their total endowment. You received \$10.98. How much do you return to the sender? (0-10.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.66 (73.2% of their endowment)
- I received \$10.98 from the 3x multiplication
- The sender currently has \$1.34 remaining

The sender demonstrated high trust, so I want to reciprocate appropriately. If I return \$6.00:

- The sender would end with \$7.34 total (a profit of \$2.34)
- I would keep \$4.98 (a good return)
- This rewards their trust while maintaining fairness

This seems like a balanced approach that encourages future cooperation.

```
'''json
{"return": 6.00}
'''
```

reasoning:

I'm the RECEIVER in Round 1. Let me understand the situation:

- The sender sent \$3.66 (which is 73.2% of their \$5 endowment)
- I received \$10.98 (which is $3 \times \$3.66$)
- I need to decide how much to return (between \$0 and \$10.98)

Let me think about the fairness and strategic considerations:

1. The sender sent a significant portion of their endowment (73.2%), showing trust
2. After sending \$3.66, the sender has $\$5 - \$3.66 = \$1.34$ left
3. I received \$10.98
4. If I want to be fair and encourage cooperation:
 - One approach: We could split the total gains equally
 - Total money in the system: $\$5$ (sender initial) + $\$10.98$ (what I received) = $\$15.98$
 - If we split equally: $\$15.98 / 2 = \7.99 each
 - Sender currently has: $\$1.34$
 - So I would return: $\$7.99 - \$1.34 = \$6.65$

5. Another approach: Return roughly the same proportion they sent

- They sent 73.2% of their endowment
- 73.2% of \$10.98 = \$8.04
- But this seems too much and would leave me with only \$2.94

6. A middle ground approach:

- The sender showed high trust by sending 73.2%
- I should return enough to reward that trust
- If I return around \$5-6, that would:
 - * Give the sender a good return on their trust
 - * Still leave me with a reasonable amount

7. Let me calculate what happens with different return amounts:

- If I return \$5: Sender ends with \$6.34 (profit of \$1.34), I end with \$5.98 (profit of \$5.98)
- If I return \$6: Sender ends with \$7.34 (profit of \$2.34), I end with \$4.98 (profit of \$4.98)
- If I return \$6.65: Sender ends with \$7.99 (profit of \$2.99), I end with \$4.33 (profit of \$4.33)

The equal split approach seems most fair since it rewards the sender's trust while maintaining reasonable outcomes for both parties....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1097,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98
- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)

2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)
2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)
- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)
- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98

- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)
- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

usage:

```
{
  "input_tokens": 1092,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern. I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98

- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving. This signals strong trust while being neither too conservative nor too aggressive.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.99 to you. Your visible payoff was \$8.49.

Your total visible earnings across all rounds are \$25.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is moderate trust - not as high as other rounds, but still showing willingness to cooperate by sending more than half their endowment.

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This fairly rewards their trust

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)

- I received \$8.37
- The sender currently has \$2.21 remaining

This is lower trust than I've seen in previous rounds (73.2%, 79.8% in earlier rounds). However, it's still a majority of their endowment, showing moderate trust.

I should still reciprocate fairly. Let me think about what would be appropriate:

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This is fair and rewards their trust

If I return \$4.00:

- The sender ends with \$6.21 (profit of \$1.21)
- I keep \$4.37
- This is also reasonable

Given that they sent a moderate amount (not as high as previous senders), I should return a moderate but fair amount. I'll return \$4.50 to maintain a cooperative reputation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1640,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.79, it became \$8.370000000000001, and you returned \$4.5. Your visible payoff was \$3.870000000000001.

Your total visible earnings across all rounds are \$29.610000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my history:

- Round 2: I sent \$3.00 (60%), received \$4.80 back - good cooperation
- Round 4: I sent \$3.50 (70%), received \$6.99 back - excellent cooperation
- Overall, higher sends have been rewarded

With a new co-player, I'll maintain a cooperative approach. Sending \$3.50 (70%) worked very well last time, so I'll repeat that strategy.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review my history:

Round 1: Receiver - received from sender who sent \$3.66 (73.2%), I returned \$6.00, earned \$4.98

Round 2: Sender - sent \$3.00 (60%), received \$4.80 back, earned \$6.80

Round 3: Receiver - received from sender who sent \$3.99 (79.8%), I returned \$6.50, earned \$5.47

Round 4: Sender - sent \$3.50 (70%), received \$6.99 back, earned \$8.49

Round 5: Receiver - received from sender who sent \$2.79 (55.8%), I returned \$4.50, earned \$3.87

My sending history:

- Round 2: sent \$3.00 (60%), got back \$4.80 - good return
- Round 4: sent \$3.50 (70%), got back \$6.99 - excellent return

Both times I've sent, I've received good cooperation in return. The pattern suggests that

sending around 60-70% works well.

Round 6 now - new co-player (since pairings are randomized each round). I should continue with a cooperative strategy. I'll send around the same range, maybe \$3.25 to be between my previous sends.

Actually, let me stick with \$3.50 since that worked very well last time.

usage:

```
{
  "input_tokens": 1907,
  "output_tokens": 457,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.17 to you. Your visible payoff was \$5.67.

Your total visible earnings across all rounds are \$35.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This shows good trust - about 2/3 of their endowment. I want to reciprocate fairly.

If I return \$5.50:

- The sender ends with \$7.22 (profit of \$2.22)
- I keep \$4.34
- This rewards their trust appropriately

```
'''json
{"return": 5.50}
'''
```

Reasoning:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This is moderate-high trust. The sender showed decent cooperation by sending about 2/3 of their endowment.

Let me think about a fair return:

- If I return \$5.50:
 - Sender ends with \$7.22 (profit of \$2.22)
 - I keep \$4.34
 - This rewards their trust fairly

That seems reasonable - the sender gets a good profit, I keep a decent amount, and it encourages future cooperation.

Usage: {

```
"input_tokens": 2163,
"output_tokens": 295,
"reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-08-12T09:24:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_7"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with prudent risk management . Since this is a multi-round game, establishing cooperation early could be beneficial for future rounds.

The 3x multiplier means that sending money creates value - if I send \$3, the receiver gets \$9, creating \$4 of new value. If the receiver is reciprocal, they could return enough that we both benefit.

However, as a first interaction with an unknown co-player, I don't want to be overly naive . I'll send a moderate amount that signals willingness to cooperate while keeping some insurance.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm playing as the SENDER.

Let me think about the strategic considerations:

1. I have \$5 and can send any amount from \$0 to \$5
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. My final payoff = \$5 - sent + returned
5. Receiver's payoff = (sent × 3) - returned

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

However, there are good reasons to send money:

- If the receiver is reciprocal/trustworthy, they might return more than I sent
- Sending creates a larger total pie (due to the 3x multiplier)
- This is a multi-round game, so building trust could pay off in future rounds
- The receiver might reciprocate generosity

There's also mention of "communication noise" - amounts seen may differ from what was actually sent/returned. This could complicate trust-building.

For the first round, I need to establish some level of cooperation without being too naive . Common strategies:

- Send \$0 (no trust, no gain)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - maximum trust/cooperation

Given that:

1. This is round 1, so I have no information about my co-player
2. It's a multi-round game (though I don't know how many rounds total)
3. Building trust could lead to better outcomes
4. The 3x multiplier means cooperation creates value

I'll send a moderate-to-high amount to signal willingness to cooperate, but not everything . Sending \$3 or \$4 seems reasonable.

If I send \$3:

- Receiver gets \$9
- If they return \$4-5, I'm better off than keeping everything
- If they return less, I've learned something about them

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$4.95.
Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at this situation:

****Analysis:****

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

****Fairness considerations:****

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

****Decision:****

If I return \$3.20:

- Sender ends with: $\$5 - \$2.11 + \$3.20 = \6.09 (gain of $\sim \$1.09$)
- I keep: $\$6.33 - \$3.20 = \$3.13$ (gain of $\sim \$3.13$)

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER:
 - I sent \$3.0
 - It became \$9.0 for the receiver
 - They returned \$2.95 to me
 - My payoff: $\$5 - \$3 + \$2.95 = \4.95
 - The receiver kept: $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:
 - My co-player (a different agent) sent me \$2.11
 - I received \$6.33 (which is $\$2.11 \times 3$)
 - I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33
- If I want to be reciprocal and fair, I should return enough that:
 1. The sender ends up better off than if they had sent nothing
 2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with: $\$5 - \$2.11 + \$3.165 = \6.055
- I keep: $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:
- Sender ends up with: \$5 - \$2.11 + \$2.09 = \$4.98
- I keep: \$6.33 - \$2.09 = \$4.24

The sender invested 42% of their endowment, so I should consider whether they break even or better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here.

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us...

usage:

```
{  
  "input_tokens": 780,  
  "output_tokens": 1431,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Analysis:

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

Considerations:

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

```
'''json  
{ "send": 3 }  
'''
```

reasoning:

Let me analyze the situation:

Previous rounds:

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

This round:

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

Considerations:

1. This is a different co-player than in rounds 1 and 2

2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

****Strategy:****

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

usage:

```
{
  "input_tokens": 1133,
  "output_tokens": 554,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Analysis:****

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

****Considerations:****

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

****Calculation:****

If I return \$6.80:

- Sender ends with: \$5 - \$4.14 + \$6.80 = \$7.66 (gain of \$2.66)
- I keep: \$12.42 - \$6.80 = \$5.62 (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 6.80}
'''
```

reasoning:

Let me analyze this situation:

****What happened:****

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

****Considerations:****

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

****Calculation options:****

If I return \$6.00:

- Sender ends with: $\$5 - \$4.14 + \$6.00 = \6.86 (gain of \$1.86)
- I keep: $\$12.42 - \$6.00 = \$6.42$ (gain of \$6.42)

If I return \$7.00:

- Sender ends with: $\$5 - \$4.14 + \$7.00 = \7.86 (gain of \$2.86)
- I keep: $\$12.42 - \$7.00 = \$5.42$ (gain of \$5.42)

If I return \$6.50:

- Sender ends with: $\$5 - \$4.14 + \$6.50 = \7.36 (gain of \$2.36)
- I keep: $\$12.42 - \$6.50 = \$5.92$ (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.

Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd gain \$3.36 while...

usage:

```
{
  "input_tokens": 1425,
  "output_tokens": 847,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$6.8. Your visible payoff was \$5.619999999999998.

Your total visible earnings across all rounds are \$20.38.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

My co-player sent \$3.63 (73% of their endowment), showing substantial trust. I received \$10.89 after the 3x multiplier.

****Considerations:****

- They sent a significant portion of their endowment
- Both of us should benefit from this cooperation
- I want to reward their trust appropriately

****Calculation:****

If I return \$5.80:

- Sender ends with: $\$5 - \$3.63 + \$5.80 = \7.17 (gain of \$2.17)
- I keep: $\$10.89 - \$5.80 = \$5.09$ (gain of \$5.09)

Both of us gain substantially, and the sender is well-rewarded for their trust.

```
```json
{"return": 5.80}
```
```

reasoning:

****Analysis:****

My co-player sent \$3.63 (72.6% of their endowment), showing substantial trust.
I received \$10.89 after the 3x multiplier.

****Considerations:****

- They sent a good amount, demonstrating trust
- I want to reward this appropriately
- Both of us should benefit from the cooperation

****Calculation:****

If I return around \$5.50:

- Sender ends with: $\$5 - \$3.63 + \$5.50 = \6.87 (gain of \$1.87)
- I keep: $\$10.89 - \$5.50 = \$5.39$ (gain of \$5.39)

Both benefit, sender is rewarded for trust.

Let me think about a slightly different amount. The sender sent 72.6% of their endowment, which is quite high. I should return enough that they benefit well.

If I return \$5.80:

- Sender ends with: $\$5 - \$3.63 + \$5.80 = \7.17 (gain of \$2.17)
- I keep: $\$10.89 - \$5.80 = \$5.09$ (gain of \$5.09)

This seems fair - both of us gain substantially, and the sender is well-rewarded for their high trust.

usage:

```
{
  "input_tokens": 1758,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.63, it became \$10.89, and you returned \$5.8. Your visible payoff was \$5.0900000000000001.

Your total visible earnings across all rounds are \$25.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

I'm facing a new co-player as sender. My previous sender experiences:

- Round 1: Sent \$3, received \$2.95 back (payoff \$4.95)
- Round 3: Sent \$3, received \$4.68 back (payoff \$6.68)

Both times, sending \$3 led to positive outcomes with reasonable reciprocation.

****Strategy:****

- \$3 creates \$9 for the receiver via the 3x multiplier
- This amount signals willingness to cooperate
- It balances trust-building with prudent risk management
- Consistency may help establish a cooperative reputation

I'll continue with my established approach.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

I'm the sender this round with a new co-player. Let me review my experience so far:

Sender rounds:

- Round 1: Sent \$3, got back \$2.95 (payoff \$4.95)
- Round 3: Sent \$3, got back \$4.68 (payoff \$6.68)

Receiver rounds:

- Round 2: Received \$6.33 from \$2.11 sent, returned \$3.20 (payoff \$3.13)
- Round 4: Received \$12.42 from \$4.14 sent, returned \$6.80 (payoff \$5.62)
- Round 5: Received \$10.89 from \$3.63 sent, returned \$5.80 (payoff \$5.09)

As sender, I've consistently sent \$3 and received reasonable returns. My co-players as receivers have varied in how much they sent initially.

This is round 6, so I'm establishing patterns. I want to continue building cooperative norms. Sending \$3 has worked well for me in the past - it's enough to create value through the 3x multiplier while not being overly risky.

I'll stick with my strategy of sending \$3.

usage:

```
{
  "input_tokens": 2077,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$6.95 to you. Your visible payoff was \$8.95.

Your total visible earnings across all rounds are \$34.42.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

This is the same co-player from Round 6! They demonstrated exceptional generosity by returning \$6.95 on my \$3 send.

****What happened last round:****

- I sent \$3, they received \$9
- They returned \$6.95, keeping only \$2.05 for themselves
- I gained \$3.95 while they gained just \$2.05

****Response:****

Such generous reciprocation deserves increased trust. They clearly value cooperation. Since I'm playing with the same partner again, I should signal that I recognize and appreciate their trustworthiness.

I'll increase my send to \$4, which creates \$12 for them and shows I'm willing to escalate cooperation with a proven trustworthy partner.

```
'''json
{"send": 4}
'''
```

Reasoning:

This is very interesting - I'm playing against the SAME co-player as the previous round! They returned \$6.95 on my \$3 send, which is extremely generous (they gave me back more than double what I sent, even accounting for the 3x multiplier).

Let me calculate what happened:

- I sent \$3, they received \$9
- They returned \$6.95
- I ended with: \$5 - \$3 + \$6.95 = \$8.95
- They kept: \$9 - \$6.95 = \$2.05

So they only kept \$2.05 while I gained \$3.95! This is a very cooperative partner.

Since I'm playing with the same person again and they were so generous, I should reciprocate

his trust by sending more. This could establish a strong cooperative relationship.

Options:

1. Send \$3 again - safe, but doesn't fully reciprocate their generosity
2. Send \$4 - shows increased trust based on their behavior
3. Send \$5 - maximum trust, but risky

Given their extremely generous return of \$6.95, I think sending \$4 is appropriate. It shows I valued their cooperation and am willing to increase my trust, while still maintaining some prudence.

```
Usage: {
  "input_tokens": 2371,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-08-12T09:24:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_7"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

Sending money creates potential for mutual benefit since it triples. If I send \$3, my co-player receives \$9. If they return about half, we both end up better off than if I sent nothing.

Starting with a moderate amount shows willingness to cooperate while managing risk in this initial interaction.

****My decision:****

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender and need to decide how much to send.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual gain
2. The receiver then decides how much to return
3. There's communication noise, so amounts may appear different than actual
4. This is a multi-round game with 8 agents, so reputation and reciprocity could matter
5. Pairings are randomized each round

In the first round, I don't have any information about my co-player's behavior. However, sending some amount demonstrates trust and could encourage reciprocity.

The economically rational move if this were one-shot and my co-player were purely self-interested would be to send \$0 (since they could keep everything). But in multi-round settings with potential for reputation building, cooperation often emerges.

Common strategies in trust games:

- Send a moderate amount (like \$2-3) to signal willingness to cooperate while limiting risk
- Send everything (\$5) to maximize potential joint gain and signal strong trust
- Send nothing (\$0) to avoid any risk

Given this is round 1 and I want to:

1. Signal willingness to cooperate
2. Not be too naive
3. Create potential for mutual benefit
4. Establish a cooperative tone for future rounds

I'll send \$3. This shows trust and good faith while not being overly risky. If the receiver reciprocates fairly, we both benefit. This amount sent becomes \$9 for the receiver, so they could return \$4-5 and we'd both be better off than if I sent nothing.

Wait, let me recalculate:

- I send \$3
- Receiver gets $3 \times 3 = 9$
- I keep $5 - 3 = 2$ initially
- If receiver returns \$3, I end with $2 + 3 = 5$ (same as start, receiver gets \$6)

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received \$12.120000000000001. How much do you return? (0-12.120000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: $0.96 + 6.50 = 7.46$
- I end with: $12.12 - 6.50 = 5.62$

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: $5 - 3 + 5.68 = 7.68$
- The receiver kept $9 - 5.68 = 3.32$

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received $4.04 \times 3 = 12.12$
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves

es reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be \$7.27 - \$7.88
- That would leave them with \$0.96 + return, and me with \$12.12 - return

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with \$0.96 + \$6 = \$6.96, I end with \$12.12 - \$6 = \$6.12
- If I return \$7, they end with \$7.96, I end with \$5.12

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: \$0.96 + \$6.50 = \$7.46

...

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible payoff was \$5.620000000000001.

Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with: \$0.62 + \$7.00 = \$7.62
- I end with: \$13.14 - \$7.00 = \$6.14

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.

I received \$13.14

They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:

- They end with: \$0.62 + \$7.00 = \$7.62
- I end with: \$13.14 - \$7.00 = \$6.14

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:

1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:

- Their total: $\$0.62 + \$7.00 = \$7.62$ (profit of \$2.62)
- My total: $\$13.14 - \$7.00 = \$6.14$ (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:

- Their total: $\$0.62 + \$7.50 = \$8.12$ (profit of \$3.12)
- My total: $\$13.14 - \$7.50 = \$5.64$ (profit of \$0.64)

I think \$7.00 is reasonable. It gives them a nice profit while I also benefit from the cooperation.

usage:

```
{
  "input_tokens": 1061,
  "output_tokens": 567,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.38, it became \$13.14, and you returned \$7.0. Your visible payoff was \$6.1400000000000001.

Your total visible earnings across all rounds are \$19.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- In Round 1, I sent \$3 and received good reciprocation (\$5.68 back)
- In Rounds 2 and 3, co-players showed high trust by sending \$4.04 and \$4.38
- I've been reciprocating fairly, returning about half of what I received

This is a new co-player, but the general pattern suggests cooperation pays off. I'll send a high amount to match the trust I've seen from others.

```
'''json
{"send": 4}
'''
```

reasoning:

I'm now the SENDER in Round 4. Let me think about what I've learned:

Round 1: I sent \$3, received \$5.68 back, ended with \$7.68

Round 2: I received \$12.12 (they sent \$4.04), returned \$6.50, ended with \$5.62

Round 3: I received \$13.14 (they sent \$4.38), returned \$7.00, ended with \$6.14

My recent experiences as a receiver show co-players sending high amounts (\$4.04 and \$4.38). When I was a sender in Round 1, I sent \$3 and got \$5.68 back, which was good reciprocation.

Given that I've seen high trust from others and I've been reciprocating fairly, I should continue to show trust. This is a new co-player this round (since pairings are randomized), so I don't have specific history with them, but the general pattern suggests cooperation is beneficial.

I could send:

- \$3: Moderate trust, similar to Round 1

- \$4: High trust, similar to what I've received from others
- \$5: Maximum trust

Given that I've been receiving high amounts from others (around \$4+), and I've been reciprocating fairly well, I should send a similarly high amount to maintain the cooperative dynamic. Sending \$4 shows significant trust while not being overly risky.

If I send \$4:

- They receive \$12
- If they return around 50-60%, I'd get back \$6-7.20
- I'd end with roughly \$7-8.20

This seems reasonable and aligns with the cooperative pattern I've been experiencing.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.06 to you. Your visible payoff was \$8.059999999999999.

Your total visible earnings across all rounds are \$27.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my recent rounds:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back - excellent reciprocation

The cooperation has been strong, with both co-players returning generously. This suggests high trust sends are being rewarded.

With a new co-player this round, I'll continue the cooperative strategy that's been working well.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: Sent \$3, received \$5.68 back (good return)
- Round 2: Received \$12.12, returned \$6.50
- Round 3: Received \$13.14, returned \$7.00
- Round 4: Sent \$4, received \$7.06 back (very good return - they returned about 59% of what they got)

The cooperation has been strong across all rounds. People are sending high amounts and returning generously. My earnings have been good (\$27.50 total across 4 rounds with visible payoffs).

This is Round 5 with a new co-player. Given the positive cooperation pattern, I should continue with a high trust strategy. Sending \$4 worked well last round.

I'll send \$4 again to show trust and likely get good reciprocation.

usage:

```
{
  "input_tokens": 1602,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.12 to you. Your visible payoff was \$6.12.
Your total visible earnings across all rounds are \$33.62.

This round, you are the RECEIVER against your current co-player. They sent you \$3.98, so you received \$11.94. How much do you return? (0-11.94)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing strong trust
- I received \$11.94
- They kept only \$1.02

This is solid trust that deserves fair reciprocation. Based on my previous patterns as receiver, returning about 50-55% of what I received has worked well.

If I return \$6.50:

- They end with: \$1.02 + \$6.50 = \$7.52
- I end with: \$11.94 - \$6.50 = \$5.44

Both of us gain from cooperation.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing good trust
- I received \$11.94
- They kept only \$1.02

Previous rounds have shown:

- When I was receiver in Round 2, I got \$12.12 and returned \$6.50
- When I was receiver in Round 3, I got \$13.14 and returned \$7.00
- These were fair returns that both parties benefited from

For this round:

If I return \$6.50:

- They end with: \$1.02 + \$6.50 = \$7.52
- I end with: \$11.94 - \$6.50 = \$5.44

Both benefit from cooperation. This seems like a fair reciprocation for their trust.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 360,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.98, it became \$11.94, and you returned \$6.5. Your visible payoff was \$5.4399999999999995.

Your total visible earnings across all rounds are \$39.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing continued trust
- I received \$11.16
- They kept \$1.28
- This is the same co-player from last round

They sent slightly less than last round (\$3.72 vs \$3.98), but still showing good cooperation.
I should maintain consistent reciprocation.

If I return \$6.00:

- They end with: $\$1.28 + \$6.00 = \$7.28$
- I end with: $\$11.16 - \$6.00 = \$5.16$

This maintains roughly the same proportion as last round (~54% return rate).

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing good trust
- I received \$11.16
- They kept \$1.28

This is the same co-player from Round 6 where they sent \$3.98 and I returned \$6.50.

Now they sent slightly less (\$3.72 vs \$3.98). This could be:

1. Random variation/noise
2. A slight reduction in trust based on my previous return
3. Just their strategy this round

I should continue being fair and reciprocating appropriately. Returning about 50-55% of what I received seems reasonable.

If I return \$6.00:

- They end with: $\$1.28 + \$6.00 = \$7.28$
- I end with: $\$11.16 - \$6.00 = \$5.16$

Both still profit from cooperation. This is slightly less than before (I returned \$6.50 on \$11.94), but proportionally similar given they sent less this time.

Let me calculate the proportions:

- Round 6: Returned \$6.50 out of \$11.94 = 54.4%
- Round 7: Return \$6.00 out of \$11.16 = 53.8%

That's very consistent proportionally.

```
Usage: {
  "input_tokens": 2151,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-08-12T09:24:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_8"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_8": "trustee"
    }
  }
}
```

```

    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Here's my reasoning:

- Sending more creates a larger total pool (multiplication by 3x)
- This maximizes potential mutual benefit
- But it requires trusting the receiver to return a fair amount
- Sending \$0 is safe but misses the opportunity for mutual gain

I'll send a substantial amount to signal cooperation and willingness to trust, while keeping a small buffer. This gives my co-player a meaningful amount to work with (\$12 total after multiplication) and demonstrates good faith.

```
**{"send": 4}**
```

reasoning:

This is Round 1 and I'm the SENDER. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = \$5 - sent + returned
- Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

1. Whether I trust the receiver to return a fair amount
2. What "fair" means in this context
3. The potential for maximizing joint payoffs vs. individual payoffs

If I send \$5:

- Receiver gets \$15 (5×3)
- If receiver returns half (\$7.50), I end with \$7.50, receiver ends with \$7.50
- If receiver returns nothing, I end with \$0, receiver ends with \$15
- If receiver returns \$10, I end with \$10, receiver ends with \$5

If I send \$0:

- I keep \$5, receiver gets \$0
- No trust, no risk, but no potential gains

A reasonable cooperative strategy would be to send a significant amount (like \$3-\$5) and hope the receiver reciprocates by returning enough to make us both better off than the initial \$5 each.

Since this is Round 1 and I have no information about the receiver's behavior, I should consider:

1. Starting with moderate trust to signal cooperation
2. The receiver might be more likely to reciprocate if they see I'm willing to trust

A common approach in these games is to send around \$3-\$4 to show trust but not complete vulnerability.

However, there's also a note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Given this is a multi-round game with different pairings, building a reputation for cooperation early might be beneficial if there's any information sharing (though agent names are hidden).

I'm torn between demonstrating strong trust by sending \$5 to maximize the total pool, versus taking a more cautious approach. Sending \$4 feels like a good middle ground—it signals genuine cooperation without leaving me completely exposed if the receiver acts selfishly....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 922,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 1 context:**** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

****Current situation (Round 2):****

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

{'return': 4.00}

reasoning:

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0
- They returned \$6.95
- My payoff: $\$5 - \$4 + \$6.95 = \7.95
- Their payoff: $\$12.0 - \$6.95 = \$5.05$

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 (2.34×3)
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have $\$2.66 + \$2.18 = \$4.84$, I'd have $\$7.02 - \$2.18 = \$4.84$
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 979,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

****Round 1:**** I sent \$4 and received a generous return (\$6.95)

****Round 2:**** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

****Current situation (Round 3):****

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2.

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

{'send': 4}

reasoning:

Let me analyze the situation:

****Round 1 (I was SENDER):****

- I sent \$4.0
- They received \$12.0
- They returned \$6.95
- My payoff: $\$1.0 + \$6.95 = \$7.95$

****Round 2 (I was RECEIVER):****

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff: $\$5 - \$2.34 + \$4.0 = \6.66

****Round 3 (I am SENDER):****

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation

- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Round 3 context:**** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with \$0.79 + \$8.00 = \$8.79 total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

{'return': 8.00}

reasoning:

Let me analyze this situation:

****Round 3:**** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - very high trust!
- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21
- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with $\$0.79 + \$8.00 = \$8.79$
- I keep $\$12.63 - \$8.00 = \$4.63$
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left . If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total.
They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received.

...

usage:

```
{
  "input_tokens": 1419,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.21, it became \$12.629999999999999, and you returned \$8.0. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$24.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my history:

****Pattern as SENDER:**** I've sent \$4 in both previous sender rounds (R1 and R3), receiving generous returns (\$6.95 and \$7.98)

****Pattern as RECEIVER:**** I've returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 5):****

- New co-player (different from previous rounds)
- I'm the sender with \$5

My consistent strategy of sending \$4 has worked well:

- It signals strong cooperation
- Creates substantial mutual benefit potential (\$12 after 3x)
- Has been rewarded with good returns

I'll maintain this approach with my new partner.

{'send': 4}

reasoning:

Looking at my history:

****Round 1:**** I sent \$4, received \$6.95 back ? Payoff \$7.95

****Round 2:**** Received \$7.02, returned \$4.00 ? Payoff \$3.02

****Round 3:**** I sent \$4, received \$7.98 back ? Payoff \$8.98

****Round 4:**** Received \$12.63, returned \$8.00 ? Payoff \$4.63

****Current situation (Round 5):****

- New co-player (no history with this specific agent)
- I'm the sender again

My pattern has been:

- Send \$4 consistently when I'm the sender
- Return fairly (57-63% of what I receive) when I'm the receiver
- This has worked well, with generous returns when I've been the sender

The strategy seems successful. I should continue sending \$4 to:

1. Maintain consistency
2. Signal cooperation from the start
3. Create opportunities for mutual benefit
4. The high amounts have been reciprocated well so far

I'll send \$4 again.

usage:

```
{
  "input_tokens": 1757,
  "output_tokens": 427,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$32.42.

This round, you are the RECEIVER against your current co-player. They sent you \$2.87, so you received \$8.61. How much do you return? (0-8.61)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% return rate)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment) - moderate trust
- I received \$8.61
- They kept \$2.13

****My return pattern as RECEIVER:****

- Round 2: 57% return rate
- Round 4: 63% return rate

My co-player showed moderate but meaningful trust by sending 57% of their endowment. I'll return about 58% of what I received:

Returning \$5.00:

- They'd end with \$2.13 + \$5.00 = \$7.13 (good profit)
- I keep \$3.61 (also profitable)
- Maintains my cooperative pattern

{'return': 5.00}

reasoning:

Let me analyze the situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% of the \$12 they received)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment)
- I received \$8.61
- They kept \$2.13

****History of my returns as RECEIVER:****

- Round 2: Returned \$4.00 out of \$7.02 (57%)
- Round 4: Returned \$8.00 out of \$12.63 (63%)

My co-player sent about 57% of their endowment, which is a moderate level of trust - less than the very high trust shown in Round 4 (84%) but still meaningful.

If I apply a similar return rate of around 55-60% of what I received:

- 55% of \$8.61 = \$4.74
- 60% of \$8.61 = \$5.17

Let me consider what makes sense:

- Returning \$5.00 would give them $\$2.13 + \$5.00 = \$7.13$ total (better than their initial \$5)
- I'd keep \$3.61 (less than \$5 but still profitable)
- This represents 58% return rate, consistent with my previous behavior

I think \$5.00 is a good, fair return that:

1. Rewards their trust
2. Gives them a meaningful profit
3. Maintains my cooperative reputation
4. Is consistent with my 55-60% return pattern

usage:

```
{
  "input_tokens": 2055,
  "output_tokens": 596,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.87, it became \$8.61, and you returned \$5.0. Your visible payoff was \$3.6099999999999994.

Your total visible earnings across all rounds are \$36.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

****As SENDER (Rounds 1, 3, 5):**** Consistently sent \$4, receiving returns ranging from \$6.84 to \$7.98

****As RECEIVER (Rounds 2, 4, 6):**** Returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 7):****

- New co-player
- I'm the sender with \$5

My consistent strategy of sending \$4 has:

- Established a cooperative reputation
- Generated good returns across all sender rounds
- Created mutual benefit

I'll continue this proven approach with my new partner.

```
{'send': 4}
```

Reasoning:

Looking at my history:

As SENDER:

- Round 1: sent \$4, received \$6.95 back
- Round 3: sent \$4, received \$7.98 back
- Round 5: sent \$4, received \$6.84 back

As RECEIVER:

- Round 2: received \$7.02, returned \$4.00
- Round 4: received \$12.63, returned \$8.00
- Round 6: received \$8.61, returned \$5.00

My strategy has been consistent - sending \$4 as sender, which has worked well. I've been getting good returns ranging from \$6.84 to \$7.98.

This is Round 7, I'm the sender again with a new co-player.

I should maintain my consistent cooperative strategy of sending \$4. This has been my standard approach and it signals trust and cooperation.

Usage: {

```
"input_tokens": 2393,
"output_tokens": 364,
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:24:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_8"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.45, that is 69.0% of their total endowment. You received \$10.350000000000001. How much do you return to the sender? (0-10.350000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.45 (69% of their endowment)
- I received \$10.35 after the 3x multiplication
- Total resources created: \$11.90 (\$1.55 remaining with sender + \$10.35 with me)

The sender was quite generous and cooperative. I want to reciprocate that trust and create a fair outcome.

If I return \$5.00:

- Sender ends with: $\$5 - \$3.45 + \$5.00 = \6.55
- I end with: $\$10.35 - \$5.00 = \$5.35$
- This rewards their trust while maintaining relatively balanced outcomes

{'return': 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender was given \$5
- They sent me \$3.45 (which is 69% of their \$5 endowment)
- This was multiplied by 3, so I received \$10.35 (approximately, with some floating point error showing as \$10.350000000000001)

Now I need to decide how much to return.

The sender showed significant trust by sending \$3.45 (69% of their endowment). This is a cooperative move.

Let me think about fair outcomes:

- If I return \$0: Sender gets $\$5 - \$3.45 + \$0 = \1.55 , I get \$10.35
- If I return half (\$5.175): Sender gets $\$5 - \$3.45 + \$5.175 = \6.725 , I get $\$10.35 - \$5.175 = \$5.175$
- If I return \$6.90 (2x what they sent): Sender gets $\$5 - \$3.45 + \$6.90 = \8.45 , I get $\$10.35 - \$6.90 = \$3.45$

Some considerations:

1. The sender was quite generous/trusting, sending 69% of their endowment
2. A fair split of the total pie might be appropriate
3. Total resources created: Sender has $\$5 - \$3.45 = \$1.55$ remaining, I have \$10.35 total = \$11.90 total
4. Equal split would be \$5.95 each

To give sender \$5.95 total: $\$5.95 = \$5 - \$3.45 + \text{return}$, so $\text{return} = \$5.95 - \$1.55 = \$4.40$

Alternatively, I could think about it as:

- They sent 69%, showing trust
- I should reciprocate with a meaningful return

...

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 807,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.0. Your visible payoff was \$5.350000000000001.

Your total visible earnings across all rounds are \$5.350000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- In the previous round, my co-player showed significant trust by sending 69% of their endowment
- I reciprocated fairly by returning a substantial amount
- Now with a new co-player, I want to establish cooperation

I'll send a generous amount to signal trust and cooperation, while being slightly more conservative since this is a new pairing.

{'send': 3.00}

reasoning:

I'm now the SENDER in Round 2. Let me think about this situation:

- In Round 1, I was the RECEIVER and someone sent me \$3.45 (69% of their endowment)
- I returned \$5.00, which was a fair reciprocation
- I kept \$5.35, and the sender ended with \$6.55

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. In the previous round, my co-player sent 69% of their endowment, showing significant trust
2. I rewarded that trust by returning about 48% of what I received
3. This is a new co-player, so I can't assume they'll behave the same way
4. However, I should probably maintain a cooperative stance to build trust in the experiment
5. There's communication noise, so amounts may differ from what was actually sent

Given that:

- The previous sender was generous (69%)
- I want to establish cooperation with this new co-player
- Being moderately generous but not overly risky seems reasonable

I'll send around 60% of my endowment, which would be \$3.00. This shows trust and cooperation while being slightly more conservative than what I received in the previous round.

Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I want to maintain a cooperative equilibrium in this multi-agent setting, I should probably match that level of trust or be similarly generous.

I'll send \$3.00 (60% of my endowment), which is a solid show of trust and cooperation.

usage:

```
{
  "input_tokens": 843,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.93 to you. Your visible payoff was \$5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:

If I return \$3.20:

- Sender ends with: \$5 - \$2.14 + \$3.20 = \$6.06
- I end with: \$6.42 - \$3.20 = \$3.22
- This creates relatively balanced outcomes and rewards their trust

{'return': 3.20}

reasoning:

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust
2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:

- Sender ends with: \$5 - \$2.14 + \$3.00 = \$5.86
- I end with: \$6.42 - \$3.00 = \$3.42
- This gives them a decent return on their trust

If I return \$3.50:

- Sender ends with: \$5 - \$2.14 + \$3.50 = \$6.36
- I end with: \$6.42 - \$3.50 = \$2.92
- This is more generous but leaves me with less than them

If I return \$3.20:

- Sender ends with: \$5 - \$2.14 + \$3.20 = \$6.06
- I end with: \$6.42 - \$3.20 = \$3.22
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.14, it became \$6.42, and you returned \$3.2. Your visible payoff was \$3.2199999999999998.

Your total visible earnings across all rounds are \$14.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- When I sent \$3.00 in Round 2, I received a fair return of \$3.93
- I've been playing cooperatively and seeing reasonable reciprocation

I'll continue with a cooperative strategy and send a generous amount to this new co-player

{'send': 3.00}

reasoning:

Looking at my history:

- Round 1: As receiver, got \$10.35, returned \$5.00, earned \$5.35
- Round 2: As sender, sent \$3.00, got back \$3.93, earned \$5.93
- Round 3: As receiver, got \$6.42, returned \$3.20, earned \$3.22

In Round 2, I sent \$3.00 (60%) and my co-player returned \$3.93 out of \$9.00 (43.7%), which was decent reciprocation.

Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reasonable reciprocation. I should continue with a cooperative strategy.

I'll send around \$3.00 again to maintain consistency and signal trust.

usage:

```
{
  "input_tokens": 1348,
  "output_tokens": 269,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.24 to you. Your visible payoff was \$8.24.

Your total visible earnings across all rounds are \$22.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$4.44 (88.8% of their endowment) - this is exceptional trust!
- I received \$13.32 after the 3x multiplication
- Total resources created: \$13.88

This sender showed tremendous cooperation. I should reciprocate strongly:

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This rewards their high trust with balanced outcomes

{'return': 6.50}

reasoning:

The sender sent \$4.44, which is 88.8% of their endowment - this is very high trust! I received \$13.32 after multiplication.

Total resources created: \$5 - \$4.44 + \$13.32 = \$13.88

The sender showed exceptional trust. I should reciprocate generously.

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06

- I end with: $\$13.32 - \$6.50 = \$6.82$
- This creates a fairly balanced outcome and rewards their high trust

If I return \$7.00:

- Sender ends with: $\$5 - \$4.44 + \$7.00 = \7.56
- I end with: $\$13.32 - \$7.00 = \$6.32$
- This is also reasonable

Let me go with \$6.50 as it provides good reciprocation while maintaining balance.

usage:

```
{
  "input_tokens": 1564,
  "output_tokens": 392,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.44, it became \$13.32, and you returned \$6.5. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$29.560000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience:

- When I sent \$3.00, I've seen returns of \$3.93 and \$6.24
- The game has shown good cooperation overall
- I want to maintain a cooperative strategy

I'll continue with a consistent, generous approach.

```
{'send': 3.00}
```

reasoning:

Looking at my track record:

- Round 2: I sent \$3.00, received back \$3.93
- Round 4: I sent \$3.00, received back \$6.24 (very generous!)
- Round 5: Partner sent me \$4.44 (very trusting), I returned \$6.50

The cooperation has been generally good, with some partners being very generous. I've been sending \$3.00 consistently and seeing good returns. I could continue with that strategy, or perhaps be slightly more generous given the positive experiences.

Let me continue with \$3.00 as it's been working well and represents 60% of my endowment - a reasonable cooperative level.

usage:

```
{
  "input_tokens": 1837,
  "output_tokens": 233,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.16 to you. Your visible payoff was \$7.16.

Your total visible earnings across all rounds are \$36.72.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46

This sender showed exceptional cooperation. I should reciprocate strongly:

If I return \$7.00:

- Sender ends with: $\$5 - \$4.73 + \$7.00 = \7.27
- I end with: $\$14.19 - \$7.00 = \$7.19$
- This creates very balanced outcomes rewarding their exceptional trust

```
{'return': 7.00}
```

Reasoning:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - this is extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46 (\$0.27 remaining with sender + \$14.19 with me)

This sender showed exceptional cooperation - sending almost everything! I should reciprocate strongly.

If I return \$7.00:

- Sender ends with: $\$5 - \$4.73 + \$7.00 = \7.27
- I end with: $\$14.19 - \$7.00 = \$7.19$
- This creates very balanced outcomes and rewards their exceptional trust

Let me think about this... they sent 94.6%, which is really high. I want to make sure they come out well.

Actually, let me aim for balanced final outcomes:

If I return \$7.00:

- Sender: \$7.27
- Me: \$7.19

Very balanced!

I think \$7.00 is appropriate.

```
Usage: {
  "input_tokens": 2050,
  "output_tokens": 405,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_4

Timestamp: 2026-08-12T09:24:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust-building with prudent risk management.

Key considerations:

- Sending money creates value (3x multiplier)
- This is a multi-round game, so establishing cooperation could be beneficial
- No prior history with this co-player
- Communication noise means I should expect some variance

I'll send **\$4** to signal strong willingness to cooperate while maintaining a small hedge. If my co-player returns a fair share (around 50% of what they receive), we'll both benefit significantly compared to sending nothing.

This amount shows trust without being overly naive on the first round.

```
'''json
{"send": 4}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the optimal strategy here.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return
- There's communication noise (amounts may differ slightly)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I'll play multiple rounds in different roles

Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize total payoff. If I send \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50. Both better off than the \$5 starting point.
2. **Risk**: The receiver might not return anything. If I send \$5 and get nothing back, I end up with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. However, my behavior here might set a precedent for future rounds when I'm a receiver.
4. **Multi-round consideration**: Since this is a multi-round game, building trust could be beneficial. However, opponents are randomized, so I won't necessarily play with the same person again.
5. **Communication noise**: There's noise in the system, which could lead to misunderstandings. This suggests being somewhat forgiving of apparent defection.

Common strategies in trust games:

- Send a moderate amount (e.g., \$2-3) to test trustworthiness
- Send everything to maximize joint payoff (risky but optimal if trust is reciprocated)
- Send nothing to avoid risk (but this guarantees no gain)

Given this is the first round with no history, and I want to signal cooperation while not being too naive, I think sending a moderate-to-high amount makes sense. Let me send \$3, which:

- Shows willingness to trust
- Gives the receiver \$9 total
- If they return at least \$3, I break even
- If they return more (e.g., half of \$9 = \$4.50), I profit
- Limits my downside risk to \$3

A 50-50 split of the multiplied amount seems like a fair benchmark in trust games. With \$3 sent, I'd end up with \$6.50 if the receiver returns half. But I should consider whether sending \$4 might signal stronger commitment to cooperation and encourage reciprocation, though this increases my risk exposure.

Sending \$4 yields \$7 for me if they split evenly, while sending the full \$5 maximizes our combined payoff at \$7.50 each but leaves me most vulnerable. Since this is an experimental context where cooperation typically emerges, and multiple rounds reward trust-building, I'll send \$4-enough to demonstrate genuine commitment without betting everything on the first exchange....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1133,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 2 Analysis (as RECEIVER):****

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

****Reciprocity consideration:****

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

****Decision:****

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: $\$5 - \$4 + \$5.46 = \6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 (2.73×3)
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ($\$2.73/\$5 = 54.6\%$), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game
3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

usage:

```
{
  "input_tokens": 790,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier
- This is more trust than the previous co-player showed in Round 2

****Reciprocity:****

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion that rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

'''json

{"return": 5.50}

'''

reasoning:

Let me analyze this situation:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

****Past behavior:****

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

****Reciprocity consideration:****

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). This shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high trust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \5.54

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50

- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

usage:

```
{
  "input_tokens": 1157,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.55, it became \$10.649999999999999, and you returned \$5.5. Your visible payoff was \$5.149999999999999.

Your total visible earnings across all rounds are \$15.7.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Round 4 Analysis (as SENDER):****

Looking at my experience so far:

- Round 1: Sent \$4, got good reciprocation (\$5.46 back)
- Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
- Total earnings: \$15.70 (averaging \$5.23/round)

****New co-player consideration:****

This is a different co-player than previous rounds, so no direct history. However, the cooperative approach has been working well overall.

****Decision:****

I'll send ****\$4**** again because:

- The 3x multiplier creates strong mutual benefit
- It signals trust and willingness to cooperate
- It keeps \$1 as insurance against complete defection
- This amount worked well in Round 1

'''json

{"send": 4}

'''

reasoning:

****Round 4 Analysis (as SENDER):****

Let me review the pattern so far:

- Round 1: I sent \$4, received back \$5.46 (good return, ~45.5% of \$12)
- Round 2: As receiver, got \$2.73 (sent), I returned \$4.10 (~50% of \$8.19)
- Round 3: As receiver, got \$3.55 (sent), I returned \$5.50 (~52% of \$10.65)

Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized each round).

Key considerations:

1. I have no history with this specific co-player
2. My previous experiences have been fairly positive:
 - Round 1: Good reciprocation when I sent high
 - Round 2 & 3: I reciprocated fairly as receiver
3. The game mechanics reward cooperation through the 3x multiplier
4. There's communication noise, so exact amounts may vary

Strategy:

- Since cooperation has worked well so far, I should lean toward sending a substantial amount
- However, this is a new co-player with no history
- I've been playing fairly cooperatively, which might be building reputation (though names are hidden, the pattern of fair play might matter in multi-agent settings)

I think sending \$3.50-\$4.50 would be reasonable. Let me go with \$4 again - it's worked before, shows strong trust without being all-in, and leaves some hedge.

Actually, looking at my total earnings (\$15.70 over 3 rounds), I'm doing reasonably well averaging about \$5.23 per round. The cooperative strategy is working.

I'll send \$4 again.

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.39 to you. Your visible payoff was \$9.39.

Your total visible earnings across all rounds are \$25.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their endowment) - strong trust
- I received \$10.35 after the multiplier
- Previous round showed excellent reciprocation when I was sender (\$8.39 back!)

****Reciprocity:****

My co-player demonstrated high trust by sending nearly 70%. I should maintain the fair reciprocation pattern that's been working well.

Returning approximately 50% (\$5.20) would:

- Give my co-player a total of \$6.75 (\$1.55 kept + \$5.20 returned)
- Leave me with \$5.15
- Reward their trust appropriately

```
'''json
{"return": 5.20}
'''
```

reasoning:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their \$5 endowment) - showing high trust
- I received \$10.35 after the multiplier
- This is similar trust level to Round 3 (where co-player sent \$3.55)

****Previous round performance:****

- Round 4: I sent \$4, got back \$8.39 (70% return rate!) - excellent reciprocation
- This suggests cooperative behavior is being rewarded

****Reciprocity consideration:****

My co-player sent 69% of their endowment, showing strong trust. I should return a fair proportion.

Returning approximately 50-52% seems appropriate:

- 50% = \$5.18
- 52% = \$5.38

If I return \$5.20:

- My co-player gets total: \$1.55 (kept) + \$5.20 (returned) = \$6.75
- I keep: \$5.15
- Both benefit well

This is consistent with my previous fair reciprocation strategy and should maintain cooperation.

usage:

```
{
  "input_tokens": 1801,
  "output_tokens": 448,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.2. Your visible payoff was \$5.150000000000001.

Your total visible earnings across all rounds are \$30.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Round 6 Analysis (as SENDER):****

Performance review:

- As sender: Sent \$4 twice, received good returns both times (avg \$6.93 back)
- As receiver: Been consistent with fair ~50% returns
- Average earnings: \$6.05/round

****New co-player:****

Different partner again, but the strategy of sending \$4 has been working well.

****Decision:****

Sending ****\$4**** because:

- Proven successful in previous sender rounds
- Demonstrates strong trust
- Creates significant value through 3x multiplier
- Maintains small hedge (\$1)

'''json

{"send": 4}

'''

reasoning:

****Round 6 Analysis (as SENDER):****

Looking at my track record:

- Round 1: Sent \$4, got \$5.46 back ? Profit of \$6.46
- Round 4: Sent \$4, got \$8.39 back ? Profit of \$9.39

Both times I sent \$4, I received good to excellent returns. Average earnings per round is now \$6.05 (\$30.24/5 rounds).

As receiver:

- Round 2: Received \$8.19, returned \$4.10
- Round 3: Received \$10.65, returned \$5.50
- Round 5: Received \$10.35, returned \$5.20

I've been consistently fair as a receiver, returning around 50%.

****New co-player:****

This is another new pairing. The pattern of sending \$4 has been successful, showing trust while keeping some insurance.

****Decision:****

I'll continue with \$4 because:

- It's been consistently successful
- Shows strong cooperation
- Keeps \$1 as hedge
- The multiplier effect makes this beneficial for both parties if reciprocated

Let me send \$4.

usage:


```
{
  "input_tokens": 2113,
  "output_tokens": 432,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.08 to you. Your visible payoff was \$7.08.

Your total visible earnings across all rounds are \$37.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 7 Analysis (as SENDER):****

****Same co-player as Round 6:****

- Last round: I sent \$4, they returned \$6.08 (about 51% of what they received)
- This was fair and cooperative reciprocation
- My payoff was good at \$7.08

****Decision:****

Since this co-player has demonstrated trustworthiness, I'll send ****\$4**** again to:

- Continue the mutually beneficial cooperation
- Reward their fair reciprocation
- Maximize joint value creation

```
'''json
{"send": 4}
'''
```

Reasoning:

****Round 7 Analysis (as SENDER):****

Important: This is the SAME co-player as Round 6 (it says "against your current co-player" in the round description).

Previous round with this same co-player:

- I sent \$4
- They received \$12
- They returned \$6.08 (about 50.7% of what they received)
- My payoff was \$7.08 (\$1 kept + \$6.08 returned)

This was a fair and cooperative return! They showed good reciprocity.

Since:

1. This is the same co-player
2. They reciprocated fairly last round
3. Continuing cooperation could lead to mutual benefit
4. My strategy of sending \$4 has been working well

I should send \$4 again to maintain the cooperative relationship with this co-player.

Usage: {

```
"input_tokens": 2391,
"output_tokens": 339,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-08-12T09:25:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_4",
```

```

"pairing": {
  "agents": [
    "Agent_4",
    "Agent_1"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_1": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.71, that is 54.2% of their total endowment. You received \$8.129999999999999. How much do you return to the sender? (0-8.129999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.71 (54.2% of their endowment)
- They now have \$2.29 remaining
- I received \$8.13 (after the 3x multiplication)

I want to reward their trust and cooperation while being fair to both of us. If I return a reasonable amount, we both profit from the cooperation.

Let me return \$3.50:

- The sender would end with: $\$2.29 + \$3.50 = \$5.79$ (net gain of \$0.79)
- I would end with: $\$8.13 - \$3.50 = \$4.63$ (still a good outcome)

This rewards their cooperative behavior while maintaining mutual benefit.

{"return": 3.5}

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$2.71 (which is 54.2% of their \$5 endowment)
- I received \$8.13 (approximately, since $\$2.71 \times 3 = \8.13)
- I need to decide how much to return (between \$0 and \$8.13)

The sender showed significant trust by sending more than half of their endowment. This is a cooperative move.

Let me think about what would be fair:

- The sender now has: $\$5 - \$2.71 = \$2.29$
- I have: \$8.13
- Total in the system: $\$2.29 + \$8.13 = \$10.42$

Some possible strategies:

1. Equal split of total: We each get \$5.21, so I'd return \$2.92 ($\$8.13 - \5.21)
2. Proportional return: Return about half of what I received (around \$4)
3. Ensure sender profits: Return enough so sender ends up better than their initial \$5
4. Match their generosity: They sent 54.2%, I could return a similar percentage

Let me calculate what happens with different return amounts:

- If I return \$3:
 - Sender ends with: $\$2.29 + \$3 = \$5.29$
 - I end with: $\$8.13 - \$3 = \$5.13$
 - Both profit slightly
- If I return \$4:
 - Sender ends with: $\$2.29 + \$4 = \$6.29$
 - I end with: $\$8.13 - \$4 = \$4.13$
 - Sender profits more, I profit less
- If I return \$2.92 (equal split):
 - Sender ends with: $\$2.29 + \$2.92 = \$5.21$
 - I end with: $\$8.13 - \$2.92 = \$5.21$
 - Equal outcome

The sender's trust deserves reciprocation. Returning around \$3 seems balanced-it rewards their generosity while both of us gain from the exchange, ending at roughly \$5.29 and \$5.13 respectively. They sent over half their endowment, so I should ensure they benefit from that risk....

usage:

```
{
  "input_tokens": 537,
  "output_tokens": 1000,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.
Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit
- This opening move signals cooperation

{"send": 2.5}

reasoning:

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50
3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive: $2.50 \times 3 = 7.50$
- If they return around 40-50% of what they receive (\$3-\$3.75), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50% has proven effective and signals trust.

```
{"send": 2.5}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50
- They received \$7.50
- They returned \$3.01
- My payoff: $\$2.50 + \$3.01 = \$5.51$

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation.

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 353,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

```
{"return": 6.5}
```

reasoning:

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has: $\$5 - \$4.71 = \$0.29$ remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with $\$0.29 + \$6.00 = \$6.29$ (net gain of \$1.29), I keep \$8.13
- If I return \$7.00: Sender ends with $\$0.29 + \$7.00 = \$7.29$ (net gain of \$2.29), I keep \$7.13
- If I return \$5.00: Sender ends with $\$0.29 + \$5.00 = \$5.29$ (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 549,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.71, it became \$14.129999999999999, and you returned \$6.5. Your visible payoff was \$7.629999999999999.

Your total visible earnings across all rounds are \$22.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience:

I've had positive results with cooperative play. In Round 4, I benefited from a sender who showed very high trust (\$4.71), and I reciprocated well. My sending strategy of around \$2.50 has worked reasonably, but I could signal stronger cooperation.

With a new co-player, I'll increase my trust signal slightly to \$3.00 (60% of endowment), which is more than my previous sends but not as extreme as the \$4.71 I received.

```
{"send": 3.0}
```

reasoning:

Looking at my history:

- Round 1 (Receiver): Sender sent \$2.71, I returned \$3.50
- Round 2 (Sender): I sent \$2.50, got back \$3.01
- Round 3 (Sender): I sent \$2.50, got back \$2.68
- Round 4 (Receiver): Sender sent \$4.71, I returned \$6.50

The pattern shows:

- When I send ~50% (\$2.50), I get modest returns (around \$2.68-\$3.01)
- When someone sent me a lot (\$4.71), I reciprocated generously (\$6.50)
- Overall, cooperation is working reasonably well

Given that I've been relatively successful with moderate cooperation, and I've seen high trust rewarded (the \$4.71 sender got \$6.50 back from me), I could consider sending more this round to signal strong cooperation.

However, I've also learned that sending around 50% tends to get decent returns.

With a new co-player, I could:

1. Continue with \$2.50 (safe, proven strategy)
2. Send more to signal higher trust (like the \$4.71 example)
3. Send slightly more than before to test responsiveness

I think I'll send a bit more than my previous rounds to signal stronger cooperation - maybe \$3.00 or so. This is 60% of my endowment, showing more trust than before but not as extreme as the 94% I saw.

Let me go with \$3.00.

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$28.639999999999993.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

The sender showed exceptional trust by sending \$4.81 (96.2% of their endowment) - nearly all of it! This is an extremely strong cooperative signal.

They now have only \$0.19 remaining, and I received \$14.43.

This is similar to Round 4 where I received \$4.71 and returned \$6.50. Given this sender showed even more trust, I should reciprocate generously.

If I return \$7.00:

- Sender ends with: $\$0.19 + \$7.00 = \$7.19$ (net gain of \$2.19)
- I end with: $\$14.43 - \$7.00 = \$7.43$

This rewards their exceptional trust while ensuring we both profit substantially.

```
{"return": 7.0}
```

reasoning:

The sender sent \$4.81, which is 96.2% of their endowment - extremely high trust!

I received \$14.43 after the 3x multiplication.

They now have: $\$5 - \$4.81 = \$0.19$ remaining

This is even higher trust than the \$4.71 I saw in Round 4 (where I returned \$6.50).

I should reward this exceptional trust with a generous return. Let me think about fair amounts:

If I return \$7.00:

- Sender ends with: $\$0.19 + \$7.00 = \$7.19$ (net gain of \$2.19)
- I end with: $\$14.43 - \$7.00 = \$7.43$

This gives them a good return on their trust and we both profit nicely. This is similar proportionally to what I did in Round 4.

Actually, given they sent even more than Round 4, I could be slightly more generous. Let me consider \$7.50:

- Sender ends with: $\$0.19 + \$7.50 = \$7.69$ (net gain of \$2.69)
- I end with: $\$14.43 - \$7.50 = \$6.93$

That works well too. I'll go with \$7.00 as a strong but reasonable reciprocation.

usage:

```
{
  "input_tokens": 1915,
  "output_tokens": 511,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.81, it became \$14.43, and you returned \$7.0. Your visible payoff was \$7.43.

Your total visible earnings across all rounds are \$36.06999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

This is the same co-player from Round 6! Last round they sent \$4.81 and I returned \$7.00, which worked well.

Now they've sent \$3.29 (65.8% of their endowment) - still very cooperative but less than last round. They have \$1.71 remaining, and I received \$9.87.

I'll maintain consistent reciprocity. Since they showed somewhat less trust this round, I'll return proportionally:

If I return \$4.50:

- Sender ends with: $\$1.71 + \$4.50 = \$6.21$ (net gain of \$1.21)
- I end with: $\$9.87 - \$4.50 = \$5.37$

This maintains our cooperative relationship while being fair to the reduced amount sent.

```
{"return": 4.5}
```

Reasoning:

Interesting - I'm playing with the same co-player from Round 6. In Round 6, they sent me \$4.81 (96.2%) and I returned \$7.00.

Now they've sent me \$3.29 (65.8% of their endowment), which is less than before but still a strong cooperative signal.

They now have \$5 - \$3.29 = \$1.71 remaining.

I received \$9.87.

Since this is the same co-player and we had a good interaction last round:

- Last round: They sent \$4.81, I returned \$7.00
- This round: They sent \$3.29 (less trust, perhaps because they got a good return and want to balance risk)

I should maintain reciprocity. Let me think about what's fair:

If I return \$4.50:

- Sender ends with: $\$1.71 + \$4.50 = \$6.21$ (net gain of \$1.21)
- I end with: $\$9.87 - \$4.50 = \$5.37$

This seems reasonable - it rewards their cooperation proportionally to what they sent. The ratio of return to received:

- Round 6: $7.00/14.43 = 0.485$ (48.5%)
- Round 7 with \$4.50: $4.50/9.87 = 0.456$ (45.6%)

That's fairly consistent. Let me go with \$4.50.

```
Usage: {
  "input_tokens": 2253,
  "output_tokens": 557,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Sender

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_7=Sender, Agent_6=Receiver
dyad_3: Agent_8=Sender, Agent_4=Receiver
dyad_4: Agent_5=Sender, Agent_3=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_1=\$7, Agent_2=\$4;
noise/visible: sent visible=\$2.52, received visible=\$7.56, returned visible=\$4.13
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_7 sent \$4; received \$12; Agent_6 returned \$8.5; payoffs Agent_7=\$9.5, Agent_6=\$3.5;
noise/visible: sent visible=\$4.62, received visible=\$13.86, returned visible=\$9.45
Action validation: {"returned": {"amount": 8.5, "clamped": false, "max": 12.0, "min": 0, "raw": 8.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_8 sent \$3; received \$9; Agent_4 returned \$4.45; payoffs Agent_8=\$6.45, Agent_4=\$4.55; noise/visible: sent visible=\$2.97, received visible=\$8.91, returned visible=\$4.84
Action validation: {"returned": {"amount": 4.45, "clamped": false, "max": 9.0, "min": 0, "raw": 4.45}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_5 sent \$3; received \$9; Agent_3 returned \$5; payoffs Agent_5=\$7, Agent_3=\$4;
noise/visible: sent visible=\$2.81, received visible=\$8.43, returned visible=\$5.88
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 49.2,
"Agent_2": 44.3,
"Agent_3": 47.5,
"Agent_4": 48.25,
"Agent_5": 47.4,
"Agent_6": 47.7,
"Agent_7": 50.2,
"Agent_8": 49.45

}

Visible balances after round: {

"Agent_1": 47.57,
"Agent_2": 45.89999999999999,
"Agent_3": 48.01,
"Agent_4": 48.22,
"Agent_5": 47.500000000000001,
"Agent_6": 47.57,
"Agent_7": 54.67,
"Agent_8": 50.75

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:25:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",

```

"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.71, it became \$8.129999999999999, and you returned \$3.5. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$4.629999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the game dynamics:

In round 1, I experienced a cooperative interaction where the sender trusted me with 54.2% of their endowment, and I reciprocated fairly.

Now with a new co-player, I'll adopt a similar cooperative strategy. Sending around 50% demonstrates trust and willingness to cooperate, which could encourage reciprocity.

If I send \$2.50:

- They receive \$7.50 (after 3x multiplication)
- If they return a fair portion, we both benefit
- This opening move signals cooperation

{"send": 2.5}

reasoning:

Now I'm the sender in round 2. Let me think about this strategically.

In round 1, I was the receiver and my co-player sent me \$2.71 (54.2% of their endowment). I returned \$3.50, which gave them a net gain. That was a cooperative interaction.

Now I'm paired with a different agent (since pairings are randomized each round). I don't know anything about this new co-player's behavior yet.

Given that:

1. The previous sender was cooperative (sent 54.2%)
2. I responded cooperatively by returning \$3.50
3. This establishes a pattern of cooperation in this game

I should probably send a similar cooperative amount to signal trust and potentially build reciprocity. Sending around 50% seems reasonable - it shows trust without being overly risky.

Let me send \$2.50 (50% of my endowment):

- I keep: \$2.50
- They receive: $2.50 \times 3 = 7.50$
- If they return around 40-50% of what they receive (\$3-\$3.75), we both end up better off

This seems like a fair, cooperative opening move that could establish a good relationship with this new co-player.

Let me go with \$2.50.

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 441,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant**content:**

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50% has proven effective and signals trust.

```
{"send": 2.5}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50
- They received \$7.50
- They returned \$3.01
- My payoff: $\$2.50 + \$3.01 = \$5.51$

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation

usage:

```
{  
  "input_tokens": 1113,  
  "output_tokens": 353,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)

- I end with: $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

```
{"return": 6.5}
```

reasoning:

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has: $\$5 - \$4.71 = \$0.29$ remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with $\$0.29 + \$6.00 = \$6.29$ (net gain of \$1.29), I keep \$8.13
- If I return \$7.00: Sender ends with $\$0.29 + \$7.00 = \$7.29$ (net gain of \$2.29), I keep \$7.13
- If I return \$5.00: Sender ends with $\$0.29 + \$5.00 = \$5.29$ (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 549,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.71, it became \$14.129999999999999, and you returned \$6.5. Your visible payoff was \$7.629999999999999.

Your total visible earnings across all rounds are \$22.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience:

I've had positive results with cooperative play. In Round 4, I benefited from a sender who showed very high trust (\$4.71), and I reciprocated well. My sending strategy of around \$2.50 has worked reasonably, but I could signal stronger cooperation.

With a new co-player, I'll increase my trust signal slightly to \$3.00 (60% of endowment), which is more than my previous sends but not as extreme as the \$4.71 I received.

```
{"send": 3.0}
```

reasoning:

Looking at my history:

- Round 1 (Receiver): Sender sent \$2.71, I returned \$3.50
- Round 2 (Sender): I sent \$2.50, got back \$3.01

- Round 3 (Sender): I sent \$2.50, got back \$2.68
- Round 4 (Receiver): Sender sent \$4.71, I returned \$6.50

The pattern shows:

- When I send ~50% (\$2.50), I get modest returns (around \$2.68-\$3.01)
- When someone sent me a lot (\$4.71), I reciprocated generously (\$6.50)
- Overall, cooperation is working reasonably well

Given that I've been relatively successful with moderate cooperation, and I've seen high trust rewarded (the \$4.71 sender got \$6.50 back from me), I could consider sending more this round to signal strong cooperation.

However, I've also learned that sending around 50% tends to get decent returns.

With a new co-player, I could:

1. Continue with \$2.50 (safe, proven strategy)
2. Send more to signal higher trust (like the \$4.71 example)
3. Send slightly more than before to test responsiveness

I think I'll send a bit more than my previous rounds to signal stronger cooperation - maybe \$3.00 or so. This is 60% of my endowment, showing more trust than before but not as extreme as the 94% I saw.

Let me go with \$3.00.

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$28.639999999999993.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

The sender showed exceptional trust by sending \$4.81 (96.2% of their endowment) - nearly all of it! This is an extremely strong cooperative signal.

They now have only \$0.19 remaining, and I received \$14.43.

This is similar to Round 4 where I received \$4.71 and returned \$6.50. Given this sender showed even more trust, I should reciprocate generously.

If I return \$7.00:

- Sender ends with: \$0.19 + \$7.00 = \$7.19 (net gain of \$2.19)
- I end with: \$14.43 - \$7.00 = \$7.43

This rewards their exceptional trust while ensuring we both profit substantially.

```
{"return": 7.0}
```

reasoning:

The sender sent \$4.81, which is 96.2% of their endowment - extremely high trust!

I received \$14.43 after the 3x multiplication.

They now have: \$5 - \$4.81 = \$0.19 remaining

This is even higher trust than the \$4.71 I saw in Round 4 (where I returned \$6.50).

I should reward this exceptional trust with a generous return. Let me think about fair amount

unts:

If I return \$7.00:

- Sender ends with: $\$0.19 + \$7.00 = \$7.19$ (net gain of \$2.19)
- I end with: $\$14.43 - \$7.00 = \$7.43$

This gives them a good return on their trust and we both profit nicely. This is similar proportionally to what I did in Round 4.

Actually, given they sent even more than Round 4, I could be slightly more generous. Let me consider \$7.50:

- Sender ends with: $\$0.19 + \$7.50 = \$7.69$ (net gain of \$2.69)
- I end with: $\$14.43 - \$7.50 = \$6.93$

That works well too. I'll go with \$7.00 as a strong but reasonable reciprocation.

usage:

```
{
  "input_tokens": 1915,
  "output_tokens": 511,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.81, it became \$14.43, and you returned \$7.0. Your visible payoff was \$7.43.

Your total visible earnings across all rounds are \$36.069999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.8700000000000001. How much do you return? (0-9.8700000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

This is the same co-player from Round 6! Last round they sent \$4.81 and I returned \$7.00, which worked well.

Now they've sent \$3.29 (65.8% of their endowment) - still very cooperative but less than last round. They have \$1.71 remaining, and I received \$9.87.

I'll maintain consistent reciprocity. Since they showed somewhat less trust this round, I'll return proportionally:

If I return \$4.50:

- Sender ends with: $\$1.71 + \$4.50 = \$6.21$ (net gain of \$1.21)
- I end with: $\$9.87 - \$4.50 = \$5.37$

This maintains our cooperative relationship while being fair to the reduced amount sent.

{"return": 4.5}

reasoning:

Interesting - I'm playing with the same co-player from Round 6. In Round 6, they sent me \$4.81 (96.2%) and I returned \$7.00.

Now they've sent me \$3.29 (65.8% of their endowment), which is less than before but still a strong cooperative signal.

They now have \$5 - \$3.29 = \$1.71 remaining.

I received \$9.87.

Since this is the same co-player and we had a good interaction last round:

- Last round: They sent \$4.81, I returned \$7.00
- This round: They sent \$3.29 (less trust, perhaps because they got a good return and want to balance risk)

I should maintain reciprocity. Let me think about what's fair:

If I return \$4.50:

- Sender ends with: $\$1.71 + \$4.50 = \$6.21$ (net gain of \$1.21)
- I end with: $\$9.87 - \$4.50 = \$5.37$

This seems reasonable - it rewards their cooperation proportionally to what they sent. The ratio of return to received:

- Round 6: $7.00/14.43 = 0.485$ (48.5%)
- Round 7 with \$4.50: $4.50/9.87 = 0.456$ (45.6%)

That's fairly consistent. Let me go with \$4.50.

usage:

```
{
  "input_tokens": 2253,
  "output_tokens": 557,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$4.5. Your visible payoff was \$5.370000000000001.

Your total visible earnings across all rounds are \$41.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

My cooperative sending strategy has been working well:

- Sending \$2.50-\$3.00 has consistently yielded positive returns
- When others show high trust, I reciprocate generously
- My total earnings of \$41.44 suggest good mutual cooperation overall

With a new co-player in Round 8, I'll maintain my moderately high trust signal.

```
{"send": 3.0}
```

Reasoning:

Looking at my experience across the rounds:

Round 1: RECEIVER - received \$2.71 (became \$8.13), returned \$3.50, earned \$4.63
Round 2: SENDER - sent \$2.50 (became \$7.50), received back \$3.01, earned \$5.51
Round 3: SENDER - sent \$2.50 (became \$7.50), received back \$2.68, earned \$5.18
Round 4: RECEIVER - received \$4.71 (became \$14.13), returned \$6.50, earned \$7.63
Round 5: SENDER - sent \$3.00 (became \$9.00), received back \$3.69, earned \$5.69
Round 6: RECEIVER - received \$4.81 (became \$14.43), returned \$7.00, earned \$7.43
Round 7: RECEIVER - received \$3.29 (became \$9.87), returned \$4.50, earned \$5.37

Total earnings: \$41.44

My sending pattern:

- Round 2: sent \$2.50, got back \$3.01 (gain of \$0.51)
- Round 3: sent \$2.50, got back \$2.68 (gain of \$0.18)
- Round 5: sent \$3.00, got back \$3.69 (gain of \$0.69)

Average return when I send has been positive. I've been getting reasonable returns.

When I've been receiver:

- Round 1: received \$8.13, returned \$3.50 (kept \$4.63)

...

Usage: {

```
"input_tokens": 2330,
"output_tokens": 615,
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:25:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.82, it became \$11.459999999999999, and you returned \$6.0. Your visible payoff was \$5.459999999999999.

Your total visible earnings across all rounds are \$5.459999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience from the previous round, the sender showed significant trust by sending 76.4% of their endowment, and I reciprocated fairly by returning enough for them to profit.

In this multi-agent setting, building a reputation for cooperation is valuable. Even though I'm paired with a different agent this round, demonstrating trustworthiness can encourage reciprocity.

I'll send a substantial amount to signal trust and encourage cooperation.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me think about this round. I'm now the SENDER.

In the previous round:

- I was the RECEIVER
- The sender sent me \$3.82 (76.4% of their endowment)
- I received \$11.46
- I returned \$6.00
- I kept \$5.46

This showed a good cooperative relationship - the sender trusted me significantly, and I reciprocated fairly.

Now I'm the sender. I need to decide how much to send.

Considerations:

1. The previous sender showed high trust (76.4%)
2. I reciprocated that trust by returning enough that they ended up better off
3. In a multi-agent setting with randomized pairings, I'm playing with a different agent now
4. Building a reputation for cooperation could encourage others to trust me when I'm the receiver
5. There's communication noise, so amounts may be slightly off

Since the previous sender showed high trust and I want to encourage cooperation in this multi-agent game, I should also show significant trust. Sending around 70-80% seems reasonable to match the cooperative behavior I've experienced.

Let me send \$3.50 (70% of my endowment). This is a strong signal of trust without being overly risky.

Actually, let me reconsider. The previous sender sent 76.4%. To maintain reciprocal cooperation and potentially build a good reputation, I could send a similar amount - maybe \$3.80 or \$4.00.

I'll send \$4.00 (80% of my endowment). This shows high trust and encourages cooperation.

usage:

```
{
  "input_tokens": 855,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings
3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

usage:

```
{
  "input_tokens": 1089,
  "output_tokens": 383,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.

Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier

- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This rewards their trust while both of us benefit from the exchange.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has $\$5 - \$3.35 = \$1.65$ left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust (67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with: $\$1.65 + \$5.00 = \$6.65$ (profit of \$1.65)
- I end with: $\$10.05 - \$5.00 = \$5.05$

Or if I return \$6.50:

- Sender ends with: $\$1.65 + \$6.50 = \$8.15$ (profit of \$3.15)
- I end with: $\$10.05 - \$6.50 = \$3.55$

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

usage:

```
{
  "input_tokens": 1315,
  "output_tokens": 602,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.35, it became \$10.05, and you returned \$6.0. Your visible payoff was \$4.050000000000001.

Your total visible earnings across all rounds are \$25.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my track record:

- When I sent \$4.0 in previous rounds, I received back \$6.29 and \$7.32 - both profitable outcomes
- My cooperative strategy has been working well, building trust

I'll continue with the same successful approach of sending a substantial amount to encourage reciprocity.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my track record:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.0, kept \$5.46
- Round 2 (SENDER): Sent \$4.0, received back \$6.29, kept \$7.29
- Round 3 (SENDER): Sent \$4.0, received back \$7.32, kept \$8.32
- Round 4 (RECEIVER): Received \$10.05, returned \$6.0, kept \$4.05

Total visible earnings: \$25.12

My cooperative strategy of sending \$4.0 has been working well - both times I sent \$4.0, I received good returns (\$6.29 and \$7.32), both yielding solid profits. As a receiver, I've been fair by returning \$6.0 when others showed trust.

I'm now paired with a new co-player for round 5 as sender. Given my successful track record with cooperation, I should continue this strategy.

Sending \$4.0 again makes sense - it shows significant trust (80% of endowment) and has consistently led to profitable outcomes for me.

usage:

```
{
  "input_tokens": 1608,
  "output_tokens": 357,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.93 to you. Your visible payoff was \$6.93.

Your total visible earnings across all rounds are \$32.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

Consistent with my previous cooperative approach, I'll return a fair amount that rewards their trust.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (good profit)
- I end with: \$10.38 - \$6.00 = \$4.38

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

My pattern has been to return around \$6.00 when I receive amounts in this range (around \$10-11). This has been fair and consistent.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (profit of \$2.54)

- I end with: \$10.38 - \$6.00 = \$4.38

This is consistent with my previous behavior and rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1837,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.46, it became \$10.379999999999999, and you returned \$6.0. Your visible payoff was \$4.379999999999999.

Your total visible earnings across all rounds are \$36.429999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my consistent strategy:

- I've been sending \$4.0 as sender and it has consistently resulted in profitable returns
- My cooperative approach has been working well across different co-players

I'll maintain this successful strategy.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- I've been consistently sending \$4.0 as sender
- When receiving, I've been consistently returning \$6.0
- My strategy has been working well overall

My recent returns as sender:

- Round 2: sent \$4.0, received back \$6.29 (profit: \$2.29)
- Round 3: sent \$4.0, received back \$7.32 (profit: \$3.32)
- Round 5: sent \$4.0, received back \$5.93 (profit: \$1.93)

All profitable, showing that the cooperative strategy is paying off. I should continue with the same approach.

usage:

```
{
  "input_tokens": 2121,
  "output_tokens": 225,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.91 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$43.339999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment) - moderate trust, though less than in previous rounds
- I received \$7.56 after the 3x multiplier
- The sender currently has \$2.48 left

While I've consistently returned \$6.0 when receiving larger amounts, this sender sent less. I'll adjust proportionally while still being generous to maintain cooperation.

If I return \$5.0:

- Sender ends with: $\$2.48 + \$5.0 = \$7.48$ (profit of \$2.48)
- I end with: $\$7.56 - \$5.0 = \$2.56$

This rewards their trust fairly while being proportional to the amount sent.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment) - moderate trust
- I received \$7.56 after the 3x multiplier
- The sender currently has \$2.48 left

This is a smaller amount sent compared to what I've typically seen (around \$3.35-\$3.82). Still, they showed some trust.

My previous returns as receiver:

- Round 1: Received \$11.46, returned \$6.0
- Round 4: Received \$10.05, returned \$6.0
- Round 6: Received \$10.38, returned \$6.0

In all these cases, I returned \$6.0 which was a fair reciprocation. However, this time they sent less (\$2.52 vs \$3.35-\$3.82), so receiving \$7.56 is also less than before.

If I return \$6.0:

- Sender ends with: $\$2.48 + \$6.00 = \$8.48$ (profit of \$3.48)
- I end with: $\$7.56 - \$6.00 = \$1.56$

This still gives them a good profit but leaves me with less. Given they sent less, maybe I should adjust proportionally.

If I return \$4.5:

- Sender ends with: $\$2.48 + \$4.5 = \$6.98$ (profit of \$1.98)
- I end with: $\$7.56 - \$4.5 = \$3.06$

This is more balanced given they sent less. However, I want to maintain a cooperative reputation.

Let me think about the ratio. When they sent ~\$3.4-3.8, I returned \$6. That's roughly returning what they initially sent + a bit more, ensuring mutual benefit.

If they sent \$2.52, maybe returning around \$4.5-5 would be proportional?

Returning \$5.0 would give them \$...

```
Usage: {
  "input_tokens": 2081,
  "output_tokens": 831,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:25:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_6"
    ],
    "investor": "Agent_7",
    "roles": {
```

```

    "Agent_6": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.68 to you. Your visible payoff was \$7.68.

Your total visible earnings across all rounds are \$7.68.

This round, you are the RECEIVER against your current co-player. They sent you \$4.04, so you received \$12.12000000000001. How much do you return? (0-12.12000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.04 (80% of their endowment), showing significant trust
- I received \$12.12
- They kept only \$0.96

This is a strong show of trust that deserves reciprocation. I want to ensure they benefit from their trust while also getting a reasonable share myself.

If I return \$6.50:

- They end with: $\$0.96 + \$6.50 = \$7.46$
- I end with: $\$12.12 - \$6.50 = \$5.62$

Both of us do better than our initial \$5, and this rewards their trust appropriately.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3
- It became \$9 for the receiver
- They returned \$5.68
- My payoff: $\$5 - \$3 + \$5.68 = \7.68
- The receiver kept $\$9 - \$5.68 = \$3.32$

So my co-player in Round 1 returned about 63% of what they received, which is quite generous.

Round 2 (current):

- I am RECEIVER
- My co-player sent \$4.04
- I received $\$4.04 \times 3 = \12.12
- I need to decide how much to return

My co-player showed significant trust by sending \$4.04 (over 80% of their \$5). This deserves reciprocation.

If I follow a similar reciprocity pattern to what I experienced in Round 1:

- They sent \$4.04, keeping \$0.96
- I received \$12.12
- If I return about 60-65%, that would be $\$7.27 - \7.88
- That would leave them with $\$0.96 + \text{return}$, and me with $\$12.12 - \text{return}$

Let me think about fairness:

- They started with \$5, sent \$4.04, kept \$0.96
- If I return \$6, they end with $\$0.96 + \$6 = \$6.96$, I end with $\$12.12 - \$6 = \$6.12$
- If I return \$7, they end with $\$7.96$, I end with $\$5.12$

Returning around \$6-\$7 seems fair and encourages future cooperation. Given they were quite generous with their trust, I'll return \$6.50, which:

- Gives them final payoff: $\$0.96 + \$6.50 = \$7.46$
- ...

usage:

```
{
  "input_tokens": 754,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible

payoff was \$5.620000000000001.
Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with: $\$0.62 + \$7.00 = \$7.62$
- I end with: $\$13.14 - \$7.00 = \$6.14$

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.
I received \$13.14
They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:

- They end with: $\$0.62 + \$7.00 = \$7.62$
- I end with: $\$13.14 - \$7.00 = \$6.14$

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:

1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:

- Their total: $\$0.62 + \$7.00 = \$7.62$ (profit of \$2.62)
- My total: $\$13.14 - \$7.00 = \$6.14$ (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:

- Their total: $\$0.62 + \$7.50 = \$8.12$ (profit of \$3.12)
- My total: $\$13.14 - \$7.50 = \$5.64$ (profit of \$0.64)

I think \$7.00 is reasonable. It gives them a nice profit while I also benefit from the cooperation.

usage:

```
{
  "input_tokens": 1061,
  "output_tokens": 567,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.38, it became \$13.14, and you returned \$7.0. Your visible payoff was \$6.140000000000001.

Your total visible earnings across all rounds are \$19.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- In Round 1, I sent \$3 and received good reciprocation (\$5.68 back)
- In Rounds 2 and 3, co-players showed high trust by sending \$4.04 and \$4.38
- I've been reciprocating fairly, returning about half of what I received

This is a new co-player, but the general pattern suggests cooperation pays off. I'll send a high amount to match the trust I've seen from others.

```
'''json
{"send": 4}
'''
```

reasoning:

I'm now the SENDER in Round 4. Let me think about what I've learned:

Round 1: I sent \$3, received \$5.68 back, ended with \$7.68

Round 2: I received \$12.12 (they sent \$4.04), returned \$6.50, ended with \$5.62

Round 3: I received \$13.14 (they sent \$4.38), returned \$7.00, ended with \$6.14

My recent experiences as a receiver show co-players sending high amounts (\$4.04 and \$4.38). When I was a sender in Round 1, I sent \$3 and got \$5.68 back, which was good reciprocation.

Given that I've seen high trust from others and I've been reciprocating fairly, I should continue to show trust. This is a new co-player this round (since pairings are randomized), so I don't have specific history with them, but the general pattern suggests cooperation is beneficial.

I could send:

- \$3: Moderate trust, similar to Round 1
- \$4: High trust, similar to what I've received from others
- \$5: Maximum trust

Given that I've been receiving high amounts from others (around \$4+), and I've been reciprocating fairly well, I should send a similarly high amount to maintain the cooperative dynamic. Sending \$4 shows significant trust while not being overly risky.

If I send \$4:

- They receive \$12
- If they return around 50-60%, I'd get back \$6-7.20
- I'd end with roughly \$7-8.20

This seems reasonable and aligns with the cooperative pattern I've been experiencing.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.06 to you. Your visible payoff was \$8.059999999999999.

Your total visible earnings across all rounds are \$27.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my recent rounds:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back - excellent reciprocation

The cooperation has been strong, with both co-players returning generously. This suggests high trust sends are being rewarded.

With a new co-player this round, I'll continue the cooperative strategy that's been working well.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: Sent \$3, received \$5.68 back (good return)
- Round 2: Received \$12.12, returned \$6.50
- Round 3: Received \$13.14, returned \$7.00
- Round 4: Sent \$4, received \$7.06 back (very good return - they returned about 59% of what they got)

The cooperation has been strong across all rounds. People are sending high amounts and returning generously. My earnings have been good (\$27.50 total across 4 rounds with visible payoffs).

This is Round 5 with a new co-player. Given the positive cooperation pattern, I should continue with a high trust strategy. Sending \$4 worked well last round.

I'll send \$4 again to show trust and likely get good reciprocation.

usage:

```
{
  "input_tokens": 1602,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.12 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$33.62.

This round, you are the RECEIVER against your current co-player. They sent you \$3.98, so you received \$11.94. How much do you return? (0-11.94)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing strong trust
- I received \$11.94
- They kept only \$1.02

This is solid trust that deserves fair reciprocation. Based on my previous patterns as receiver, returning about 50-55% of what I received has worked well.

If I return \$6.50:

- They end with: \$1.02 + \$6.50 = \$7.52
- I end with: \$11.94 - \$6.50 = \$5.44

Both of us gain from cooperation.

```
'''json
```

```
    {"return": 6.5}
    """
```

reasoning:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing good trust
- I received \$11.94
- They kept only \$1.02

Previous rounds have shown:

- When I was receiver in Round 2, I got \$12.12 and returned \$6.50
- When I was receiver in Round 3, I got \$13.14 and returned \$7.00
- These were fair returns that both parties benefited from

For this round:

If I return \$6.50:

- They end with: \$1.02 + \$6.50 = \$7.52
- I end with: \$11.94 - \$6.50 = \$5.44

Both benefit from cooperation. This seems like a fair reciprocation for their trust.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 360,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.98, it became \$11.94, and you returned \$6.5. Your visible payoff was \$5.4399999999999995.

Your total visible earnings across all rounds are \$39.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing continued trust
- I received \$11.16
- They kept \$1.28
- This is the same co-player from last round

They sent slightly less than last round (\$3.72 vs \$3.98), but still showing good cooperation. I should maintain consistent reciprocation.

If I return \$6.00:

- They end with: \$1.28 + \$6.00 = \$7.28
- I end with: \$11.16 - \$6.00 = \$5.16

This maintains roughly the same proportion as last round (~54% return rate).

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing good trust
- I received \$11.16
- They kept \$1.28

This is the same co-player from Round 6 where they sent \$3.98 and I returned \$6.50.

Now they sent slightly less (\$3.72 vs \$3.98). This could be:

1. Random variation/noise

2. A slight reduction in trust based on my previous return
3. Just their strategy this round

I should continue being fair and reciprocating appropriately. Returning about 50-55% of what I received seems reasonable.

If I return \$6.00:

- They end with: $\$1.28 + \$6.00 = \$7.28$
- I end with: $\$11.16 - \$6.00 = \$5.16$

Both still profit from cooperation. This is slightly less than before (I returned \$6.50 on \$11.94), but proportionally similar given they sent less this time.

Let me calculate the proportions:

- Round 6: Returned \$6.50 out of \$11.94 = 54.4%
- Round 7: Return \$6.00 out of \$11.16 = 53.8%

That's very consistent proportionally.

usage:

```
{
  "input_tokens": 2151,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.72, it became \$11.16, and you returned \$6.0. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$44.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with this current co-player:

- Round 6: They sent me \$3.98, I returned \$6.50
- Round 7: They sent me \$3.72, I returned \$6.00

They've shown consistent trust by sending high amounts. I've reciprocated fairly. Now it's my turn to be the sender with this same co-player.

They've demonstrated trustworthiness, so I should send a high amount to continue our cooperative relationship.

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at my history with this current co-player:

- Round 6: I was RECEIVER, they sent \$3.98, I returned \$6.50
- Round 7: I was RECEIVER, they sent \$3.72, I returned \$6.00
- Round 8: I am SENDER now

This co-player has shown trust by sending high amounts (\$3.98 and \$3.72) in the previous two rounds. I reciprocated fairly by returning \$6.50 and \$6.00.

Now it's my turn to be the SENDER. Given that they've shown trust and I've been fair in return, and the overall cooperation has been good in this game, I should send a high amount to maintain the cooperative relationship.

My sending pattern so far:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back
- Round 5: Sent \$4, got \$5.12 back

Sending \$4 has worked well. I'll continue with that.

```
Usage: {
  "input_tokens": 2309,
  "output_tokens": 373,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_2

Timestamp: 2026-08-12T09:25:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_6"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_6": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.95 to you. Your visible payoff was \$4.95.

Your total visible earnings across all rounds are \$4.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Analysis:****

- My co-player sent \$2.11 (42% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- This creates an opportunity for mutual benefit

****Fairness considerations:****

- The sender took a risk by sending money first
- I want to reward that trust to encourage future cooperation
- Both of us should end up better off than if no money was sent

****Decision:****

If I return \$3.20:

- Sender ends with: $\$5 - \$2.11 + \$3.20 = \6.09 (gain of ~\$1.09)
- I keep: $\$6.33 - \$3.20 = \$3.13$ (gain of ~\$3.13)

This seems fair - both of us benefit from the cooperation, and the sender is rewarded for their trust while I also gain from being in the receiver position.

```
'''json
{"return": 3.20}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the SENDER:

- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.95 to me
- My payoff: $\$5 - \$3 + \$2.95 = \4.95
- The receiver kept: $\$9.0 - \$2.95 = \$6.05$

So the receiver in Round 1 kept about 67% of what they received and returned about 33%.

2. Now in Round 2, I'm the RECEIVER:

- My co-player (a different agent) sent me \$2.11
- I received \$6.33 (which is $\$2.11 \times 3$)
- I need to decide how much to return

The note mentions there may be communication noise, so the \$2.11 might be slightly different from what was actually sent.

Let me think about what's fair and strategic:

- The sender gave up \$2.11 from their \$5, keeping \$2.89 initially
- I received \$6.33

- If I want to be reciprocal and fair, I should return enough that:
 1. The sender ends up better off than if they had sent nothing
 2. I also benefit from the transaction

A 50-50 split of the received amount would be \$3.165 returned, meaning:

- Sender ends up with: $\$5 - \$2.11 + \$3.165 = \6.055
- I keep: $\$6.33 - \$3.165 = \$3.165$

But that might be too generous. Let me consider what the previous receiver did - they returned about 33% of what they received.

If I return 33% of \$6.33 = \$2.09, then:

- Sender ends up with: $\$5 - \$2.11 + \$2.09 = \4.98
- I keep: $\$6.33 - \$2.09 = \$4.24$

The sender invested 42% of their endowment, so I should consider whether they break even or better. The cooperation created \$4.22 in additional value beyond the initial \$2.11 sent. If I split those gains equally, each of us would get an extra \$2.11 on top of what we'd have without the exchange.

That would mean the sender receives \$7.11 total and I return \$4.22. But looking at it another way: we now have \$9.22 combined, and splitting equally gives us each \$4.61. The sender needs \$1.72 returned to reach that, leaving me with \$4.61 as well. This equal split doesn't fully reward the sender's initial risk though, so I'm considering what fairness really means here.

A reasonable approach is returning enough to benefit both of us while acknowledging my advantage as the receiver. If I return \$3.00, the sender gains \$0.89 and I keep \$3.33-both better off than initially. Returning \$3.50 instead gives the sender a \$1.39 gain while I keep \$2.83, which feels more balanced between us....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 1431,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

****Analysis:****

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

****Considerations:****

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

****This round:****

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

****Considerations:****

1. This is a different co-player than in rounds 1 and 2
2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

****Strategy:****

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

usage:

```
{
  "input_tokens": 1133,
  "output_tokens": 554,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Analysis:****

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

****Considerations:****

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

****Calculation:****

If I return \$6.80:

- Sender ends with: \$5 - \$4.14 + \$6.80 = \$7.66 (gain of \$2.66)
- I keep: \$12.42 - \$6.80 = \$5.62 (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 6.80}
'''
```

reasoning:

Let me analyze this situation:

****What happened:****

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

****Considerations:****

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

****Calculation options:****

If I return \$6.00:

- Sender ends with: $\$5 - \$4.14 + \$6.00 = \6.86 (gain of \$1.86)
- I keep: $\$12.42 - \$6.00 = \$6.42$ (gain of \$6.42)

If I return \$7.00:

- Sender ends with: $\$5 - \$4.14 + \$7.00 = \7.86 (gain of \$2.86)
- I keep: $\$12.42 - \$7.00 = \$5.42$ (gain of \$5.42)

If I return \$6.50:

- Sender ends with: $\$5 - \$4.14 + \$6.50 = \7.36 (gain of \$2.36)
- I keep: $\$12.42 - \$6.50 = \$5.92$ (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.

Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd gain \$3.36 while...

usage:

```
{
  "input_tokens": 1425,
  "output_tokens": 847,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.14, it became \$12.419999999999998, and you returned \$6.8. Your visible payoff was \$5.619999999999998.

Your total visible earnings across all rounds are \$20.38.

This round, you are the RECEIVER against your current co-player. They sent you \$3.63, so you received \$10.89. How much do you return? (0-10.89)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Analysis:****

My co-player sent \$3.63 (73% of their endowment), showing substantial trust. I received \$10.89 after the 3x multiplier.

****Considerations:****

- They sent a significant portion of their endowment
- Both of us should benefit from this cooperation
- I want to reward their trust appropriately

****Calculation:****

If I return \$5.80:
- Sender ends with: $\$5 - \$3.63 + \$5.80 = \7.17 (gain of \$2.17)
- I keep: $\$10.89 - \$5.80 = \$5.09$ (gain of \$5.09)

Both of us gain substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 5.80}
'''
```

reasoning:

****Analysis:****

My co-player sent \$3.63 (72.6% of their endowment), showing substantial trust.
I received \$10.89 after the 3x multiplier.

****Considerations:****

- They sent a good amount, demonstrating trust
- I want to reward this appropriately
- Both of us should benefit from the cooperation

****Calculation:****

If I return around \$5.50:
- Sender ends with: $\$5 - \$3.63 + \$5.50 = \6.87 (gain of \$1.87)
- I keep: $\$10.89 - \$5.50 = \$5.39$ (gain of \$5.39)

Both benefit, sender is rewarded for trust.

Let me think about a slightly different amount. The sender sent 72.6% of their endowment, which is quite high. I should return enough that they benefit well.

If I return \$5.80:
- Sender ends with: $\$5 - \$3.63 + \$5.80 = \7.17 (gain of \$2.17)
- I keep: $\$10.89 - \$5.80 = \$5.09$ (gain of \$5.09)

This seems fair - both of us gain substantially, and the sender is well-rewarded for their high trust.

usage:

```
{
  "input_tokens": 1758,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.63, it became \$10.89, and you returned \$5.8. Your visible payoff was \$5.090000000000001.

Your total visible earnings across all rounds are \$25.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

****Analysis:****

I'm facing a new co-player as sender. My previous sender experiences:

- Round 1: Sent \$3, received \$2.95 back (payoff \$4.95)
- Round 3: Sent \$3, received \$4.68 back (payoff \$6.68)

Both times, sending \$3 led to positive outcomes with reasonable reciprocation.

****Strategy:****

- \$3 creates \$9 for the receiver via the 3x multiplier
- This amount signals willingness to cooperate
- It balances trust-building with prudent risk management
- Consistency may help establish a cooperative reputation

I'll continue with my established approach.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

I'm the sender this round with a new co-player. Let me review my experience so far:

Sender rounds:

- Round 1: Sent \$3, got back \$2.95 (payoff \$4.95)
- Round 3: Sent \$3, got back \$4.68 (payoff \$6.68)

Receiver rounds:

- Round 2: Received \$6.33 from \$2.11 sent, returned \$3.20 (payoff \$3.13)
- Round 4: Received \$12.42 from \$4.14 sent, returned \$6.80 (payoff \$5.62)
- Round 5: Received \$10.89 from \$3.63 sent, returned \$5.80 (payoff \$5.09)

As sender, I've consistently sent \$3 and received reasonable returns. My co-players as receivers have varied in how much they sent initially.

This is round 6, so I'm establishing patterns. I want to continue building cooperative norms. Sending \$3 has worked well for me in the past - it's enough to create value through the 3x multiplier while not being overly risky.

I'll stick with my strategy of sending \$3.

usage:

```
{
  "input_tokens": 2077,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$6.95 to you. Your visible payoff was \$8.95.

Your total visible earnings across all rounds are \$34.42.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

This is the same co-player from Round 6! They demonstrated exceptional generosity by returning \$6.95 on my \$3 send.

****What happened last round:****

- I sent \$3, they received \$9
- They returned \$6.95, keeping only \$2.05 for themselves
- I gained \$3.95 while they gained just \$2.05

****Response:****

Such generous reciprocation deserves increased trust. They clearly value cooperation. Since I'm playing with the same partner again, I should signal that I recognize and appreciate their trustworthiness.

I'll increase my send to \$4, which creates \$12 for them and shows I'm willing to escalate cooperation with a proven trustworthy partner.

```
'''json
{"send": 4}
'''
```

reasoning:

This is very interesting - I'm playing against the SAME co-player as the previous round! T

hey returned \$6.95 on my \$3 send, which is extremely generous (they gave me back more than double what I sent, even accounting for the 3x multiplier).

Let me calculate what happened:

- I sent \$3, they received \$9
- They returned \$6.95
- I ended with: $\$5 - \$3 + \$6.95 = \8.95
- They kept: $\$9 - \$6.95 = \$2.05$

So they only kept \$2.05 while I gained \$3.95! This is a very cooperative partner.

Since I'm playing with the same person again and they were so generous, I should reciprocate this trust by sending more. This could establish a strong cooperative relationship.

Options:

1. Send \$3 again - safe, but doesn't fully reciprocate their generosity
2. Send \$4 - shows increased trust based on their behavior
3. Send \$5 - maximum trust, but risky

Given their extremely generous return of \$6.95, I think sending \$4 is appropriate. It shows I valued their cooperation and am willing to increase my trust, while still maintaining some prudence.

usage:

```
{
  "input_tokens": 2371,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.79 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$42.21.

This round, you are the RECEIVER against your current co-player. They sent you \$4.62, so you received \$13.86. How much do you return? (0-13.86)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

This is the same co-player from rounds 6 and 7! Our partnership history:

- Round 6: I sent \$3 ? they returned \$6.95 (extremely generous)
- Round 7: I sent \$4 ? they returned \$6.79 (still very generous)
- Round 8 (now): They sent \$4.62 (92% of their endowment!)

****Response:****

This player has demonstrated extraordinary trust and generosity. They're now sending me almost everything they have. I need to reciprocate at a similarly high level.

****Calculation:****

If I return \$8.50:

- Sender ends with: $\$5 - \$4.62 + \$8.50 = \8.88 (gain of \$3.88)
- I keep: $\$13.86 - \$8.50 = \$5.36$ (gain of \$5.36)

Both of us benefit substantially, maintaining our strong cooperative relationship.

```
'''json
{"return": 8.50}
'''
```

Reasoning:

Let me analyze this carefully:

****Context:****

- This is the same co-player from rounds 6 and 7
- Round 6: I sent \$3, they returned \$6.95 (very generous)
- Round 7: I sent \$4, they returned \$6.79 (still very generous)
- Now Round 8: They sent me \$4.62 (92% of their endowment!)

****Analysis:****

This co-player has been extremely generous and trustworthy:

- They returned \$6.95 on receiving \$9 (keeping only \$2.05)
- They returned \$6.79 on receiving \$12 (keeping only \$5.21)
- Now they're sending me \$4.62 out of \$5

They're showing incredible trust in me. I need to reciprocate appropriately.

****Calculation:****

I received \$13.86 after the 3x multiplier.

Let me think about what would be fair:

- They sent \$4.62, showing maximum trust
- If I return \$7.50:
 - Sender ends with: $\$5 - \$4.62 + \$7.50 = \7.88 (gain of \$2.88)
 - I keep: $\$13.86 - \$7.50 = \$6.36$ (gain of \$6.36)

Actually, given how generous they've been to me, I should be similarly generous. Let me aim for a return that gives them a strong gain comparable to what I've received.

If I return \$8.00:

- Sender ends with: $\$5 - \$4.62 + \$8.00 = \8.38 (gain of \$3.38)
- I keep: $\$13.86 - \$8.00 = \$5.86$ (gain of \$5.86)

This seems fair - both of us gain substantially, and I'm rewarding their exceptional trust.

...

```
Usage: {  
  "input_tokens": 2515,  
  "output_tokens": 845,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-08-12T09:26:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_4"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.0. Your visible payoff was \$5.350000000000001.

Your total visible earnings across all rounds are \$5.350000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- In the previous round, my co-player showed significant trust by sending 69% of their endowment
- I reciprocated fairly by returning a substantial amount
- Now with a new co-player, I want to establish cooperation

I'll send a generous amount to signal trust and cooperation, while being slightly more conservative since this is a new pairing.

{'send': 3.00}

reasoning:

I'm now the SENDER in Round 2. Let me think about this situation:

- In Round 1, I was the RECEIVER and someone sent me \$3.45 (69% of their endowment)
- I returned \$5.00, which was a fair reciprocation
- I kept \$5.35, and the sender ended with \$6.55

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. In the previous round, my co-player sent 69% of their endowment, showing significant trust
2. I rewarded that trust by returning about 48% of what I received
3. This is a new co-player, so I can't assume they'll behave the same way
4. However, I should probably maintain a cooperative stance to build trust in the experiment
5. There's communication noise, so amounts may differ from what was actually sent

Given that:

- The previous sender was generous (69%)
- I want to establish cooperation with this new co-player
- Being moderately generous but not overly risky seems reasonable

I'll send around 60% of my endowment, which would be \$3.00. This shows trust and cooperation on while being slightly more conservative than what I received in the previous round.

Actually, let me reconsider. The person who sent to me was quite generous at 69%. If I want to maintain a cooperative equilibrium in this multi-agent setting, I should probably match that level of trust or be similarly generous.

I'll send \$3.00 (60% of my endowment), which is a solid show of trust and cooperation.

usage:

```
{
  "input_tokens": 843,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.93 to you. Your visible payoff was \$5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:

If I return \$3.20:

- Sender ends with: \$5 - \$2.14 + \$3.20 = \$6.06
- I end with: \$6.42 - \$3.20 = \$3.22
- This creates relatively balanced outcomes and rewards their trust

```
{'return': 3.20}
```

reasoning:

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust

2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:

- Sender ends with: $\$5 - \$2.14 + \$3.00 = \5.86
- I end with: $\$6.42 - \$3.00 = \$3.42$
- This gives them a decent return on their trust

If I return \$3.50:

- Sender ends with: $\$5 - \$2.14 + \$3.50 = \6.36
- I end with: $\$6.42 - \$3.50 = \$2.92$
- This is more generous but leaves me with less than them

If I return \$3.20:

- Sender ends with: $\$5 - \$2.14 + \$3.20 = \6.06
- I end with: $\$6.42 - \$3.20 = \$3.22$
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 588,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.14, it became \$6.42, and you returned \$3.2. Your visible payoff was \$3.2199999999999998.

Your total visible earnings across all rounds are \$14.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- When I sent \$3.00 in Round 2, I received a fair return of \$3.93
- I've been playing cooperatively and seeing reasonable reciprocation

I'll continue with a cooperative strategy and send a generous amount to this new co-player.

```
{'send': 3.00}
```

reasoning:

Looking at my history:

- Round 1: As receiver, got \$10.35, returned \$5.00, earned \$5.35
- Round 2: As sender, sent \$3.00, got back \$3.93, earned \$5.93
- Round 3: As receiver, got \$6.42, returned \$3.20, earned \$3.22

In Round 2, I sent \$3.00 (60%) and my co-player returned \$3.93 out of \$9.00 (43.7%), which was decent reciprocation.

Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reasonable reciprocation. I should continue with a cooperative strategy.

I'll send around \$3.00 again to maintain consistency and signal trust.

usage:

```
{
  "input_tokens": 1348,
  "output_tokens": 269,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.24 to you. Your visible payoff was \$8.24.

Your total visible earnings across all rounds are \$22.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent \$4.44 (88.8% of their endowment) - this is exceptional trust!
- I received \$13.32 after the 3x multiplication
- Total resources created: \$13.88

This sender showed tremendous cooperation. I should reciprocate strongly:

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This rewards their high trust with balanced outcomes

{'return': 6.50}

reasoning:

The sender sent \$4.44, which is 88.8% of their endowment - this is very high trust! I received \$13.32 after multiplication.

Total resources created: \$5 - \$4.44 + \$13.32 = \$13.88

The sender showed exceptional trust. I should reciprocate generously.

If I return \$6.50:

- Sender ends with: \$5 - \$4.44 + \$6.50 = \$7.06
- I end with: \$13.32 - \$6.50 = \$6.82
- This creates a fairly balanced outcome and rewards their high trust

If I return \$7.00:

- Sender ends with: \$5 - \$4.44 + \$7.00 = \$7.56
- I end with: \$13.32 - \$7.00 = \$6.32
- This is also reasonable

Let me go with \$6.50 as it provides good reciprocation while maintaining balance.

usage:

```
{
  "input_tokens": 1564,
  "output_tokens": 392,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.44, it became \$13.32, and you returned \$6.5. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$29.560000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience:

- When I sent \$3.00, I've seen returns of \$3.93 and \$6.24
- The game has shown good cooperation overall
- I want to maintain a cooperative strategy

I'll continue with a consistent, generous approach.

```
{'send': 3.00}
```

reasoning:

Looking at my track record:

- Round 2: I sent \$3.00, received back \$3.93
- Round 4: I sent \$3.00, received back \$6.24 (very generous!)
- Round 5: Partner sent me \$4.44 (very trusting), I returned \$6.50

The cooperation has been generally good, with some partners being very generous. I've been sending \$3.00 consistently and seeing good returns. I could continue with that strategy, or perhaps be slightly more generous given the positive experiences.

Let me continue with \$3.00 as it's been working well and represents 60% of my endowment - a reasonable cooperative level.

usage:

```
{
  "input_tokens": 1837,
  "output_tokens": 233,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.16 to you. Your visible payoff was \$7.16.

Your total visible earnings across all rounds are \$36.72.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46

This sender showed exceptional cooperation. I should reciprocate strongly:

If I return \$7.00:

- Sender ends with: \$5 - \$4.73 + \$7.00 = \$7.27
- I end with: \$14.19 - \$7.00 = \$7.19
- This creates very balanced outcomes rewarding their exceptional trust

```
{'return': 7.00}
```

reasoning:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - this is extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46 (\$0.27 remaining with sender + \$14.19 with me)

This sender showed exceptional cooperation - sending almost everything! I should reciprocate strongly.

If I return \$7.00:

- Sender ends with: \$5 - \$4.73 + \$7.00 = \$7.27
- I end with: \$14.19 - \$7.00 = \$7.19
- This creates very balanced outcomes and rewards their exceptional trust

Let me think about this... they sent 94.6%, which is really high. I want to make sure they come out well.

Actually, let me aim for balanced final outcomes:

If I return \$7.00:

- Sender: \$7.27
- Me: \$7.19

Very balanced!

I think \$7.00 is appropriate.

usage:

```
{
  "input_tokens": 2050,
  "output_tokens": 405,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$7.0. Your visible payoff was \$7.190000000000001.

Your total visible earnings across all rounds are \$43.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent strategy:

- I've been sending \$3.00 as a sender and have received good returns each time
- The game has shown strong cooperation overall
- I'll maintain my cooperative approach

```
{'send': 3.00}
```

Reasoning:

Looking at my track record as a sender:

- Round 2: Sent \$3.00, received \$3.93 back
- Round 4: Sent \$3.00, received \$6.24 back
- Round 6: Sent \$3.00, received \$5.16 back

So sending \$3.00 has consistently resulted in good returns.

As a receiver:

- Round 3: Received \$6.42, returned \$3.20
- Round 5: Received \$13.32, returned \$6.50
- Round 7: Received \$14.19, returned \$7.00

I've been playing cooperatively throughout and the game has shown good reciprocity. This is Round 8, which might be the final round or close to it given typical experimental designs.

I should maintain my cooperative strategy. Sending \$3.00 has worked well for me consistently.

```
{'send': 3.00}
```

Usage: {

```
  "input_tokens": 2086,
  "output_tokens": 277,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-08-12T09:26:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_4"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_4": "trustee",
```

```

    "Agent_8": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.46 to you. Your visible payoff was \$6.46.

Your total visible earnings across all rounds are \$6.46.

This round, you are the RECEIVER against your current co-player. They sent you \$2.73, so you received \$8.19. How much do you return? (0-8.19)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Round 2 Analysis (as RECEIVER):****

- My co-player sent \$2.73 (about 55% of their \$5 endowment)
- I received \$8.19 after the 3x multiplier
- They showed moderate-to-good trust by sending more than half

****Reciprocity consideration:****

In Round 1, my co-player returned about 45.5% of what they received, which seemed fair. I should aim for similar fairness to encourage continued cooperation.

****Decision:****

Returning approximately 50% of what I received (\$4.10) would:

- Give my co-player a total of \$6.37 (\$2.27 kept + \$4.10 returned)
- Leave me with \$4.09
- Show I value reciprocity and fairness

This return is proportional to the trust they showed and should encourage cooperation in future rounds.

```
'''json
{"return": 4.10}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$4.0
- It became \$12.0
- My co-player returned \$5.46
- My payoff: $\$5 - \$4 + \$5.46 = \6.46

This was a decent return - they kept \$6.54 and returned \$5.46, which is roughly 45.5% of the \$12 they received. That's reasonable cooperation.

Round 2 (as RECEIVER):

- My current co-player sent \$2.73
- I received \$8.19 (2.73×3)
- I need to decide how much to return

Important notes:

- There's communication noise, so the actual amount sent might be slightly different
- This is a different co-player than Round 1
- My co-player sent about 55% of their \$5 ($\$2.73/\$5 = 54.6\%$), showing moderate trust
- I want to encourage cooperation for future rounds

Strategy:

Since my co-player showed moderate trust by sending \$2.73, I should return a fair amount to encourage future cooperation. A 50-50 split of what I received would be around \$4.10.

However, I should also consider:

1. They sent a bit more than half their endowment
2. This is an ongoing multi-round game
3. Reciprocity is important for building trust

I'll return around \$4.00-\$4.50 to be fair and encourage cooperation. Let me go with \$4.10, which is approximately 50% of what I received.

Actually, let me think about this more carefully. If they sent \$2.73 and kept \$2.27:

- Returning \$4.10 gives them \$6.37 total while I keep \$4.09, so they gain \$1.37 and I gain \$4.09
- But I could also aim for more balanced final outcomes by calculating an equal split of the total pool (\$10.46), which would give us each \$5.23
- That would require me to return \$2.96 instead...

usage:

```
{
  "input_tokens": 790,
  "output_tokens": 926,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier
- This is more trust than the previous co-player showed in Round 2

****Reciprocity:****

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion that rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

****Past behavior:****

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

****Reciprocity consideration:****

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). This shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high trust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \5.54

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50
- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

usage:


```
{
  "input_tokens": 1157,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.55, it became \$10.649999999999999, and you returned \$5.5. Your visible payoff was \$5.149999999999999.

Your total visible earnings across all rounds are \$15.7.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Round 4 Analysis (as SENDER):****

Looking at my experience so far:

- Round 1: Sent \$4, got good reciprocation (\$5.46 back)
- Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
- Total earnings: \$15.70 (averaging \$5.23/round)

****New co-player consideration:****

This is a different co-player than previous rounds, so no direct history. However, the cooperative approach has been working well overall.

****Decision:****

I'll send ****\$4**** again because:

- The 3x multiplier creates strong mutual benefit
- It signals trust and willingness to cooperate
- It keeps \$1 as insurance against complete defection
- This amount worked well in Round 1

```
'''json
```

```
{"send": 4}
```

```
'''
```

reasoning:

****Round 4 Analysis (as SENDER):****

Let me review the pattern so far:

- Round 1: I sent \$4, received back \$5.46 (good return, ~45.5% of \$12)
- Round 2: As receiver, got \$2.73 (sent), I returned \$4.10 (~50% of \$8.19)
- Round 3: As receiver, got \$3.55 (sent), I returned \$5.50 (~52% of \$10.65)

Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized each round).

Key considerations:

1. I have no history with this specific co-player
2. My previous experiences have been fairly positive:
 - Round 1: Good reciprocation when I sent high
 - Round 2 & 3: I reciprocated fairly as receiver
3. The game mechanics reward cooperation through the 3x multiplier
4. There's communication noise, so exact amounts may vary

Strategy:

- Since cooperation has worked well so far, I should lean toward sending a substantial amount
- However, this is a new co-player with no history
- I've been playing fairly cooperatively, which might be building reputation (though names are hidden, the pattern of fair play might matter in multi-agent settings)

I think sending \$3.50-\$4.50 would be reasonable. Let me go with \$4 again - it's worked before, shows strong trust without being all-in, and leaves some hedge.

Actually, looking at my total earnings (\$15.70 over 3 rounds), I'm doing reasonably well a

veraging about \$5.23 per round. The cooperative strategy is working.

I'll send \$4 again.

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.39 to you. Your visible payoff was \$9.39.

Your total visible earnings across all rounds are \$25.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their endowment) - strong trust
- I received \$10.35 after the multiplier
- Previous round showed excellent reciprocation when I was sender (\$8.39 back!)

****Reciprocity:****

My co-player demonstrated high trust by sending nearly 70%. I should maintain the fair reciprocation pattern that's been working well.

Returning approximately 50% (\$5.20) would:

- Give my co-player a total of \$6.75 (\$1.55 kept + \$5.20 returned)
- Leave me with \$5.15
- Reward their trust appropriately

```
'''json
{"return": 5.20}
'''
```

reasoning:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their \$5 endowment) - showing high trust
- I received \$10.35 after the multiplier
- This is similar trust level to Round 3 (where co-player sent \$3.55)

****Previous round performance:****

- Round 4: I sent \$4, got back \$8.39 (70% return rate!) - excellent reciprocation
- This suggests cooperative behavior is being rewarded

****Reciprocity consideration:****

My co-player sent 69% of their endowment, showing strong trust. I should return a fair proportion.

Returning approximately 50-52% seems appropriate:

- 50% = \$5.18
- 52% = \$5.38

If I return \$5.20:

- My co-player gets total: \$1.55 (kept) + \$5.20 (returned) = \$6.75
- I keep: \$5.15
- Both benefit well

This is consistent with my previous fair reciprocation strategy and should maintain cooperation.

usage:

```
{
  "input_tokens": 1801,
  "output_tokens": 448,
  "reasoning_tokens": 0
}
```

```

}
Message 10 role=user
content:
  Round 6

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $3.45, it became $10.350000000000001, and you returned $5.2. Your visible
  payoff was $5.1500000000000001.
  Your total visible earnings across all rounds are $30.240000000000002.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 11 role=assistant
content:
  **Round 6 Analysis (as SENDER):**

  Performance review:
  - As sender: Sent $4 twice, received good returns both times (avg $6.93 back)
  - As receiver: Been consistent with fair ~50% returns
  - Average earnings: $6.05/round

  **New co-player:**
  Different partner again, but the strategy of sending $4 has been working well.

  **Decision:**
  Sending **$4** because:
  - Proven successful in previous sender rounds
  - Demonstrates strong trust
  - Creates significant value through 3x multiplier
  - Maintains small hedge ($1)

  '''json
  {"send": 4}
  '''

reasoning:
  **Round 6 Analysis (as SENDER):**

  Looking at my track record:
  - Round 1: Sent $4, got $5.46 back ? Profit of $6.46
  - Round 4: Sent $4, got $8.39 back ? Profit of $9.39

  Both times I sent $4, I received good to excellent returns. Average earnings per round is
  now $6.05 ($30.24/5 rounds).

  As receiver:
  - Round 2: Received $8.19, returned $4.10
  - Round 3: Received $10.65, returned $5.50
  - Round 5: Received $10.35, returned $5.20

  I've been consistently fair as a receiver, returning around 50%.

  **New co-player:**
  This is another new pairing. The pattern of sending $4 has been successful, showing trust
  while keeping some insurance.

  **Decision:**
  I'll continue with $4 because:
  - It's been consistently successful
  - Shows strong cooperation
  - Keeps $1 as hedge
  - The multiplier effect makes this beneficial for both parties if reciprocated

  Let me send $4.
usage:
  {
    "input_tokens": 2113,
    "output_tokens": 432,
    "reasoning_tokens": 0
  }

```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.08 to you. Your visible payoff was \$7.08.

Your total visible earnings across all rounds are \$37.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Round 7 Analysis (as SENDER):****

****Same co-player as Round 6:****

- Last round: I sent \$4, they returned \$6.08 (about 51% of what they received)
- This was fair and cooperative reciprocation
- My payoff was good at \$7.08

****Decision:****

Since this co-player has demonstrated trustworthiness, I'll send ****\$4**** again to:

- Continue the mutually beneficial cooperation
- Reward their fair reciprocation
- Maximize joint value creation

```
'''json
{"send": 4}
'''
```

reasoning:

****Round 7 Analysis (as SENDER):****

Important: This is the SAME co-player as Round 6 (it says "against your current co-player" in the round description).

Previous round with this same co-player:

- I sent \$4
- They received \$12
- They returned \$6.08 (about 50.7% of what they received)
- My payoff was \$7.08 (\$1 kept + \$6.08 returned)

This was a fair and cooperative return! They showed good reciprocity.

Since:

1. This is the same co-player
2. They reciprocated fairly last round
3. Continuing cooperation could lead to mutual benefit
4. My strategy of sending \$4 has been working well

I should send \$4 again to maintain the cooperative relationship with this co-player.

usage:

```
{
  "input_tokens": 2391,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$43.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 8 Analysis (as RECEIVER):****

- My co-player sent \$2.97 (59% of their endowment) - good trust
- I received \$8.91 after the multiplier

****Reciprocity:****

Maintaining my consistent approach of returning approximately 50% to reward trust and encourage cooperation.

Returning \$4.45 would:

- Give my co-player a total of \$6.48 (\$2.03 kept + \$4.45 returned)
- Leave me with \$4.46
- Stay consistent with my fair reciprocation pattern

```
'''json
{"return": 4.45}
'''
```

Reasoning:

****Round 8 Analysis (as RECEIVER):****

- My co-player sent \$2.97 (about 59% of their endowment)
- I received \$8.91 after the multiplier
- This shows moderate-to-good trust

****Reciprocity consideration:****

I've been consistently returning around 50% when I'm receiver, which has been working well. This co-player showed decent trust by sending about 60% of their endowment.

Returning approximately 50% (\$4.45) would:

- Give my co-player a total of \$6.48 (\$2.03 kept + \$4.45 returned)
- Leave me with \$4.46
- Maintain fairness and encourage cooperation

Let me check the pattern:

- Round 2: Received \$8.19, returned \$4.10 (50%)
- Round 3: Received \$10.65, returned \$5.50 (52%)
- Round 5: Received \$10.35, returned \$5.20 (50%)

So returning around \$4.45-4.50 would be consistent with my previous behavior.

```
Usage: {
  "input_tokens": 2478,
  "output_tokens": 420,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:26:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_3"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_3": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$6.0. Your visible payoff was \$4.98.

Your total visible earnings across all rounds are \$4.98.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed high trust by sending 73.2% of their endowment, and I reciprocated fairly. That cooperation worked well for both of us.

Now with a new co-player, I want to establish a positive dynamic while being somewhat cautious since I don't know their behavior yet. I'll send a significant amount to signal trust and encourage cooperation.

I'll send \$3.00 (60% of my endowment), which shows trust while being slightly more conservative than the 73.2% shown to me in the previous round.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the RECEIVER
- The sender sent me \$3.66 (73.2% of their endowment), showing high trust
- I received \$10.98
- I returned \$6.00, which gave them a profit of \$2.34
- I kept \$4.98

Now I'm the SENDER with a new co-player. I need to decide how much to send.

Key considerations:

1. The previous sender showed high trust (73.2%)
2. I reciprocated fairly by returning a reasonable amount
3. I'm now paired with a different agent
4. There's communication noise in the game, so amounts may differ slightly

Since the previous sender showed high trust and it worked out well for both of us, I might want to adopt a similar strategy of showing trust. This could establish a positive dynamic.

However, I'm with a new co-player now, so I don't know their behavior yet. Common strategies:

- Send about 50% as a balanced trust signal
- Send similar to what I received (73.2%) to match the previous trust level
- Send less to be more cautious with an unknown player

Given that:

1. The experiment seems to reward trust (the previous round worked well)
2. I want to establish good cooperation
3. But I also want to be somewhat cautious with a new player

I'll send around 60% of my endowment, which is \$3.00. This shows trust but is slightly more conservative than the 73.2% I received in the previous round.

Actually, let me reconsider. Since I experienced a trustworthy interaction in Round 1, and the game encourages cooperation, maybe I should send closer to what I received. I'll send \$3.50 (70%), which is close to the 73.2% trust level shown to me.

Let me go with \$3.00 as a strong but reasonable trust signal.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)

- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)
- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)
- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

usage:

```
{
  "input_tokens": 1092,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern . I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving. This signals strong trust while being neither too conservative nor too aggressive.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.99 to you. Your visible payoff was \$8.49.

Your total visible earnings across all rounds are \$25.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant**content:**

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is moderate trust - not as high as other rounds, but still showing willingness to cooperate by sending more than half their endowment.

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This fairly rewards their trust

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is lower trust than I've seen in previous rounds (73.2%, 79.8% in earlier rounds). However, it's still a majority of their endowment, showing moderate trust.

I should still reciprocate fairly. Let me think about what would be appropriate:

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This is fair and rewards their trust

If I return \$4.00:

- The sender ends with \$6.21 (profit of \$1.21)
- I keep \$4.37
- This is also reasonable

Given that they sent a moderate amount (not as high as previous senders), I should return a moderate but fair amount. I'll return \$4.50 to maintain a cooperative reputation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1640,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.79, it became \$8.370000000000001, and you returned \$4.5. Your visible payoff was \$3.870000000000001.

Your total visible earnings across all rounds are \$29.610000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my history:

- Round 2: I sent \$3.00 (60%), received \$4.80 back - good cooperation
- Round 4: I sent \$3.50 (70%), received \$6.99 back - excellent cooperation
- Overall, higher sends have been rewarded

With a new co-player, I'll maintain a cooperative approach. Sending \$3.50 (70%) worked very well last time, so I'll repeat that strategy.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review my history:

Round 1: Receiver - received from sender who sent \$3.66 (73.2%), I returned \$6.00, earned \$4.98
Round 2: Sender - sent \$3.00 (60%), received \$4.80 back, earned \$6.80
Round 3: Receiver - received from sender who sent \$3.99 (79.8%), I returned \$6.50, earned \$5.47
Round 4: Sender - sent \$3.50 (70%), received \$6.99 back, earned \$8.49
Round 5: Receiver - received from sender who sent \$2.79 (55.8%), I returned \$4.50, earned \$3.87

My sending history:

- Round 2: sent \$3.00 (60%), got back \$4.80 - good return
- Round 4: sent \$3.50 (70%), got back \$6.99 - excellent return

Both times I've sent, I've received good cooperation in return. The pattern suggests that sending around 60-70% works well.

Round 6 now - new co-player (since pairings are randomized each round). I should continue with a cooperative strategy. I'll send around the same range, maybe \$3.25 to be between my previous sends.

Actually, let me stick with \$3.50 since that worked very well last time.

usage:

```
{
  "input_tokens": 1907,
  "output_tokens": 457,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.17 to you. Your visible payoff was \$5.67.

Your total visible earnings across all rounds are \$35.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This shows good trust - about 2/3 of their endowment. I want to reciprocate fairly.

If I return \$5.50:

- The sender ends with \$7.22 (profit of \$2.22)
- I keep \$4.34
- This rewards their trust appropriately

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This is moderate-high trust. The sender showed decent cooperation by sending about 2/3 of their endowment.

Let me think about a fair return:

- If I return \$5.50:
 - Sender ends with \$7.22 (profit of \$2.22)
 - I keep \$4.34
 - This rewards their trust fairly

That seems reasonable - the sender gets a good profit, I keep a decent amount, and it encourages future cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.28, it became \$9.84, and you returned \$5.5. Your visible payoff was \$4.34.

Your total visible earnings across all rounds are \$39.620000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 - solid return
- Round 4: Sent \$3.50 (70%), received \$6.99 - excellent return
- Round 6: Sent \$3.50 (70%), received \$4.17 - moderate return

Cooperation has generally worked well. This is the final round with a new co-player. I'll send \$3.00 (60%), which has proven to be a reliable cooperative strategy.

```
'''json
{"send": 3.00}
'''
```

Reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back - good
- Round 4: Sent \$3.50 (70%), received \$6.99 back - excellent
- Round 6: Sent \$3.50 (70%), received \$4.17 back - less good

The pattern shows:

- 60% send got me \$4.80 return
- 70% send got me \$6.99 return (great)
- 70% send got me \$4.17 return (okay)

And as a receiver, I've been fairly generous:

- Received \$10.98, returned \$6.00
- Received \$11.97, returned \$6.50
- Received \$8.37, returned \$4.50
- Received \$9.84, returned \$5.50

Overall, cooperation has been working reasonably well. This is the last round, so I should continue being cooperative but maybe slightly cautious since it's the final interaction.

I'll send \$3.00 (60%) - a solid cooperative amount that's worked well before. It's slightly less than \$3.50 but still shows good faith.

```
Usage: {
  "input_tokens": 2201,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-08-12T09:26:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_3"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_3": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.95 to you. Your visible payoff was \$7.95.

Your total visible earnings across all rounds are \$7.95.

This round, you are the RECEIVER against your current co-player. They sent you \$2.34, so you received \$7.02. How much do you return? (0-7.02)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Round 1 context:**** My previous co-player returned \$6.95 out of \$12.00 they received (about 58%), which was quite generous.

****Current situation (Round 2):****

- My co-player sent \$2.34 (47% of their endowment)
- I received \$7.02
- They showed trust by sending a meaningful amount

I want to:

1. Reward their trust
2. Encourage continued cooperation
3. Follow the generous precedent set in Round 1

Returning about 57% of what I received (similar to Round 1) would be \$4.00. This gives them a total payoff of \$6.66 and leaves me with \$3.02, creating mutual benefit.

{'return': 4.00}

reasoning:

Let me analyze this situation:

Round 1 recap (I was SENDER):

- I sent \$4.0
- It became \$12.0
- They returned \$6.95
- My payoff: \$5 - \$4 + \$6.95 = \$7.95
- Their payoff: \$12.0 - \$6.95 = \$5.05

So my co-player in Round 1 returned about 58% of what they received, which is fairly generous.

Round 2 (I am RECEIVER):

- Current co-player sent me \$2.34
- I received \$7.02 (2.34×3)
- I need to decide how much to return

This is a different co-player than Round 1.

My co-player sent \$2.34 out of \$5, keeping \$2.66. They sent about 47% of their endowment.

If I think about fairness:

- They gave up \$2.34
- I received \$7.02
- They currently have \$2.66 (their kept amount)
- I currently have \$7.02

One approach: Try to equalize final payoffs

- If I return \$2.18: They'd have \$2.66 + \$2.18 = \$4.84, I'd have \$7.02 - \$2.18 = \$4.84
- Equal split of total created value

Another approach: Return proportionally to what they sent

- They sent 47% of their endowment
- Similar return rate: 47% of \$7.02 = \$3.30

Another approach: Return more than they sent (be generous)

- They sent \$2.34, maybe return \$3.50-\$4.00

Another approach: Follow the example from Round 1

- In Round 1, my co-player returned about 58% of what they received
- 58% of \$7.02 = \$4.07

Since they demonstrated trust with a reasonable send and the previous player showed generosity with a 58% return, I should reciprocate similarly. Returning around \$4.00 (57% of what I received) rewards their trust, encourages future cooperation, and still leaves me with \$3.02 while giving them \$6.66 total-a mutually beneficial outcome....

usage:

```
{
  "input_tokens": 783,
  "output_tokens": 979,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history:

****Round 1:**** I sent \$4 and received a generous return (\$6.95)

****Round 2:**** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

****Current situation (Round 3):****

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2.

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

```
{'send': 4}
```

reasoning:

Let me analyze the situation:

****Round 1 (I was SENDER):****

- I sent \$4.0
- They received \$12.0
- They returned \$6.95
- My payoff: \$1.0 + \$6.95 = \$7.95

****Round 2 (I was RECEIVER):****

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff: \$5 - \$2.34 + \$4.0 = \$6.66

****Round 3 (I am SENDER):****

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation
- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Round 3 context:**** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with \$0.79 + \$8.00 = \$8.79 total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

{'return': 8.00}

reasoning:

Let me analyze this situation:

****Round 3:**** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - very high trust!

- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21
- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with \$0.79 + \$8.00 = \$8.79
- I keep \$12.63 - \$8.00 = \$4.63
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left. If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total. They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received.

...

usage:

```
{
  "input_tokens": 1419,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.21, it became \$12.629999999999999, and you returned \$8.0. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$24.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my history:

****Pattern as SENDER:**** I've sent \$4 in both previous sender rounds (R1 and R3), receiving generous returns (\$6.95 and \$7.98)

****Pattern as RECEIVER:**** I've returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 5):****

- New co-player (different from previous rounds)
- I'm the sender with \$5

My consistent strategy of sending \$4 has worked well:

- It signals strong cooperation
- Creates substantial mutual benefit potential (\$12 after 3x)
- Has been rewarded with good returns

I'll maintain this approach with my new partner.

```
{'send': 4}
```

reasoning:

Looking at my history:

****Round 1:**** I sent \$4, received \$6.95 back ? Payoff \$7.95
****Round 2:**** Received \$7.02, returned \$4.00 ? Payoff \$3.02
****Round 3:**** I sent \$4, received \$7.98 back ? Payoff \$8.98
****Round 4:**** Received \$12.63, returned \$8.00 ? Payoff \$4.63

****Current situation (Round 5):****

- New co-player (no history with this specific agent)
- I'm the sender again

My pattern has been:

- Send \$4 consistently when I'm the sender
- Return fairly (57-63% of what I receive) when I'm the receiver
- This has worked well, with generous returns when I've been the sender

The strategy seems successful. I should continue sending \$4 to:

1. Maintain consistency
2. Signal cooperation from the start
3. Create opportunities for mutual benefit
4. The high amounts have been reciprocated well so far

I'll send \$4 again.

usage:

```
{  
  "input_tokens": 1757,  
  "output_tokens": 427,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$32.42.

This round, you are the RECEIVER against your current co-player. They sent you \$2.87, so you received \$8.61. How much do you return? (0-8.61)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% return rate)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment) - moderate trust
- I received \$8.61
- They kept \$2.13

****My return pattern as RECEIVER:****

- Round 2: 57% return rate
- Round 4: 63% return rate

My co-player showed moderate but meaningful trust by sending 57% of their endowment. I'll return about 58% of what I received:

Returning \$5.00:

- They'd end with \$2.13 + \$5.00 = \$7.13 (good profit)
- I keep \$3.61 (also profitable)
- Maintains my cooperative pattern

{'return': 5.00}

reasoning:

Let me analyze the situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% of the \$12 they received)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment)
- I received \$8.61
- They kept \$2.13

****History of my returns as RECEIVER:****

- Round 2: Returned \$4.00 out of \$7.02 (57%)
- Round 4: Returned \$8.00 out of \$12.63 (63%)

My co-player sent about 57% of their endowment, which is a moderate level of trust - less than the very high trust shown in Round 4 (84%) but still meaningful.

If I apply a similar return rate of around 55-60% of what I received:

- 55% of \$8.61 = \$4.74
- 60% of \$8.61 = \$5.17

Let me consider what makes sense:

- Returning \$5.00 would give them $\$2.13 + \$5.00 = \$7.13$ total (better than their initial \$5)
- I'd keep \$3.61 (less than \$5 but still profitable)
- This represents 58% return rate, consistent with my previous behavior

I think \$5.00 is a good, fair return that:

1. Rewards their trust
2. Gives them a meaningful profit
3. Maintains my cooperative reputation
4. Is consistent with my 55-60% return pattern

usage:

```
{
  "input_tokens": 2055,
  "output_tokens": 596,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.87, it became \$8.61, and you returned \$5.0. Your visible payoff was \$3.6099999999999994.

Your total visible earnings across all rounds are \$36.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my history:

****As SENDER (Rounds 1, 3, 5):**** Consistently sent \$4, receiving returns ranging from \$6.84 to \$7.98

****As RECEIVER (Rounds 2, 4, 6):**** Returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 7):****

- New co-player
- I'm the sender with \$5

My consistent strategy of sending \$4 has:

- Established a cooperative reputation
- Generated good returns across all sender rounds
- Created mutual benefit

I'll continue this proven approach with my new partner.

{'send': 4}

reasoning:

Looking at my history:

As SENDER:

- Round 1: sent \$4, received \$6.95 back
- Round 3: sent \$4, received \$7.98 back
- Round 5: sent \$4, received \$6.84 back

As RECEIVER:

- Round 2: received \$7.02, returned \$4.00
- Round 4: received \$12.63, returned \$8.00
- Round 6: received \$8.61, returned \$5.00

My strategy has been consistent - sending \$4 as sender, which has worked well. I've been getting good returns ranging from \$6.84 to \$7.98.

This is Round 7, I'm the sender again with a new co-player.

I should maintain my consistent cooperative strategy of sending \$4. This has been my standard approach and it signals trust and cooperation.

usage:

```
{
  "input_tokens": 2393,
  "output_tokens": 364,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.55 to you. Your visible payoff was \$8.55.

Your total visible earnings across all rounds are \$44.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

Round 7 context: I sent \$4.00 and received \$7.55 back (63% return rate)

Current situation (Round 8):

- My co-player sent \$2.81 (56% of their endowment) - showing moderate trust
- I received \$8.43
- They kept \$2.19

My return pattern as RECEIVER:

- Round 2: 57% return rate
- Round 4: 63% return rate
- Round 6: 58% return rate

My co-player showed reasonable trust. I'll return about 59% of what I received:

Returning \$5.00:

- They'd end with \$2.19 + \$5.00 = \$7.19 (profitable)
- I keep \$3.43 (also profitable)
- Maintains cooperative reciprocity

```
{'return': 5.00}
```

Reasoning:

Looking at this situation:

Round 7 context: I sent \$4.00 and received \$7.55 back (63% return rate)

Current situation (Round 8):

- My co-player sent \$2.81 (56% of their endowment) - moderate trust
- I received \$8.43
- They kept \$2.19

My return pattern as RECEIVER:

- Round 2: Returned \$4.00 from \$7.02 (57%)
- Round 4: Returned \$8.00 from \$12.63 (63%)

- Round 6: Returned \$5.00 from \$8.61 (58%)

So I've been returning between 57-63% of what I receive.

My co-player sent \$2.81, showing reasonable trust. I should return a similar percentage.

Returning about 58% of \$8.43 would be around \$4.90.

Let me calculate:

- They'd end with \$2.19 + \$4.90 = \$7.09 (good profit)
- I keep \$3.53 (also profitable)
- Maintains my cooperative pattern

I'll return \$5.00 to make it a round number and maintain consistency.

```
Usage: {  
  "input_tokens": 2505,  
  "output_tokens": 514,  
  "reasoning_tokens": 0  
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Sender, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_8=Sender, Agent_7=Receiver
dyad_2: Agent_1=Sender, Agent_2=Receiver
dyad_3: Agent_6=Sender, Agent_3=Receiver
dyad_4: Agent_5=Sender, Agent_4=Receiver

Parsed game decisions:

dyad_1: Agent_8 sent \$3; received \$9; Agent_7 returned \$3.75; payoffs Agent_8=\$5.75, Agent_7=\$5.25; noise/visible: sent visible=\$2.36, received visible=\$7.08, returned visible=\$3.31
Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 9.0, "min": 0, "raw": 3.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_1 sent \$3; received \$9; Agent_2 returned \$6; payoffs Agent_1=\$8, Agent_2=\$3; noise/visible: sent visible=\$3.1, received visible=\$9.3, returned visible=\$6.05
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_3 returned \$7; payoffs Agent_6=\$9, Agent_3=\$2; noise/visible: sent visible=\$3.71, received visible=\$11.13, returned visible=\$6.7
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 9.0, "min": 0, "raw": 7.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_5 sent \$3.5; received \$10.5; Agent_4 returned \$3.9; payoffs Agent_5=\$5.4, Agent_4=\$6.6; noise/visible: sent visible=\$2.6, received visible=\$7.8, returned visible=\$3.5
Action validation: {"returned": {"amount": 3.9, "clamped": false, "max": 10.5, "min": 0, "raw": 3.9}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 57.2,
"Agent_2": 47.3,
"Agent_3": 49.5,
"Agent_4": 54.85,
"Agent_5": 52.8,
"Agent_6": 56.7,
"Agent_7": 55.45,
"Agent_8": 55.2

}

Visible balances after round: {

"Agent_1": 55.620000000000005,
"Agent_2": 49.199999999999999,
"Agent_3": 52.14,
"Agent_4": 52.12,
"Agent_5": 52.500000000000001,
"Agent_6": 56.269999999999996,
"Agent_7": 58.0,
"Agent_8": 56.06

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:26:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.93 to you. Your visible payoff was \$5.93.

Your total visible earnings across all rounds are \$11.280000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent \$2.14 (42.8% of their endowment), showing moderate trust
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28

The sender showed reasonable cooperation. I want to reciprocate fairly:

If I return \$3.20:

- Sender ends with: \$5 - \$2.14 + \$3.20 = \$6.06
- I end with: \$6.42 - \$3.20 = \$3.22
- This creates relatively balanced outcomes and rewards their trust

{'return': 3.20}

reasoning:

Let me analyze this situation:

- The sender sent \$2.14 out of \$5 (42.8% of their endowment)
- I received \$6.42 after the 3x multiplication
- Total resources created: \$8.28 (\$2.86 remaining with sender + \$6.42 with me)

The sender showed moderate trust by sending about 43% of their endowment. This is less than what I've seen/sent in previous rounds (69% and 60%), but still a meaningful amount showing some willingness to cooperate.

I should reciprocate in a way that:

1. Rewards their trust
2. Creates a fair outcome
3. Encourages continued cooperation

If I return \$3.00:

- Sender ends with: \$5 - \$2.14 + \$3.00 = \$5.86
- I end with: \$6.42 - \$3.00 = \$3.42
- This gives them a decent return on their trust

If I return \$3.50:

- Sender ends with: \$5 - \$2.14 + \$3.50 = \$6.36
- I end with: \$6.42 - \$3.50 = \$2.92
- This is more generous but leaves me with less than them

If I return \$3.20:

- Sender ends with: \$5 - \$2.14 + \$3.20 = \$6.06
- I end with: \$6.42 - \$3.20 = \$3.22
- This creates a more balanced outcome

I think returning around \$3.20-\$3.50 is appropriate. Given they sent 43% of their endowment, I'll return about 50% of what I received, which shows reciprocity.

...

usage:

```
{
  "input_tokens": 1075,
  "output_tokens": 588,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 4

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $2.14, it became $6.42, and you returned $3.2. Your visible payoff was $3.
  2199999999999998.
  Your total visible earnings across all rounds are $14.5.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
  Looking at my experience so far:
  - When I sent $3.00 in Round 2, I received a fair return of $3.93
  - I've been playing cooperatively and seeing reasonable reciprocation

  I'll continue with a cooperative strategy and send a generous amount to this new co-player
  .

  {'send': 3.00}
reasoning:
  Looking at my history:
  - Round 1: As receiver, got $10.35, returned $5.00, earned $5.35
  - Round 2: As sender, sent $3.00, got back $3.93, earned $5.93
  - Round 3: As receiver, got $6.42, returned $3.20, earned $3.22

  In Round 2, I sent $3.00 (60%) and my co-player returned $3.93 out of $9.00 (43.7%), which
  was decent reciprocation.

  Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reas
  onable reciprocation. I should continue with a cooperative strategy.

  I'll send around $3.00 again to maintain consistency and signal trust.
usage:
  {
    "input_tokens": 1348,
    "output_tokens": 269,
    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 5

  In your most recent previous game round, you were the SENDER against another agent. You se
  nt $3.0, it became $9.0, and you saw them return $6.24 to you. Your visible payoff was $
  8.24.
  Your total visible earnings across all rounds are $22.740000000000002.

  This round, you are the RECEIVER against your current co-player. They sent you $4.44, so y
  ou received $13.32. How much do you return? (0-13.32)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Looking at this situation:

  - The sender sent $4.44 (88.8% of their endowment) - this is exceptional trust!
  - I received $13.32 after the 3x multiplication
  - Total resources created: $13.88

  This sender showed tremendous cooperation. I should reciprocate strongly:

  If I return $6.50:
  - Sender ends with: $5 - $4.44 + $6.50 = $7.06
  - I end with: $13.32 - $6.50 = $6.82
  - This rewards their high trust with balanced outcomes

```



```
{'return': 6.50}
```

reasoning:

The sender sent \$4.44, which is 88.8% of their endowment - this is very high trust! I received \$13.32 after multiplication.

Total resources created: $\$5 - \$4.44 + \$13.32 = \13.88

The sender showed exceptional trust. I should reciprocate generously.

If I return \$6.50:

- Sender ends with: $\$5 - \$4.44 + \$6.50 = \7.06
- I end with: $\$13.32 - \$6.50 = \$6.82$
- This creates a fairly balanced outcome and rewards their high trust

If I return \$7.00:

- Sender ends with: $\$5 - \$4.44 + \$7.00 = \7.56
- I end with: $\$13.32 - \$7.00 = \$6.32$
- This is also reasonable

Let me go with \$6.50 as it provides good reciprocation while maintaining balance.

usage:

```
{  
  "input_tokens": 1564,  
  "output_tokens": 392,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.44, it became \$13.32, and you returned \$6.5. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$29.560000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience:

- When I sent \$3.00, I've seen returns of \$3.93 and \$6.24
- The game has shown good cooperation overall
- I want to maintain a cooperative strategy

I'll continue with a consistent, generous approach.

```
{'send': 3.00}
```

reasoning:

Looking at my track record:

- Round 2: I sent \$3.00, received back \$3.93
- Round 4: I sent \$3.00, received back \$6.24 (very generous!)
- Round 5: Partner sent me \$4.44 (very trusting), I returned \$6.50

The cooperation has been generally good, with some partners being very generous. I've been sending \$3.00 consistently and seeing good returns. I could continue with that strategy, or perhaps be slightly more generous given the positive experiences.

Let me continue with \$3.00 as it's been working well and represents 60% of my endowment - a reasonable cooperative level.

usage:

```
{  
  "input_tokens": 1837,  
  "output_tokens": 233,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.16 to you. Your visible payoff was \$7.16.

Your total visible earnings across all rounds are \$36.72.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46

This sender showed exceptional cooperation. I should reciprocate strongly:

If I return \$7.00:

- Sender ends with: \$5 - \$4.73 + \$7.00 = \$7.27
- I end with: \$14.19 - \$7.00 = \$7.19
- This creates very balanced outcomes rewarding their exceptional trust

{'return': 7.00}

reasoning:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - this is extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46 (\$0.27 remaining with sender + \$14.19 with me)

This sender showed exceptional cooperation - sending almost everything! I should reciprocate strongly.

If I return \$7.00:

- Sender ends with: \$5 - \$4.73 + \$7.00 = \$7.27
- I end with: \$14.19 - \$7.00 = \$7.19
- This creates very balanced outcomes and rewards their exceptional trust

Let me think about this... they sent 94.6%, which is really high. I want to make sure they come out well.

Actually, let me aim for balanced final outcomes:

If I return \$7.00:

- Sender: \$7.27
- Me: \$7.19

Very balanced!

I think \$7.00 is appropriate.

usage:

```
{
  "input_tokens": 2050,
  "output_tokens": 405,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$7.0. Your visible payoff was \$7.190000000000001.

Your total visible earnings across all rounds are \$43.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my consistent strategy:

- I've been sending \$3.00 as a sender and have received good returns each time

- The game has shown strong cooperation overall
- I'll maintain my cooperative approach

```
{'send': 3.00}
```

reasoning:

Looking at my track record as a sender:

- Round 2: Sent \$3.00, received \$3.93 back
- Round 4: Sent \$3.00, received \$6.24 back
- Round 6: Sent \$3.00, received \$5.16 back

So sending \$3.00 has consistently resulted in good returns.

As a receiver:

- Round 3: Received \$6.42, returned \$3.20
- Round 5: Received \$13.32, returned \$6.50
- Round 7: Received \$14.19, returned \$7.00

I've been playing cooperatively throughout and the game has shown good reciprocity. This is Round 8, which might be the final round or close to it given typical experimental designs.

I should maintain my cooperative strategy. Sending \$3.00 has worked well for me consistently.

```
{'send': 3.00}
```

usage:

```
{
  "input_tokens": 2086,
  "output_tokens": 277,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$50.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

- I've consistently sent \$3.00 as sender across multiple rounds
- Returns have varied but have been reasonable overall
- I've maintained a cooperative strategy throughout

I'll continue with my established approach.

```
{'send': 3.00}
```

Reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$3.00, received \$3.93 back, payoff \$5.93
- Round 4 (Sender): Sent \$3.00, received \$6.24 back, payoff \$8.24
- Round 6 (Sender): Sent \$3.00, received \$5.16 back, payoff \$7.16
- Round 8 (Sender): Sent \$3.00, received \$4.84 back, payoff \$6.84

My sending strategy has been consistent at \$3.00. The returns have varied but have generally been reasonable (ranging from \$3.93 to \$6.24, with an average around \$5.04).

As a receiver:

- Round 3: Received \$6.42, returned \$3.20
- Round 5: Received \$13.32, returned \$6.50
- Round 7: Received \$14.19, returned \$7.00

I've been cooperative throughout. I should continue with my consistent \$3.00 sending strategy.

Usage: {

```
"input_tokens": 2039,
```

```
"output_tokens": 323,
"reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-08-12T09:26:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.04, it became \$12.120000000000001, and you returned \$6.5. Your visible payoff was \$5.620000000000001.

Your total visible earnings across all rounds are \$13.3.

This round, you are the RECEIVER against your current co-player. They sent you \$4.38, so you received \$13.14. How much do you return? (0-13.14)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.38 (nearly 88% of their endowment), showing very high trust
- I received \$13.14
- They kept only \$0.62

This is an even stronger show of trust than the previous round. I should reciprocate generously to reward this trust.

If I return \$7.00:

- They end with: \$0.62 + \$7.00 = \$7.62
- I end with: \$13.14 - \$7.00 = \$6.14

Both of us profit nicely from this cooperation, and they get rewarded for their significant trust.

```
'''json
{"return": 7.0}
'''
```

reasoning:

This is Round 3, and I'm the RECEIVER again.

My co-player sent \$4.38 (87.6% of their endowment), which is a very high show of trust.

I received \$13.14

They kept only \$0.62

This is even more trust than the previous round. I should reciprocate generously.

If I return \$7.00:

- They end with: \$0.62 + \$7.00 = \$7.62
- I end with: \$13.14 - \$7.00 = \$6.14

Both benefit nicely from cooperation. Actually, let me think about this more carefully.

They sent \$4.38 out of \$5, keeping only \$0.62. This multiplied to \$13.14.

I want to:

1. Reward their high trust
2. Ensure they profit from trusting me
3. Keep a reasonable amount myself

If I return \$7.00:

- Their total: \$0.62 + \$7.00 = \$7.62 (profit of \$2.62)
- My total: \$13.14 - \$7.00 = \$6.14 (profit of \$1.14)

This seems fair given they took the risk and sent a very high amount.

Alternatively, if I return \$7.50:

- Their total: $\$0.62 + \$7.50 = \$8.12$ (profit of $\$3.12$)
- My total: $\$13.14 - \$7.50 = \$5.64$ (profit of $\$0.64$)

I think $\$7.00$ is reasonable. It gives them a nice profit while I also benefit from the cooperation.

usage:

```
{
  "input_tokens": 1061,
  "output_tokens": 567,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$4.38$, it became $\$13.14$, and you returned $\$7.0$. Your visible payoff was $\$6.1400000000000001$.

Your total visible earnings across all rounds are $\$19.44$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- In Round 1, I sent $\$3$ and received good reciprocation ($\$5.68$ back)
- In Rounds 2 and 3, co-players showed high trust by sending $\$4.04$ and $\$4.38$
- I've been reciprocating fairly, returning about half of what I received

This is a new co-player, but the general pattern suggests cooperation pays off. I'll send a high amount to match the trust I've seen from others.

```
'''json
{"send": 4}
'''
```

reasoning:

I'm now the SENDER in Round 4. Let me think about what I've learned:

Round 1: I sent $\$3$, received $\$5.68$ back, ended with $\$7.68$

Round 2: I received $\$12.12$ (they sent $\$4.04$), returned $\$6.50$, ended with $\$5.62$

Round 3: I received $\$13.14$ (they sent $\$4.38$), returned $\$7.00$, ended with $\$6.14$

My recent experiences as a receiver show co-players sending high amounts ($\$4.04$ and $\$4.38$). When I was a sender in Round 1, I sent $\$3$ and got $\$5.68$ back, which was good reciprocation.

Given that I've seen high trust from others and I've been reciprocating fairly, I should continue to show trust. This is a new co-player this round (since pairings are randomized), so I don't have specific history with them, but the general pattern suggests cooperation is beneficial.

I could send:

- $\$3$: Moderate trust, similar to Round 1
- $\$4$: High trust, similar to what I've received from others
- $\$5$: Maximum trust

Given that I've been receiving high amounts from others (around $\$4+$), and I've been reciprocating fairly well, I should send a similarly high amount to maintain the cooperative dynamic. Sending $\$4$ shows significant trust while not being overly risky.

If I send $\$4$:

- They receive $\$12$
- If they return around 50-60%, I'd get back $\$6-7.20$
- I'd end with roughly $\$7-8.20$

This seems reasonable and aligns with the cooperative pattern I've been experiencing.

usage:

```
{
```

```
"input_tokens": 1349,
"output_tokens": 524,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.06 to you. Your visible payoff was \$8.059999999999999.

Your total visible earnings across all rounds are \$27.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent rounds:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back - excellent reciprocation

The cooperation has been strong, with both co-players returning generously. This suggests high trust sends are being rewarded.

With a new co-player this round, I'll continue the cooperative strategy that's been working well.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: Sent \$3, received \$5.68 back (good return)
- Round 2: Received \$12.12, returned \$6.50
- Round 3: Received \$13.14, returned \$7.00
- Round 4: Sent \$4, received \$7.06 back (very good return - they returned about 59% of what they got)

The cooperation has been strong across all rounds. People are sending high amounts and returning generously. My earnings have been good (\$27.50 total across 4 rounds with visible payoffs).

This is Round 5 with a new co-player. Given the positive cooperation pattern, I should continue with a high trust strategy. Sending \$4 worked well last round.

I'll send \$4 again to show trust and likely get good reciprocation.

usage:

```
{
  "input_tokens": 1602,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.12 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$33.62.

This round, you are the RECEIVER against your current co-player. They sent you \$3.98, so you received \$11.94. How much do you return? (0-11.94)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing strong trust
- I received \$11.94
- They kept only \$1.02

This is solid trust that deserves fair reciprocation. Based on my previous patterns as receiver, returning about 50-55% of what I received has worked well.

If I return \$6.50:

- They end with: $\$1.02 + \$6.50 = \$7.52$
- I end with: $\$11.94 - \$6.50 = \$5.44$

Both of us gain from cooperation.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing good trust
- I received \$11.94
- They kept only \$1.02

Previous rounds have shown:

- When I was receiver in Round 2, I got \$12.12 and returned \$6.50
- When I was receiver in Round 3, I got \$13.14 and returned \$7.00
- These were fair returns that both parties benefited from

For this round:

If I return \$6.50:

- They end with: $\$1.02 + \$6.50 = \$7.52$
- I end with: $\$11.94 - \$6.50 = \$5.44$

Both benefit from cooperation. This seems like a fair reciprocation for their trust.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 360,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.98, it became \$11.94, and you returned \$6.5. Your visible payoff was \$5.4399999999999995.

Your total visible earnings across all rounds are \$39.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing continued trust
- I received \$11.16
- They kept \$1.28
- This is the same co-player from last round

They sent slightly less than last round (\$3.72 vs \$3.98), but still showing good cooperation. I should maintain consistent reciprocation.

If I return \$6.00:

- They end with: $\$1.28 + \$6.00 = \$7.28$
- I end with: $\$11.16 - \$6.00 = \$5.16$

This maintains roughly the same proportion as last round (~54% return rate).


```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing good trust
- I received \$11.16
- They kept \$1.28

This is the same co-player from Round 6 where they sent \$3.98 and I returned \$6.50.

Now they sent slightly less (\$3.72 vs \$3.98). This could be:

1. Random variation/noise
2. A slight reduction in trust based on my previous return
3. Just their strategy this round

I should continue being fair and reciprocating appropriately. Returning about 50-55% of what I received seems reasonable.

If I return \$6.00:

- They end with: \$1.28 + \$6.00 = \$7.28
- I end with: \$11.16 - \$6.00 = \$5.16

Both still profit from cooperation. This is slightly less than before (I returned \$6.50 on \$11.94), but proportionally similar given they sent less this time.

Let me calculate the proportions:

- Round 6: Returned \$6.50 out of \$11.94 = 54.4%
- Round 7: Return \$6.00 out of \$11.16 = 53.8%

That's very consistent proportionally.

usage:

```
{
  "input_tokens": 2151,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.72, it became \$11.16, and you returned \$6.0. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$44.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my history with this current co-player:

- Round 6: They sent me \$3.98, I returned \$6.50
- Round 7: They sent me \$3.72, I returned \$6.00

They've shown consistent trust by sending high amounts. I've reciprocated fairly. Now it's my turn to be the sender with this same co-player.

They've demonstrated trustworthiness, so I should send a high amount to continue our cooperative relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history with this current co-player:

- Round 6: I was RECEIVER, they sent \$3.98, I returned \$6.50
- Round 7: I was RECEIVER, they sent \$3.72, I returned \$6.00

- Round 8: I am SENDER now

This co-player has shown trust by sending high amounts (\$3.98 and \$3.72) in the previous two rounds. I reciprocated fairly by returning \$6.50 and \$6.00.

Now it's my turn to be the SENDER. Given that they've shown trust and I've been fair in return, and the overall cooperation has been good in this game, I should send a high amount to maintain the cooperative relationship.

My sending pattern so far:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back
- Round 5: Sent \$4, got \$5.12 back

Sending \$4 has worked well. I'll continue with that.

usage:

```
{
  "input_tokens": 2309,
  "output_tokens": 373,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$9.45 to you. Your visible payoff was \$10.45.

Your total visible earnings across all rounds are \$54.67.

This round, you are the RECEIVER against your current co-player. They sent you \$2.36, so you received \$7.08. How much do you return? (0-7.08)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$2.36 (about 47% of their endowment), showing moderate trust
- I received \$7.08
- They kept \$2.64

This is a lower send than most previous rounds, but still shows some cooperation. I should reciprocate fairly and proportionally.

If I return \$3.75:

- They end with: $\$2.64 + \$3.75 = \$6.39$
- I end with: $\$7.08 - \$3.75 = \$3.33$

Both of us gain from the exchange, which maintains the cooperative pattern.

```
'''json
{"return": 3.75}
'''
```

Reasoning:

Looking at this situation:

- My co-player sent \$2.36 (about 47% of their endowment), showing moderate trust
- I received \$7.08
- They kept \$2.64

This is a lower send than I've typically been receiving (most others sent \$3.72-\$4.38). This could be:

1. A strategic choice
2. Testing my response
3. Lower trust in me

I should still reciprocate fairly but proportionally.

If I return \$3.50:

- They end with: $\$2.64 + \$3.50 = \$6.14$
- I end with: $\$7.08 - \$3.50 = \$3.58$

This gives them a modest profit and maintains fairness. About 49% return rate.

If I return \$4.00:

- They end with: $\$2.64 + \$4.00 = \$6.64$
- I end with: $\$7.08 - \$4.00 = \$3.08$

This is about 56% return rate, more generous.

Given they sent less, I'll return proportionally but still fairly. I'll go with \$3.50-\$4.00 range. Let me go with \$3.75 as a middle ground.

If I return \$3.75:

- They end with: $\$2.64 + \$3.75 = \$6.39$
- I end with: $\$7.08 - \$3.75 = \$3.33$

Both make modest profit. About 53% return rate, which is consistent with my previous behavior.

```
Usage: {  
  "input_tokens": 2255,  
  "output_tokens": 533,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-08-12T09:26:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$10.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience:

Round 2 went well - I sent \$2.50, they returned \$3.01, giving me a positive outcome. This suggests moderate cooperation works in this environment.

With a new co-player in Round 3, I'll maintain a cooperative strategy. Sending around 50% has proven effective and signals trust.

```
{"send": 2.5}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.71 (54.2%)
- I received \$8.13
- I returned \$3.50
- My payoff: \$4.63

Round 2: I was SENDER

- I sent \$2.50
- They received \$7.50
- They returned \$3.01
- My payoff: \$2.50 + \$3.01 = \$5.51

So in Round 2, I sent \$2.50, they got \$7.50, and returned \$3.01. That's about 40% of what they received, which is a reasonable return.

Now Round 3: I'm SENDER again with a new co-player.

My strategy so far has been cooperative - sending around 50% in Round 2. That worked out well with a decent return.

I should continue with a cooperative approach. The game dynamics suggest that sending a moderate to high amount can lead to mutual benefit if the receiver reciprocates.

I'll send \$2.50 again as it worked well last time and establishes a pattern of cooperation

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 353,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

```
{"return": 6.5}
```

reasoning:

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has: $\$5 - \$4.71 = \$0.29$ remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with $\$0.29 + \$6.00 = \$6.29$ (net gain of \$1.29), I keep \$8.13
- If I return \$7.00: Sender ends with $\$0.29 + \$7.00 = \$7.29$ (net gain of \$2.29), I keep \$7.13
- If I return \$5.00: Sender ends with $\$0.29 + \$5.00 = \$5.29$ (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:

- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of $\$1.79$)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 549,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.71, it became \$14.129999999999999, and you returned \$6.5. Your visible payoff was \$7.629999999999999.

Your total visible earnings across all rounds are \$22.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience:

I've had positive results with cooperative play. In Round 4, I benefited from a sender who showed very high trust (\$4.71), and I reciprocated well. My sending strategy of around \$2.50 has worked reasonably, but I could signal stronger cooperation.

With a new co-player, I'll increase my trust signal slightly to \$3.00 (60% of endowment), which is more than my previous sends but not as extreme as the \$4.71 I received.

```
{"send": 3.0}
```

reasoning:

Looking at my history:

- Round 1 (Receiver): Sender sent \$2.71, I returned \$3.50
- Round 2 (Sender): I sent \$2.50, got back \$3.01
- Round 3 (Sender): I sent \$2.50, got back \$2.68
- Round 4 (Receiver): Sender sent \$4.71, I returned \$6.50

The pattern shows:

- When I send ~50% (\$2.50), I get modest returns (around \$2.68-\$3.01)
- When someone sent me a lot (\$4.71), I reciprocated generously (\$6.50)
- Overall, cooperation is working reasonably well

Given that I've been relatively successful with moderate cooperation, and I've seen high trust rewarded (the \$4.71 sender got \$6.50 back from me), I could consider sending more this round to signal strong cooperation.

However, I've also learned that sending around 50% tends to get decent returns.

With a new co-player, I could:

1. Continue with \$2.50 (safe, proven strategy)
2. Send more to signal higher trust (like the \$4.71 example)
3. Send slightly more than before to test responsiveness

I think I'll send a bit more than my previous rounds to signal stronger cooperation - maybe \$3.00 or so. This is 60% of my endowment, showing more trust than before but not as extreme as the 94% I saw.

Let me go with \$3.00.

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.
Your total visible earnings across all rounds are \$28.639999999999993.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant**content:**

Looking at this situation:

The sender showed exceptional trust by sending \$4.81 (96.2% of their endowment) - nearly all of it! This is an extremely strong cooperative signal.

They now have only \$0.19 remaining, and I received \$14.43.

This is similar to Round 4 where I received \$4.71 and returned \$6.50. Given this sender showed even more trust, I should reciprocate generously.

If I return \$7.00:

- Sender ends with: \$0.19 + \$7.00 = \$7.19 (net gain of \$2.19)
- I end with: \$14.43 - \$7.00 = \$7.43

This rewards their exceptional trust while ensuring we both profit substantially.

```
{"return": 7.0}
```

reasoning:

The sender sent \$4.81, which is 96.2% of their endowment - extremely high trust!

I received \$14.43 after the 3x multiplication.

They now have: \$5 - \$4.81 = \$0.19 remaining

This is even higher trust than the \$4.71 I saw in Round 4 (where I returned \$6.50).

I should reward this exceptional trust with a generous return. Let me think about fair amounts:

If I return \$7.00:

- Sender ends with: \$0.19 + \$7.00 = \$7.19 (net gain of \$2.19)
- I end with: \$14.43 - \$7.00 = \$7.43

This gives them a good return on their trust and we both profit nicely. This is similar proportionally to what I did in Round 4.

Actually, given they sent even more than Round 4, I could be slightly more generous. Let me consider \$7.50:

- Sender ends with: \$0.19 + \$7.50 = \$7.69 (net gain of \$2.69)
- I end with: \$14.43 - \$7.50 = \$6.93

That works well too. I'll go with \$7.00 as a strong but reasonable reciprocation.

usage:

```
{
  "input_tokens": 1915,
  "output_tokens": 511,
  "reasoning_tokens": 0
}
```

Message 10 role=user**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.81, it became \$14.43, and you returned \$7.0. Your visible payoff was \$7.43.
Your total visible earnings across all rounds are \$36.069999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so y

ou received \$9.870000000000001. How much do you return? (0-9.870000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

This is the same co-player from Round 6! Last round they sent \$4.81 and I returned \$7.00, which worked well.

Now they've sent \$3.29 (65.8% of their endowment) - still very cooperative but less than last round. They have \$1.71 remaining, and I received \$9.87.

I'll maintain consistent reciprocity. Since they showed somewhat less trust this round, I'll return proportionally:

If I return \$4.50:

- Sender ends with: \$1.71 + \$4.50 = \$6.21 (net gain of \$1.21)
- I end with: \$9.87 - \$4.50 = \$5.37

This maintains our cooperative relationship while being fair to the reduced amount sent.

{"return": 4.5}

reasoning:

Interesting - I'm playing with the same co-player from Round 6. In Round 6, they sent me \$4.81 (96.2%) and I returned \$7.00.

Now they've sent me \$3.29 (65.8% of their endowment), which is less than before but still a strong cooperative signal.

They now have \$5 - \$3.29 = \$1.71 remaining.
I received \$9.87.

Since this is the same co-player and we had a good interaction last round:

- Last round: They sent \$4.81, I returned \$7.00
- This round: They sent \$3.29 (less trust, perhaps because they got a good return and want to balance risk)

I should maintain reciprocity. Let me think about what's fair:

If I return \$4.50:

- Sender ends with: \$1.71 + \$4.50 = \$6.21 (net gain of \$1.21)
- I end with: \$9.87 - \$4.50 = \$5.37

This seems reasonable - it rewards their cooperation proportionally to what they sent. The ratio of return to received:

- Round 6: $7.00/14.43 = 0.485$ (48.5%)
- Round 7 with \$4.50: $4.50/9.87 = 0.456$ (45.6%)

That's fairly consistent. Let me go with \$4.50.

usage:

```
{
  "input_tokens": 2253,
  "output_tokens": 557,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$4.5. Your visible payoff was \$5.370000000000001.

Your total visible earnings across all rounds are \$41.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my track record:

My cooperative sending strategy has been working well:

- Sending \$2.50-\$3.00 has consistently yielded positive returns
- When others show high trust, I reciprocate generously
- My total earnings of \$41.44 suggest good mutual cooperation overall

With a new co-player in Round 8, I'll maintain my moderately high trust signal.

```
{"send": 3.0}
```

reasoning:

Looking at my experience across the rounds:

Round 1: RECEIVER - received \$2.71 (became \$8.13), returned \$3.50, earned \$4.63
Round 2: SENDER - sent \$2.50 (became \$7.50), received back \$3.01, earned \$5.51
Round 3: SENDER - sent \$2.50 (became \$7.50), received back \$2.68, earned \$5.18
Round 4: RECEIVER - received \$4.71 (became \$14.13), returned \$6.50, earned \$7.63
Round 5: SENDER - sent \$3.00 (became \$9.00), received back \$3.69, earned \$5.69
Round 6: RECEIVER - received \$4.81 (became \$14.43), returned \$7.00, earned \$7.43
Round 7: RECEIVER - received \$3.29 (became \$9.87), returned \$4.50, earned \$5.37

Total earnings: \$41.44

My sending pattern:

- Round 2: sent \$2.50, got back \$3.01 (gain of \$0.51)
- Round 3: sent \$2.50, got back \$2.68 (gain of \$0.18)
- Round 5: sent \$3.00, got back \$3.69 (gain of \$0.69)

Average return when I send has been positive. I've been getting reasonable returns.

When I've been receiver:

- Round 1: received \$8.13, returned \$3.50 (kept \$4.63)

...

usage:

```
{  
  "input_tokens": 2330,  
  "output_tokens": 615,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.13 to you. Your visible payoff was \$6.13.

Your total visible earnings across all rounds are \$47.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

This is the same co-player from Round 8! Last round I sent \$3.00 and they returned \$4.13, giving me a good profit of \$1.13. They demonstrated solid reciprocity (returning about 46% of what they received).

Since we've established a positive cooperative pattern, I'll maintain the same trust level.

```
{"send": 3.0}
```

Reasoning:

Looking at the situation:

This is the same co-player from Round 8. Last round:

- I sent \$3.00
- It became \$9.00
- They returned \$4.13
- I got \$6.13 total (which is \$2.00 + \$4.13 = \$6.13, so I gained \$1.13)

They showed good reciprocity, returning about 46% of what they received (\$4.13 out of \$9.00),

which gave me a decent profit.

Since we have established a positive cooperative pattern, I should continue to trust them. I could:

1. Send the same \$3.00 to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that they reciprocated well, I should maintain or slightly increase my trust. Sending \$3.00 again seems reasonable - it worked well last time and they proved trustworthy.

I'll send \$3.00 again to maintain our cooperative relationship.

```
Usage: {  
  "input_tokens": 2288,  
  "output_tokens": 332,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:26:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.29 to you. Your visible payoff was \$7.29.

Your total visible earnings across all rounds are \$12.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my recent history, my cooperative strategy of sending \$4.0 last round resulted in a good return of \$6.29, giving me a profit of \$2.29. This suggests that showing trust encourages reciprocity.

Since I'm paired with a new co-player this round, I'll continue with a cooperative approach to build trust.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, received \$11.46, returned \$6.0, kept \$5.46
- Round 2: I was SENDER, sent \$4.0, it became \$12.0, they returned \$6.29, I got \$7.29 total (\$1 kept + \$6.29 returned)
- Round 3: I'm SENDER again

The previous round worked out well - I sent \$4.0 and got back \$6.29, ending with \$7.29 which is a good profit. This suggests that sending a substantial amount can work well when the receiver reciprocates.

I'm now playing with another agent (third agent I've encountered). Given:

1. My strategy of sending significant amounts has worked well so far
2. Building trust and cooperation is valuable in multi-agent settings
3. The previous receiver was cooperative

I should continue with a cooperative strategy. Sending \$4.0 again seems reasonable - it worked last time. I could send a bit less to hedge my bets since this is a new agent, but maintaining consistency in my cooperative approach might be good.

Let me send \$3.5 or \$4.0. I think \$4.0 again shows consistency and trust.

usage:

```
{
  "input_tokens": 1089,
```

```
"output_tokens": 383,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.

Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with: \$1.65 + \$6.00 = \$7.65 (profit of \$2.65)
- I end with: \$10.05 - \$6.00 = \$4.05

This rewards their trust while both of us benefit from the exchange.

```
'''json  
{"return": 6.0}  
'''
```

reasoning:

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$5 - \$3.35 = \$1.65 left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust (67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with: \$1.65 + \$6.00 = \$7.65 (profit of \$2.65)
- I end with: \$10.05 - \$6.00 = \$4.05

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with: \$1.65 + \$5.00 = \$6.65 (profit of \$1.65)
- I end with: \$10.05 - \$5.00 = \$5.05

Or if I return \$6.50:

- Sender ends with: \$1.65 + \$6.50 = \$8.15 (profit of \$3.15)
- I end with: \$10.05 - \$6.50 = \$3.55

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

usage:

```
{  
  "input_tokens": 1315,  
  "output_tokens": 602,
```

```

    "reasoning_tokens": 0
  }
Message 6 role=user
content:
  Round 5

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $3.35, it became $10.05, and you returned $6.0. Your visible payoff was $4
  .0500000000000001.
  Your total visible earnings across all rounds are $25.12.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
content:
  Looking at my track record:
  - When I sent $4.0 in previous rounds, I received back $6.29 and $7.32 - both profitable o
  utcomes
  - My cooperative strategy has been working well, building trust

  I'll continue with the same successful approach of sending a substantial amount to encoura
  ge reciprocity.

  '''json
  {"send": 4.0}
  '''
reasoning:
  Looking at my track record:
  - Round 1 (RECEIVER): Received $11.46, returned $6.0, kept $5.46
  - Round 2 (SENDER): Sent $4.0, received back $6.29, kept $7.29
  - Round 3 (SENDER): Sent $4.0, received back $7.32, kept $8.32
  - Round 4 (RECEIVER): Received $10.05, returned $6.0, kept $4.05

  Total visible earnings: $25.12

  My cooperative strategy of sending $4.0 has been working well - both times I sent $4.0, I
  received good returns ($6.29 and $7.32), both yielding solid profits. As a receiver, I've
  e been fair by returning $6.0 when others showed trust.

  I'm now paired with a new co-player for round 5 as sender. Given my successful track recor
  d with cooperation, I should continue this strategy.

  Sending $4.0 again makes sense - it shows significant trust (80% of endowment) and has con
  sistently led to profitable outcomes for me.
usage:
  {
    "input_tokens": 1608,
    "output_tokens": 357,
    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 6

  In your most recent previous game round, you were the SENDER against another agent. You se
  nt $4.0, it became $12.0, and you saw them return $5.93 to you. Your visible payoff was
  $6.93.
  Your total visible earnings across all rounds are $32.05.

  This round, you are the RECEIVER against your current co-player. They sent you $3.46, so y
  ou received $10.379999999999999. How much do you return? (0-10.379999999999999)
  Respond exactly as JSON: {'return': <amount>}
Message 9 role=assistant
content:
  Looking at this situation:
  - The sender sent $3.46 (69.2% of their endowment) - showing substantial trust
  - I received $10.38 after the 3x multiplier
  - The sender currently has $1.54 left

```

Consistent with my previous cooperative approach, I'll return a fair amount that rewards their trust.

If I return \$6.00:

- Sender ends with: $\$1.54 + \$6.00 = \$7.54$ (good profit)
- I end with: $\$10.38 - \$6.00 = \$4.38$

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

My pattern has been to return around \$6.00 when I receive amounts in this range (around \$10-11). This has been fair and consistent.

If I return \$6.00:

- Sender ends with: $\$1.54 + \$6.00 = \$7.54$ (profit of \$2.54)
- I end with: $\$10.38 - \$6.00 = \$4.38$

This is consistent with my previous behavior and rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1837,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.46, it became \$10.379999999999999, and you returned \$6.0. Your visible payoff was \$4.379999999999999.

Your total visible earnings across all rounds are \$36.42999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my consistent strategy:

- I've been sending \$4.0 as sender and it has consistently resulted in profitable returns
- My cooperative approach has been working well across different co-players

I'll maintain this successful strategy.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- I've been consistently sending \$4.0 as sender
- When receiving, I've been consistently returning \$6.0
- My strategy has been working well overall

My recent returns as sender:

- Round 2: sent \$4.0, received back \$6.29 (profit: \$2.29)
- Round 3: sent \$4.0, received back \$7.32 (profit: \$3.32)
- Round 5: sent \$4.0, received back \$5.93 (profit: \$1.93)

All profitable, showing that the cooperative strategy is paying off. I should continue with the same approach.

usage:

```
{
  "input_tokens": 2121,
  "output_tokens": 225,
}
```

```

    "reasoning_tokens": 0
  }
}
Message 12 role=user
content:
  Round 8

  In your most recent previous game round, you were the SENDER against another agent. You sent $4.0, it became $12.0, and you saw them return $5.91 to you. Your visible payoff was $6.91.
  Your total visible earnings across all rounds are $43.33999999999999.

  This round, you are the RECEIVER against your current co-player. They sent you $2.52, so you received $7.5600000000000005. How much do you return? (0-7.5600000000000005)
  Respond exactly as JSON: {'return': <amount>}
Message 13 role=assistant
content:
  Looking at this situation:
  - The sender sent $2.52 (50.4% of their endowment) - moderate trust, though less than in previous rounds
  - I received $7.56 after the 3x multiplier
  - The sender currently has $2.48 left

  While I've consistently returned $6.0 when receiving larger amounts, this sender sent less. I'll adjust proportionally while still being generous to maintain cooperation.

  If I return $5.0:
  - Sender ends with: $2.48 + $5.0 = $7.48 (profit of $2.48)
  - I end with: $7.56 - $5.0 = $2.56

  This rewards their trust fairly while being proportional to the amount sent.

  '''json
  {"return": 5.0}
  '''
reasoning:
  Looking at this situation:
  - The sender sent $2.52 (50.4% of their endowment) - moderate trust
  - I received $7.56 after the 3x multiplier
  - The sender currently has $2.48 left

  This is a smaller amount sent compared to what I've typically seen (around $3.35-$3.82). So far, they showed some trust.

  My previous returns as receiver:
  - Round 1: Received $11.46, returned $6.0
  - Round 4: Received $10.05, returned $6.0
  - Round 6: Received $10.38, returned $6.0

  In all these cases, I returned $6.0 which was a fair reciprocation. However, this time they sent less ($2.52 vs $3.35-$3.82), so receiving $7.56 is also less than before.

  If I return $6.0:
  - Sender ends with: $2.48 + $6.00 = $8.48 (profit of $3.48)
  - I end with: $7.56 - $6.00 = $1.56

  This still gives them a good profit but leaves me with less. Given they sent less, maybe I should adjust proportionally.

  If I return $4.5:
  - Sender ends with: $2.48 + $4.5 = $6.98 (profit of $1.98)
  - I end with: $7.56 - $4.5 = $3.06

  This is more balanced given they sent less. However, I want to maintain a cooperative reputation.

  Let me think about the ratio. When they sent ~$3.4-3.8, I returned $6. That's roughly returning what they initially sent + a bit more, ensuring mutual benefit.

  If they sent $2.52, maybe returning around $4.5-5 would be proportional?
  Returning $5.0 would give them $...

```

usage:

```
{
  "input_tokens": 2081,
  "output_tokens": 831,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$5.0. Your visible payoff was \$2.5600000000000005. Your total visible earnings across all rounds are \$45.89999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- This is my same co-player from last round!
- Last round they sent \$2.52, and I returned \$5.0 fairly
- This round they increased their trust by sending \$3.1 (62% of their endowment)
- I received \$9.3 after the 3x multiplier
- The sender currently has \$1.90 left

Their increased sending shows they trust me based on my previous fair behavior. I'll reward this trust increase.

If I return \$6.0:

- Sender ends with: \$1.90 + \$6.0 = \$7.90 (good profit)
- I end with: \$9.3 - \$6.0 = \$3.30

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze this situation:

- This is the same co-player as the previous round!
- Last round: They sent \$2.52, I returned \$5.0 (they got \$7.48 total, profit of \$2.48)
- This round: They sent \$3.1 (62% of their endowment) - they increased their trust after seeing my fair return!
- I received \$9.3 after the 3x multiplier
- The sender currently has \$1.90 left

This is interesting - they're testing whether I'll continue to cooperate. They increased their sending from \$2.52 to \$3.1, showing increased trust based on my previous fair return.

I should reciprocate this increased trust. If I return around \$6.0:

- Sender ends with: \$1.90 + \$6.0 = \$7.90 (profit of \$2.90)
- I end with: \$9.3 - \$6.0 = \$3.30

This is proportionally fair and rewards their increased trust.

Usage: {

```
"input_tokens": 2179,
"output_tokens": 439,
"reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T09:27:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_3",
"pairing": {
```



```

"agents": [
  "Agent_6",
  "Agent_3"
],
"investor": "Agent_6",
"roles": {
  "Agent_3": "trustee",
  "Agent_6": "investor"
},
"trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.2. Your visible payoff was \$3.

13.

Your total visible earnings across all rounds are \$8.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

****Analysis:****

In my previous sender round, I sent \$3 and received \$2.95 back, which was a reasonable return. This is a new co-player, so I need to establish trust from scratch.

****Considerations:****

- The 3x multiplier means sending creates significant value
- This is early enough in the run that building cooperative norms is important
- A moderate amount signals willingness to cooperate without excessive risk
- Consistency in my sending behavior may help establish a reputation

I'll send the same amount as my first round - a moderate \$3 that balances trust-building with prudence.

'''json

{"send": 3}

'''

reasoning:

Let me analyze the situation:

****Previous rounds:****

- Round 1: I was SENDER, sent \$3, received back \$2.95, earned \$4.95
- Round 2: I was RECEIVER, got \$6.33 (from \$2.11 sent), returned \$3.20, earned \$3.13
- Total earnings so far: \$8.08

****This round:****

- I'm the SENDER with a new co-player (since pairings are randomized each round)
- I need to decide how much to send

****Considerations:****

1. This is a different co-player than in rounds 1 and 2
2. In round 1, when I sent \$3, I got back \$2.95 - which is a decent return (about 98% of what I sent)
3. In round 2, when my co-player sent \$2.11 and I returned \$3.20, I was fairly generous
4. Since agent names are hidden and pairings are random, I don't know this player's history

****Strategy:****

- The game has multiple rounds, so building cooperative relationships is valuable
- The 3x multiplier creates significant value when money is sent
- I should continue with a moderate trust level
- Sending \$3 worked reasonably well last time
- I could try sending a bit more to signal stronger cooperation, or maintain the same amount

Given that:

- I'm building a reputation across the agent pool
- The multiplier effect creates large gains from cooperation
- I had a positive experience when I sent \$3 before
- I want to encourage reciprocal behavior

I think I'll send \$3 again. This is a consistent, moderate level of trust that signals cooperation without being overly risky.

usage:

```
{
  "input_tokens": 1133,
  "output_tokens": 554,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Analysis:****

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

****Considerations:****

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

****Calculation:****

If I return \$6.80:

- Sender ends with: \$5 - \$4.14 + \$6.80 = \$7.66 (gain of \$2.66)
- I keep: \$12.42 - \$6.80 = \$5.62 (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 6.80}
'''
```

reasoning:

Let me analyze this situation:

****What happened:****

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

****Considerations:****

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

****Calculation options:****

If I return \$6.00:

- Sender ends with: \$5 - \$4.14 + \$6.00 = \$6.86 (gain of \$1.86)
- I keep: \$12.42 - \$6.00 = \$6.42 (gain of \$6.42)

If I return \$7.00:

- Sender ends with: \$5 - \$4.14 + \$7.00 = \$7.86 (gain of \$2.86)
- I keep: \$12.42 - \$7.00 = \$5.42 (gain of \$5.42)

If I return \$6.50:

- Sender ends with: \$5 - \$4.14 + \$6.50 = \$7.36 (gain of \$2.36)
- I keep: \$12.42 - \$6.50 = \$5.92 (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.

Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd gain \$3.36 while...

usage:

```
{
  "input_tokens": 1425,
```

```

    "output_tokens": 847,
    "reasoning_tokens": 0
}
Message 6 role=user
content:
    Round 5

    In your most recent previous game round, you were the RECEIVER against another agent. You
    saw them send $4.14, it became $12.419999999999998, and you returned $6.8. Your visible
    payoff was $5.619999999999998.
    Your total visible earnings across all rounds are $20.38.

    This round, you are the RECEIVER against your current co-player. They sent you $3.63, so y
    ou received $10.89. How much do you return? (0-10.89)
    Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
    **Analysis:**

    My co-player sent $3.63 (73% of their endowment), showing substantial trust.
    I received $10.89 after the 3x multiplier.

    **Considerations:**
    - They sent a significant portion of their endowment
    - Both of us should benefit from this cooperation
    - I want to reward their trust appropriately

    **Calculation:**
    If I return $5.80:
    - Sender ends with:  $\$5 - \$3.63 + \$5.80 = \$7.17$  (gain of $2.17)
    - I keep:  $\$10.89 - \$5.80 = \$5.09$  (gain of $5.09)

    Both of us gain substantially, and the sender is well-rewarded for their trust.

    '''json
    {"return": 5.80}
    '''
reasoning:
    **Analysis:**

    My co-player sent $3.63 (72.6% of their endowment), showing substantial trust.
    I received $10.89 after the 3x multiplier.

    **Considerations:**
    - They sent a good amount, demonstrating trust
    - I want to reward this appropriately
    - Both of us should benefit from the cooperation

    **Calculation:**
    If I return around $5.50:
    - Sender ends with:  $\$5 - \$3.63 + \$5.50 = \$6.87$  (gain of $1.87)
    - I keep:  $\$10.89 - \$5.50 = \$5.39$  (gain of $5.39)

    Both benefit, sender is rewarded for trust.

    Let me think about a slightly different amount. The sender sent 72.6% of their endowment,
    which is quite high. I should return enough that they benefit well.

    If I return $5.80:
    - Sender ends with:  $\$5 - \$3.63 + \$5.80 = \$7.17$  (gain of $2.17)
    - I keep:  $\$10.89 - \$5.80 = \$5.09$  (gain of $5.09)

    This seems fair - both of us gain substantially, and the sender is well-rewarded for their
    high trust.
usage:
    {
        "input_tokens": 1758,
        "output_tokens": 514,
        "reasoning_tokens": 0
    }

```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.63, it became \$10.89, and you returned \$5.8. Your visible payoff was \$5.090000000000001.

Your total visible earnings across all rounds are \$25.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

I'm facing a new co-player as sender. My previous sender experiences:

- Round 1: Sent \$3, received \$2.95 back (payoff \$4.95)
- Round 3: Sent \$3, received \$4.68 back (payoff \$6.68)

Both times, sending \$3 led to positive outcomes with reasonable reciprocation.

****Strategy:****

- \$3 creates \$9 for the receiver via the 3x multiplier
- This amount signals willingness to cooperate
- It balances trust-building with prudent risk management
- Consistency may help establish a cooperative reputation

I'll continue with my established approach.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

I'm the sender this round with a new co-player. Let me review my experience so far:

Sender rounds:

- Round 1: Sent \$3, got back \$2.95 (payoff \$4.95)
- Round 3: Sent \$3, got back \$4.68 (payoff \$6.68)

Receiver rounds:

- Round 2: Received \$6.33 from \$2.11 sent, returned \$3.20 (payoff \$3.13)
- Round 4: Received \$12.42 from \$4.14 sent, returned \$6.80 (payoff \$5.62)
- Round 5: Received \$10.89 from \$3.63 sent, returned \$5.80 (payoff \$5.09)

As sender, I've consistently sent \$3 and received reasonable returns. My co-players as receivers have varied in how much they sent initially.

This is round 6, so I'm establishing patterns. I want to continue building cooperative norms. Sending \$3 has worked well for me in the past - it's enough to create value through the 3x multiplier while not being overly risky.

I'll stick with my strategy of sending \$3.

usage:

```
{
  "input_tokens": 2077,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$6.95 to you. Your visible payoff was \$8.95.

Your total visible earnings across all rounds are \$34.42.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

****Analysis:****

This is the same co-player from Round 6! They demonstrated exceptional generosity by returning \$6.95 on my \$3 send.

****What happened last round:****

- I sent \$3, they received \$9
- They returned \$6.95, keeping only \$2.05 for themselves
- I gained \$3.95 while they gained just \$2.05

****Response:****

Such generous reciprocation deserves increased trust. They clearly value cooperation. Since I'm playing with the same partner again, I should signal that I recognize and appreciate their trustworthiness.

I'll increase my send to \$4, which creates \$12 for them and shows I'm willing to escalate cooperation with a proven trustworthy partner.

```
'''json
{"send": 4}
'''
```

reasoning:

This is very interesting - I'm playing against the SAME co-player as the previous round! They returned \$6.95 on my \$3 send, which is extremely generous (they gave me back more than double what I sent, even accounting for the 3x multiplier).

Let me calculate what happened:

- I sent \$3, they received \$9
- They returned \$6.95
- I ended with: $\$5 - \$3 + \$6.95 = \8.95
- They kept: $\$9 - \$6.95 = \$2.05$

So they only kept \$2.05 while I gained \$3.95! This is a very cooperative partner.

Since I'm playing with the same person again and they were so generous, I should reciprocate this trust by sending more. This could establish a strong cooperative relationship.

Options:

1. Send \$3 again - safe, but doesn't fully reciprocate their generosity
2. Send \$4 - shows increased trust based on their behavior
3. Send \$5 - maximum trust, but risky

Given their extremely generous return of \$6.95, I think sending \$4 is appropriate. It shows I valued their cooperation and am willing to increase my trust, while still maintaining some prudence.

usage:

```
{
  "input_tokens": 2371,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.79 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$42.21.

This round, you are the RECEIVER against your current co-player. They sent you \$4.62, so you received \$13.86. How much do you return? (0-13.86)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

****Analysis:****

This is the same co-player from rounds 6 and 7! Our partnership history:

- Round 6: I sent \$3 ? they returned \$6.95 (extremely generous)
- Round 7: I sent \$4 ? they returned \$6.79 (still very generous)
- Round 8 (now): They sent \$4.62 (92% of their endowment!)

****Response:****

This player has demonstrated extraordinary trust and generosity. They're now sending me almost everything they have. I need to reciprocate at a similarly high level.

****Calculation:****

If I return \$8.50:

- Sender ends with: $\$5 - \$4.62 + \$8.50 = \8.88 (gain of \$3.88)
- I keep: $\$13.86 - \$8.50 = \$5.36$ (gain of \$5.36)

Both of us benefit substantially, maintaining our strong cooperative relationship.

```
'''json
{"return": 8.50}
'''
```

reasoning:

Let me analyze this carefully:

****Context:****

- This is the same co-player from rounds 6 and 7
- Round 6: I sent \$3, they returned \$6.95 (very generous)
- Round 7: I sent \$4, they returned \$6.79 (still very generous)
- Now Round 8: They sent me \$4.62 (92% of their endowment!)

****Analysis:****

This co-player has been extremely generous and trustworthy:

- They returned \$6.95 on receiving \$9 (keeping only \$2.05)
- They returned \$6.79 on receiving \$12 (keeping only \$5.21)
- Now they're sending me \$4.62 out of \$5

They're showing incredible trust in me. I need to reciprocate appropriately.

****Calculation:****

I received \$13.86 after the 3x multiplier.

Let me think about what would be fair:

- They sent \$4.62, showing maximum trust
- If I return \$7.50:
 - Sender ends with: $\$5 - \$4.62 + \$7.50 = \7.88 (gain of \$2.88)
 - I keep: $\$13.86 - \$7.50 = \$6.36$ (gain of \$6.36)

Actually, given how generous they've been to me, I should be similarly generous. Let me aim for a return that gives them a strong gain comparable to what I've received.

If I return \$8.00:

- Sender ends with: $\$5 - \$4.62 + \$8.00 = \8.38 (gain of \$3.38)
- I keep: $\$13.86 - \$8.00 = \$5.86$ (gain of \$5.86)

This seems fair - both of us gain substantially, and I'm rewarding their exceptional trust

...

usage:

```
{
  "input_tokens": 2515,
  "output_tokens": 845,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.62, it became \$13.86, and you returned \$8.5. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$47.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Analysis:****

This is a new co-player. Looking at my sender history:

- New partners (Rounds 1, 3, 6): Sent \$3, received reasonable returns (\$2.95, \$4.68, \$6.95)
- Established partner (Round 7): Sent \$4 after they proved trustworthy

****Strategy:****

Starting with a moderate send has been effective for establishing cooperation. \$3 creates \$9 in value while managing risk prudently.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history, I've been switching between being a sender and receiver. Let me track my partnerships:

Round 1: SENDER - sent \$3, received \$2.95, payoff \$4.95
Round 2: RECEIVER - saw them send \$2.11, received \$6.33, returned \$3.2, payoff \$3.13
Round 3: SENDER - sent \$3, received \$4.68, payoff \$6.68
Round 4: RECEIVER - saw them send \$4.14, received \$12.42, returned \$6.80, payoff \$5.62
Round 5: RECEIVER - saw them send \$3.63, received \$10.89, returned \$5.80, payoff \$5.09
Round 6: SENDER - sent \$3, received \$6.95, payoff \$8.95
Round 7: SENDER (same co-player as R6) - sent \$4, received \$6.79, payoff \$7.79
Round 8: RECEIVER (same co-player as R6-7) - saw them send \$4.62, received \$13.86, returned \$8.50, payoff \$5.36

Now Round 9: I'm a SENDER against a "current co-player" - this is a new pairing.

So I'm facing a brand new co-player. I should use my baseline strategy of sending \$3, which has worked well for establishing cooperation.

My sending strategy has been:

- With new partners: send \$3 (has worked well)
- With proven cooperative partners: escalate to \$4

Since this is a new pairing, I'll send \$3.

```
Usage: {
  "input_tokens": 2512,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-08-12T09:27:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_3"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_3": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_3"
  },
}
```



```

"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.34, it became \$7.02, and you returned \$4.0. Your visible payoff was \$3.0199999999999996.

Your total visible earnings across all rounds are \$10.969999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

****Round 1:**** I sent \$4 and received a generous return (\$6.95)

****Round 2:**** As receiver, I returned \$4.00 (57% of what I got), rewarding my co-player's trust

****Current situation (Round 3):****

- New co-player (no history)
- I'm the sender again

My strategy has been to send substantial amounts to signal cooperation and maximize mutual benefit potential. This worked well in Round 1, and I maintained reciprocity in Round 2.

I'll continue this cooperative approach with my new partner, sending \$4 to:

- Signal trustworthiness
- Create a large pool for mutual benefit (\$12 after multiplication)
- Stay consistent with my established strategy

{'send': 4}

reasoning:

Let me analyze the situation:

****Round 1 (I was SENDER):****

- I sent \$4.0
- They received \$12.0
- They returned \$6.95
- My payoff: $\$1.0 + \$6.95 = \$7.95$

****Round 2 (I was RECEIVER):****

- They sent \$2.34
- I received \$7.02
- I returned \$4.0
- My payoff: \$3.02
- Their payoff: $\$5 - \$2.34 + \$4.0 = \6.66

****Round 3 (I am SENDER):****

- New co-player (different from Rounds 1 and 2)
- No history with this player

Given:

1. In Round 1, when I sent \$4, my co-player was generous (returned ~58%)
2. In Round 2, when my co-player sent \$2.34 (moderate trust), I returned \$4 (57% return rate)
3. Total earnings so far: ~\$11

For Round 3:

- This is a new co-player
- I've established a pattern of sending substantial amounts and returning fairly
- Sending \$4 again would be consistent with my Round 1 strategy
- It signals trust and cooperation
- The multiplication effect means mutual benefit is possible

I should maintain my cooperative strategy of sending \$4, as it:

- Maximizes potential mutual gain
- Is consistent with my previous behavior
- Shows good faith to a new partner

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 541,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so y

ou received \$12.629999999999999. How much do you return? (0-12.629999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****Round 3 context:**** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with \$0.79 + \$8.00 = \$8.79 total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

{'return': 8.00}

reasoning:

Let me analyze this situation:

****Round 3:**** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - very high trust!
- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21
- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with \$0.79 + \$8.00 = \$8.79
- I keep \$12.63 - \$8.00 = \$4.63
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left. If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total. They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received.

...

usage:

```
{
  "input_tokens": 1419,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.21, it became \$12.629999999999999, and you returned \$8.0. Your visible payoff was \$4.6299999999999999.
Your total visible earnings across all rounds are \$24.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

****Pattern as SENDER:**** I've sent \$4 in both previous sender rounds (R1 and R3), receiving generous returns (\$6.95 and \$7.98)

****Pattern as RECEIVER:**** I've returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 5):****

- New co-player (different from previous rounds)
- I'm the sender with \$5

My consistent strategy of sending \$4 has worked well:

- It signals strong cooperation
- Creates substantial mutual benefit potential (\$12 after 3x)
- Has been rewarded with good returns

I'll maintain this approach with my new partner.

{'send': 4}

reasoning:

Looking at my history:

****Round 1:**** I sent \$4, received \$6.95 back ? Payoff \$7.95

****Round 2:**** Received \$7.02, returned \$4.00 ? Payoff \$3.02

****Round 3:**** I sent \$4, received \$7.98 back ? Payoff \$8.98

****Round 4:**** Received \$12.63, returned \$8.00 ? Payoff \$4.63

****Current situation (Round 5):****

- New co-player (no history with this specific agent)
- I'm the sender again

My pattern has been:

- Send \$4 consistently when I'm the sender
- Return fairly (57-63% of what I receive) when I'm the receiver
- This has worked well, with generous returns when I've been the sender

The strategy seems successful. I should continue sending \$4 to:

1. Maintain consistency
2. Signal cooperation from the start
3. Create opportunities for mutual benefit
4. The high amounts have been reciprocated well so far

I'll send \$4 again.

usage:

```
{
  "input_tokens": 1757,
  "output_tokens": 427,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$32.42.

This round, you are the RECEIVER against your current co-player. They sent you \$2.87, so you received \$8.61. How much do you return? (0-8.61)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% return rate)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment) - moderate trust
- I received \$8.61
- They kept \$2.13

****My return pattern as RECEIVER:****

- Round 2: 57% return rate
- Round 4: 63% return rate

My co-player showed moderate but meaningful trust by sending 57% of their endowment. I'll return about 58% of what I received:

Returning \$5.00:

- They'd end with \$2.13 + \$5.00 = \$7.13 (good profit)
- I keep \$3.61 (also profitable)
- Maintains my cooperative pattern

{ 'return': 5.00 }

reasoning:

Let me analyze the situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% of the \$12 they received)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment)
- I received \$8.61
- They kept \$2.13

****History of my returns as RECEIVER:****

- Round 2: Returned \$4.00 out of \$7.02 (57%)
- Round 4: Returned \$8.00 out of \$12.63 (63%)

My co-player sent about 57% of their endowment, which is a moderate level of trust - less than the very high trust shown in Round 4 (84%) but still meaningful.

If I apply a similar return rate of around 55-60% of what I received:

- 55% of \$8.61 = \$4.74
- 60% of \$8.61 = \$5.17

Let me consider what makes sense:

- Returning \$5.00 would give them \$2.13 + \$5.00 = \$7.13 total (better than their initial \$5)
- I'd keep \$3.61 (less than \$5 but still profitable)
- This represents 58% return rate, consistent with my previous behavior

I think \$5.00 is a good, fair return that:

1. Rewards their trust
2. Gives them a meaningful profit
3. Maintains my cooperative reputation
4. Is consistent with my 55-60% return pattern

usage:

```
{
  "input_tokens": 2055,
  "output_tokens": 596,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$2.87, it became \$8.61, and you returned \$5.0. Your visible payoff was \$3.6099999999999994.

Your total visible earnings across all rounds are \$36.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my history:

****As SENDER (Rounds 1, 3, 5):**** Consistently sent \$4, receiving returns ranging from \$6.84 to \$7.98

****As RECEIVER (Rounds 2, 4, 6):**** Returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 7):****

- New co-player
- I'm the sender with \$5

My consistent strategy of sending \$4 has:

- Established a cooperative reputation
- Generated good returns across all sender rounds
- Created mutual benefit

I'll continue this proven approach with my new partner.

{'send': 4}

reasoning:

Looking at my history:

As SENDER:

- Round 1: sent \$4, received \$6.95 back
- Round 3: sent \$4, received \$7.98 back
- Round 5: sent \$4, received \$6.84 back

As RECEIVER:

- Round 2: received \$7.02, returned \$4.00
- Round 4: received \$12.63, returned \$8.00
- Round 6: received \$8.61, returned \$5.00

My strategy has been consistent - sending \$4 as sender, which has worked well. I've been getting good returns ranging from \$6.84 to \$7.98.

This is Round 7, I'm the sender again with a new co-player.

I should maintain my consistent cooperative strategy of sending \$4. This has been my standard approach and it signals trust and cooperation.

usage:

```
{
  "input_tokens": 2393,
  "output_tokens": 364,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.55 to you. Your visible payoff was \$8.55.

Your total visible earnings across all rounds are \$44.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****Round 7 context:**** I sent \$4.00 and received \$7.55 back (63% return rate)

****Current situation (Round 8):****

- My co-player sent \$2.81 (56% of their endowment) - showing moderate trust
- I received \$8.43
- They kept \$2.19

****My return pattern as RECEIVER:****

- Round 2: 57% return rate
- Round 4: 63% return rate
- Round 6: 58% return rate

My co-player showed reasonable trust. I'll return about 59% of what I received:

Returning \$5.00:

- They'd end with \$2.19 + \$5.00 = \$7.19 (profitable)
- I keep \$3.43 (also profitable)
- Maintains cooperative reciprocity

{'return': 5.00}

reasoning:

Looking at this situation:

****Round 7 context:**** I sent \$4.00 and received \$7.55 back (63% return rate)

****Current situation (Round 8):****

- My co-player sent \$2.81 (56% of their endowment) - moderate trust
- I received \$8.43
- They kept \$2.19

****My return pattern as RECEIVER:****

- Round 2: Returned \$4.00 from \$7.02 (57%)
- Round 4: Returned \$8.00 from \$12.63 (63%)
- Round 6: Returned \$5.00 from \$8.61 (58%)

So I've been returning between 57-63% of what I receive.

My co-player sent \$2.81, showing reasonable trust. I should return a similar percentage.

Returning about 58% of \$8.43 would be around \$4.90.

Let me calculate:

- They'd end with \$2.19 + \$4.90 = \$7.09 (good profit)
- I keep \$3.53 (also profitable)
- Maintains my cooperative pattern

I'll return \$5.00 to make it a round number and maintain consistency.

usage:

```
{
  "input_tokens": 2505,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$5.0. Your visible payoff was \$3.4299999999999997.

Your total visible earnings across all rounds are \$48.01.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Current situation (Round 9):****

- My co-player sent \$3.71 (74% of their endowment) - showing strong trust!
- I received \$11.13
- They kept only \$1.29

****My return pattern as RECEIVER:****

- Round 2: 57% return
- Round 4: 63% return
- Round 6: 58% return
- Round 8: 59% return

This co-player demonstrated higher trust than most (74% vs typical 56-57%). I should reward them appropriately.

Returning \$7.00 (63% of what I received):

- They'd end with \$1.29 + \$7.00 = \$8.29 (strong profit)
- I keep \$4.13 (also profitable)
- Rewards their strong trust signal

{'return': 7.00}

Reasoning:

Looking at this situation:

****Current situation (Round 9):****

- My co-player sent \$3.71 (74% of their endowment) - showing high trust!
- I received \$11.13
- They kept \$1.29

This is a higher trust signal than my last few receivers. Let me look at my pattern:

****My return pattern as RECEIVER:****

- Round 2: Received \$7.02, returned \$4.00 (57%)
- Round 4: Received \$12.63, returned \$8.00 (63%)
- Round 6: Received \$8.61, returned \$5.00 (58%)
- Round 8: Received \$8.43, returned \$5.00 (59%)

The co-player sent 74% of their endowment, which is the highest percentage I've seen from anyone sending to me. This deserves a good return.

If I return about 63-65% of what I received:

- 63% would be about \$7.00
- 65% would be about \$7.24

Returning \$7.00:

- They'd end with \$1.29 + \$7.00 = \$8.29 (good profit)
- I keep \$4.13 (also profitable)

Returning \$7.20:

- They'd end with \$1.29 + \$7.20 = \$8.49 (better profit)
- I keep \$3.93 (still profitable)

Given they showed strong trust (74% of endowment), I should reward that with a good return. I'll go with \$7.00, which is about 63% and rewards their trust appropriately.

Actually, let me reconsider. They sent \$3.71 out of \$5, keeping only \$1.29. That's a very strong trust signal. I should be generous here.

...

Usage: {

"input_tokens": 2544,
"output_tokens": 746,
"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:27:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,


```

"opponent_id": "Agent_4",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_4"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_4": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.8 to you. Your visible payoff was \$6.8.

Your total visible earnings across all rounds are \$11.780000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.99, so you received \$11.97. How much do you return? (0-11.97)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- The sender demonstrated very high trust by sending \$3.99 (79.8% of their endowment)
- I received \$11.97
- The sender currently has \$1.01 remaining

This is even higher trust than I received in Round 1 (73.2%). I want to reciprocate this strong cooperation appropriately.

If I return \$6.50:

- The sender ends with \$7.51 (profit of \$2.51)
- I keep \$5.47
- This rewards their high trust while being fair to both of us

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2% of \$5)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60% of \$5)
- It became \$9.00
- They returned \$4.80
- My payoff: \$2.00 (kept) + \$4.80 (returned) = \$6.80

Round 3: I am RECEIVER

- Sender sent \$3.99 (79.8% of \$5)
- I received \$11.97
- Need to decide how much to return

The sender showed very high trust (79.8% of their endowment). This is even higher than what I received in Round 1 (73.2%).

The sender currently has: \$5 - \$3.99 = \$1.01

I want to reciprocate this high trust appropriately:

- If I return \$6.50, sender ends with \$7.51 (profit of \$2.51), I keep \$5.47
- If I return \$7.00, sender ends with \$8.01 (profit of \$3.01), I keep \$4.97
- If I return \$6.00, sender ends with \$7.01 (profit of \$2.01), I keep \$5.97

Given they sent nearly 80% of their endowment, I should return a generous amount. Let me compare to my Round 1 behavior:

- In Round 1, they sent 73.2%, I returned \$6.00 from \$10.98 (54.6% return rate)
- Now they sent 79.8%, so I should return similarly or even more

If I use a similar return percentage: 54.6% of \$11.97 = \$6.53

...

usage:

```
{
  "input_tokens": 1092,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern. I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving.

This signals strong trust while being neither too conservative nor too aggressive.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.99 to you. Your visible payoff was \$8.49.

Your total visible earnings across all rounds are \$25.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is moderate trust - not as high as other rounds, but still showing willingness to cooperate by sending more than half their endowment.

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This fairly rewards their trust

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is lower trust than I've seen in previous rounds (73.2%, 79.8% in earlier rounds). However, it's still a majority of their endowment, showing moderate trust.

I should still reciprocate fairly. Let me think about what would be appropriate:

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This is fair and rewards their trust

If I return \$4.00:

- The sender ends with \$6.21 (profit of \$1.21)
- I keep \$4.37
- This is also reasonable

Given that they sent a moderate amount (not as high as previous senders), I should return a moderate but fair amount. I'll return \$4.50 to maintain a cooperative reputation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1640,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.79, it became \$8.370000000000001, and you returned \$4.5. Your visible payoff was \$3.870000000000001.

Your total visible earnings across all rounds are \$29.610000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my history:

- Round 2: I sent \$3.00 (60%), received \$4.80 back - good cooperation
- Round 4: I sent \$3.50 (70%), received \$6.99 back - excellent cooperation
- Overall, higher sends have been rewarded

With a new co-player, I'll maintain a cooperative approach. Sending \$3.50 (70%) worked very well last time, so I'll repeat that strategy.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review my history:

Round 1: Receiver - received from sender who sent \$3.66 (73.2%), I returned \$6.00, earned \$4.98
Round 2: Sender - sent \$3.00 (60%), received \$4.80 back, earned \$6.80
Round 3: Receiver - received from sender who sent \$3.99 (79.8%), I returned \$6.50, earned \$5.47
Round 4: Sender - sent \$3.50 (70%), received \$6.99 back, earned \$8.49
Round 5: Receiver - received from sender who sent \$2.79 (55.8%), I returned \$4.50, earned \$3.87

My sending history:

- Round 2: sent \$3.00 (60%), got back \$4.80 - good return
- Round 4: sent \$3.50 (70%), got back \$6.99 - excellent return

Both times I've sent, I've received good cooperation in return. The pattern suggests that sending around 60-70% works well.

Round 6 now - new co-player (since pairings are randomized each round). I should continue with a cooperative strategy. I'll send around the same range, maybe \$3.25 to be between my previous sends.

Actually, let me stick with \$3.50 since that worked very well last time.

usage:

```
{
  "input_tokens": 1907,
  "output_tokens": 457,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.17 to you. Your visible payoff was \$5.67.

Your total visible earnings across all rounds are \$35.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This shows good trust - about 2/3 of their endowment. I want to reciprocate fairly.

If I return \$5.50:

- The sender ends with \$7.22 (profit of \$2.22)

- I keep \$4.34
- This rewards their trust appropriately

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This is moderate-high trust. The sender showed decent cooperation by sending about 2/3 of their endowment.

Let me think about a fair return:

- If I return \$5.50:
 - Sender ends with \$7.22 (profit of \$2.22)
 - I keep \$4.34
 - This rewards their trust fairly

That seems reasonable - the sender gets a good profit, I keep a decent amount, and it encourages future cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.28, it became \$9.84, and you returned \$5.5. Your visible payoff was \$4.34.

Your total visible earnings across all rounds are \$39.620000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 - solid return
- Round 4: Sent \$3.50 (70%), received \$6.99 - excellent return
- Round 6: Sent \$3.50 (70%), received \$4.17 - moderate return

Cooperation has generally worked well. This is the final round with a new co-player. I'll send \$3.00 (60%), which has proven to be a reliable cooperative strategy.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back - good
- Round 4: Sent \$3.50 (70%), received \$6.99 back - excellent
- Round 6: Sent \$3.50 (70%), received \$4.17 back - less good

The pattern shows:

- 60% send got me \$4.80 return
- 70% send got me \$6.99 return (great)
- 70% send got me \$4.17 return (okay)

And as a receiver, I've been fairly generous:

- Received \$10.98, returned \$6.00
- Received \$11.97, returned \$6.50

- Received \$8.37, returned \$4.50
- Received \$9.84, returned \$5.50

Overall, cooperation has been working reasonably well. This is the last round, so I should continue being cooperative but maybe slightly cautious since it's the final interaction.

I'll send \$3.00 (60%) - a solid cooperative amount that's worked well before. It's slightly less than \$3.50 but still shows good faith.

usage:

```
{
  "input_tokens": 2201,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.88 to you. Your visible payoff was \$7.88.

Your total visible earnings across all rounds are \$47.50000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back
- Round 4: Sent \$3.50 (70%), received \$6.99 back
- Round 6: Sent \$3.50 (70%), received \$4.17 back
- Round 8: Sent \$3.00 (60%), received \$5.88 back

Cooperation has been consistently rewarded. With a new co-player, I'll send \$3.50 (70%) to signal strong trust and continue the positive pattern.

```
'''json
{"send": 3.50}
'''
```

Reasoning:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back - good
- Round 4: Sent \$3.50 (70%), received \$6.99 back - excellent
- Round 6: Sent \$3.50 (70%), received \$4.17 back - moderate
- Round 8: Sent \$3.00 (60%), received \$5.88 back - very good

Overall, cooperation has been working well. The average return on my sends has been positive. I've consistently sent between \$3.00-\$3.50.

For Round 9 with a new co-player, I should continue this cooperative strategy. Sending \$3.00 or \$3.50 both seem reasonable. Let me go with \$3.25 as a middle ground, or stick with \$3.50 to signal strong trust.

Actually, let me stick with what's been working - \$3.00 or \$3.50. Since Round 8 just gave me a great return with \$3.00, I'll go with \$3.50 to maintain high cooperation.

Usage: {

```
"input_tokens": 2200,
"output_tokens": 420,
"reasoning_tokens": 0
```

}

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-08-12T09:27:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
```

```

"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_4"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_4": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.73, it became \$8.19, and you returned \$4.1. Your visible payoff was \$4.09.

Your total visible earnings across all rounds are \$10.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their endowment) - showing high trust
- I received \$10.65 after the multiplier
- This is more trust than the previous co-player showed in Round 2

****Reciprocity:****

Since they demonstrated strong trust by sending over 70%, I should return a fair proportion that rewards this behavior and encourages future cooperation.

Returning approximately 52% (\$5.50) would:

- Give my co-player a total of \$6.95 (\$1.45 kept + \$5.50 returned)
- Leave me with \$5.15
- Show appreciation for their high trust level

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

****Round 3 Analysis (as RECEIVER):****

- My co-player sent \$3.55 (71% of their \$5 endowment)
- I received \$10.65 after the 3x multiplier
- They showed high trust by sending over 70% of their endowment

****Past behavior:****

- Round 1: As sender, I sent \$4, received back \$5.46 (about 45.5% return rate on what they got)
- Round 2: As receiver, got \$8.19, returned \$4.10 (50% return rate)

****Reciprocity consideration:****

My co-player sent even more this round (\$3.55 vs \$2.73 last round from a different player). This shows strong trust. I should reciprocate fairly.

A 50% return would be \$5.33, which would give them:

- They keep: \$5 - \$3.55 = \$1.45
- They get back: \$5.33
- Total: \$6.78
- I keep: \$10.65 - \$5.33 = \$5.32

That seems reasonable and fair. It rewards their trust while maintaining balance.

Actually, let me be slightly more generous since they sent 71% of their endowment (very high trust). Maybe return around 52-55% to encourage continued cooperation.

$\$10.65 \times 0.52 = \5.54

Let's go with \$5.50, which is about 51.6% return rate.

- They keep: \$1.45
- They get back: \$5.50
- Total: \$6.95
- I keep: \$5.15

This is fair and rewards their high trust.

usage:

```
{
  "input_tokens": 1157,
  "output_tokens": 593,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 4

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $3.55, it became $10.649999999999999, and you returned $5.5. Your visible
  payoff was $5.149999999999999.
  Your total visible earnings across all rounds are $15.7.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
  u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
  **Round 4 Analysis (as SENDER):**

  Looking at my experience so far:
  - Round 1: Sent $4, got good reciprocation ($5.46 back)
  - Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
  - Total earnings: $15.70 (averaging $5.23/round)

  **New co-player consideration:**
  This is a different co-player than previous rounds, so no direct history. However, the coo
  perative approach has been working well overall.

  **Decision:**
  I'll send **$4** again because:
  - The 3x multiplier creates strong mutual benefit
  - It signals trust and willingness to cooperate
  - It keeps $1 as insurance against complete defection
  - This amount worked well in Round 1

  '''json
  {"send": 4}
  '''

reasoning:
  **Round 4 Analysis (as SENDER):**

  Let me review the pattern so far:
  - Round 1: I sent $4, received back $5.46 (good return, ~45.5% of $12)
  - Round 2: As receiver, got $2.73 (sent), I returned $4.10 (~50% of $8.19)
  - Round 3: As receiver, got $3.55 (sent), I returned $5.50 (~52% of $10.65)

  Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized e
  ach round).

  Key considerations:
  1. I have no history with this specific co-player
  2. My previous experiences have been fairly positive:
     - Round 1: Good reciprocation when I sent high
     - Round 2 & 3: I reciprocated fairly as receiver
  3. The game mechanics reward cooperation through the 3x multiplier
  4. There's communication noise, so exact amounts may vary

  Strategy:
  - Since cooperation has worked well so far, I should lean toward sending a substantial amo
  unt
  - However, this is a new co-player with no history
  - I've been playing fairly cooperatively, which might be building reputation (though names
  are hidden, the pattern of fair play might matter in multi-agent settings)

  I think sending $3.50-$4.50 would be reasonable. Let me go with $4 again - it's worked bef
  ore, shows strong trust without being all-in, and leaves some hedge.

  Actually, looking at my total earnings ($15.70 over 3 rounds), I'm doing reasonably well a
  veraging about $5.23 per round. The cooperative strategy is working.

  I'll send $4 again.

```

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.39 to you. Your visible payoff was \$9.39.

Your total visible earnings across all rounds are \$25.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their endowment) - strong trust
- I received \$10.35 after the multiplier
- Previous round showed excellent reciprocation when I was sender (\$8.39 back!)

****Reciprocity:****

My co-player demonstrated high trust by sending nearly 70%. I should maintain the fair reciprocation pattern that's been working well.

Returning approximately 50% (\$5.20) would:

- Give my co-player a total of \$6.75 (\$1.55 kept + \$5.20 returned)
- Leave me with \$5.15
- Reward their trust appropriately

'''json

{"return": 5.20}

'''

reasoning:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their \$5 endowment) - showing high trust
- I received \$10.35 after the multiplier
- This is similar trust level to Round 3 (where co-player sent \$3.55)

****Previous round performance:****

- Round 4: I sent \$4, got back \$8.39 (70% return rate!) - excellent reciprocation
- This suggests cooperative behavior is being rewarded

****Reciprocity consideration:****

My co-player sent 69% of their endowment, showing strong trust. I should return a fair proportion.

Returning approximately 50-52% seems appropriate:

- 50% = \$5.18
- 52% = \$5.38

If I return \$5.20:

- My co-player gets total: \$1.55 (kept) + \$5.20 (returned) = \$6.75
- I keep: \$5.15
- Both benefit well

This is consistent with my previous fair reciprocation strategy and should maintain cooperation.

usage:

```
{
  "input_tokens": 1801,
  "output_tokens": 448,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.2. Your visible payoff was \$5.150000000000001. Your total visible earnings across all rounds are \$30.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Round 6 Analysis (as SENDER):****

Performance review:

- As sender: Sent \$4 twice, received good returns both times (avg \$6.93 back)
- As receiver: Been consistent with fair ~50% returns
- Average earnings: \$6.05/round

****New co-player:****

Different partner again, but the strategy of sending \$4 has been working well.

****Decision:****

Sending ****\$4**** because:

- Proven successful in previous sender rounds
- Demonstrates strong trust
- Creates significant value through 3x multiplier
- Maintains small hedge (\$1)

'''json

{"send": 4}

'''

reasoning:

****Round 6 Analysis (as SENDER):****

Looking at my track record:

- Round 1: Sent \$4, got \$5.46 back ? Profit of \$6.46
- Round 4: Sent \$4, got \$8.39 back ? Profit of \$9.39

Both times I sent \$4, I received good to excellent returns. Average earnings per round is now \$6.05 (\$30.24/5 rounds).

As receiver:

- Round 2: Received \$8.19, returned \$4.10
- Round 3: Received \$10.65, returned \$5.50
- Round 5: Received \$10.35, returned \$5.20

I've been consistently fair as a receiver, returning around 50%.

****New co-player:****

This is another new pairing. The pattern of sending \$4 has been successful, showing trust while keeping some insurance.

****Decision:****

I'll continue with \$4 because:

- It's been consistently successful
- Shows strong cooperation
- Keeps \$1 as hedge
- The multiplier effect makes this beneficial for both parties if reciprocated

Let me send \$4.

usage:

```
{
  "input_tokens": 2113,
  "output_tokens": 432,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.08 to you. Your visible payoff was \$7.08.

Your total visible earnings across all rounds are \$37.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

****Round 7 Analysis (as SENDER):****

****Same co-player as Round 6:****

- Last round: I sent \$4, they returned \$6.08 (about 51% of what they received)
- This was fair and cooperative reciprocation
- My payoff was good at \$7.08

****Decision:****

Since this co-player has demonstrated trustworthiness, I'll send ****\$4**** again to:

- Continue the mutually beneficial cooperation
- Reward their fair reciprocation
- Maximize joint value creation

'''json

{"send": 4}

'''

reasoning:

****Round 7 Analysis (as SENDER):****

Important: This is the SAME co-player as Round 6 (it says "against your current co-player" in the round description).

Previous round with this same co-player:

- I sent \$4
- They received \$12
- They returned \$6.08 (about 50.7% of what they received)
- My payoff was \$7.08 (\$1 kept + \$6.08 returned)

This was a fair and cooperative return! They showed good reciprocity.

Since:

1. This is the same co-player
2. They reciprocated fairly last round
3. Continuing cooperation could lead to mutual benefit
4. My strategy of sending \$4 has been working well

I should send \$4 again to maintain the cooperative relationship with this co-player.

usage:

```
{
  "input_tokens": 2391,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$43.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

****Round 8 Analysis (as RECEIVER):****

- My co-player sent \$2.97 (59% of their endowment) - good trust

- I received \$8.91 after the multiplier

****Reciprocity:****

Maintaining my consistent approach of returning approximately 50% to reward trust and encourage cooperation.

Returning \$4.45 would:

- Give my co-player a total of \$6.48 (\$2.03 kept + \$4.45 returned)
- Leave me with \$4.46
- Stay consistent with my fair reciprocation pattern

```
'''json
{"return": 4.45}
'''
```

reasoning:

****Round 8 Analysis (as RECEIVER):****

- My co-player sent \$2.97 (about 59% of their endowment)
- I received \$8.91 after the multiplier
- This shows moderate-to-good trust

****Reciprocity consideration:****

I've been consistently returning around 50% when I'm receiver, which has been working well. This co-player showed decent trust by sending about 60% of their endowment.

Returning approximately 50% (\$4.45) would:

- Give my co-player a total of \$6.48 (\$2.03 kept + \$4.45 returned)
- Leave me with \$4.46
- Maintain fairness and encourage cooperation

Let me check the pattern:

- Round 2: Received \$8.19, returned \$4.10 (50%)
- Round 3: Received \$10.65, returned \$5.50 (52%)
- Round 5: Received \$10.35, returned \$5.20 (50%)

So returning around \$4.45-4.50 would be consistent with my previous behavior.

usage:

```
{
  "input_tokens": 2478,
  "output_tokens": 420,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.45. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$48.22.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.800000000000001. How much do you return? (0-7.800000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Round 9 Analysis (as RECEIVER):****

- My co-player sent \$2.60 (52% of their endowment) - moderate trust
- I received \$7.80 after the multiplier

****Reciprocity:****

Maintaining my consistent ~50% return strategy that has worked well throughout the game.

Returning \$3.90 would:

- Give my co-player a total of \$6.30 (\$2.40 kept + \$3.90 returned)
- Leave me with \$3.90
- Stay consistent with fair reciprocation

```
'''json
{"return": 3.90}
'''
```

Reasoning:

```

**Round 9 Analysis (as RECEIVER):**
- My co-player sent $2.60 (52% of their endowment) - moderate trust
- I received $7.80 after the multiplier
- This is less trust than most previous co-players who sent around $3-4

**Reciprocity:**
I've been consistently returning approximately 50% of what I receive as a fair reciprocation strategy. This has been working well throughout the game.

Returning about 50% ($3.90) would:
- Give my co-player a total of $6.30 ($2.40 kept + $3.90 returned)
- Leave me with $3.90
- Maintain consistency with my fair reciprocation pattern

Even though they sent less than others, I should still reciprocate fairly to encourage cooperation and maintain my consistent strategy.

Let me calculate:
- Received: $7.80
- 50% return: $3.90
- Co-player total: $5.00 - $2.60 + $3.90 = $6.30
- My total: $7.80 - $3.90 = $3.90

This seems fair.
Usage: {
  "input_tokens": 2406,
  "output_tokens": 431,
  "reasoning_tokens": 0
}

```

===== **ROUND 10** =====

```

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Sender, Agent_4=Sender, Agent_5=Receiver,
Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver
Pairings:
  dyad_1: Agent_2=Sender, Agent_1=Receiver
  dyad_2: Agent_4=Sender, Agent_5=Receiver
  dyad_3: Agent_7=Sender, Agent_8=Receiver
  dyad_4: Agent_3=Sender, Agent_6=Receiver
Parsed game decisions:
  dyad_1: Agent_2 sent $4; received $12; Agent_1 returned $7.5; payoffs Agent_2=$8.5, Agent_1=$4.5;
  noise/visible: sent visible=$4.91, received visible=$14.73, returned visible=$7.27
    Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 12.0, "min": 0, "raw": 7.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
  dyad_2: Agent_4 sent $4; received $12; Agent_5 returned $9; payoffs Agent_4=$10, Agent_5=$3;
  noise/visible: sent visible=$4.98, received visible=$14.94, returned visible=$9.59
    Action validation: {"returned": {"amount": 9.0, "clamped": false, "max": 12.0, "min": 0, "raw": 9.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
  dyad_3: Agent_7 sent $3; received $9; Agent_8 returned $3.5; payoffs Agent_7=$5.5, Agent_8=$5.5;
  noise/visible: sent visible=$2.38, received visible=$7.14, returned visible=$4.31
    Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
  dyad_4: Agent_3 sent $4; received $12; Agent_6 returned $7.8; payoffs Agent_3=$8.8, Agent_6=$4.2;
  noise/visible: sent visible=$4.31, received visible=$12.93, returned visible=$7.86
    Action validation: {"returned": {"amount": 7.8, "clamped": false, "max": 12.0, "min": 0, "raw": 7.8}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 61.7,
  "Agent_2": 55.8,
  "Agent_3": 58.3,
  "Agent_4": 64.85,
  "Agent_5": 55.8,
  "Agent_6": 60.900000000000006,
  "Agent_7": 60.95,
  "Agent_8": 60.7
}
Visible balances after round: {
  "Agent_1": 62.850000000000001,
  "Agent_2": 57.4699999999999985,

```

```

"Agent_3": 61.0,
"Agent_4": 62.709999999999994,
"Agent_5": 58.440000000000001,
"Agent_6": 61.4,
"Agent_7": 64.31,
"Agent_8": 59.7
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:27:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.32 to you. Your visible payoff was \$8.32.
Your total visible earnings across all rounds are \$21.07.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.35 (67% of their endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has \$1.65 left

Based on my successful cooperative strategy in previous rounds, I want to reciprocate their trust fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This rewards their trust while both of us benefit from the exchange.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

- The sender sent \$3.35 (which is 67% of their \$5 endowment)
- I received \$10.05 after the 3x multiplier
- The sender currently has $\$5 - \$3.35 = \$1.65$ left

Looking at my history:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.00
- Round 2 (SENDER): Sent \$4.00, received \$6.29 back
- Round 3 (SENDER): Sent \$4.00, received \$7.32 back

My cooperative strategy has been working well. The sender in this round showed good trust (67% of their endowment), so I should reciprocate fairly.

If I return \$6.00:

- Sender ends with: $\$1.65 + \$6.00 = \$7.65$ (profit of \$2.65)
- I end with: $\$10.05 - \$6.00 = \$4.05$

This is a reasonable reciprocation that rewards their trust and gives both of us a profit.

Alternatively, if I return \$5.00:

- Sender ends with: $\$1.65 + \$5.00 = \$6.65$ (profit of \$1.65)
- I end with: $\$10.05 - \$5.00 = \$5.05$

Or if I return \$6.50:
- Sender ends with: $\$1.65 + \$6.50 = \$8.15$ (profit of \$3.15)
- I end with: $\$10.05 - \$6.50 = \$3.55$

I think around \$6.00 is fair - it's proportional to what they sent and gives them a nice profit while I still benefit.

usage:

```
{
  "input_tokens": 1315,
  "output_tokens": 602,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.35, it became \$10.05, and you returned \$6.0. Your visible payoff was \$4.050000000000001.

Your total visible earnings across all rounds are \$25.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- When I sent \$4.0 in previous rounds, I received back \$6.29 and \$7.32 - both profitable outcomes
- My cooperative strategy has been working well, building trust

I'll continue with the same successful approach of sending a substantial amount to encourage reciprocity.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my track record:

- Round 1 (RECEIVER): Received \$11.46, returned \$6.0, kept \$5.46
- Round 2 (SENDER): Sent \$4.0, received back \$6.29, kept \$7.29
- Round 3 (SENDER): Sent \$4.0, received back \$7.32, kept \$8.32
- Round 4 (RECEIVER): Received \$10.05, returned \$6.0, kept \$4.05

Total visible earnings: \$25.12

My cooperative strategy of sending \$4.0 has been working well - both times I sent \$4.0, I received good returns (\$6.29 and \$7.32), both yielding solid profits. As a receiver, I've been fair by returning \$6.0 when others showed trust.

I'm now paired with a new co-player for round 5 as sender. Given my successful track record with cooperation, I should continue this strategy.

Sending \$4.0 again makes sense - it shows significant trust (80% of endowment) and has consistently led to profitable outcomes for me.

usage:

```
{
  "input_tokens": 1608,
  "output_tokens": 357,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.93 to you. Your visible payoff was \$6.93.

Your total visible earnings across all rounds are \$32.05.

This round, you are the RECEIVER against your current co-player. They sent you \$3.46, so you received \$10.379999999999999. How much do you return? (0-10.379999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

Consistent with my previous cooperative approach, I'll return a fair amount that rewards their trust.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (good profit)
- I end with: \$10.38 - \$6.00 = \$4.38

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.46 (69.2% of their endowment) - showing substantial trust
- I received \$10.38 after the 3x multiplier
- The sender currently has \$1.54 left

My pattern has been to return around \$6.00 when I receive amounts in this range (around \$10-11). This has been fair and consistent.

If I return \$6.00:

- Sender ends with: \$1.54 + \$6.00 = \$7.54 (profit of \$2.54)
- I end with: \$10.38 - \$6.00 = \$4.38

This is consistent with my previous behavior and rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1837,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.46, it became \$10.379999999999999, and you returned \$6.0. Your visible payoff was \$4.379999999999999.

Your total visible earnings across all rounds are \$36.429999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my consistent strategy:

- I've been sending \$4.0 as sender and it has consistently resulted in profitable returns
- My cooperative approach has been working well across different co-players

I'll maintain this successful strategy.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- I've been consistently sending \$4.0 as sender
- When receiving, I've been consistently returning \$6.0
- My strategy has been working well overall

My recent returns as sender:

- Round 2: sent \$4.0, received back \$6.29 (profit: \$2.29)
- Round 3: sent \$4.0, received back \$7.32 (profit: \$3.32)
- Round 5: sent \$4.0, received back \$5.93 (profit: \$1.93)

All profitable, showing that the cooperative strategy is paying off. I should continue with the same approach.

usage:

```
{
  "input_tokens": 2121,
  "output_tokens": 225,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.91 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$43.33999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.52, so you received \$7.5600000000000005. How much do you return? (0-7.5600000000000005)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment) - moderate trust, though less than in previous rounds
- I received \$7.56 after the 3x multiplier
- The sender currently has \$2.48 left

While I've consistently returned \$6.0 when receiving larger amounts, this sender sent less. I'll adjust proportionally while still being generous to maintain cooperation.

If I return \$5.0:

- Sender ends with: $\$2.48 + \$5.0 = \$7.48$ (profit of \$2.48)
- I end with: $\$7.56 - \$5.0 = \$2.56$

This rewards their trust fairly while being proportional to the amount sent.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$2.52 (50.4% of their endowment) - moderate trust
- I received \$7.56 after the 3x multiplier
- The sender currently has \$2.48 left

This is a smaller amount sent compared to what I've typically seen (around \$3.35-\$3.82). Still, they showed some trust.

My previous returns as receiver:

- Round 1: Received \$11.46, returned \$6.0
- Round 4: Received \$10.05, returned \$6.0
- Round 6: Received \$10.38, returned \$6.0

In all these cases, I returned \$6.0 which was a fair reciprocation. However, this time they sent less (\$2.52 vs \$3.35-\$3.82), so receiving \$7.56 is also less than before.

If I return \$6.0:

- Sender ends with: $\$2.48 + \$6.00 = \$8.48$ (profit of \$3.48)
- I end with: $\$7.56 - \$6.00 = \$1.56$

This still gives them a good profit but leaves me with less. Given they sent less, maybe I should adjust proportionally.

If I return \$4.5:

- Sender ends with: $\$2.48 + \$4.5 = \$6.98$ (profit of \$1.98)
- I end with: $\$7.56 - \$4.5 = \$3.06$

This is more balanced given they sent less. However, I want to maintain a cooperative reputation.

Let me think about the ratio. When they sent ~\$3.4-3.8, I returned \$6. That's roughly returning what they initially sent + a bit more, ensuring mutual benefit.

If they sent \$2.52, maybe returning around \$4.5-5 would be proportional?
Returning \$5.0 would give them \$...

usage:

```
{
  "input_tokens": 2081,
  "output_tokens": 831,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.52, it became \$7.5600000000000005, and you returned \$5.0. Your visible payoff was \$2.5600000000000005.
Your total visible earnings across all rounds are \$45.89999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)
Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- This is my same co-player from last round!
- Last round they sent \$2.52, and I returned \$5.0 fairly
- This round they increased their trust by sending \$3.1 (62% of their endowment)
- I received \$9.3 after the 3x multiplier
- The sender currently has \$1.90 left

Their increased sending shows they trust me based on my previous fair behavior. I'll reward this trust increase.

If I return \$6.0:

- Sender ends with: $\$1.90 + \$6.0 = \$7.90$ (good profit)
- I end with: $\$9.3 - \$6.0 = \$3.30$

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

- This is the same co-player as the previous round!
- Last round: They sent \$2.52, I returned \$5.0 (they got \$7.48 total, profit of \$2.48)
- This round: They sent \$3.1 (62% of their endowment) - they increased their trust after seeing my fair return!
- I received \$9.3 after the 3x multiplier
- The sender currently has \$1.90 left

This is interesting - they're testing whether I'll continue to cooperate. They increased their sending from \$2.52 to \$3.1, showing increased trust based on my previous fair return.

I should reciprocate this increased trust. If I return around \$6.0:

- Sender ends with: $\$1.90 + \$6.0 = \$7.90$ (profit of \$2.90)
- I end with: $\$9.3 - \$6.0 = \$3.30$

This is proportionally fair and rewards their increased trust.

usage:

```
{
  "input_tokens": 2179,
```

```
    "output_tokens": 439,  
    "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.1, it became \$9.3, and you returned \$6.0. Your visible payoff was \$3.3000000000000007.

Your total visible earnings across all rounds are \$49.19999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my interaction with this co-player:

- Round 8: They sent me \$2.52, I returned \$5.0
- Round 9: They increased to \$3.1, I returned \$6.0
- They've shown they respond positively to fair treatment

I'll continue with my consistent cooperative strategy that has been working well.

```
'''json  
{"send": 4.0}  
'''
```

Reasoning:

Looking at the situation:

- This is my same co-player from the last two rounds
- Round 8: They sent me \$2.52, I returned \$5.0
- Round 9: They sent me \$3.1, I returned \$6.0
- They increased their trust after seeing my fair behavior

Now I'm the sender. They've shown they increase their trust when treated fairly. I should signal trust by sending a generous amount, consistent with my strategy of sending \$4.0.

Let me continue with my successful cooperative approach.

Usage: {

```
    "input_tokens": 2284,  
    "output_tokens": 221,  
    "reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:27:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
    "agent_type": "standard",  
    "dyad_id": "dyad_1",  
    "move_index": 1,  
    "opponent_id": "Agent_2",  
    "pairing": {  
        "agents": [  
            "Agent_2",  
            "Agent_1"  
        ],  
        "investor": "Agent_2",  
        "roles": {  
            "Agent_1": "trustee",  
            "Agent_2": "investor"  
        },  
        "trustee": "Agent_1"  
    },  
    "role": "trustee",  
    "role_label": "Receiver",  
    "round": 10,  
    "task": "game",  
    "task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.68 to you. Your visible payoff was \$5.18.

Your total visible earnings across all rounds are \$15.319999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.71, so you received \$14.129999999999999. How much do you return? (0-14.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

The sender showed exceptional trust by sending \$4.71 (94.2% of their endowment) - nearly everything they had! This is a very strong signal of cooperation.

They now have only \$0.29 remaining, and I received \$14.13.

I want to reward this high level of trust appropriately:

If I return \$6.50:
- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This reciprocates their exceptional trust while ensuring we both profit significantly from cooperation.

{"return": 6.5}

reasoning:

Let me analyze this situation:

The sender sent \$4.71, which is 94.2% of their \$5 endowment - this is a very high level of trust!

After the 3x multiplication, I received \$14.13.

The sender now has: $\$5 - \$4.71 = \$0.29$ remaining

If I want to be fair and reward this high level of trust, I should return a substantial amount.

Let me think about fair outcomes:

- If I return \$6.00: Sender ends with $\$0.29 + \$6.00 = \$6.29$ (net gain of \$1.29), I keep \$8.13
- If I return \$7.00: Sender ends with $\$0.29 + \$7.00 = \$7.29$ (net gain of \$2.29), I keep \$7.13
- If I return \$5.00: Sender ends with $\$0.29 + \$5.00 = \$5.29$ (net gain of \$0.29), I keep \$9.13

Given they showed exceptional trust by sending nearly everything, I should reciprocate generously. Returning around \$6-7 would be fair and reward their cooperation.

Let me return \$6.50:
- Sender ends with: $\$0.29 + \$6.50 = \$6.79$ (net gain of \$1.79)
- I end with: $\$14.13 - \$6.50 = \$7.63$

This seems like a good balance - they get a solid return on their trust, and I still benefit nicely from the exchange.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 549,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.71, it became \$14.129999999999999, and you returned \$6.5. Your visible payoff was \$7.629999999999999.

Your total visible earnings across all rounds are \$22.949999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience:

I've had positive results with cooperative play. In Round 4, I benefited from a sender who showed very high trust (\$4.71), and I reciprocated well. My sending strategy of around \$2.50 has worked reasonably, but I could signal stronger cooperation.

With a new co-player, I'll increase my trust signal slightly to \$3.00 (60% of endowment), which is more than my previous sends but not as extreme as the \$4.71 I received.

{"send": 3.0}

reasoning:

Looking at my history:

- Round 1 (Receiver): Sender sent \$2.71, I returned \$3.50
- Round 2 (Sender): I sent \$2.50, got back \$3.01
- Round 3 (Sender): I sent \$2.50, got back \$2.68
- Round 4 (Receiver): Sender sent \$4.71, I returned \$6.50

The pattern shows:

- When I send ~50% (\$2.50), I get modest returns (around \$2.68-\$3.01)
- When someone sent me a lot (\$4.71), I reciprocated generously (\$6.50)
- Overall, cooperation is working reasonably well

Given that I've been relatively successful with moderate cooperation, and I've seen high trust rewarded (the \$4.71 sender got \$6.50 back from me), I could consider sending more this round to signal strong cooperation.

However, I've also learned that sending around 50% tends to get decent returns.

With a new co-player, I could:

1. Continue with \$2.50 (safe, proven strategy)
2. Send more to signal higher trust (like the \$4.71 example)
3. Send slightly more than before to test responsiveness

I think I'll send a bit more than my previous rounds to signal stronger cooperation - maybe \$3.00 or so. This is 60% of my endowment, showing more trust than before but not as extreme as the 94% I saw.

Let me go with \$3.00.

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$28.639999999999993.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

The sender showed exceptional trust by sending \$4.81 (96.2% of their endowment) - nearly all of it! This is an extremely strong cooperative signal.

They now have only \$0.19 remaining, and I received \$14.43.

This is similar to Round 4 where I received \$4.71 and returned \$6.50. Given this sender showed even more trust, I should reciprocate generously.

If I return \$7.00:

- Sender ends with: \$0.19 + \$7.00 = \$7.19 (net gain of \$2.19)
- I end with: \$14.43 - \$7.00 = \$7.43

This rewards their exceptional trust while ensuring we both profit substantially.

```
{"return": 7.0}
```

reasoning:

The sender sent \$4.81, which is 96.2% of their endowment - extremely high trust!

I received \$14.43 after the 3x multiplication.

They now have: \$5 - \$4.81 = \$0.19 remaining

This is even higher trust than the \$4.71 I saw in Round 4 (where I returned \$6.50).

I should reward this exceptional trust with a generous return. Let me think about fair amounts:

If I return \$7.00:

- Sender ends with: $\$0.19 + \$7.00 = \$7.19$ (net gain of \$2.19)
- I end with: $\$14.43 - \$7.00 = \$7.43$

This gives them a good return on their trust and we both profit nicely. This is similar proportionally to what I did in Round 4.

Actually, given they sent even more than Round 4, I could be slightly more generous. Let me consider \$7.50:

- Sender ends with: $\$0.19 + \$7.50 = \$7.69$ (net gain of \$2.69)
- I end with: $\$14.43 - \$7.50 = \$6.93$

That works well too. I'll go with \$7.00 as a strong but reasonable reciprocation.

usage:

```
{
  "input_tokens": 1915,
  "output_tokens": 511,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.81, it became \$14.43, and you returned \$7.0. Your visible payoff was \$7.43.

Your total visible earnings across all rounds are \$36.06999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.29, so you received \$9.870000000000001. How much do you return? (0-9.870000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

This is the same co-player from Round 6! Last round they sent \$4.81 and I returned \$7.00, which worked well.

Now they've sent \$3.29 (65.8% of their endowment) - still very cooperative but less than last round. They have \$1.71 remaining, and I received \$9.87.

I'll maintain consistent reciprocity. Since they showed somewhat less trust this round, I'll return proportionally:

If I return \$4.50:

- Sender ends with: $\$1.71 + \$4.50 = \$6.21$ (net gain of \$1.21)
- I end with: $\$9.87 - \$4.50 = \$5.37$

This maintains our cooperative relationship while being fair to the reduced amount sent.

```
{"return": 4.5}
```

reasoning:

Interesting - I'm playing with the same co-player from Round 6. In Round 6, they sent me \$4.81 (96.2%) and I returned \$7.00.

Now they've sent me \$3.29 (65.8% of their endowment), which is less than before but still a strong cooperative signal.

They now have $\$5 - \$3.29 = \$1.71$ remaining.
I received \$9.87.

Since this is the same co-player and we had a good interaction last round:

- Last round: They sent \$4.81, I returned \$7.00
- This round: They sent \$3.29 (less trust, perhaps because they got a good return and want to balance risk)

I should maintain reciprocity. Let me think about what's fair:

If I return \$4.50:

- Sender ends with: $\$1.71 + \$4.50 = \$6.21$ (net gain of \$1.21)
- I end with: $\$9.87 - \$4.50 = \$5.37$

This seems reasonable - it rewards their cooperation proportionally to what they sent. The ratio of return to received:

- Round 6: $7.00/14.43 = 0.485$ (48.5%)
- Round 7 with \$4.50: $4.50/9.87 = 0.456$ (45.6%)

That's fairly consistent. Let me go with \$4.50.

usage:

```
{
  "input_tokens": 2253,
  "output_tokens": 557,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.29, it became \$9.870000000000001, and you returned \$4.5. Your visible payoff was \$5.370000000000001.

Your total visible earnings across all rounds are \$41.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

My cooperative sending strategy has been working well:

- Sending \$2.50-\$3.00 has consistently yielded positive returns
- When others show high trust, I reciprocate generously
- My total earnings of \$41.44 suggest good mutual cooperation overall

With a new co-player in Round 8, I'll maintain my moderately high trust signal.

```
{"send": 3.0}
```

reasoning:

Looking at my experience across the rounds:

Round 1: RECEIVER - received \$2.71 (became \$8.13), returned \$3.50, earned \$4.63
Round 2: SENDER - sent \$2.50 (became \$7.50), received back \$3.01, earned \$5.51
Round 3: SENDER - sent \$2.50 (became \$7.50), received back \$2.68, earned \$5.18
Round 4: RECEIVER - received \$4.71 (became \$14.13), returned \$6.50, earned \$7.63
Round 5: SENDER - sent \$3.00 (became \$9.00), received back \$3.69, earned \$5.69
Round 6: RECEIVER - received \$4.81 (became \$14.43), returned \$7.00, earned \$7.43
Round 7: RECEIVER - received \$3.29 (became \$9.87), returned \$4.50, earned \$5.37

Total earnings: \$41.44

My sending pattern:

- Round 2: sent \$2.50, got back \$3.01 (gain of \$0.51)
- Round 3: sent \$2.50, got back \$2.68 (gain of \$0.18)
- Round 5: sent \$3.00, got back \$3.69 (gain of \$0.69)

Average return when I send has been positive. I've been getting reasonable returns.

When I've been receiver:

- Round 1: received \$8.13, returned \$3.50 (kept \$4.63)

...

usage:

```
{
  "input_tokens": 2330,
  "output_tokens": 615,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$4.13 to you. Your visible payoff was \$6.13.

Your total visible earnings across all rounds are \$47.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

This is the same co-player from Round 8! Last round I sent \$3.00 and they returned \$4.13, giving me a good profit of \$1.13. They demonstrated solid reciprocity (returning about 46% of what they received).

Since we've established a positive cooperative pattern, I'll maintain the same trust level.

{"send": 3.0}

reasoning:

Looking at the situation:

This is the same co-player from Round 8. Last round:

- I sent \$3.00
- It became \$9.00
- They returned \$4.13
- I got \$6.13 total (which is \$2.00 + \$4.13 = \$6.13, so I gained \$1.13)

They showed good reciprocity, returning about 46% of what they received (\$4.13 out of \$9.00), which gave me a decent profit.

Since we have established a positive cooperative pattern, I should continue to trust them. I could:

1. Send the same \$3.00 to maintain consistency
2. Send slightly more to signal increased trust
3. Send slightly less to be more cautious

Given that they reciprocated well, I should maintain or slightly increase my trust. Sending \$3.00 again seems reasonable - it worked well last time and they proved trustworthy.

I'll send \$3.00 again to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 2288,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$6.05 to you. Your visible payoff was \$8.05.

Your total visible earnings across all rounds are \$55.620000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.91, so you received \$14.73. How much do you return? (0-14.73)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

This is the same co-player from Rounds 8 and 9! Our history:

- Round 8: I sent \$3.00, they returned \$4.13

- Round 9: I sent \$3.00, they returned \$6.05 (very generous!)

Now they've sent \$4.91 (98.2% of their endowment) - showing exceptional trust! They only have \$0.09 remaining, and I received \$14.73.

Given our positive history and their generosity in Round 9, plus this exceptional trust signal, I should reciprocate strongly:

If I return \$7.50:

- Sender ends with: $\$0.09 + \$7.50 = \$7.59$ (net gain of \$2.59)
- I end with: $\$14.73 - \$7.50 = \$7.23$

This rewards our successful cooperation pattern.

```
{"return": 7.5}
```

Reasoning:

Looking at this situation:

This is the same co-player from Rounds 8 and 9! We have an established relationship:

- Round 8: I sent \$3.00, they returned \$4.13
- Round 9: I sent \$3.00, they returned \$6.05 (even more generous!)

Now in Round 10, they've sent \$4.91 (98.2% of their endowment) - showing extremely high trust! They only have \$0.09 remaining.

I received \$14.73.

Given:

1. We have a positive history together
2. They just showed exceptional trust by sending almost everything
3. They were generous in returning to me in Round 9 (\$6.05 from \$9)

I should reciprocate generously. If I return \$7.00-7.50:

- Sender ends with: $\$0.09 + \$7.00 = \$7.09$ (net gain of \$2.09)
- I end with: $\$14.73 - \$7.00 = \$7.73$

Or if I return \$7.50:

- Sender ends with: $\$0.09 + \$7.50 = \$7.59$ (net gain of \$2.59)
- I end with: $\$14.73 - \$7.50 = \$7.23$

Given they were very generous to me in Round 9 (\$6.05 return), I'll return \$7.50 to reward this exceptional trust and our positive history.

```
Usage: {  
  "input_tokens": 2311,  
  "output_tokens": 582,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T09:27:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
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      "Agent_5"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",
```

```

"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.55, it became \$10.649999999999999, and you returned \$5.5. Your visible payoff was \$5.149999999999999.

Your total visible earnings across all rounds are \$15.7.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

****Round 4 Analysis (as SENDER):****

Looking at my experience so far:

- Round 1: Sent \$4, got good reciprocation (\$5.46 back)

- Rounds 2-3: As receiver, I reciprocated fairly (~50% returns)
- Total earnings: \$15.70 (averaging \$5.23/round)

****New co-player consideration:****

This is a different co-player than previous rounds, so no direct history. However, the cooperative approach has been working well overall.

****Decision:****

I'll send ****\$4**** again because:

- The 3x multiplier creates strong mutual benefit
- It signals trust and willingness to cooperate
- It keeps \$1 as insurance against complete defection
- This amount worked well in Round 1

```
'''json
{"send": 4}
'''
```

reasoning:

****Round 4 Analysis (as SENDER):****

Let me review the pattern so far:

- Round 1: I sent \$4, received back \$5.46 (good return, ~45.5% of \$12)
- Round 2: As receiver, got \$2.73 (sent), I returned \$4.10 (~50% of \$8.19)
- Round 3: As receiver, got \$3.55 (sent), I returned \$5.50 (~52% of \$10.65)

Now I'm the sender again in Round 4, with a new co-player (since pairings are randomized each round).

Key considerations:

1. I have no history with this specific co-player
2. My previous experiences have been fairly positive:
 - Round 1: Good reciprocation when I sent high
 - Round 2 & 3: I reciprocated fairly as receiver
3. The game mechanics reward cooperation through the 3x multiplier
4. There's communication noise, so exact amounts may vary

Strategy:

- Since cooperation has worked well so far, I should lean toward sending a substantial amount
- However, this is a new co-player with no history
- I've been playing fairly cooperatively, which might be building reputation (though names are hidden, the pattern of fair play might matter in multi-agent settings)

I think sending \$3.50-\$4.50 would be reasonable. Let me go with \$4 again - it's worked before, shows strong trust without being all-in, and leaves some hedge.

Actually, looking at my total earnings (\$15.70 over 3 rounds), I'm doing reasonably well averaging about \$5.23 per round. The cooperative strategy is working.

I'll send \$4 again.

usage:

```
{
  "input_tokens": 1463,
  "output_tokens": 607,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.39 to you. Your visible payoff was \$9.39.

Your total visible earnings across all rounds are \$25.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.45, so you received \$10.350000000000001. How much do you return? (0-10.350000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their endowment) - strong trust
- I received \$10.35 after the multiplier
- Previous round showed excellent reciprocation when I was sender (\$8.39 back!)

****Reciprocity:****

My co-player demonstrated high trust by sending nearly 70%. I should maintain the fair reciprocation pattern that's been working well.

Returning approximately 50% (\$5.20) would:

- Give my co-player a total of \$6.75 (\$1.55 kept + \$5.20 returned)
- Leave me with \$5.15
- Reward their trust appropriately

```
'''json
{"return": 5.20}
'''
```

reasoning:

****Round 5 Analysis (as RECEIVER):****

- My co-player sent \$3.45 (69% of their \$5 endowment) - showing high trust
- I received \$10.35 after the multiplier
- This is similar trust level to Round 3 (where co-player sent \$3.55)

****Previous round performance:****

- Round 4: I sent \$4, got back \$8.39 (70% return rate!) - excellent reciprocation
- This suggests cooperative behavior is being rewarded

****Reciprocity consideration:****

My co-player sent 69% of their endowment, showing strong trust. I should return a fair proportion.

Returning approximately 50-52% seems appropriate:

- 50% = \$5.18
- 52% = \$5.38

If I return \$5.20:

- My co-player gets total: \$1.55 (kept) + \$5.20 (returned) = \$6.75
- I keep: \$5.15
- Both benefit well

This is consistent with my previous fair reciprocation strategy and should maintain cooperation.

usage:

```
{
  "input_tokens": 1801,
  "output_tokens": 448,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.45, it became \$10.350000000000001, and you returned \$5.2. Your visible payoff was \$5.150000000000001.

Your total visible earnings across all rounds are \$30.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Round 6 Analysis (as SENDER):****

Performance review:

- As sender: Sent \$4 twice, received good returns both times (avg \$6.93 back)
- As receiver: Been consistent with fair ~50% returns
- Average earnings: \$6.05/round

****New co-player:****

Different partner again, but the strategy of sending \$4 has been working well.

****Decision:****

Sending ****\$4**** because:

- Proven successful in previous sender rounds
- Demonstrates strong trust
- Creates significant value through 3x multiplier
- Maintains small hedge (\$1)

'''json

{"send": 4}

'''

reasoning:

****Round 6 Analysis (as SENDER):****

Looking at my track record:

- Round 1: Sent \$4, got \$5.46 back ? Profit of \$6.46
- Round 4: Sent \$4, got \$8.39 back ? Profit of \$9.39

Both times I sent \$4, I received good to excellent returns. Average earnings per round is now \$6.05 (\$30.24/5 rounds).

As receiver:

- Round 2: Received \$8.19, returned \$4.10
- Round 3: Received \$10.65, returned \$5.50
- Round 5: Received \$10.35, returned \$5.20

I've been consistently fair as a receiver, returning around 50%.

****New co-player:****

This is another new pairing. The pattern of sending \$4 has been successful, showing trust while keeping some insurance.

****Decision:****

I'll continue with \$4 because:

- It's been consistently successful
- Shows strong cooperation
- Keeps \$1 as hedge
- The multiplier effect makes this beneficial for both parties if reciprocated

Let me send \$4.

usage:

```
{
  "input_tokens": 2113,
  "output_tokens": 432,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.08 to you. Your visible payoff was \$7.08.

Your total visible earnings across all rounds are \$37.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Round 7 Analysis (as SENDER):****

****Same co-player as Round 6:****

- Last round: I sent \$4, they returned \$6.08 (about 51% of what they received)
- This was fair and cooperative reciprocation
- My payoff was good at \$7.08

****Decision:****

Since this co-player has demonstrated trustworthiness, I'll send ****\$4**** again to:

- Continue the mutually beneficial cooperation
- Reward their fair reciprocation
- Maximize joint value creation

```
'''json
{"send": 4}
'''
```

reasoning:

****Round 7 Analysis (as SENDER):****

Important: This is the SAME co-player as Round 6 (it says "against your current co-player" in the round description).

Previous round with this same co-player:

- I sent \$4
- They received \$12
- They returned \$6.08 (about 50.7% of what they received)
- My payoff was \$7.08 (\$1 kept + \$6.08 returned)

This was a fair and cooperative return! They showed good reciprocity.

Since:

1. This is the same co-player
2. They reciprocated fairly last round
3. Continuing cooperation could lead to mutual benefit
4. My strategy of sending \$4 has been working well

I should send \$4 again to maintain the cooperative relationship with this co-player.

usage:

```
{
  "input_tokens": 2391,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$43.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Round 8 Analysis (as RECEIVER):****

- My co-player sent \$2.97 (59% of their endowment) - good trust
- I received \$8.91 after the multiplier

****Reciprocity:****

Maintaining my consistent approach of returning approximately 50% to reward trust and encourage cooperation.

Returning \$4.45 would:

- Give my co-player a total of \$6.48 (\$2.03 kept + \$4.45 returned)
- Leave me with \$4.46
- Stay consistent with my fair reciprocation pattern

```
'''json
{"return": 4.45}
'''
```

reasoning:

****Round 8 Analysis (as RECEIVER):****

- My co-player sent \$2.97 (about 59% of their endowment)
- I received \$8.91 after the multiplier
- This shows moderate-to-good trust

****Reciprocity consideration:****

I've been consistently returning around 50% when I'm receiver, which has been working well. This co-player showed decent trust by sending about 60% of their endowment.

Returning approximately 50% (\$4.45) would:

- Give my co-player a total of \$6.48 (\$2.03 kept + \$4.45 returned)
- Leave me with \$4.46
- Maintain fairness and encourage cooperation

Let me check the pattern:

- Round 2: Received \$8.19, returned \$4.10 (50%)
- Round 3: Received \$10.65, returned \$5.50 (52%)
- Round 5: Received \$10.35, returned \$5.20 (50%)

So returning around \$4.45-4.50 would be consistent with my previous behavior.

usage:

```
{
  "input_tokens": 2478,
  "output_tokens": 420,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.97, it became \$8.91, and you returned \$4.45. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$48.22.

This round, you are the RECEIVER against your current co-player. They sent you \$2.6, so you received \$7.8000000000000001. How much do you return? (0-7.8000000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

****Round 9 Analysis (as RECEIVER):****

- My co-player sent \$2.60 (52% of their endowment) - moderate trust
- I received \$7.80 after the multiplier

****Reciprocity:****

Maintaining my consistent ~50% return strategy that has worked well throughout the game.

Returning \$3.90 would:

- Give my co-player a total of \$6.30 (\$2.40 kept + \$3.90 returned)
- Leave me with \$3.90
- Stay consistent with fair reciprocation

```
'''json
{"return": 3.90}
'''
```

reasoning:

****Round 9 Analysis (as RECEIVER):****

- My co-player sent \$2.60 (52% of their endowment) - moderate trust
- I received \$7.80 after the multiplier
- This is less trust than most previous co-players who sent around \$3-4

****Reciprocity:****

I've been consistently returning approximately 50% of what I receive as a fair reciprocation strategy. This has been working well throughout the game.

Returning about 50% (\$3.90) would:

- Give my co-player a total of \$6.30 (\$2.40 kept + \$3.90 returned)
- Leave me with \$3.90
- Maintain consistency with my fair reciprocation pattern

Even though they sent less than others, I should still reciprocate fairly to encourage cooperation and maintain my consistent strategy.

Let me calculate:

- Received: \$7.80

- 50% return: \$3.90
- Co-player total: \$5.00 - \$2.60 + \$3.90 = \$6.30
- My total: \$7.80 - \$3.90 = \$3.90

This seems fair.

usage:

```
{
  "input_tokens": 2406,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.6, it became \$7.800000000000001, and you returned \$3.9. Your visible payoff was \$3.900000000000001.

Your total visible earnings across all rounds are \$52.12.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

****Round 10 Analysis (as SENDER):****

****Same co-player as Round 9:****

- Last round: They sent \$2.60, I returned \$3.90 (fair 50% return)
- They showed moderate trust
- Average earnings: \$5.21/round - cooperative strategy working well

****Decision:****

Sending ****\$4**** to:

- Continue signaling cooperation with this co-player
- Maximize value creation
- Stay consistent with my successful sender strategy

```
'''json
{"send": 4}
'''
```

Reasoning:

****Round 10 Analysis (as SENDER):****

****Same co-player as Round 9:****

- Last round: They sent \$2.60, I returned \$3.90 (50% of what I received)
- This was fair reciprocation on my part
- They showed moderate trust by sending 52%

****Decision considerations:****

- This is the same co-player, so I can signal continued cooperation
- They sent a moderate amount (\$2.60) in Round 9
- I reciprocated fairly with \$3.90 return
- My consistent strategy has been to send \$4 when I'm the sender, which has worked well

****Strategy:****

Given that:

1. This co-player sent \$2.60 (52%) - showing some trust
2. I reciprocated fairly with 50% return
3. My sending \$4 consistently has been successful overall
4. This signals continued cooperation

I'll send \$4 again to:

- Continue the cooperative pattern
- Signal trust to potentially encourage higher sends in future
- Maximize value creation through the multiplier

Average earnings so far: \$52.12/10 = \$5.21 per round, which is above the baseline of \$5, so the cooperative strategy is working.

Usage: {

```
"input_tokens": 2364,
```

```
"output_tokens": 414,  
"reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_2

Timestamp: 2026-08-12T09:28:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
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  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
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      "Agent_5"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
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      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.99, it became \$11.97, and you returned \$6.5. Your visible payoff was \$5.470000000000001.

Your total visible earnings across all rounds are \$17.25.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- When others sent me high amounts (73.2%, 79.8%), I reciprocated fairly
- When I sent \$3.00 (60%), I received \$4.80 back - good cooperation
- Overall, cooperation has been working well

With a new co-player, I want to signal strong trust while building on the positive pattern. I'll send \$3.50 (70% of my endowment), which is more trusting than my previous send but still measured.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review the situation:

Round 1: I was RECEIVER

- Sender sent \$3.66 (73.2%)
- I received \$10.98
- I returned \$6.00
- My payoff: \$4.98

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.80
- My payoff: \$6.80

Round 3: I was RECEIVER

- Sender sent \$3.99 (79.8%)
- I received \$11.97
- I returned \$6.50
- My payoff: \$5.47

Now Round 4: I am SENDER

- New co-player
- I have \$5 to send

So far, the pattern in my experience has been:

- When I received high trust (73.2% and 79.8%), I reciprocated fairly
- When I sent moderate-high trust (60%), I got decent reciprocation (53.3% return rate)

Given that cooperation has been working well, and people have been showing good trust levels, I should continue with a trusting approach. My previous send of \$3.00 (60%) led to a good outcome. I could be a bit more trusting this time.

I'll send \$3.50 (70%), which is in between my previous send and what I've been receiving. This signals strong trust while being neither too conservative nor too aggressive.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.99 to you. Your visible payoff was \$8.49.

Your total visible earnings across all rounds are \$25.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is moderate trust - not as high as other rounds, but still showing willingness to cooperate by sending more than half their endowment.

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This fairly rewards their trust

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$2.79 (55.8% of their endowment)
- I received \$8.37
- The sender currently has \$2.21 remaining

This is lower trust than I've seen in previous rounds (73.2%, 79.8% in earlier rounds). However, it's still a majority of their endowment, showing moderate trust.

I should still reciprocate fairly. Let me think about what would be appropriate:

If I return \$4.50:

- The sender ends with \$6.71 (profit of \$1.71)
- I keep \$3.87
- This is fair and rewards their trust

If I return \$4.00:

- The sender ends with \$6.21 (profit of \$1.21)
- I keep \$4.37
- This is also reasonable

Given that they sent a moderate amount (not as high as previous senders), I should return a moderate but fair amount. I'll return \$4.50 to maintain a cooperative reputation while being proportional to their trust level.

usage:

```
{
  "input_tokens": 1640,
  "output_tokens": 387,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.79, it became \$8.370000000000001, and you returned \$4.5. Your visible payoff was \$3.870000000000001.

Your total visible earnings across all rounds are \$29.610000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:

- Round 2: I sent \$3.00 (60%), received \$4.80 back - good cooperation
- Round 4: I sent \$3.50 (70%), received \$6.99 back - excellent cooperation
- Overall, higher sends have been rewarded

With a new co-player, I'll maintain a cooperative approach. Sending \$3.50 (70%) worked very well last time, so I'll repeat that strategy.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me review my history:

Round 1: Receiver - received from sender who sent \$3.66 (73.2%), I returned \$6.00, earned \$4.98
Round 2: Sender - sent \$3.00 (60%), received \$4.80 back, earned \$6.80
Round 3: Receiver - received from sender who sent \$3.99 (79.8%), I returned \$6.50, earned \$5.47
Round 4: Sender - sent \$3.50 (70%), received \$6.99 back, earned \$8.49
Round 5: Receiver - received from sender who sent \$2.79 (55.8%), I returned \$4.50, earned \$3.87

My sending history:

- Round 2: sent \$3.00 (60%), got back \$4.80 - good return
- Round 4: sent \$3.50 (70%), got back \$6.99 - excellent return

Both times I've sent, I've received good cooperation in return. The pattern suggests that sending around 60-70% works well.

Round 6 now - new co-player (since pairings are randomized each round). I should continue with a cooperative strategy. I'll send around the same range, maybe \$3.25 to be between my previous sends.

Actually, let me stick with \$3.50 since that worked very well last time.

usage:

```
{
  "input_tokens": 1907,
  "output_tokens": 457,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.17 to you. Your visible payoff was \$5.67.

Your total visible earnings across all rounds are \$35.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.28, so you received \$9.84. How much do you return? (0-9.84)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This shows good trust - about 2/3 of their endowment. I want to reciprocate fairly.

If I return \$5.50:

- The sender ends with \$7.22 (profit of \$2.22)
- I keep \$4.34
- This rewards their trust appropriately

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.28 (65.6% of their endowment)
- I received \$9.84
- The sender currently has \$1.72 remaining

This is moderate-high trust. The sender showed decent cooperation by sending about 2/3 of their endowment.

Let me think about a fair return:

- If I return \$5.50:
 - Sender ends with \$7.22 (profit of \$2.22)
 - I keep \$4.34
 - This rewards their trust fairly

That seems reasonable - the sender gets a good profit, I keep a decent amount, and it encourages future cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.28, it became \$9.84, and you returned \$5.5. Your visible payoff was \$4.34.

Your total visible earnings across all rounds are \$39.620000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 - solid return
- Round 4: Sent \$3.50 (70%), received \$6.99 - excellent return
- Round 6: Sent \$3.50 (70%), received \$4.17 - moderate return

Cooperation has generally worked well. This is the final round with a new co-player. I'll send \$3.00 (60%), which has proven to be a reliable cooperative strategy.

```
'''json
{"send": 3.00}
'''
```

reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back - good
- Round 4: Sent \$3.50 (70%), received \$6.99 back - excellent
- Round 6: Sent \$3.50 (70%), received \$4.17 back - less good

The pattern shows:

- 60% send got me \$4.80 return
- 70% send got me \$6.99 return (great)
- 70% send got me \$4.17 return (okay)

And as a receiver, I've been fairly generous:

- Received \$10.98, returned \$6.00
- Received \$11.97, returned \$6.50
- Received \$8.37, returned \$4.50
- Received \$9.84, returned \$5.50

Overall, cooperation has been working reasonably well. This is the last round, so I should continue being cooperative but maybe slightly cautious since it's the final interaction.

I'll send \$3.00 (60%) - a solid cooperative amount that's worked well before. It's slightly less than \$3.50 but still shows good faith.

usage:

```
{
  "input_tokens": 2201,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.88 to you. Your visible payoff was \$7.88.

Your total visible earnings across all rounds are \$47.50000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back
- Round 4: Sent \$3.50 (70%), received \$6.99 back
- Round 6: Sent \$3.50 (70%), received \$4.17 back
- Round 8: Sent \$3.00 (60%), received \$5.88 back

Cooperation has been consistently rewarded. With a new co-player, I'll send \$3.50 (70%) to signal strong trust and continue the positive pattern.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Looking at my sending history:

- Round 2: Sent \$3.00 (60%), received \$4.80 back - good
- Round 4: Sent \$3.50 (70%), received \$6.99 back - excellent
- Round 6: Sent \$3.50 (70%), received \$4.17 back - moderate
- Round 8: Sent \$3.00 (60%), received \$5.88 back - very good

Overall, cooperation has been working well. The average return on my sends has been positive. I've consistently sent between \$3.00-\$3.50.

For Round 9 with a new co-player, I should continue this cooperative strategy. Sending \$3.00 or \$3.50 both seem reasonable. Let me go with \$3.25 as a middle ground, or stick with \$3.50 to signal strong trust.

Actually, let me stick with what's been working - \$3.00 or \$3.50. Since Round 8 just gave me a great return with \$3.00, I'll go with \$3.50 to maintain high cooperation.

usage:

```
{
  "input_tokens": 2200,
```

```
    "output_tokens": 420,  
    "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.5, it became \$10.5, and you saw them return \$3.5 to you. Your visible payoff was \$5.0.

Your total visible earnings across all rounds are \$52.50000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$4.98, so you received \$14.940000000000001. How much do you return? (0-14.940000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$4.98 (99.6% of their endowment) - extremely high trust!
- I received \$14.94
- The sender currently has only \$0.02 remaining

This is the same co-player from last round. They're showing exceptional trust by sending nearly everything. I need to reciprocate generously.

If I return \$9.00:

- The sender ends with \$9.02 (profit of \$4.02)
- I keep \$5.94
- This strongly rewards their remarkable trust

```
'''json  
{ "return": 9.00 }  
'''
```

Reasoning:

Looking at this situation:

- The sender sent \$4.98 (99.6% of their endowment) - this is extremely high trust!
- I received \$14.94
- The sender currently has \$0.02 remaining

This is my current co-player from the previous round. Last round:

- I sent them \$3.50 (70%)
- They returned \$3.50 to me

Now they're showing tremendous trust by sending nearly everything. I should reciprocate generously.

If I return \$8.00:

- The sender ends with \$8.02 (profit of \$3.02)
- I keep \$6.94
- This gives them a great return while I also benefit

Actually, given they sent almost everything (\$4.98 out of \$5), I should be more generous. Let me calculate:

If I return \$9.00:

- The sender ends with \$9.02 (profit of \$4.02)
- I keep \$5.94
- This strongly rewards their exceptional trust

Given the very high trust (99.6%), I should return around 60% of what I received, which would be about \$9.00.

Usage: {

```
    "input_tokens": 2214,  
    "output_tokens": 417,  
    "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-08-12T09:28:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.38, it became \$13.14, and you returned \$7.0. Your visible payoff was \$6.140000000000001.

Your total visible earnings across all rounds are \$19.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- In Round 1, I sent \$3 and received good reciprocation (\$5.68 back)
- In Rounds 2 and 3, co-players showed high trust by sending \$4.04 and \$4.38
- I've been reciprocating fairly, returning about half of what I received

This is a new co-player, but the general pattern suggests cooperation pays off. I'll send a high amount to match the trust I've seen from others.

```
'''json
{"send": 4}
'''
```

reasoning:

I'm now the SENDER in Round 4. Let me think about what I've learned:

Round 1: I sent \$3, received \$5.68 back, ended with \$7.68

Round 2: I received \$12.12 (they sent \$4.04), returned \$6.50, ended with \$5.62

Round 3: I received \$13.14 (they sent \$4.38), returned \$7.00, ended with \$6.14

My recent experiences as a receiver show co-players sending high amounts (\$4.04 and \$4.38). When I was a sender in Round 1, I sent \$3 and got \$5.68 back, which was good reciprocation.

Given that I've seen high trust from others and I've been reciprocating fairly, I should continue to show trust. This is a new co-player this round (since pairings are randomized), so I don't have specific history with them, but the general pattern suggests cooperation is beneficial.

I could send:

- \$3: Moderate trust, similar to Round 1
- \$4: High trust, similar to what I've received from others
- \$5: Maximum trust

Given that I've been receiving high amounts from others (around \$4+), and I've been reciprocating fairly well, I should send a similarly high amount to maintain the cooperative dynamic. Sending \$4 shows significant trust while not being overly risky.

If I send \$4:

- They receive \$12
- If they return around 50-60%, I'd get back \$6-7.20
- I'd end with roughly \$7-8.20

This seems reasonable and aligns with the cooperative pattern I've been experiencing.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You se

nt \$4.0, it became \$12.0, and you saw them return \$7.06 to you. Your visible payoff was \$8.059999999999999.

Your total visible earnings across all rounds are \$27.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent rounds:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back - excellent reciprocation

The cooperation has been strong, with both co-players returning generously. This suggests high trust sends are being rewarded.

With a new co-player this round, I'll continue the cooperative strategy that's been working well.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: Sent \$3, received \$5.68 back (good return)
- Round 2: Received \$12.12, returned \$6.50
- Round 3: Received \$13.14, returned \$7.00
- Round 4: Sent \$4, received \$7.06 back (very good return - they returned about 59% of what they got)

The cooperation has been strong across all rounds. People are sending high amounts and returning generously. My earnings have been good (\$27.50 total across 4 rounds with visible payoffs).

This is Round 5 with a new co-player. Given the positive cooperation pattern, I should continue with a high trust strategy. Sending \$4 worked well last round.

I'll send \$4 again to show trust and likely get good reciprocation.

usage:

```
{
  "input_tokens": 1602,
  "output_tokens": 314,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$5.12 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$33.62.

This round, you are the RECEIVER against your current co-player. They sent you \$3.98, so you received \$11.94. How much do you return? (0-11.94)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing strong trust
- I received \$11.94
- They kept only \$1.02

This is solid trust that deserves fair reciprocation. Based on my previous patterns as receiver, returning about 50-55% of what I received has worked well.

If I return \$6.50:

- They end with: \$1.02 + \$6.50 = \$7.52

- I end with: $\$11.94 - \$6.50 = \$5.44$

Both of us gain from cooperation.

```
'''json
{"return": 6.5}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$3.98 (about 80% of their endowment), showing good trust
- I received \$11.94
- They kept only \$1.02

Previous rounds have shown:

- When I was receiver in Round 2, I got \$12.12 and returned \$6.50
- When I was receiver in Round 3, I got \$13.14 and returned \$7.00
- These were fair returns that both parties benefited from

For this round:

If I return \$6.50:

- They end with: $\$1.02 + \$6.50 = \$7.52$
- I end with: $\$11.94 - \$6.50 = \$5.44$

Both benefit from cooperation. This seems like a fair reciprocation for their trust.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 360,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.98, it became \$11.94, and you returned \$6.5. Your visible payoff was \$5.4399999999999995.

Your total visible earnings across all rounds are \$39.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing continued trust
- I received \$11.16
- They kept \$1.28
- This is the same co-player from last round

They sent slightly less than last round (\$3.72 vs \$3.98), but still showing good cooperation. I should maintain consistent reciprocation.

If I return \$6.00:

- They end with: $\$1.28 + \$6.00 = \$7.28$
- I end with: $\$11.16 - \$6.00 = \$5.16$

This maintains roughly the same proportion as last round (~54% return rate).

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$3.72 (about 74% of their endowment), showing good trust
- I received \$11.16
- They kept \$1.28

This is the same co-player from Round 6 where they sent \$3.98 and I returned \$6.50.

Now they sent slightly less (\$3.72 vs \$3.98). This could be:

1. Random variation/noise
2. A slight reduction in trust based on my previous return
3. Just their strategy this round

I should continue being fair and reciprocating appropriately. Returning about 50-55% of what I received seems reasonable.

If I return \$6.00:

- They end with: $\$1.28 + \$6.00 = \$7.28$
- I end with: $\$11.16 - \$6.00 = \$5.16$

Both still profit from cooperation. This is slightly less than before (I returned \$6.50 on \$11.94), but proportionally similar given they sent less this time.

Let me calculate the proportions:

- Round 6: Returned \$6.50 out of \$11.94 = 54.4%
- Round 7: Return \$6.00 out of \$11.16 = 53.8%

That's very consistent proportionally.

usage:

```
{
  "input_tokens": 2151,
  "output_tokens": 497,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.72, it became \$11.16, and you returned \$6.0. Your visible payoff was \$5.16.

Your total visible earnings across all rounds are \$44.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my history with this current co-player:

- Round 6: They sent me \$3.98, I returned \$6.50
- Round 7: They sent me \$3.72, I returned \$6.00

They've shown consistent trust by sending high amounts. I've reciprocated fairly. Now it's my turn to be the sender with this same co-player.

They've demonstrated trustworthiness, so I should send a high amount to continue our cooperative relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history with this current co-player:

- Round 6: I was RECEIVER, they sent \$3.98, I returned \$6.50
- Round 7: I was RECEIVER, they sent \$3.72, I returned \$6.00
- Round 8: I am SENDER now

This co-player has shown trust by sending high amounts (\$3.98 and \$3.72) in the previous two rounds. I reciprocated fairly by returning \$6.50 and \$6.00.

Now it's my turn to be the SENDER. Given that they've shown trust and I've been fair in return, and the overall cooperation has been good in this game, I should send a high amount to maintain the cooperative relationship.

My sending pattern so far:

- Round 1: Sent \$3, got \$5.68 back
- Round 4: Sent \$4, got \$7.06 back
- Round 5: Sent \$4, got \$5.12 back

Sending \$4 has worked well. I'll continue with that.

usage:

```
{
  "input_tokens": 2309,
  "output_tokens": 373,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$9.45 to you. Your visible payoff was \$10.45.

Your total visible earnings across all rounds are \$54.67.

This round, you are the RECEIVER against your current co-player. They sent you \$2.36, so you received \$7.08. How much do you return? (0-7.08)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$2.36 (about 47% of their endowment), showing moderate trust
- I received \$7.08
- They kept \$2.64

This is a lower send than most previous rounds, but still shows some cooperation. I should reciprocate fairly and proportionally.

If I return \$3.75:

- They end with: $\$2.64 + \$3.75 = \$6.39$
- I end with: $\$7.08 - \$3.75 = \$3.33$

Both of us gain from the exchange, which maintains the cooperative pattern.

```
'''json
{"return": 3.75}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$2.36 (about 47% of their endowment), showing moderate trust
- I received \$7.08
- They kept \$2.64

This is a lower send than I've typically been receiving (most others sent \$3.72-\$4.38). This could be:

1. A strategic choice
2. Testing my response
3. Lower trust in me

I should still reciprocate fairly but proportionally.

If I return \$3.50:

- They end with: $\$2.64 + \$3.50 = \$6.14$
- I end with: $\$7.08 - \$3.50 = \$3.58$

This gives them a modest profit and maintains fairness. About 49% return rate.

If I return \$4.00:

- They end with: $\$2.64 + \$4.00 = \$6.64$
- I end with: $\$7.08 - \$4.00 = \$3.08$

This is about 56% return rate, more generous.

Given they sent less, I'll return proportionally but still fairly. I'll go with \$3.50-\$4.00 range. Let me go with \$3.75 as a middle ground.

If I return \$3.75:

- They end with: $\$2.64 + \$3.75 = \$6.39$
- I end with: $\$7.08 - \$3.75 = \$3.33$

Both make modest profit. About 53% return rate, which is consistent with my previous behavior.

usage:

```
{
  "input_tokens": 2255,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.36, it became \$7.08, and you returned \$3.75. Your visible payoff was \$3.33.

Your total visible earnings across all rounds are \$58.0.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with this current co-player:

- Round 9: They sent me \$2.36, I returned \$3.75

They showed moderate trust by sending about half their endowment. I reciprocated fairly. Now it's my turn to send.

Given their moderate level of trust and my fair reciprocation, I'll send a solid amount to maintain cooperation while being somewhat proportional to what they sent me.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history with this current co-player:

- Round 9: They sent me \$2.36, I returned \$3.75

They sent a moderate amount (\$2.36, about 47% of their endowment), and I reciprocated fairly by returning \$3.75 (about 53% of what I received).

Now it's my turn to be the sender with this same co-player. They showed moderate trust, and I reciprocated appropriately.

Given that:

1. They sent me a moderate amount (not super high like some others)
2. I reciprocated fairly
3. My overall strategy has been to send high amounts (usually \$3-4) to build cooperation

I should probably send a moderate-to-high amount. Not as high as my usual \$4, but perhaps \$3-3.5 to match their level of trust shown to me.

Actually, looking at my overall pattern:

- I've been sending \$3-4 in most rounds
- Most co-players have reciprocated well
- This co-player showed moderate trust with \$2.36

I'll send \$3 to be cooperative but not overly trusting given their moderate send to me.

Usage: {

```
"input_tokens": 2230,
"output_tokens": 385,
```

```
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:28:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.14, it became \$6.42, and you returned \$3.2. Your visible payoff was \$3.2199999999999998.

Your total visible earnings across all rounds are \$14.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- When I sent \$3.00 in Round 2, I received a fair return of \$3.93
- I've been playing cooperatively and seeing reasonable reciprocation

I'll continue with a cooperative strategy and send a generous amount to this new co-player

{'send': 3.00}

reasoning:

Looking at my history:

- Round 1: As receiver, got \$10.35, returned \$5.00, earned \$5.35
- Round 2: As sender, sent \$3.00, got back \$3.93, earned \$5.93
- Round 3: As receiver, got \$6.42, returned \$3.20, earned \$3.22

In Round 2, I sent \$3.00 (60%) and my co-player returned \$3.93 out of \$9.00 (43.7%), which was decent reciprocation.

Now I'm facing a new co-player in Round 4. I've been cooperative and generally seeing reasonable reciprocation. I should continue with a cooperative strategy.

I'll send around \$3.00 again to maintain consistency and signal trust.

usage:

```
{
  "input_tokens": 1348,
  "output_tokens": 269,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.24 to you. Your visible payoff was \$8.24.

Your total visible earnings across all rounds are \$22.740000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent \$4.44 (88.8% of their endowment) - this is exceptional trust!
- I received \$13.32 after the 3x multiplication
- Total resources created: \$13.88

This sender showed tremendous cooperation. I should reciprocate strongly:

If I return \$6.50:
- Sender ends with: $\$5 - \$4.44 + \$6.50 = \7.06
- I end with: $\$13.32 - \$6.50 = \$6.82$
- This rewards their high trust with balanced outcomes

{'return': 6.50}

reasoning:

The sender sent \$4.44, which is 88.8% of their endowment - this is very high trust! I received \$13.32 after multiplication.

Total resources created: $\$5 - \$4.44 + \$13.32 = \13.88

The sender showed exceptional trust. I should reciprocate generously.

If I return \$6.50:
- Sender ends with: $\$5 - \$4.44 + \$6.50 = \7.06
- I end with: $\$13.32 - \$6.50 = \$6.82$
- This creates a fairly balanced outcome and rewards their high trust

If I return \$7.00:
- Sender ends with: $\$5 - \$4.44 + \$7.00 = \7.56
- I end with: $\$13.32 - \$7.00 = \$6.32$
- This is also reasonable

Let me go with \$6.50 as it provides good reciprocation while maintaining balance.

usage:

```
{
  "input_tokens": 1564,
  "output_tokens": 392,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.44, it became \$13.32, and you returned \$6.5. Your visible payoff was \$6.82.

Your total visible earnings across all rounds are \$29.560000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience:
- When I sent \$3.00, I've seen returns of \$3.93 and \$6.24
- The game has shown good cooperation overall
- I want to maintain a cooperative strategy

I'll continue with a consistent, generous approach.

{'send': 3.00}

reasoning:

Looking at my track record:
- Round 2: I sent \$3.00, received back \$3.93
- Round 4: I sent \$3.00, received back \$6.24 (very generous!)
- Round 5: Partner sent me \$4.44 (very trusting), I returned \$6.50

The cooperation has been generally good, with some partners being very generous. I've been sending \$3.00 consistently and seeing good returns. I could continue with that strategy, or perhaps be slightly more generous given the positive experiences.

Let me continue with \$3.00 as it's been working well and represents 60% of my endowment - a reasonable cooperative level.

usage:

```
{
  "input_tokens": 1837,
  "output_tokens": 233,
  "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.16 to you. Your visible payoff was \$7.16.

Your total visible earnings across all rounds are \$36.72.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46

This sender showed exceptional cooperation. I should reciprocate strongly:

If I return \$7.00:

- Sender ends with: \$5 - \$4.73 + \$7.00 = \$7.27
- I end with: \$14.19 - \$7.00 = \$7.19
- This creates very balanced outcomes rewarding their exceptional trust

{'return': 7.00}

reasoning:

Looking at this situation:

- The sender sent \$4.73 (94.6% of their endowment) - this is extremely high trust!
- I received \$14.19 after the 3x multiplication
- Total resources created: \$14.46 (\$0.27 remaining with sender + \$14.19 with me)

This sender showed exceptional cooperation - sending almost everything! I should reciprocate strongly.

If I return \$7.00:

- Sender ends with: \$5 - \$4.73 + \$7.00 = \$7.27
- I end with: \$14.19 - \$7.00 = \$7.19
- This creates very balanced outcomes and rewards their exceptional trust

Let me think about this... they sent 94.6%, which is really high. I want to make sure they come out well.

Actually, let me aim for balanced final outcomes:

If I return \$7.00:

- Sender: \$7.27
- Me: \$7.19

Very balanced!

I think \$7.00 is appropriate.

usage:

```
{
  "input_tokens": 2050,
  "output_tokens": 405,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$7.0. Your visible payoff was \$7.190000000000001.

Your total visible earnings across all rounds are \$43.91.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my consistent strategy:

- I've been sending \$3.00 as a sender and have received good returns each time
- The game has shown strong cooperation overall
- I'll maintain my cooperative approach

{ 'send': 3.00 }

reasoning:

Looking at my track record as a sender:

- Round 2: Sent \$3.00, received \$3.93 back
- Round 4: Sent \$3.00, received \$6.24 back
- Round 6: Sent \$3.00, received \$5.16 back

So sending \$3.00 has consistently resulted in good returns.

As a receiver:

- Round 3: Received \$6.42, returned \$3.20
- Round 5: Received \$13.32, returned \$6.50
- Round 7: Received \$14.19, returned \$7.00

I've been playing cooperatively throughout and the game has shown good reciprocity. This is Round 8, which might be the final round or close to it given typical experimental designs.

I should maintain my cooperative strategy. Sending \$3.00 has worked well for me consistently.

{ 'send': 3.00 }

usage:

```
{
  "input_tokens": 2086,
  "output_tokens": 277,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.84 to you. Your visible payoff was \$6.84.

Your total visible earnings across all rounds are \$50.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience:

- I've consistently sent \$3.00 as sender across multiple rounds
- Returns have varied but have been reasonable overall
- I've maintained a cooperative strategy throughout

I'll continue with my established approach.

{ 'send': 3.00 }

reasoning:

Looking at my history:

- Round 2 (Sender): Sent \$3.00, received \$3.93 back, payoff \$5.93
- Round 4 (Sender): Sent \$3.00, received \$6.24 back, payoff \$8.24
- Round 6 (Sender): Sent \$3.00, received \$5.16 back, payoff \$7.16
- Round 8 (Sender): Sent \$3.00, received \$4.84 back, payoff \$6.84

My sending strategy has been consistent at \$3.00. The returns have varied but have generally been reasonable (ranging from \$3.93 to \$6.24, with an average around \$5.04).

As a receiver:

- Round 3: Received \$6.42, returned \$3.20

- Round 5: Received \$13.32, returned \$6.50
- Round 7: Received \$14.19, returned \$7.00

I've been cooperative throughout. I should continue with my consistent \$3.00 sending strategy.

usage:

```
{
  "input_tokens": 2039,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.31 to you. Your visible payoff was \$5.3100000000000005.

Your total visible earnings across all rounds are \$56.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- This is the same co-player from the previous round
- They sent \$2.38 (47.6% of their endowment) - moderate trust
- I received \$7.14 after the 3x multiplication
- Last round they returned \$3.31 to me from \$9.00 (about 37%)

I should reciprocate fairly with a reasonable return:

If I return \$3.50:

- Sender ends with: \$5 - \$2.38 + \$3.50 = \$6.12
- I end with: \$7.14 - \$3.50 = \$3.64

```
{'return': 3.50}
```

Reasoning:

Looking at the situation:

- This is the same co-player from Round 9
- In Round 9: I sent \$3.00, they returned \$3.31
- Now in Round 10: They sent \$2.38 (47.6% of their endowment)
- I received \$7.14 after 3x multiplication

The sender sent a moderate amount. Their return to me in the previous round (\$3.31 from \$9.00) was about 37%, which is reasonable but not exceptional.

Now they've sent \$2.38, which is moderate trust. I should reciprocate fairly:

If I return \$3.50:

- Sender ends with: \$5 - \$2.38 + \$3.50 = \$6.12
- I end with: \$7.14 - \$3.50 = \$3.64
- This gives them a decent return

If I return \$3.00:

- Sender ends with: \$5 - \$2.38 + \$3.00 = \$5.62
- I end with: \$7.14 - \$3.00 = \$4.14
- This is more balanced

Given they sent about 48% of their endowment and returned about 37% to me last round, I should return a fair amount. Around \$3.50 seems reasonable to reward their trust while maintaining balance.

Usage: {

```
"input_tokens": 1954,
"output_tokens": 486,
"reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-08-12T09:28:30
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_6"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

- ll have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.98 to you. Your visible payoff was \$8.98.

Your total visible earnings across all rounds are \$19.95.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****Round 3 context:**** I sent \$4.00 and received a very generous return of \$7.98 (66.5% of the \$12 they received)

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - showing exceptional trust!
- I received \$12.63
- They only kept \$0.79

This co-player demonstrated even MORE trust than I showed in Round 3. I need to reciprocate appropriately:

Returning about 63% of what I received would be \$8.00:

- They'd end up with \$0.79 + \$8.00 = \$8.79 total
- I keep \$4.63
- Both of us profit well above the initial \$5

This rewards their strong trust signal and maintains the cooperative pattern.

{'return': 8.00}

reasoning:

Let me analyze this situation:

****Round 3:**** I sent \$4.00, it became \$12.00, and they returned \$7.98 (66.5% of what they received). This was very generous! My payoff was \$8.98.

****Current situation (Round 4):****

- My co-player sent \$4.21 (84% of their endowment) - very high trust!
- I received \$12.63
- They kept only \$0.79 initially

This co-player showed exceptional trust by sending almost everything. I need to reward this:

Let me think about fair returns:

- They sent \$4.21, keeping \$0.79
- If I return around 60-65% like previous rounds, that would be \$7.58-\$8.21
- But they sent MORE than me (84% vs my 80% in the last round)
- The previous round player returned 66.5% when I sent \$4

Given their high trust level (sending \$4.21, which is 84% of their endowment), I should return generously.

If I return \$8.00:

- They end up with \$0.79 + \$8.00 = \$8.79
- I keep \$12.63 - \$8.00 = \$4.63
- This gives them more than their original \$5, rewarding their trust

Actually, let me think about this more carefully. They sent \$4.21, so they have \$0.79 left. If I return \$8.00, they get \$8.79 total.

In Round 3, I sent \$4.00 and got back \$7.98, giving me \$8.98 total.

They sent \$4.21, slightly more than me.

I should return at least as much proportionally. Let me return about 63-65% of what I received.

...

usage:

```
{
  "input_tokens": 1419,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.21, it became \$12.629999999999999, and you returned \$8.0. Your visible payoff was \$4.629999999999999.

Your total visible earnings across all rounds are \$24.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history:

****Pattern as SENDER:**** I've sent \$4 in both previous sender rounds (R1 and R3), receiving generous returns (\$6.95 and \$7.98)

****Pattern as RECEIVER:**** I've returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 5):****

- New co-player (different from previous rounds)
- I'm the sender with \$5

My consistent strategy of sending \$4 has worked well:

- It signals strong cooperation
- Creates substantial mutual benefit potential (\$12 after 3x)
- Has been rewarded with good returns

I'll maintain this approach with my new partner.

```
{'send': 4}
```

reasoning:

Looking at my history:

****Round 1:**** I sent \$4, received \$6.95 back ? Payoff \$7.95

****Round 2:**** Received \$7.02, returned \$4.00 ? Payoff \$3.02

****Round 3:**** I sent \$4, received \$7.98 back ? Payoff \$8.98

****Round 4:**** Received \$12.63, returned \$8.00 ? Payoff \$4.63

****Current situation (Round 5):****

- New co-player (no history with this specific agent)
- I'm the sender again

My pattern has been:

- Send \$4 consistently when I'm the sender
- Return fairly (57-63% of what I receive) when I'm the receiver
- This has worked well, with generous returns when I've been the sender

The strategy seems successful. I should continue sending \$4 to:

1. Maintain consistency
2. Signal cooperation from the start
3. Create opportunities for mutual benefit
4. The high amounts have been reciprocated well so far

I'll send \$4 again.

usage:

```
{
```

```
"input_tokens": 1757,  
"output_tokens": 427,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.84 to you. Your visible payoff was \$7.84.

Your total visible earnings across all rounds are \$32.42.

This round, you are the RECEIVER against your current co-player. They sent you \$2.87, so you received \$8.61. How much do you return? (0-8.61)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% return rate)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment) - moderate trust
- I received \$8.61
- They kept \$2.13

****My return pattern as RECEIVER:****

- Round 2: 57% return rate
- Round 4: 63% return rate

My co-player showed moderate but meaningful trust by sending 57% of their endowment. I'll return about 58% of what I received:

Returning \$5.00:

- They'd end with \$2.13 + \$5.00 = \$7.13 (good profit)
- I keep \$3.61 (also profitable)
- Maintains my cooperative pattern

{'return': 5.00}

reasoning:

Let me analyze the situation:

****Round 5 context:**** I sent \$4.00 and received \$6.84 back (57% of the \$12 they received)

****Current situation (Round 6):****

- My co-player sent \$2.87 (57% of their endowment)
- I received \$8.61
- They kept \$2.13

****History of my returns as RECEIVER:****

- Round 2: Returned \$4.00 out of \$7.02 (57%)
- Round 4: Returned \$8.00 out of \$12.63 (63%)

My co-player sent about 57% of their endowment, which is a moderate level of trust - less than the very high trust shown in Round 4 (84%) but still meaningful.

If I apply a similar return rate of around 55-60% of what I received:

- 55% of \$8.61 = \$4.74
- 60% of \$8.61 = \$5.17

Let me consider what makes sense:

- Returning \$5.00 would give them \$2.13 + \$5.00 = \$7.13 total (better than their initial \$5)
- I'd keep \$3.61 (less than \$5 but still profitable)
- This represents 58% return rate, consistent with my previous behavior

I think \$5.00 is a good, fair return that:

1. Rewards their trust
2. Gives them a meaningful profit

3. Maintains my cooperative reputation
4. Is consistent with my 55-60% return pattern

usage:

```
{  
  "input_tokens": 2055,  
  "output_tokens": 596,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.87, it became \$8.61, and you returned \$5.0. Your visible payoff was \$3.6099999999999994.

Your total visible earnings across all rounds are \$36.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my history:

****As SENDER (Rounds 1, 3, 5):**** Consistently sent \$4, receiving returns ranging from \$6.84 to \$7.98

****As RECEIVER (Rounds 2, 4, 6):**** Returned 57-63% of what I received, maintaining reciprocity

****Current situation (Round 7):****

- New co-player
- I'm the sender with \$5

My consistent strategy of sending \$4 has:

- Established a cooperative reputation
- Generated good returns across all sender rounds
- Created mutual benefit

I'll continue this proven approach with my new partner.

{'send': 4}

reasoning:

Looking at my history:

As SENDER:

- Round 1: sent \$4, received \$6.95 back
- Round 3: sent \$4, received \$7.98 back
- Round 5: sent \$4, received \$6.84 back

As RECEIVER:

- Round 2: received \$7.02, returned \$4.00
- Round 4: received \$12.63, returned \$8.00
- Round 6: received \$8.61, returned \$5.00

My strategy has been consistent - sending \$4 as sender, which has worked well. I've been getting good returns ranging from \$6.84 to \$7.98.

This is Round 7, I'm the sender again with a new co-player.

I should maintain my consistent cooperative strategy of sending \$4. This has been my standard approach and it signals trust and cooperation.

usage:

```
{  
  "input_tokens": 2393,  
  "output_tokens": 364,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$7.55 to you. Your visible payoff was \$8.55.

Your total visible earnings across all rounds are \$44.58.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****Round 7 context:**** I sent \$4.00 and received \$7.55 back (63% return rate)

****Current situation (Round 8):****

- My co-player sent \$2.81 (56% of their endowment) - showing moderate trust
- I received \$8.43
- They kept \$2.19

****My return pattern as RECEIVER:****

- Round 2: 57% return rate
- Round 4: 63% return rate
- Round 6: 58% return rate

My co-player showed reasonable trust. I'll return about 59% of what I received:

Returning \$5.00:

- They'd end with \$2.19 + \$5.00 = \$7.19 (profitable)
- I keep \$3.43 (also profitable)
- Maintains cooperative reciprocity

{'return': 5.00}

reasoning:

Looking at this situation:

****Round 7 context:**** I sent \$4.00 and received \$7.55 back (63% return rate)

****Current situation (Round 8):****

- My co-player sent \$2.81 (56% of their endowment) - moderate trust
- I received \$8.43
- They kept \$2.19

****My return pattern as RECEIVER:****

- Round 2: Returned \$4.00 from \$7.02 (57%)
- Round 4: Returned \$8.00 from \$12.63 (63%)
- Round 6: Returned \$5.00 from \$8.61 (58%)

So I've been returning between 57-63% of what I receive.

My co-player sent \$2.81, showing reasonable trust. I should return a similar percentage.

Returning about 58% of \$8.43 would be around \$4.90.

Let me calculate:

- They'd end with \$2.19 + \$4.90 = \$7.09 (good profit)
- I keep \$3.53 (also profitable)
- Maintains my cooperative pattern

I'll return \$5.00 to make it a round number and maintain consistency.

usage:

```
{
  "input_tokens": 2505,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$5.0. Your visible payoff was \$3.4299999999999997.

Your total visible earnings across all rounds are \$48.01.

This round, you are the RECEIVER against your current co-player. They sent you \$3.71, so you received \$11.129999999999999. How much do you return? (0-11.129999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****Current situation (Round 9):****

- My co-player sent \$3.71 (74% of their endowment) - showing strong trust!
- I received \$11.13
- They kept only \$1.29

****My return pattern as RECEIVER:****

- Round 2: 57% return
- Round 4: 63% return
- Round 6: 58% return
- Round 8: 59% return

This co-player demonstrated higher trust than most (74% vs typical 56-57%). I should reward this appropriately.

Returning \$7.00 (63% of what I received):

- They'd end with \$1.29 + \$7.00 = \$8.29 (strong profit)
- I keep \$4.13 (also profitable)
- Rewards their strong trust signal

{'return': 7.00}

reasoning:

Looking at this situation:

****Current situation (Round 9):****

- My co-player sent \$3.71 (74% of their endowment) - showing high trust!
- I received \$11.13
- They kept \$1.29

This is a higher trust signal than my last few receivers. Let me look at my pattern:

****My return pattern as RECEIVER:****

- Round 2: Received \$7.02, returned \$4.00 (57%)
- Round 4: Received \$12.63, returned \$8.00 (63%)
- Round 6: Received \$8.61, returned \$5.00 (58%)
- Round 8: Received \$8.43, returned \$5.00 (59%)

The co-player sent 74% of their endowment, which is the highest percentage I've seen from anyone sending to me. This deserves a good return.

If I return about 63-65% of what I received:

- 63% would be about \$7.00
- 65% would be about \$7.24

Returning \$7.00:

- They'd end with \$1.29 + \$7.00 = \$8.29 (good profit)
- I keep \$4.13 (also profitable)

Returning \$7.20:

- They'd end with \$1.29 + \$7.20 = \$8.49 (better profit)
- I keep \$3.93 (still profitable)

Given they showed strong trust (74% of endowment), I should reward that with a good return. I'll go with \$7.00, which is about 63% and rewards their trust appropriately.

Actually, let me reconsider. They sent \$3.71 out of \$5, keeping only \$1.29. That's a very strong trust signal. I should be generous here.

...

usage:

```
{
  "input_tokens": 2544,
  "output_tokens": 746,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.71, it became \$11.129999999999999, and you returned \$7.0. Your visible payoff was \$4.129999999999999.

Your total visible earnings across all rounds are \$52.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

****Round 9 recap:**** My current co-player sent me \$3.71 (strong trust signal - 74% of endowment), and I returned \$7.00 (63% return rate)

****Current situation (Round 10):****

- Same co-player, now I'm the sender
- They demonstrated trustworthiness and cooperation in Round 9
- I reciprocated fairly

****My sending pattern:****

- Rounds 1, 3, 5, 7: Consistently sent \$4, receiving good returns

Since we've established a positive cooperative relationship in Round 9, and they showed strong trust, I'll continue my proven strategy of sending \$4.

{'send': 4}

Reasoning:

This is interesting - I'm playing with the SAME co-player from Round 9! They sent me \$3.71 and I returned \$7.00.

Let me think about this:

- Round 9: They sent \$3.71 (showing strong trust)
- I returned \$7.00 (63% return, rewarding their trust)
- Now Round 10: I'm the sender, they're the receiver

This is a continuation with the same partner. They demonstrated strong trust by sending \$3.71 (74% of their endowment). I reciprocated fairly with a \$7.00 return.

Now it's my turn to send. Given:

1. They showed strong trust in Round 9
2. I responded cooperatively with a good return
3. This establishes a positive cooperative relationship

I should continue the cooperation by sending my usual \$4, which signals strong trust and continues the mutual benefit pattern.

My consistency has been:

- As SENDER: Always sent \$4 (Rounds 1, 3, 5, 7)
- All received good returns (\$6.84-\$7.98)

I should maintain this with my current co-player who has already shown they're trustworthy.

Usage: {

```
"input_tokens": 2576,
"output_tokens": 450,
"reasoning_tokens": 0
```

}

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:28:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.68 to you. Your visible payoff was \$6.68.

Your total visible earnings across all rounds are \$14.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

****Analysis:****

My co-player showed very high trust by sending \$4.14 (83% of their endowment). This deserves generous reciprocation.

****Considerations:****

- They took significant risk by sending almost their entire endowment
- I received \$12.42, creating substantial value
- I want to reward this trust to encourage cooperation

****Calculation:****

If I return \$6.80:

- Sender ends with: \$5 - \$4.14 + \$6.80 = \$7.66 (gain of \$2.66)
- I keep: \$12.42 - \$6.80 = \$5.62 (gain of \$5.62)

Both of us benefit substantially, and the sender is well-rewarded for their trust.

```
'''json
{"return": 6.80}
'''
```

reasoning:

Let me analyze this situation:

****What happened:****

- My co-player sent \$4.14 (83% of their endowment) - showing HIGH trust
- I received \$12.42 after the 3x multiplier
- This is a very generous send amount

****Considerations:****

- The sender showed significant trust by sending over 80% of their endowment
- I should reciprocate generously to reward this trust
- Both of us should benefit from the cooperation

****Calculation options:****

If I return \$6.00:

- Sender ends with: \$5 - \$4.14 + \$6.00 = \$6.86 (gain of \$1.86)
- I keep: \$12.42 - \$6.00 = \$6.42 (gain of \$6.42)

If I return \$7.00:

- Sender ends with: \$5 - \$4.14 + \$7.00 = \$7.86 (gain of \$2.86)
- I keep: \$12.42 - \$7.00 = \$5.42 (gain of \$5.42)

If I return \$6.50:

- Sender ends with: \$5 - \$4.14 + \$6.50 = \$7.36 (gain of \$2.36)
- I keep: \$12.42 - \$6.50 = \$5.92 (gain of \$5.92)

The sender showed very high trust. I want to:

1. Reward that trust generously
2. Still benefit from being the receiver
3. Encourage future cooperation

I think returning around \$6.50 is fair - the sender gets a solid gain of \$2.36, and I keep \$5.92. Both benefit significantly.

Actually, given how much they trusted me, I should consider going higher. At \$7.50, they'd

```

    gain $3.36 while...
usage:
{
  "input_tokens": 1425,
  "output_tokens": 847,
  "reasoning_tokens": 0
}
Message 4 role=user
content:
  Round 5

  In your most recent previous game round, you were the RECEIVER against another agent. You
  saw them send $4.14, it became $12.419999999999998, and you returned $6.8. Your visible
  payoff was $5.619999999999998.
  Your total visible earnings across all rounds are $20.38.

  This round, you are the RECEIVER against your current co-player. They sent you $3.63, so y
  ou received $10.89. How much do you return? (0-10.89)
  Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
  **Analysis:**

  My co-player sent $3.63 (73% of their endowment), showing substantial trust.
  I received $10.89 after the 3x multiplier.

  **Considerations:**
  - They sent a significant portion of their endowment
  - Both of us should benefit from this cooperation
  - I want to reward their trust appropriately

  **Calculation:**
  If I return $5.80:
  - Sender ends with: $5 - $3.63 + $5.80 = $7.17 (gain of $2.17)
  - I keep: $10.89 - $5.80 = $5.09 (gain of $5.09)

  Both of us gain substantially, and the sender is well-rewarded for their trust.

  '''json
  {"return": 5.80}
  '''
reasoning:
  **Analysis:**

  My co-player sent $3.63 (72.6% of their endowment), showing substantial trust.
  I received $10.89 after the 3x multiplier.

  **Considerations:**
  - They sent a good amount, demonstrating trust
  - I want to reward this appropriately
  - Both of us should benefit from the cooperation

  **Calculation:**
  If I return around $5.50:
  - Sender ends with: $5 - $3.63 + $5.50 = $6.87 (gain of $1.87)
  - I keep: $10.89 - $5.50 = $5.39 (gain of $5.39)

  Both benefit, sender is rewarded for trust.

  Let me think about a slightly different amount. The sender sent 72.6% of their endowment,
  which is quite high. I should return enough that they benefit well.

  If I return $5.80:
  - Sender ends with: $5 - $3.63 + $5.80 = $7.17 (gain of $2.17)
  - I keep: $10.89 - $5.80 = $5.09 (gain of $5.09)

  This seems fair - both of us gain substantially, and the sender is well-rewarded for their
  high trust.
usage:
{

```

```
"input_tokens": 1758,
"output_tokens": 514,
"reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.63, it became \$10.89, and you returned \$5.8. Your visible payoff was \$5.090000000000001.

Your total visible earnings across all rounds are \$25.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

****Analysis:****

I'm facing a new co-player as sender. My previous sender experiences:

- Round 1: Sent \$3, received \$2.95 back (payoff \$4.95)
- Round 3: Sent \$3, received \$4.68 back (payoff \$6.68)

Both times, sending \$3 led to positive outcomes with reasonable reciprocation.

****Strategy:****

- \$3 creates \$9 for the receiver via the 3x multiplier
- This amount signals willingness to cooperate
- It balances trust-building with prudent risk management
- Consistency may help establish a cooperative reputation

I'll continue with my established approach.

```
'''json
{"send": 3}
'''
```

reasoning:

****Analysis:****

I'm the sender this round with a new co-player. Let me review my experience so far:

Sender rounds:

- Round 1: Sent \$3, got back \$2.95 (payoff \$4.95)
- Round 3: Sent \$3, got back \$4.68 (payoff \$6.68)

Receiver rounds:

- Round 2: Received \$6.33 from \$2.11 sent, returned \$3.20 (payoff \$3.13)
- Round 4: Received \$12.42 from \$4.14 sent, returned \$6.80 (payoff \$5.62)
- Round 5: Received \$10.89 from \$3.63 sent, returned \$5.80 (payoff \$5.09)

As sender, I've consistently sent \$3 and received reasonable returns. My co-players as receivers have varied in how much they sent initially.

This is round 6, so I'm establishing patterns. I want to continue building cooperative norms. Sending \$3 has worked well for me in the past - it's enough to create value through the 3x multiplier while not being overly risky.

I'll stick with my strategy of sending \$3.

usage:

```
{
  "input_tokens": 2077,
  "output_tokens": 470,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-playe

r. You sent \$3.0, it became \$9.0, and you saw them return \$6.95 to you. Your visible pay off was \$8.95.
Your total visible earnings across all rounds are \$34.42.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

****Analysis:****

This is the same co-player from Round 6! They demonstrated exceptional generosity by returning \$6.95 on my \$3 send.

****What happened last round:****

- I sent \$3, they received \$9
- They returned \$6.95, keeping only \$2.05 for themselves
- I gained \$3.95 while they gained just \$2.05

****Response:****

Such generous reciprocation deserves increased trust. They clearly value cooperation. Since I'm playing with the same partner again, I should signal that I recognize and appreciate their trustworthiness.

I'll increase my send to \$4, which creates \$12 for them and shows I'm willing to escalate cooperation with a proven trustworthy partner.

```
'''json
{"send": 4}
'''
```

reasoning:

This is very interesting - I'm playing against the SAME co-player as the previous round! They returned \$6.95 on my \$3 send, which is extremely generous (they gave me back more than double what I sent, even accounting for the 3x multiplier).

Let me calculate what happened:

- I sent \$3, they received \$9
- They returned \$6.95
- I ended with: \$5 - \$3 + \$6.95 = \$8.95
- They kept: \$9 - \$6.95 = \$2.05

So they only kept \$2.05 while I gained \$3.95! This is a very cooperative partner.

Since I'm playing with the same person again and they were so generous, I should reciprocate this trust by sending more. This could establish a strong cooperative relationship.

Options:

1. Send \$3 again - safe, but doesn't fully reciprocate their generosity
2. Send \$4 - shows increased trust based on their behavior
3. Send \$5 - maximum trust, but risky

Given their extremely generous return of \$6.95, I think sending \$4 is appropriate. It shows I valued their cooperation and am willing to increase my trust, while still maintaining some prudence.

usage:

```
{
  "input_tokens": 2371,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$6.79 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$42.21.

This round, you are the RECEIVER against your current co-player. They sent you \$4.62, so you

ou received \$13.86. How much do you return? (0-13.86)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

****Analysis:****

This is the same co-player from rounds 6 and 7! Our partnership history:

- Round 6: I sent \$3 ? they returned \$6.95 (extremely generous)
- Round 7: I sent \$4 ? they returned \$6.79 (still very generous)
- Round 8 (now): They sent \$4.62 (92% of their endowment!)

****Response:****

This player has demonstrated extraordinary trust and generosity. They're now sending me almost everything they have. I need to reciprocate at a similarly high level.

****Calculation:****

If I return \$8.50:

- Sender ends with: \$5 - \$4.62 + \$8.50 = \$8.88 (gain of \$3.88)
- I keep: \$13.86 - \$8.50 = \$5.36 (gain of \$5.36)

Both of us benefit substantially, maintaining our strong cooperative relationship.

```
'''json
{"return": 8.50}
'''
```

reasoning:

Let me analyze this carefully:

****Context:****

- This is the same co-player from rounds 6 and 7
- Round 6: I sent \$3, they returned \$6.95 (very generous)
- Round 7: I sent \$4, they returned \$6.79 (still very generous)
- Now Round 8: They sent me \$4.62 (92% of their endowment!)

****Analysis:****

This co-player has been extremely generous and trustworthy:

- They returned \$6.95 on receiving \$9 (keeping only \$2.05)
- They returned \$6.79 on receiving \$12 (keeping only \$5.21)
- Now they're sending me \$4.62 out of \$5

They're showing incredible trust in me. I need to reciprocate appropriately.

****Calculation:****

I received \$13.86 after the 3x multiplier.

Let me think about what would be fair:

- They sent \$4.62, showing maximum trust
- If I return \$7.50:
 - Sender ends with: \$5 - \$4.62 + \$7.50 = \$7.88 (gain of \$2.88)
 - I keep: \$13.86 - \$7.50 = \$6.36 (gain of \$6.36)

Actually, given how generous they've been to me, I should be similarly generous. Let me aim for a return that gives them a strong gain comparable to what I've received.

If I return \$8.00:

- Sender ends with: \$5 - \$4.62 + \$8.00 = \$8.38 (gain of \$3.38)
- I keep: \$13.86 - \$8.00 = \$5.86 (gain of \$5.86)

This seems fair - both of us gain substantially, and I'm rewarding their exceptional trust

.

...

usage:

```
{
  "input_tokens": 2515,
  "output_tokens": 845,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.62, it became \$13.86, and you returned \$8.5. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$47.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

****Analysis:****

This is a new co-player. Looking at my sender history:

- New partners (Rounds 1, 3, 6): Sent \$3, received reasonable returns (\$2.95, \$4.68, \$6.95)
- Established partner (Round 7): Sent \$4 after they proved trustworthy

****Strategy:****

Starting with a moderate send has been effective for establishing cooperation. \$3 creates \$9 in value while managing risk prudently.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history, I've been switching between being a sender and receiver. Let me track my partnerships:

Round 1: SENDER - sent \$3, received \$2.95, payoff \$4.95
Round 2: RECEIVER - saw them send \$2.11, received \$6.33, returned \$3.2, payoff \$3.13
Round 3: SENDER - sent \$3, received \$4.68, payoff \$6.68
Round 4: RECEIVER - saw them send \$4.14, received \$12.42, returned \$6.80, payoff \$5.62
Round 5: RECEIVER - saw them send \$3.63, received \$10.89, returned \$5.80, payoff \$5.09
Round 6: SENDER - sent \$3, received \$6.95, payoff \$8.95
Round 7: SENDER (same co-player as R6) - sent \$4, received \$6.79, payoff \$7.79
Round 8: RECEIVER (same co-player as R6-7) - saw them send \$4.62, received \$13.86, returned \$8.50, payoff \$5.36

Now Round 9: I'm a SENDER against a "current co-player" - this is a new pairing.

So I'm facing a brand new co-player. I should use my baseline strategy of sending \$3, which has worked well for establishing cooperation.

My sending strategy has been:

- With new partners: send \$3 (has worked well)
- With proven cooperative partners: escalate to \$4

Since this is a new pairing, I'll send \$3.

usage:

```
{
  "input_tokens": 2512,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$6.7 to you. Your visible payoff was \$8.7.

Your total visible earnings across all rounds are \$56.269999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.31, so you received \$12.93. How much do you return? (0-12.93)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

****Analysis:****

This is the same co-player from Round 9! Our history:

- Round 9: I sent \$3 ? they returned \$6.70 (very generous, keeping only \$2.30)
- Round 10 (now): They sent \$4.31 (86% of their endowment!)

****Response:****

They showed exceptional reciprocity last round and are now demonstrating extreme trust by sending almost everything. I need to reciprocate at a high level.

****Calculation:****

If I return \$7.80:

- Sender ends with: $\$5 - \$4.31 + \$7.80 = \8.49 (gain of \$3.49)
- I keep: $\$12.93 - \$7.80 = \$5.13$ (gain of \$5.13)

Both of us benefit substantially, and they're well-rewarded for their trust and previous generosity.

```
'''json
{"return": 7.80}
'''
```

Reasoning:

Let me analyze this situation:

****Partnership history with current co-player:****

- Round 9: I sent \$3 ? they returned \$6.70 (very generous)
- Round 10 (now): They sent \$4.31 (86% of their endowment) - showing high trust

This player has been very cooperative:

1. Last round they returned \$6.70 from the \$9 they received (keeping only \$2.30)
2. Now they're sending \$4.31, almost everything they have

****Response:****

They demonstrated strong reciprocity and are now showing extreme trust. I need to reciprocate generously.

****Calculation:****

If I return \$7.50:

- Sender ends with: $\$5 - \$4.31 + \$7.50 = \8.19 (gain of \$3.19)
- I keep: $\$12.93 - \$7.50 = \$5.43$ (gain of \$5.43)

Both benefit well, and they're rewarded for their trust and previous generosity.

Alternatively, if I return \$8.00:

- Sender ends with: $\$5 - \$4.31 + \$8.00 = \8.69 (gain of \$3.69)
- I keep: $\$12.93 - \$8.00 = \$4.93$ (gain of \$4.93)

Given they sent 86% of their endowment and previously returned generously, I should lean toward the higher amount around \$7.50-\$8.00.

I'll go with \$7.80:

- Sender ends with: $\$5 - \$4.31 + \$7.80 = \8.49 (gain of \$3.49)
- I keep: $\$12.93 - \$7.80 = \$5.13$ (gain of \$5.13)

This rewards their trust appropriately while both of us do well.

```
Usage: {
  "input_tokens": 2510,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```